

SU(2) Yang–Mills theory in four dimensions via Balaban’s renormalisation group: construction, axioms and non-triviality

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Abstract

We study four-dimensional SU(2) Yang–Mills theory with Wilson’s lattice action on periodic tori, through the smeared action density at separated points. Assuming the proof inputs declared clause by clause in Section 9, running Balaban’s renormalisation group with complex sources gives the following deductions: two-sided bounds on the running coupling and its exact one-loop law; correlation bounds uniform in the cutoff, and in the volume for test functions supported in balls; continuum limits on tori and on $\mathbb{R}^4$, the latter with Osterwalder–Schrader regularity, reflection positivity, symmetry, and translation and hypercubic invariance, hence a Hilbert space with a vacuum and a non-negative Hamiltonian; and non-triviality uniformly in the volume: at separation $\mathbf{d}$ the connected two-point function equals $\frac{3}{2}\zeta'^2 g(\mathbf{d})^4 \mathcal{K}$ up to a relative error at most one half, with $\zeta' \in [\frac{2}{3}, \frac{4}{3}]$, $g(\mathbf{d})$ the running coupling at the scale $\mathbf{d}$ and $\mathcal{K}$ an explicit free-field kernel, and two-sided bounds of this form hold in every first-passage limit on tori and in every limit on $\mathbb{R}^4$. On each torus the limits form a real-analytic one-parameter family labelled by the renormalised coupling; a displayed hypothesis, the vanishing of the flux of the lattice rotation current through far spheres, gives O(4) invariance. Without that hypothesis, rotation invariance holds at short distance to all orders in the running coupling: for two and three points, rotating a configuration of size $\mathbf{d}$ changes the connected correlation functions by at most a constant times $\mathbf{d}^2$, respectively $\mathbf{d}$, times the product of the suprema of the test functions, uniformly in the cutoff and the volume and with no long-distance input (for two points this is a relative error $O(\mathbf{d}^2 \log^2(1/\mathbf{d}))$, below $e^{-c/g(\mathbf{d})^2}$ for some $c > 0$), while exact invariance fails at every scale on every periodic torus. The statements up to non-triviality, except the convergence without subsequences on a fixed torus, are complete deductions from Balaban’s theorems as cited (those of his statements that we use as he asserts them are listed in Section 9.1) and from a companion account described in Section 9; the statements on the one-parameter family and on O(4) invariance, and the short-distance rotation statement, also rest on supporting statements of ours that we do not claim as proved, two of them not yet written, together with one clause to be added to a definition, listed in Sections 7 and 9.

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1 Introduction, setting, hypotheses and statement

1.1 Introduction

The theory and the observable. This paper concerns pure Yang–Mills theory in four Euclidean dimensions with gauge group $SU(2)$, regularised by Wilson’s lattice action on a periodic torus, and the continuum limit of its correlation functions. We work in the renormalisation-group framework of Balaban [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11], which describes the effective theory at every scale between the lattice spacing and a fixed physical length, with bounds uniform in the lattice spacing, as long as the running coupling stays small.

For a non-abelian gauge group the field strength $F_{\mu\nu}$ is not gauge invariant, so a statement of Osterwalder–Schrader type [21, 22] about the curvature has to be a statement about gauge-invariant local functions of it. We treat one of them, the Yang–Mills action density $\text{tr } F_{\mu\nu} F_{\mu\nu}$, in its lattice form: for a real function f on the torus,

$$\mathcal{O}(f) = \sum_p f(x(p)) [1 - \text{Re tr } U(\partial p)],$$

the sum running over all plaquettes p of the lattice, $x(p)$ the barycentre of p (Subsection 1.2). The action density has canonical dimension four. Its two-point function is of order $|x - y|^{-8}$ at short distances, up to logarithmic factors (clause (iv) of Theorem 0 makes this precise), and is not locally integrable. All correlation functions in this paper are therefore evaluated on test functions with pairwise disjoint supports, and nothing is asserted at coinciding points.

What is constructed. Theorem 0 is stated under the hypotheses (H1)–(H7) of Subsection 1.3: one existential lower bound $L_0(N)$ on the blocking factor L , never evaluated; Balaban’s auxiliary parameters, chosen in one fixed order, every condition on them one-sided or a window that earlier one-sided conditions make non-empty; a lower bound on the volume; and small couplings chosen last. The parameter hypotheses (H1)–(H7) impose no dynamical property—no bound on the behaviour of the running coupling, no convergence property, no clustering, no mass gap. The technical proof inputs of Section 9 are additionally assumed in the conditional theorem below. Clause (vi) has the further displayed hypothesis, the vanishing of the flux of the lattice rotation current through far

spheres (implied by a decay of stress-tensor correlations), which is a hypothesis of clause (vi) only. In brief:

- (0) the running coupling exists at every scale between the cutoff and the reference length, and its inverse square obeys two-sided linear flow bounds, uniformly in the cutoff; for $N = 2$ and small coupling it obeys the exact one-loop law, with an error that is $o(s)$ over s steps;
- (i) Balaban's ultraviolet stability theorem [11, Theorem 1] holds, its one hypothesis being supplied by (0);
- (iii) the connected correlation functions of $n \geq 2$ smeared action densities with separated supports are bounded uniformly in the cutoff and, for test functions supported in balls, uniformly in the volume and the position;
- (ii) they converge as the cutoff is removed—on each torus without passing to subsequences along bare couplings tuned at the reference length, on $\mathbb{R}^4$ along subsequences of cutoffs and volumes—to limits that satisfy, in the numbering of [21], the regularity condition E0 in a stated form, the invariance E1 under translations and the hypercubic group, the reflection positivity E2 and the symmetry E3, whence a Hilbert space with a vacuum vector, a non-negative Hamiltonian and spatial momenta;
- (iv) the connected two-point function at separation d equals $\zeta'^2(N^2 - 1)/2$ times an explicit positive free kernel times the fourth power of the running coupling at the scale d , up to a relative error at most one half, where $\zeta' \in [\frac{2}{3}, \frac{4}{3}]$ is the derivative of the running shift of Theorem 0(iii)(c) at that scale and tends to 1 with the coupling; this holds uniformly in the cutoff, the bare coupling, the volume and the position, and the resulting two-sided bounds pass to every first-passage limit on tori and to every limit on $\mathbb{R}^4$;
- (v) on each torus the continuum limits obtained from admissible bare couplings form one real-analytic one-parameter family, the parameter being the renormalised coupling at the reference length;
- (vi) if the flux of the lattice rotation (angular-momentum) current through far spheres vanishes along one sequence of radii, uniformly in the cutoff—as it does, for instance, when one far-away insertion of the stress tensor decorrelates from the action densities faster than the inverse fourth power of the distance—then every limit on $\mathbb{R}^4$ taken along cutoffs that grow fast enough compared with the volume is invariant under the full rotation group $O(4)$.

All clauses are stated for $SU(2)$; clauses (0)(a)–(c) and (i) are written for $SU(N)$, every $N \geq 2$, in the companion account, as described in Remark 1.4.

In what sense the limit is not trivial. For an elementary field, the non-vanishing of the truncated four-point function shows that a theory is not a generalised free field, because a generalised free field has vanishing truncated functions beyond the second. For a composite field such as the action density this test is empty. For free gluons the action density is the Wick square of a Gaussian field, and its truncated functions of every even order are non-zero at non-coinciding points (the odd ones vanish identically in the continuum; on the lattice, and at overlapping supports, every order is non-zero). At four points this free term is the one-loop box, which in perturbation theory is also the leading term of the Yang–Mills four-point function, with the eighth power of the running coupling in front. What distinguishes the constructed theory from every free theory, as far as these correlation functions are concerned, is that the coupling read off a gauge-invariant correlation runs. Clause (iv)

shows that the connected two-point function at separation d , divided by an explicit scale-free kernel, is strictly positive at every separation below a threshold depending on the coupling only, and is bounded above and below by fixed multiples of $(\log(1/d))^{-2}$ as $d \downarrow 0$; with clause (0)(d), the inverse square of the coupling read off this correlation has logarithmic growth between two fixed multiples of the one-loop Yang–Mills coefficient, in the $\liminf/\limsup$ sense of Remark 2.15. Consequently every first-passage limit on a torus and every limit on $\mathbb{R}^4$ is non-trivial in the following precise sense: there is a centred action-density vector with strictly positive reflected norm, so the reconstructed Hilbert space has dimension at least two; the limit is not scale invariant for any assignment of a scaling dimension to the action density; it is reproduced by no free Maxwell field at any charge; and the coupling read off the two-point function at separation d is strictly positive and tends to zero logarithmically as $d \downarrow 0$. We do not claim that the limit is interacting in any further sense, and we do not assert non-Gaussianity, which for a Wick-square composite would in any case be the wrong criterion.

What is not claimed. We do not claim uniqueness of the infinite-volume limit (the limits on $\mathbb{R}^4$ are taken along subsequences of volumes), uniqueness of the vacuum, the cluster property, a mass gap, the growth condition of [22] in the number of points (hence field operators or Wightman functions), anything at coinciding points, invariance under all rotations without the hypothesis of clause (vi), statements about Wilson loops or about Balaban’s block-averaged variables, or any statement for $SU(N)$, $N \geq 3$ (Remark 1.4). The traceless stress tensor enters only through clause (vi): its hypothesis concerns the flux, through far spheres, of the lattice rotation current, whose leading part is built from the lattice stress tensor; a sufficient condition for it concerns one insertion of the lattice stress tensor among action densities; and its proof controls such insertions through the lattice normalisation fixed in Section 8 and through the one-loop running of their coefficients, which in the traceless-diagonal channel is the running of the inverse coupling; a cutoff-uniform bound for the stress tensor in its physical normalisation is not claimed. Correlations of several stress tensors, the topological density $\text{tr } F \wedge F$, the remaining components of $F \otimes F$, and Wilson loops are not treated. Remark 1.3 gives the complete list.

Relation to Balaban’s programme. Balaban’s papers construct, for lattice gauge theories, the propagators, averaging operations, regular configurations and background fields of a block-spin renormalisation group [1, 2, 3, 4, 5, 6]; the effective actions in the small-field approximation, together with a coupling-constant renormalisation in four dimensions [7, 8]; the complete effective densities, large-field regions included [9]; and the renormalisation of the large-field expressions [10, 11]. The conclusion is the ultraviolet stability theorem [11, Theorem 1]: if the sequence of effective couplings is contained in an interval $(0, \gamma]$ with γ small, the effective densities have the form, and satisfy the bounds, described in [9, §2]. Balaban explains that this hypothesis would follow from the coupling-constant theorem [7, Theorem 2], whose perturbative proof “has not been published yet”, and he describes the analysis of expectation values of physical observables as problems which “deserve detailed analysis and further publication” [11, Introduction]. Clause (0) supplies the coupling hypothesis without that theorem: the flow is run from the bare coupling and stopped at the first scale at which the running inverse coupling falls below a prescribed value, and two-sided linear bounds on the flow are obtained along the way (Section 2), from the inputs listed in Section 9. The remaining clauses carry the construction to the correlation functions of a local gauge-invariant observable: a correlation theorem, in which Balaban’s induction is run with complex sources for the smeared action density (Section 3); volume-free bounds; continuum limits with their Osterwalder–Schrader structure; the two-point law and its transfer to all volumes; uniqueness

up to one real parameter on each torus; and rotation invariance under a hypothesis. Balaban’s block-averaged variables are used only inside the proofs; no clause of Theorem 0 concerns the laws of block variables.

Relation to other work. Wilson’s lattice action [25] and the reflection positivity of lattice gauge theories [23] (see also [24]) are the starting point of every construction of this kind. For pure Yang–Mills theory in four dimensions the ultraviolet problem was treated by Balaban in the series cited above and, in a continuum regularisation with an infrared cutoff, by Magnen, Rivasseau and Sénéor [20]; Federbush developed a phase-cell approach (from 1986 on) [16]. Dimock has given a detailed exposition of Balaban’s renormalisation-group methods for a scalar model (2013–2014) [14], and Brydges, Dimock and Hurd developed related estimates for renormalisation-group transformations (1998) [12]. In two dimensions the continuum Yang–Mills measure was constructed by Driver (1989) [15] and by Gross, King and Sengupta (1989) [18]; Chatterjee (2019) [13] surveys probabilistic approaches to lattice Yang–Mills theory. The problem of constructing quantum Yang–Mills theory in four dimensions and of proving that it has a mass gap is described by Jaffe and Witten (2006) [19]; the present paper does not address the mass gap.

What is proved where. The proof of Theorem 0 occupies Sections 2–8, in the order (0), (i), (iii), (ii), (iv), (v), (vi). Where a statement is used without proof we say exactly one of the following: that it is taken from Balaban’s papers, with the reference; that it is proved in the companion account—the detailed exposition of the construction on which this paper is based, together with the complete proofs to which it refers, whose parts are cited by keys of the form [T-] and are described in Section 9—in which case we give its statement and the structure of its proof; or that we state it and defer its proof. Section 9 lists, clause by clause, every statement of the last two kinds. For a statement of the third kind we do not claim that a complete written proof exists, here or in the companion account. Accordingly each clause of Theorem 0 is proved in this paper as an implication from the statements listed under its heading in Section 9, and from nothing else. The clauses are thus deduced from Balaban’s theorems as cited—the statements of his that we use as he asserts them, without a full proof in print, are listed with their places in Subsection 9.1—with declared readings where his text is ambiguous and two corrections of his text: the sign of a pair of boundary terms in [9, (3.15)], on p. 269 (Remark 2.2(2)), which affects no estimate, and the treatment of the field tA_k asserted on [9, p. 276] (Section 3); and from the companion account. At $N = 2$ no statement of ours on which clauses (0)–(iv) rest is deferred: each of them is proved in this paper or in the companion account, except that the convergence without subsequences in (ii-T-a), and the identification of first-passage limits in (ii-T-b), are deduced from clause (v) in Section 7 and have its status. Clause (v) is proved as an implication from the uniqueness theorem of the companion account; its supporting statements are proved in the companion account except five, which we do not claim as proved: three are written, with independent verification in progress, and two are not yet written; Sections 7 and 9 list them, together with one clause to be added to a definition. The volume-transfer argument of Section 6, which carries clause (iv) from one torus to all volumes and to $\mathbb{R}^4$, is new; its proofs are given in full, and its inputs are clauses (0) and (i), the correlation theorem together with the reflection-positivity and chessboard lemmas of Section 3, and the two-point law on one torus. Clause (vi) is in addition conditional on its displayed hypothesis, about which Theorem 0 asserts nothing. The deduction of its conclusion from that hypothesis is proved in the companion account as an implication of the same kind as clauses (iii)–(v): beneath it lie Balaban’s theorems as cited, the two-cutoff comparison theorem of clause (v) as an implication, with the supporting statements listed under (v), and one proposition on the class form of the densities, used as a reading.

Apart from the statements listed under clause (v), which it shares through the two-cutoff comparison, every statement of ours beneath (vi) is written out in the companion account.

1.2 Setting and conventions

Gauge group, torus and lattice. The gauge group is $G = \text{SU}(N)$ and the dimension is $d = 4$; we keep N in the notation, and Theorem 0 is stated for $N = 2$ (for $N \geq 3$ see Remark 1.4). Fix an odd integer L (the blocking factor), an integer m (the torus exponent) and an integer $K \geq 0$ (the number of renormalisation steps). The torus of side $2L^m$ is $\mathbb{T}_m := \mathbb{R}^4 / 2L^m \mathbb{Z}^4$, and the lattice spacing at cutoff K is $\varepsilon := L^{-K}$. The bare sites are the points of $\varepsilon(\mathbb{Z} + \frac{1}{2})^4$ modulo $2L^m \mathbb{Z}^4$. A bond is an ordered pair $b = \langle x, y \rangle$ of bare sites at distance ε ; a plaquette p is an elementary square of the lattice, ∂p its boundary, a closed path of four bonds, and $x(p) \in \mathbb{T}_m$ its barycentre. A gauge field is a map U from bonds to G with $U(\langle y, x \rangle) = U(\langle x, y \rangle)^{-1}$, and dU is the product of the normalised Haar measures over one orientation of each bond. If ∂p is traversed as $b_1 b_2 b_3 b_4$, then $U(\partial p) := U(b_1)U(b_2)U(b_3)U(b_4)$. The trace is normalised by $\text{tr } \mathbf{1} = 1$, so that $1 - \text{Re tr } U(\partial p) \in [0, 2]$ depends only on the unoriented plaquette. We write $x = (x^0, x^1, x^2, x^3)$ for the coordinates, x^0 being the Euclidean time. Logarithms are natural.

Units, levels and depths. Lengths are measured in physical units, in which the torus has side $2L^m$ and the bare spacing is $\varepsilon = L^{-K}$. For $0 \leq k \leq K$ the block lattice of level k is $L^{k-K}(\mathbb{Z} + \frac{1}{2})^4$ modulo $2L^m \mathbb{Z}^4$, of spacing L^{k-K} ; we say that it lies at depth $s := K - k$. The bare lattice lies at depth K ; the lattice of level K , of spacing 1, lies at depth 0. We call the latter the reference lattice and its spacing, the unit of length, the reference length.

Wilson action and the state at cutoff K . For a real or complex function c on $\mathbb{T}_m$ put

$$A(U) := \sum_p [1 - \text{Re tr } U(\partial p)], \quad A(c, U) := \sum_p c(x(p)) [1 - \text{Re tr } U(\partial p)], \quad (1.1)$$

the sums running over all bare plaquettes of all six orientations. For a bare coupling $g_0 > 0$ the bare density is $\exp[-g_0^{-2} A(U)] dU$, and for a bounded Borel function F of the gauge field

$$\langle F \rangle_{K,m} := \frac{\int F(U) e^{-g_0^{-2} A(U)} dU}{\int e^{-g_0^{-2} A(U)} dU};$$

the dependence on g_0 is not shown in the notation. Balaban writes the bare density as $\rho_0 = \exp[-g_0^{-2} A - E]$ with a normalisation constant E [9, §0]; the constant cancels from all expectations.

The smeared action density. For a real Lipschitz function f on $\mathbb{T}_m$ put $\mathcal{O}(f) := A(f, U)$, that is,

$$\mathcal{O}(f) = \sum_p f(x(p)) [1 - \text{Re tr } U(\partial p)], \quad (1.2)$$

the sum running over all bare plaquettes of all six orientations: the gauge-invariant action density, smeared with the weight f . A function $f \in C_c^1(\mathbb{R}^4)$ whose support lies in a ball of radius r with $2r < L^m$ is regarded as a function on $\mathbb{T}_m$ through the quotient map $\mathbb{R}^4 \rightarrow \mathbb{T}_m$. With $M = L^{m'}$ Balaban's cube size (one of the parameters of (H3)) and $\text{Lip } f$ the Lipschitz constant of f in physical units, we use the norm

$$\|f\|_{\sharp} := \max\{\|f\|_{\infty}, 40M \text{Lip } f\}, \quad (1.3)$$

which controls a test function through its supremum and one derivative only. For a radius $\varrho > 0$ the class of complex sources is $\mathcal{W} := \{w : \mathbb{T}_m \rightarrow \mathbb{C} \text{ Lipschitz} : \|w\|_{\sharp} < \varrho\}$. Since $A(c, U)$ is linear in c , $-g_0^{-2}A(U) + \mathcal{O}(w) = -A(g_0^{-2} - w, U)$: a source shifts the bare inverse coupling locally.

Schwinger functions. For real Lipschitz $f_1, \dots, f_n$ on $\mathbb{T}_m$ and $z \in \mathbb{C}^n$ put

$$Z_{K,m}(z) := \int \exp \left[-g_0^{-2}A(U) + \sum_{i=1}^n z_i \mathcal{O}(f_i) \right] dU, \quad (1.4)$$

an entire function of z with $Z_{K,m}(0) > 0$, and

$$S_{n;K,m}^T(f_1, \dots, f_n) := \partial_{z_1} \cdots \partial_{z_n} \log Z_{K,m}(z) \Big|_{z=0} = \langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_{K,m}^{\text{conn}}, \quad (1.5)$$

the connected (truncated) correlation function. It is defined at finite K for every tuple; the assertions of Theorem 0 concern $n \geq 2$ and tuples whose supports are pairwise at positive distance. The centred Schwinger functions are the moments determined by the connected ones,

$$S_q(f_1, \dots, f_q) := \sum_{\pi} \prod_{B \in \pi} S_{|B|}^T((f_i)_{i \in B}), \quad S_0 := 1, \quad S_1 := 0, \quad (1.6)$$

the sum running over the partitions π of $\{1, \dots, q\}$ into blocks of at least two elements (with the indices K, m , or ∞ , of the connected functions carried along); they are the moments of the centred fields $\mathcal{O}(f) - \langle \mathcal{O}(f) \rangle$. These are the Schwinger functions of Theorem 0.

Symmetries. W_4 is the hypercubic group, the $2^4 \cdot 4!$ signed permutations of the four coordinate axes, and $W_3 \subset W_4$ the subgroup acting on the spatial coordinates (x^1, x^2, x^3) only. The group $\mathbb{R}^4 \rtimes W_4$ of translations and hypercubic transformations acts on functions on $\mathbb{R}^4$ by $f \mapsto f \circ \tau^{-1}$, $\tau \in \mathbb{R}^4 \rtimes W_4$; the lattice translations and W_4 act on $\mathbb{T}_m$ and on the bare lattice. For a coordinate hyperplane $P = \{x^\mu = c\}$, θ_P is the reflection in P and $\theta_P f := f \circ \theta_P$. We write $(\mathbb{T}_m)_{\neq}^n$ and $(\mathbb{R}^4)_{\neq}^n$ for the sets of n -tuples of pairwise distinct points.

Running couplings and flow constants. Balaban's renormalisation group is run forward from the bare coupling:

$$x_0 := g_0^{-2}, \quad x_{k+1} := x_k - \beta_{k+1}, \quad g_k := x_k^{-1/2} \quad (x_k > 0). \quad (1.7)$$

This is the orientation $g_k^{-2} = g_{k+1}^{-2} + \beta_{k+1}$ of [7, (0.18), (0.20)]. Here β_{k+1} is the coefficient defined in [7, (1.22)] and identified by Balaban with [9, (3.61)], computed from the complete regular part produced by step $k+1$ —the term of the Gaussian normaliser and the fluctuation integral, the latter including the large-field remainders that Balaban incorporates into the effective action—summed over the localisation domains of the infinite lattice. The numbers x_k are the running inverse couplings. The flow constants are

$$c_0 := \frac{11N^2}{12\pi^2} \log L, \quad c_2 = c_2(L, N) \geq 0, \quad c_5 := 2c_2,$$

where c_2 is the constant of the one-loop lemma (the sum of the Gaussian parts of the first k increments differs from kc_0 by at most c_2 , for every k ; Section 2), together with the slopes $\lambda_{\sharp}$ and $B_{\sharp}$ of clause (0), which depend only on L , the parameters of (H3), N and γ and satisfy

$$\lambda_{\sharp} \geq \underline{\lambda} := c_0/4, \quad B_{\sharp} \leq \bar{B} := C_{\beta} + 1,$$

C_β being Balaban's per-step constant for the coupling increments (one of the parameters of (H3)). A threshold that reads the flow is chosen through the triple $(\lambda, c_5, \bar{B})$, which is fixed before γ . Further,

$$g_-(g) := (g^{-2} + B_\#)^{-1/2}, \quad K(g_0, g) := \min\{k \geq 0 : x_k \leq g^{-2} + B_\#\}, \quad (1.8)$$

$$R(g') := \min\{L^i : i \in \mathbb{N}, L^i \geq (\log g'^{-2})^r\}, \quad m_{\mathbb{T}}(g') := 11 + m' + \log_L R(g'),$$

where r is a fixed integer exponent and $M = L^{m'}$. The index $K(g_0, g)$ is the first passage of the running inverse coupling below $g^{-2} + B_\#$. Theorem 0 stops the flow there and takes $K = K(g_0, g)$ as the number of steps, so that the running coupling at the reference lattice lies in $[g_-(g), g)$ (clause (0)).

One-loop normalisation. The coupling is normalised so that its inverse square multiplies the Wilson action with the trace normalised to one: $g^{-2} = 2N/g_{\text{std}}^2$, where g_{std} is the coupling in the standard normalisation, in which the action at small spacing reduces to $(4g_{\text{std}}^2)^{-1} \int F_{\mu\nu}^a F_{\mu\nu}^a d^4x$ for generators normalised by $\text{Tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$. In standard units the inverse square of the coupling grows by $11N/(24\pi^2) = \beta_0/(8\pi^2)$ per e-fold of the scale, $\beta_0 = 11N/3$; the factor $2N$ between this coefficient and the coefficient $11N^2/(12\pi^2)$ used throughout is this normalisation. Accordingly c_0 is the one-loop growth of x_k per renormalisation step, and the one-loop running law for the inverse square of the coupling at the length d reads

$$x(d) = \frac{11N^2}{12\pi^2} \log(1/d) (1 + o(1)) \quad (d \downarrow 0);$$

clause (0)(d) proves this law for the running coupling of the construction (Section 2). The number

$$b_0 := \frac{N^2 - 1}{2},$$

half the dimension of the Lie algebra of $\text{SU}(N)$, is the leading coefficient of the two-point function of the action density in clause (iv).

Test functions for the two-point law. For $d > 0$ and a shape constant $\vartheta \in (0, 1]$, $\mathfrak{T}(d; \vartheta)$ is the set of pairs (f_1, f_2) of non-negative Lipschitz functions $f_i \not\equiv 0$ on $\mathbb{T}_m$ (or on $\mathbb{R}^4$) such that, for $i = 1, 2$, $\text{supp } f_i$ lies in a ball of radius d , $\int f_i \geq \vartheta \|f_i\|_\infty d^4$ and $\text{Lip } f_i \leq \|f_i\|_\infty / (\vartheta d)$, and such that the supports obey the separation window

$$d \leq \text{dist}(\text{supp } f_1, \text{supp } f_2) \leq 6d.$$

The window is a condition on the test functions, on the side of the conclusion; it is not a hypothesis on the theory.

Balaban's construction. The proofs use Balaban's construction, which we recall here only as far as is needed to state clause (i); Section 2 gives details and references. The renormalisation transformation $\mathbf{T}$ maps a density on the gauge fields of one level to a density on the block fields of the next level, by integrating over the fields with prescribed block averages; the averaging operation is that of [7, (0.4), (0.12)] (see [7, (0.13)] and [9, (0.1)]), and we write $\mathcal{M}$ for the averaging map from one level to the next and V_k for the block field of level k . After each transformation Balaban applies the large-field operation $\mathbf{R}$ [9, 10, 11], which changes the densities on large-field regions without changing their integrals. The effective densities are

$$\rho_0 := e^{-g_0^{-2}A-E}, \quad \rho_{k+1} := \mathbf{R}\mathbf{T}\rho_k \quad (0 \leq k < K),$$

and the objects defined are the positive measures $\rho_k dV_k$ on the block fields of level k . Balaban writes them as sums $\rho_k dV_k = \sum_h \tau_h$ of positive measures τ_h over histories h , i.e. over admissible sequences of large-field domains [9, (2.18)]. The background configuration $U_k = U_k(V_k)$ is the configuration of minimal action with block averages V_k [6]. The set $\mathcal{D}_j$ of localisation domains of level j consists of the finite connected unions of cubes of Balaban's partitions of level j [7], [8, §§1–2]; $d_j(X)$ is Balaban's size of $X \in \mathcal{D}_j$ [7], and $d_{j,Z}(X)$ his length of X relative to a large-field region Z [11, (1.67)]. The domain $U_j^c(X, \alpha_0, \alpha_1)$ of complex configurations is that of [7, (1.11)–(1.16)], with the radii α_0, α_1 of (H3), read hereditarily: Balaban's regularity conditions are imposed also on every subregion of the towers of subdomains of X used in the construction. Finally N_k is the number of sites of the lattice of level k , χ_k the indicator that no large-field region is present at level k , and κ Balaban's decay rate (one of the parameters of (H3)).

1.3 Hypotheses

Exactly the following are assumed. No two-sided condition is imposed on L , m , K , g_0 , g or a running coupling; the finitely many windows of (H2)–(H3) confine auxiliary parameters and are made non-empty by one-sided conditions on earlier choices. No dynamical property is assumed: no bound on the behaviour of the running coupling, no convergence property, no clustering, no mass gap.

- (H1) Pure SU(2) Yang–Mills theory in dimension $d = 4$, with Wilson action, on the periodic torus $\mathbb{T}_m$ of side $2L^m$ with lattice spacing L^{-K} ; we keep N in the notation ($N = 2$), and the case $N \geq 3$ is discussed in Remark 1.4.
- (H2) L is odd and $L \geq L_0(N)$, where $L_0(N)$ is one existential constant, never evaluated. A finite list of integer exponents and of numbers depending only on d and N is fixed first; $L_0(N)$ is then the least odd integer above finitely many lower bounds on L that read only these data. Every condition on L is a lower bound, and no lower bound on L reads κ , M or a flow constant.
- (H3) After L , Balaban's auxiliary parameters $\mathbf{p}$ are chosen in one fixed order. Among them are the decay rates κ, κ_1 ; the cube sizes $M_1 < M_2 = L^2 M_1 < M = L^{m'}$, powers of L ; the constants A_1, A_0, C_0, C_1 ; the radii $\alpha_1, \alpha_0, \alpha_2, \alpha_4$; the constant C_β ; and the radii $\varepsilon_0, \varepsilon_1$ of [7]. Each parameter is determined by earlier ones, or bounded on one side by a positive function of earlier ones, or, for finitely many of them, confined to a window that the conditions on earlier choices make non-empty (for instance $M_1 < M_2 = L^2 M_1 < M$, made non-empty by the lower bound on M). Then the thresholds γ and γ_H are chosen; the condition $\gamma \leq (8c_2 + 4)^{-1/2}$ is among the upper bounds on γ , and $\gamma_H \leq (\gamma^{-2} + c_5 + 1)^{-1/2} < \gamma$. No condition bounds L , M , M_1 , κ , κ_1 , q , p_0 , K , m , an age, a depth or a number of steps from above, or any coupling from below.
- (H4) $0 < g \leq \gamma_H$.
- (H5) $m \geq m_{\mathbb{T}}(g_-(g))$. This is a lower bound on the volume only; it makes Balaban's coarsest cubes fit in the torus at every scale.
- (H6) $0 < g_0 \leq g_-(g)$, with no lower bound, and $K = K(g_0, g)$.
- (H7) For clause (0)(d) and clauses (ii)–(vi) only: a radius $\varrho > 0$, the radius of the disc of complex sources, is fixed after γ_H ; then $g \leq \gamma_J(L, \mathbf{p}, \gamma; \varrho) \leq \gamma_H$; inside clause (iv) further

$g \leq \gamma_W^{\mathbb{T}}(L, \mathbf{p}, \gamma, \vartheta; \varrho) \leq \gamma_J$; inside clause (vi) further $g \leq \gamma_{vi} \leq \gamma_J$; inside clause (v) and for (ii- $\mathbb{T}$ -a), $g \leq \gamma_U \leq \gamma_J$.

Among the integer exponents fixed first in (H2) are Balaban's exponent $q = q_0 = q_1$, the power of the logarithm in his analyticity radii [9, (2.28)], the exponent p_0 of his small-field threshold $A_0(\log g^{-2})^{p_0}$ [10, §0], and the exponent r of (H5); each of them is bounded from below only, except for finitely many exponents confined to windows that lower bounds on other exponents make non-empty. Among these lower bounds is

$$q \geq 21 p_0 + 1, \quad (1.9)$$

on an exponent that Balaban's text leaves free (his q_0, q_1 are "integers greater than 1"); it is used in the second-order extraction estimate beneath clause (iv) (Section 5), and nothing bounds q from above. An age is the number of renormalisation steps since a large-field region was created; a depth is as in Subsection 1.2.

Order of the constants. The quantifiers of Theorem 0 come in the order

$$\begin{aligned} \exists (\text{exponents and numbers depending on } d, N) \quad \exists L_0(N) \quad \forall L \geq L_0 \text{ odd} \\ \exists \mathbf{p} \text{ (in one fixed order)} \quad \exists \gamma \quad \exists \gamma_H \\ \forall g \in (0, \gamma_H] \quad \forall m \geq m_{\mathbb{T}}(g_-(g)) \quad \forall g_0 \in (0, g_-(g)] : \quad K := K(g_0, g). \end{aligned}$$

They are modified inside clauses only, each time by an upper bound on a coupling, chosen last, or by a one-sided lower bound:

- in (0)(d), (iii), and in (ii) apart from (ii- $\mathbb{T}$ -a), the segment $\forall g \in (0, \gamma_H]$ becomes $\forall \varrho > 0 \exists \gamma_J \in (0, \gamma_H] \forall g \in (0, \gamma_J]$;
- in (vi) it becomes $\forall \varrho > 0 \exists \gamma_J \exists \gamma_{vi} \in (0, \gamma_J] \forall g \in (0, \gamma_{vi}]$, and the limits on $\mathbb{R}^4$ are taken along sequences obeying the lower bound (1.21) on the cutoff;
- in (iv) it becomes $\forall \varrho > 0 \exists \gamma_J \forall \vartheta \in (0, 1] \exists \gamma_W^{\mathbb{T}} \in (0, \gamma_J] \forall g \in (0, \gamma_W^{\mathbb{T}}]$;
- in (v) it becomes $\forall \varrho > 0 \exists \gamma_J \exists \gamma_U \in (0, \gamma_J] \forall g \in (0, \gamma_U]$, followed by an integer m_0 that fixes the torus, with (H5) for that torus, and by the depth n ; the bare inverse coupling is then tuned, $g_0^{-2} = a_n - \hat{w}$, instead of being stopped at first passage, and the first-passage quantifier over g_0 returns only in (v-c)(2) and in (ii- $\mathbb{T}$ -b); in (v-f) the radius obeys in addition the lower bound (1.20), and the threshold is a smaller γ_{Λ} ;
- in (ii- $\mathbb{T}$ -a) the threshold is γ_U , with the tuned bare couplings of (v).

In every other respect the quantifiers over m and g_0 follow unchanged. The thresholds $\gamma_J, \gamma_W^{\mathbb{T}}, \gamma_{vi}, \gamma_U, \gamma_{\Lambda}$ are existential and are chosen after everything they depend on; replacing one of them by a smaller positive number preserves every statement, so a clause stated in the regime of another may use a smaller threshold than that other clause. The radius ϱ is not one of the parameters: nothing up to and including γ_H reads it.

The shape of the conditions. Every condition met in the clauses and in their proofs is a lower bound on L , on a parameter, on m , on a depth, on the distance $K - s$ of a depth s from the cutoff, or on the depth n of clause (v); a determination of a parameter by earlier ones, or an upper bound on it by a positive function of earlier ones, at its own place in the order; a window, made non-empty by lower bounds, for finitely many parameters, for an exponent fixed before L , or for a quantity of the flow determined at scale k from $(x_i)_{i \leq k}$; an upper bound on a threshold γ , γ_H , γ_J , $\gamma_W^{\mathbb{T}}$, γ_{vi} , γ_U , γ_A , chosen last; the lower bound (1.21) on the cutoff in (vi); the lower bound $\varrho \geq 6(B_{\sharp} + 1)$ in (v-c)(2) and the lower bound (1.20) in (v-f); or a condition on the test functions (a free lower bound $d > 0$ on the separation of their supports; for (iv) the class $\mathfrak{T}(d; \vartheta)$, an upper bound on d by a threshold depending on g only, and, for the transfer statement, scale $d \leq 1/40$). None is an upper bound on L , M , M_1 , κ , κ_1 , q , p_0 , K or m , on an age, a depth or a number of steps, and none is a lower bound on g , g_0 or a running coupling.

Consistency of the order. That the order (H3) has no cycle is proved in the companion account, and so is the non-emptiness of the windows for the exponents and numbers fixed before L . That every remaining choice can be made, given the earlier ones, is proved there as well. At the constants of two kernel estimates for the fluctuation operators, which are fixed after M_1 and before M by formulas in L and M_1 , this uses that the two kernel estimates hold with these constants, for every cube size M , on the cells of the one family of cells over which the effective densities are decomposed. This independence of M is proved in the companion account: directly for the kernel estimates on the whole lattice, and, for the local version that is used, as an implication from the operator theorem at free current of Section 2. Among the inputs of that theorem is the structural statement on the creation of large-field regions (statement (C4) of Section 2), which is proved in the companion account as a combinatorial lemma from Balaban's definitions (Section 9).

1.4 The theorem

Clause (iii) is stated before clause (ii) because (ii) uses it. Within each clause, the constants and thresholds depend only on the quantities named; in particular none of them depends on K , g_0 or the position of the supports unless this is said.

Proof status of this local revision. Theorem 0 is conditional on the proof inputs declared for its individual clauses in Section 9, as well as on (H1)–(H7) where stipulated. Proofs attributed to the companion account are not reproduced here and were not supplied with the source reviewed for this revision; this is an availability and verification boundary, not a claim that those proofs are false. Separately, the author explicitly withholds proved status from (U1), (U3) and (U5), identifies (U4) and (U6) as not yet written, and leaves one definition clause to be added (Sections 7.5 and 9). Consequently clause (v), (ii-T-a), the final identification sentence of (ii-T-b), clause (vi), and the short-distance rotation results remain conditional on those unfinished inputs. This revision supplies only the local corrections documented in the accompanying repair record; it does not supply any missing multiscale estimate.

Theorem 0 (Conditional on the declared proof inputs). *Let $N = 2$ and let $L_0 = L_0(N)$ be as in (H2). Let $L \geq L_0$ be odd, and let $\mathbf{p}$, γ and γ_H be chosen after L in the order of (H3). Assume (H1)–(H6), and (H7) where a clause says so. Assume also, for each clause asserted below, the proof inputs listed under that clause in Section 9. The constants in (0)(a)–(c) and in (i) depend only on L , $\mathbf{p}$, γ and N , and on the scale k only through g_k ; they never depend on K , m , g_0 or the position,*

except for a ratio $\max\{1, (x_j/x_n)^2\}$ of two running inverse couplings that enters the bounds of (i-a). Then the following hold.

(0) The coupling flow.

(a) K is a well-defined non-negative integer, and

$$\frac{g_0^{-2} - g^{-2}}{B_\sharp} - 1 \leq K \leq 1 + \frac{g_0^{-2} - g^{-2} - B_\sharp + c_5}{\lambda_\sharp}; \quad (1.10)$$

in particular $K \rightarrow \infty$ as $g_0 \downarrow 0$.

(b) $g_k \in (0, g) \subset (0, \gamma_H]$ for every $k \leq K$, and $g_K \in [g_-(g), g)$.

(c) The running inverse couplings obey the two-sided flow bounds

$$\lambda_\sharp(j-i) - c_5 \leq x_i - x_j \leq B_\sharp(j-i), \quad 0 \leq i \leq j \leq K, \quad (\text{FLOW}\pm)$$

and the K renormalisation steps on $\mathbb{T}_m$ at spacing L^{-K} produce exactly the couplings $g_0, \dots, g_K$, which do not depend on the torus. Consequently, along any sequence $g_0^{(i)} \downarrow 0$, with $K_i := K(g_0^{(i)}, g)$, for every depth s and every i with $K_i \geq s$,

$$x_{K_i-s} \in (g^{-2} + \max\{0, \lambda_\sharp s - c_5\}, g^{-2} + B_\sharp(s+1)],$$

and along a subsequence $g_{K_i-s} \rightarrow g_{\infty,s}$ for every $s \geq 0$, with $g_{\infty,s}^{-2} \in [g^{-2} + \max\{0, \lambda_\sharp s - c_5\}, g^{-2} + B_\sharp(s+1)]$.

(d) (Exact one-loop running; $N = 2$; under (H7), for $g \leq \gamma_J$.) There are a function e on $(0, \gamma_J]$ with $e(g') \rightarrow 0$ as $g' \downarrow 0$, explicit non-negative numbers v_k , and a function Δ on the non-negative integers with $\Delta(s)/s \rightarrow 0$ as $s \rightarrow \infty$, all depending only on L , N and $\mathfrak{p}$ (the v_k also on the couplings), such that for every $g_0 \in (0, g_-(g)]$, with $K = K(g_0, g)$, every m , and all $0 \leq i < j \leq K$,

$$|x_i - x_j - (j-i)c_0| \leq 2c_2 + \sum_{k=i}^{j-1} [C_\beta e(g_k) + v_k], \quad c_0 = \frac{11N^2}{12\pi^2} \log L, \quad (1.11)$$

and in particular $|x_{K-s} - x_K - sc_0| \leq \Delta(s)$ for $0 \leq s \leq K$. In words: the inverse square of the running coupling grows by c_0 per renormalisation step towards short distances, up to an error whose Gaussian part is bounded cumulatively, over any number of steps, by $2c_2$, and whose remaining part vanishes with the coupling; over s steps the error is $o(s)$, uniformly in the cutoff and in the volume. This is the one-loop law of Yang–Mills theory with its universal first coefficient ($11N/3$ in the standard normalisation; Subsection 1.2), obtained as a consequence of the control of the flow by the correlation theorem of Section 3, not assumed.

(i) Ultraviolet stability. Items (i-a)–(i-c) rest on [11, Theorem 1], whose one hypothesis—that all running couplings lie in its coupling interval—is supplied by (0): (i-a) is the representation of that theorem, read as described in Section 2, and (i-b), (i-c) are derived from it in the companion account (structure of the proofs in Section 2). Item (i-d) is an exact identity of the construction (proof in the companion account; structure in Section 2); its last sentence, on sourced densities, is part of the correlation theorem of Section 3. Balaban’s block variables are the vehicle of the proofs; no statement about their laws is a clause of this theorem. For every $k \leq K$:

(i-a) ρ_k has the representation [9, (2.18)]: a sum, over admissible histories, of large-field operations applied to characteristic functions, times the exponential of an effective action. The effective action is $-A(g_k^{-2}(\cdot), U_k)$ plus localised terms of every level $j \leq k$; the coupling $g_k^{-2}(\cdot)$ depends on the position only near large-field regions [9, (2.24)]. Each localised term, with domain $X \in \mathcal{D}_j$, is gauge invariant, analytic on $U_j^c(X, \alpha_0, \alpha_1)$ read hereditarily, and bounded by a constant times $e^{-\kappa d_j(X)}$, with the following exception. A boundary term whose domain contains a large-field region Z of the second class (in Balaban's terminology [11]) obeys, if it was created beside no older such region, the bound [11, (1.99)], which decays in the relative length $d_{j,Z}(X)$; every other such term is bounded by a constant times an exponential decaying in a tree length $d^\theta(X) \leq d_{j,Z}(X)$, defined in the companion account, and no lower bound of $d^\theta(X)$ in terms of $d_{j,Z}(X)$ is asserted. Balaban's displays are read in the form forced by his own definitions.

(i-b) There are constants $E_- = E_-(L, \mathbf{p}, N)$, independent of g_k , and $E_+ \leq C(L, \mathbf{p}, \gamma, N)(1 + \log g_k^{-2})$ such that, as measures on real fields,

$$\chi_k e^{-g_k^{-2} A(U_k) - E_- N_k} dV_k \leq \rho_k dV_k,$$

and, after integration only, $\int \rho_k dV_k \leq e^{E_+ N_k}$. The pointwise upper bound [11, (0.1)] is not asserted.

(i-c) $\int \rho_k dV_k = \int \rho_0 dU$.

(i-d) (Exact insertion.) For $0 \leq s \leq K$ and every bounded Borel function F of the block field of level $K - s$, writing $\rho_{K-s} dV_{K-s} = \sum_h \tau_h$ as in (i-a),

$$\begin{aligned} \langle F(\mathcal{M}^{K-s} U) \rangle_{K,m} &= \frac{\sum_h \int F^{(h)} d\tau_h}{\sum_h \int d\tau_h}, \\ F^{(h)} &= \mathbb{E}[F(\mathcal{M}^{K-s} U) \mid h, V_{K-s}], \\ |F^{(h)}| &\leq \sup |F|, \end{aligned}$$

the conditional expectation being taken for Balaban's procedure run as a Markov chain from the Wilson measure, with every outcome of every decomposition of unity kept. For $N = 2$ and under (H7) the construction applies also to the sourced bare density $\rho_0^w := e^{-A(g_0^{-2} - w, U) - E}$ with $w = \sum_i z_i f_i \in \mathcal{W}$: the sourced densities ρ_k^w keep the format of (i-a), with the coefficient of the Wilson term shifted by the running shift of (iii)(c), and $\int \rho_k^w dV_k = \int \rho_0^w dU$.

(iii) Correlation bounds uniform in the cutoff. ($N = 2$; under (H7), for $g \leq \gamma_J$.)

(iii)(a) (Torus bound.) For $n \geq 2$ and real Lipschitz $f_1, \dots, f_n$ on $\mathbb{T}_m$ whose supports are pairwise at distance at least $\mathbf{d} > 0$,

$$|S_{n;K,m}^T(f_1, \dots, f_n)| \leq C_n(\mathbf{d}) \prod_{i=1}^n \|f_i\|_\sharp, \quad (1.12)$$

where $C_n(\mathbf{d})$ depends only on $L, \mathbf{p}, \gamma, g, m, \varrho, n$ and $\mathbf{d}$ —not on K, g_0 or the position of the supports. It grows with the volume through a factor $Q(m) \geq 1$, the volume factor of this bound, which depends on $m, g, \varrho, L, \mathbf{p}$ and γ but not on K, g_0 or the position (Section 3, Theorem 3.1(e)).

- (iii)(b) (Per-ball bound, free of the volume.) For $n \geq 2$, $R > 0$ and $\mathbf{d} > 0$ there is a constant $C_n^\infty(\mathbf{d}; R)$, depending only on L , $\mathbf{p}$, γ , g , ϱ , n , $\mathbf{d}$ and R —not on K , g_0 , m or the position—and non-increasing in $\mathbf{d}$, such that for every m with $L^m \geq \ell(n, R, g)$, every $g_0 \in (0, g_-(g)]$, and all real Lipschitz $f_1, \dots, f_n$, each supported in a ball of radius R , with supports pairwise at distance at least $\mathbf{d}$,

$$|S_{n;K,m}^T(f_1, \dots, f_n)| \leq C_n^\infty(\mathbf{d}; R) \prod_{i=1}^n \|f_i\|_\#; \quad (1.13)$$

here $\ell(n, R, g)$ is the least power of L that is at least $\max\{8n(R+1), L^{m_\mathbb{T}(g_-(g))}\}$, a lower bound on the volume. For fixed n and R , $C_n^\infty(\mathbf{d}; R)$ grows at most like $\mathbf{d}^{-4} \log(1/\mathbf{d})$ as $\mathbf{d} \downarrow 0$ (Sections 3 and 4); no decay in $\mathbf{d}$ is asserted.

- (iii)(c) (Running shift.) For $\zeta \in \mathbb{C}$ with $|\zeta| < 2\varrho$, consider the bare density with g_0^{-2} replaced by $g_0^{-2} - \zeta$ (the constant source $w \equiv \zeta$), all auxiliary choices of Balaban's procedure being frozen at their values for $\zeta = 0$. Let $b_{k+1}(\zeta)$ be the coefficient of $[\gamma, (1.22)]$, $[9, (3.61)]$ for step $k+1$ of this run, so that $b_{k+1}(0) = \beta_{k+1}$, and put

$$\zeta_0 := \zeta, \quad \zeta_{k+1} := \zeta_k + b_{k+1}(\zeta) - b_{k+1}(0). \quad (1.14)$$

Then, for $k \leq K$, ζ_k is analytic on $\{|\zeta| < 2\varrho\}$ and real for real ζ , $|\zeta_k - \zeta| \leq |\zeta|/3$ there, and $\sum_{k < K} \text{Lip}_{\bar{D}(0, \varrho)} b_{k+1} \leq \frac{2}{3}$. Hence $\zeta'_k := \partial_\zeta \zeta_k(0) \in [\frac{2}{3}, \frac{4}{3}]$, because $\zeta_k - \zeta$ vanishes at $\zeta = 0$ and has modulus at most $|\zeta|/3$, so that its derivative at 0 has modulus at most $\frac{1}{3}$. The number ζ_k is the shift of the inverse coupling of level k produced by the shift ζ of the bare inverse coupling.

(ii) Continuum limits; invariance; reflection positivity; reconstruction. ($N = 2$; under (H7), for $g \leq \gamma_J$; for (ii-T-a) in addition $g \leq \gamma_U$.)

- (ii-T-a) (Tori: full convergence along the tuned bare couplings.) Let $m_0 \geq m_\mathbb{T}(g_-(g))$ and $\mathbb{T} := \mathbb{T}_{m_0}$, lengths being measured in units of the reference lattice of the tuned runs of clause (v). For $n \geq 0$ and $x \in \mathbb{C}$, let $S^n(x; f_1, \dots, f_\mathbf{p})$ be the connected $\mathbf{p}$ -point function (1.5) on $\mathbb{T}$ at bare spacing L^{-n} and bare inverse coupling x (that is, (1.5) with K replaced by n and g_0^{-2} by x), defined whenever the partition function does not vanish near $z = 0$; for the complex values of x used below, that it is defined is part of the assertion; let a_n be the tuned bare inverse couplings of (v-a), and put $\varrho_2 := \varrho/2$. There is $v_w > 0$, depending on $\mathbb{T}$ but not on n , with the following property. For every $\mathbf{p} \geq 2$, all real Lipschitz $f_1, \dots, f_\mathbf{p}$ on $\mathbb{T}$ with supports pairwise at distance at least $\mathbf{d} > 0$, and every $\hat{w} \in \mathbb{C}$ with $|\text{Re } \hat{w}| \leq \varrho_2/2$ and $|\text{Im } \hat{w}| < v_w$,

$$S^n(a_n - \hat{w}; f_1, \dots, f_\mathbf{p}) \longrightarrow S^{(\hat{w})}(f_1, \dots, f_\mathbf{p}) \quad (n \rightarrow \infty)$$

without passing to subsequences, uniformly in $\hat{w}$, and

$$|S^n(a_n - \hat{w}; \vec{f}) - S^{(\hat{w})}(\vec{f})| \leq C n^{-1/2} (\log n)^{p_0+1} \prod_i \|f_i\|_\#$$

for $n \geq n_0$, where n_0 and C do not depend on n or $\hat{w}$, and C depends on L , $\mathbf{p}$, g , $\mathbf{p}$, ϱ , $\mathbf{d}$, the supports and the torus (Section 7); in the renormalised labelling of (v-d) the rate is exponential. For real $\hat{w}$ the limits $S^{(\hat{w})}$ are multilinear, symmetric, invariant under all translations of $\mathbb{T}$ (for C^1 test functions) and under W_4 , reflection positive, for the centred functions on separated test functions, at every hyperplane $\{x^\mu = c\}$, and obey the bound (1.12).

(ii-T-b) (Tori: first passage.) Let $m \geq m_{\mathbb{T}}(g_-(g))$ and $g_0^{(i)} \in (0, g_-(g)]$ with $K_i := K(g_0^{(i)}, g) \rightarrow \infty$. Along a subsequence, for every $n \geq 2$ and every n -tuple of real C^1 functions on $\mathbb{T}_m$ with pairwise disjoint supports simultaneously, $S_{n;K_i,m}^T(f_1, \dots, f_n) \rightarrow S_{n;m}^T(f_1, \dots, f_n)$. Each $S_{n;m}^T$ extends to a unique distribution of locally finite order on $(\mathbb{T}_m)_{\neq}^n$, which is symmetric and invariant under all translations of $\mathbb{T}_m$ and under W_4 ; the centred functions $S_{q;m}$ built from them by (1.6) are reflection positive at every hyperplane $\{x^\mu = c\}$, in the form (1.15). If $\varrho \geq 6(B_{\sharp} + 1)$ and g lies below the threshold of $(v-c)(2)$, every such subsequential limit on $\mathbb{T}_m$ is one of the $S^{(\hat{w})}$ of (ii-T-a), $\hat{w}$ real, $|\hat{w}| \leq \varrho_2/2$.

(ii- $\mathbb{R}^4$) (Infinite volume along growing volumes.) Let $g_0^{(j)} \in (0, g_-(g)]$ with $K_j := K(g_0^{(j)}, g) \rightarrow \infty$, and let $m_j \geq m_{\mathbb{T}}(g_-(g))$ with $m_j \rightarrow \infty$. Along one joint subsequence, for every $n \geq 2$ and all real $f_1, \dots, f_n \in C_c^1(\mathbb{R}^4)$ with pairwise disjoint supports,

$$S_{n;K_j,m_j}^T(f_1, \dots, f_n) \longrightarrow S_{n;\infty}^T(f_1, \dots, f_n).$$

The limits $S_{n;\infty}^T$ are real, symmetric, multilinear on tuples with pairwise disjoint supports, invariant under $\mathbb{R}^4 \rtimes W_4$ exactly, and obey (1.13) with the same constant $C_n^\infty(\mathbf{d}; R)$; each extends to a distribution of locally finite order on $(\mathbb{R}^4)_{\neq}^n$. Moreover each $S_{n;\infty}^T$, and each centred function $S_{q;\infty}$ built from them by (1.6), extends uniquely to a continuous linear functional on ${}^0\mathcal{S}(\mathbb{R}^{4n})$ (resp. ${}^0\mathcal{S}(\mathbb{R}^{4q})$), the space of Schwartz functions that vanish together with all their derivatives on the diagonals $\{x_a = x_b\}$, with the topology induced by the Schwartz space (Section 4, Theorem 4.12, where these extensions are constructed). The centred Schwinger functions $S_{q;\infty}$ are reflection positive at every coordinate hyperplane $P = \{x^\mu = c\}$, in the form of [21]:

$$\sum_{k,l} \overline{c_k} c_l S_{n_k+n_l;\infty}(\theta_P f_1^{(k)}, \dots, \theta_P f_{n_k}^{(k)}, f_1^{(l)}, \dots, f_{n_l}^{(l)}) \geq 0 \quad (1.15)$$

for all finite families of complex numbers c_k and of tuples $(f_1^{(k)}, \dots, f_{n_k}^{(k)})$, $n_k \geq 0$, of real C_c^1 functions with pairwise disjoint supports contained in one open half-space bounded by P (the empty tuple standing for the constant 1). In the numbering of [21], every such limit therefore satisfies the regularity condition E0 in the form just stated ($S_{q;\infty} \in {}^0\mathcal{S}'(\mathbb{R}^{4q})$ for every q , at non-coinciding points), the invariance E1 under all translations and under the hypercubic group, the reflection positivity E2 at every coordinate hyperplane, and the symmetry E3; the growth condition of [22] in the number of points is not asserted, so no field operators are claimed, and the cluster property E4 is not asserted. The Osterwalder–Schrader reconstruction [21], applied to the linear span of 1 and of the products of centred action densities with pairwise disjoint compact supports in the open half-space $\{x^0 > 0\}$, yields: a separable Hilbert space $\mathcal{H}^{\text{curv}}$; a unit vector Ω ; a strongly continuous semigroup e^{-tH} , $t \geq 0$, of positive self-adjoint contractions, with $H \geq 0$ and $H\Omega = 0$; a strongly continuous unitary representation of the spatial translations $\mathbb{R}^3$, fixing Ω , whose generators $\vec{P} = (P_1, P_2, P_3)$ commute strongly with H , the joint spectrum of $(H, \vec{P})$ lying in $[0, \infty) \times \mathbb{R}^3$; and a unitary representation of W_3 .

(iv) Non-triviality: the two-point law of the action density, uniformly in the volume. ($N = 2$; in the regime of (iii).) For every $\vartheta \in (0, 1]$ there are: a threshold $\gamma_W^\mathbb{T} = \gamma_W^\mathbb{T}(L, \mathbf{p}, \gamma, \vartheta; \varrho) \in (0, \gamma_J]$; a function $C_W(s) \geq 1$ of the depth s , non-decreasing and polylogarithmic in $\bar{x}_s := g^{-2} + B_{\sharp}(s+1)$; non-negative integers $s_W(g)$, $s_{\text{vol}}^*(g)$ and $s_T(g)$ which, besides L , $\mathbf{p}$, γ , ϑ and ϱ , depend

on g only (s_W non-decreasing in g^{-2} and of order $\log_L \log g^{-2}$, s_T at most polylogarithmic in g^{-2}); an integer $k_W = k_W(L) \geq 1$; constants $0 < c_K \leq C_K$; and, for each g , a constant $C_T(g)$ —none of them depending on K , g_0 , m , $\mathbf{d}$ or the position—with the following properties. For $\mathbf{d} > 0$ put

$$s(\mathbf{d}) := \min\{s \geq 0 : C_W(s)L^{-s} \leq \mathbf{d}\};$$

$s(\mathbf{d})$ does not depend on K and increases to ∞ as $\mathbf{d} \downarrow 0$. Consider the conditions

$$\max\{s_W(g), s_{\text{vol}}^*(g), s_T(g)\} \leq s(\mathbf{d}), \quad (\text{F-depth})$$

$$K - s(\mathbf{d}) \geq k_W. \quad (\text{F-UV})$$

Condition (F-depth) is a lower bound on the depth $s(\mathbf{d})$, equivalently an upper bound $\mathbf{d} < \mathbf{d}_W(g)$ on the separation by a threshold that depends on g only; we do not claim that it can be removed. Condition (F-UV) asks that at least k_W renormalisation steps separate the bare lattice from the lattice at depth $s(\mathbf{d})$.

The kernel. Let $\mathcal{A}$ be the real free lattice gauge field at unit coupling on the bare lattice of a torus, taken modulo gauge transformations and harmonic modes; let Q be the linearised block-averaging map from the bare lattice to the lattice of level $K - s$; and let $F_B(p)$ be the lattice curvature, on the bare plaquette p , of the conditional expectation $\mathbb{E}[\mathcal{A} | Q\mathcal{A}]$. Put

$$\mathcal{K}_s(f_1, f_2) := \sum_{p, q} f_1(x(p)) f_2(x(q)) (\mathbb{E}[F_B(p)F_B(q)])^2,$$

the sum running over bare plaquettes, and let $\mathcal{K}_s^*$ denote this kernel formed on the smallest torus of the family, $\mathbb{T}_{m_{\mathbb{T}}(g_-(g))}$, test pairs being transported to it. Multiplied by b_0 , its continuum counterpart is $3(N^2 - 1)\pi^{-4} \iint f_1(x)f_2(y)|x - y|^{-8} dx dy$. The kernel formed on $\mathbb{T}_m$ itself differs from $\mathcal{K}_s^*$ by a free-field correction for the torus; both kernels obey the two-sided bound below and have the same continuum counterpart.

The two-point law. For every $g \leq \gamma_W^{\mathbb{T}}$, every $g_0 \in (0, g_-(g)]$ with $K = K(g_0, g)$, every $m \geq m_{\mathbb{T}}(g_-(g))$ —that is, (H5) and nothing more on m —, every $\mathbf{d}$ satisfying (F-depth) and (F-UV), with $s := s(\mathbf{d})$, and every $(f_1, f_2) \in \mathfrak{T}(\mathbf{d}; \vartheta)$ on $\mathbb{T}_m$,

$$S_{2;K,m}^T(f_1, f_2) = \zeta_{K-s}'^2 b_0 g_{K-s}^4 \mathcal{K}_s^*(f_1, f_2) (1 + R_{K,m}(f_1, f_2)), \quad |R_{K,m}(f_1, f_2)| \leq \frac{1}{2}, \quad (1.16)$$

where $\zeta_{K-s}' \in [\frac{2}{3}, \frac{4}{3}]$ is the derivative at zero of the running shift of (iii)(c); moreover $|\zeta_{K-s}' - 1| \leq C_{\zeta} g_{K-s} (\log g_{K-s}^{-2})^{p_0}$, with C_{ζ} depending only on L , $\mathfrak{p}$, γ and ϱ (Section 3); and

$$c_K \|f_1\|_{\infty} \|f_2\|_{\infty} \leq \mathcal{K}_s^*(f_1, f_2) \leq C_K \|f_1\|_{\infty} \|f_2\|_{\infty}.$$

Hence, uniformly in K , g_0 , m and the position,

$$\frac{2}{9} b_0 g_{K-s}^4 \mathcal{K}_s^*(f_1, f_2) \leq S_{2;K,m}^T(f_1, f_2) \leq \frac{8}{3} b_0 g_{K-s}^4 \mathcal{K}_s^*(f_1, f_2), \quad (1.17)$$

in particular $S_{2;K,m}^T(f_1, f_2) > 0$, and, by (0),

$$\frac{\frac{2}{9} b_0 c_K \|f_1\|_{\infty} \|f_2\|_{\infty}}{(g^{-2} + B_{\sharp}(s+1))^2} \leq S_{2;K,m}^T(f_1, f_2) \leq \frac{\frac{8}{3} b_0 C_K \|f_1\|_{\infty} \|f_2\|_{\infty}}{(g^{-2} + \lambda_{\sharp}s - c_5)^2}. \quad (1.18)$$

Volume transfer. For every $g \leq \gamma_W^{\mathbb{T}}$, g_0 and K as above, all $m \geq m' \geq m_{\mathbb{T}}(g_-(g))$, every $\mathbf{d} \in (0, \frac{1}{40}]$ with $\varepsilon \leq \mathbf{d}$, and all real Lipschitz f_1, f_2 on $\mathbb{R}^4$ —of any sign and in any mutual position—with $\text{supp } f_1 \cup \text{supp } f_2$ contained in one closed ball of radius $5\mathbf{d}$ and $\text{Lip } f_i \leq \|f_i\|_{\infty}/(\vartheta\mathbf{d})$, regarded as functions on $\mathbb{T}_m$ and on $\mathbb{T}_{m'}$,

$$|S_{2;K,m}^T(f_1, f_2) - S_{2;K,m'}^T(f_1, f_2)| \leq C_T(g) \mathbf{d}^2 \|f_1\|_{\infty} \|f_2\|_{\infty}. \quad (1.19)$$

Consequences. (1) Let $S_{2;\infty}^T$ be a limit of (ii- $\mathbb{R}^4$), with $g \leq \gamma_W^T$. For every $\mathbf{d}$ satisfying (F-depth), with $s := s(\mathbf{d})$, and every pair $(f_1, f_2) \in \mathfrak{T}(\mathbf{d}; \vartheta)$ of $C_c^1(\mathbb{R}^4)$ functions, (1.18) holds with $S_{2;K,m}^T$ replaced by $S_{2;\infty}^T$; the same holds for the limits of (ii-T-b). (2) In every limit of (ii-T-b) and of (ii- $\mathbb{R}^4$), there is a centred action-density vector with strictly positive reflected norm, so that the reconstructed Hilbert space has dimension at least two ($\dim \mathcal{H}^{\text{curv}} \geq 2$ on $\mathbb{R}^4$; on a torus, for the space obtained in the same way, Corollary 6.25); the limit is not scale invariant under any assignment of a scaling dimension to the action density; it is reproduced by no free Maxwell field at any charge; and the coupling read off the two-point function at separation $\mathbf{d}$ is strictly positive and tends to zero logarithmically as $\mathbf{d} \downarrow 0$, at the two-sided rate (FLOW $\pm$). (3) With (0)(d): the running coupling $g(\mathbf{d}) := g_{K-s(\mathbf{d})}$ satisfies $g(\mathbf{d})^{-2} = \frac{11N^2}{12\pi^2} \log(1/\mathbf{d}) (1 + o(1))$ as $\mathbf{d} \downarrow 0$, uniformly in the cutoff; and the charge g_{phys} read off the two-point function, defined by $S_{2;K,m}^T(f_1, f_2) = b_0 g_{\text{phys}}^4 \mathcal{K}_s^*(f_1, f_2)$, satisfies $g_{\text{phys}}^4/g(\mathbf{d})^4 \in [\frac{2}{9}, \frac{8}{3}]$, so that

$$g_{\text{phys}}^{-2} \in \left[\left(\frac{3}{8}\right)^{1/2}, \left(\frac{9}{2}\right)^{1/2} \right] \cdot \frac{11N^2}{12\pi^2} \log(1/\mathbf{d}) (1 + o(1)) \quad (\mathbf{d} \downarrow 0) :$$

the inverse square of the charge read off the two-point function has logarithmic growth between these fixed multiples of the one-loop Yang–Mills coefficient (in the normalisation of Subsection 1.2), in the liminf/limsup sense of Remark 2.15; no derivative or individual-increment bound for that charge is asserted.

(v) **Uniqueness up to one real parameter, the renormalised coupling at the reference length.** ($N = 2$; under (H7): for every $\varrho > 0$ there is $\gamma_U \in (0, \gamma_J]$ such that the following holds for every $g \leq \gamma_U$; we write $X := g^{-2}$ and $\lambda_U := c_0/2$.) In this clause the bare inverse coupling is tuned instead of being stopped at first passage. For a depth $n \geq 0$, the n -step recursion (1.7) is run from a bare inverse coupling x_0 , with the increments computed on the infinite lattice and with the coupling-dependent choices of Balaban’s procedure fixed in terms of X ; the run is admissible if its couplings stay in the range in which the construction and the correlation theorem apply (Section 7). Lengths are measured in units of the reference lattice of the run, and $\mathbb{T}$, $S^n(x; \cdot)$, ϱ_2 , v_w and $S^{(\hat{w})}$ are as in (ii-T-a).

(v-a) (Exact pin; asymptotic freedom of the tuned bare coupling.) For every $n \geq 0$ there is exactly one real a_n , depending on L , $\mathbf{p}$, g and n and on no torus, such that the admissible n -step run started at $x_0 = a_n$ ends at $x_n = X$ exactly. Moreover

$$\begin{aligned} \lambda_U n - 2c_2 &\leq a_n - X \leq 3\lambda_U n + 2c_2, \\ \lambda_U(n-l) - c' &\leq a_n - a_l \leq 3\lambda_U(n-l) + c' \quad (0 \leq l \leq n), \end{aligned}$$

with a constant c' independent of n and l ; no monotonicity from one step to the next is asserted.

(v-b) (The one-parameter family of limits.) The limits $S^{(\hat{w})}(\vec{f})$ of (ii-T-a) exist for $|\text{Re } \hat{w}| \leq \varrho_2/2$, $|\text{Im } \hat{w}| < v_w$. For fixed separated Lipschitz $\vec{f}$, the map $\hat{w} \mapsto S^{(\hat{w})}(\vec{f})$ is real-analytic on $(-\varrho_2/2, \varrho_2/2)$, holomorphic on the open rectangle $\{|\text{Re } \hat{w}| < \varrho_2/2, |\text{Im } \hat{w}| < v_w\}$ —holomorphy is asserted on this rectangle only—and real for real $\hat{w}$. The depth-to-depth coupling map $\mathbf{n}_n$, defined by the requirement that the run of depth $n+1$ from $a_{n+1} - \mathbf{n}_n(\hat{w})$ and the run of depth n from $a_n - \hat{w}$ have the same coupling at the reference scale, is holomorphic on the disc $D(0, \varrho)$, $\mathbf{n}_n(0) = 0$, and $|\mathbf{n}_n(\hat{w}) - \hat{w}| \leq \Upsilon_n |\hat{w}|$ with $\Upsilon_n \leq C n^{-3/2} (\log n)^{p_0+1}$, so that $\sum_n \Upsilon_n < \infty$. The derivative $\mathcal{N}_n$ of the running shift at the reference scale, at the tuned point, lies in $[\frac{2}{3}, \frac{4}{3}]$, satisfies $\mathcal{N}_{n+1} \mathbf{n}'_n(0) = \mathcal{N}_n$, and $(\mathcal{N}_n)$ converges. The numbers a_n , the

maps $\mathbf{n}_n$, the numbers $\mathcal{N}_n$, Υ_n and every constant are independent of the torus, which enters only through v_w , through the constant in the rate of (ii-T-a), and through a lower bound n_0 on n .

- (v-c) (Exhaustion.) (1) For every real sequence (a'_n) with $|a'_n - a_n| \leq \varrho_2/2$, $S^n(a'_n; \vec{f}) - S^{(a_n - a'_n)}(\vec{f}) \rightarrow 0$ as $n \rightarrow \infty$. Hence two such sequences whose offsets $a_n - a'_n$ converge to the same $\hat{w}$ have the same limit $S^{(\hat{w})}$, and such a sequence has convergent correlation functions, without passing to subsequences, if and only if $S^{(a_n - a'_n)}(\vec{f})$ converges—in particular whenever its offset converges. (2) If moreover $\varrho \geq 6(B_{\sharp} + 1)$ and g lies below a further threshold in $(0, \gamma_U]$, then every bare inverse coupling whose first-passage run stops at depth K lies within $\varrho_2/2$ of a_K , and every subsequential limit of (ii-T-b) on $\mathbb{T}$ is one of the $S^{(\hat{w})}$, $\hat{w}$ real, $|\hat{w}| \leq \varrho_2/2$. (3) The set of continuum limits on $\mathbb{T}$ obtained from sequences of bare inverse couplings in the tube $\{|x - a_n| \leq \varrho_2/2\}$ is exactly $\{S^{(\hat{w})} : \hat{w} \text{ real}, |\hat{w}| \leq \varrho_2/2\}$.
- (v-d) (The marginal coordinate is the renormalised coupling.) (1) Re-file each large-field remainder into the regular part of the effective action at the level at which its partition cube is a single cell, let $\hat{x}$ be the inverse coupling at the reference scale of the run so re-filed, and label the family by the renormalised coordinate $t := X - \hat{x}$: for $|t| \leq \varrho_2$ there is exactly one real bare inverse coupling $\hat{a}_n(t)$ whose re-filed run of depth n is admissible with $\hat{x} = X - t$. Normalise the $\mathbf{p}$ -point functions by $\hat{N}_n(t)^{\mathbf{p}}$, where $\hat{N}_n(t)$ is the derivative of $\hat{x}$ with respect to the bare inverse coupling at $\hat{a}_n(t)$. Then the normalised family labelled by t converges at the exponential rate $n^c \rho^{n/2}$, with $\rho \in (0, 1)$ a contraction constant of the flow. The bare offset $\hat{w}$ and the renormalised coordinate t parametrise the same family, at each depth real-analytically and with slope in $[\frac{2}{3}, \frac{4}{3}]$ near 0. (2) Two admissible sequences on $\mathbb{T}$ whose renormalised inverse couplings at the reference scale converge to the same value have the same limit. (3) (Up to scale.) The data of a lattice theory of the family are (m_0, y) , the torus exponent and the reference inverse coupling. The exact L -adic change of units

$$S_{\mathbf{p}; K, m}^T[x_0](f_1, \dots, f_{\mathbf{p}}) = S_{\mathbf{p}; K-s, m+s}^T[x_0](f_1(L^{-s} \cdot), \dots, f_{\mathbf{p}}(L^{-s} \cdot)),$$

in which $[x_0]$ indicates the bare inverse coupling—the same plaquettes, read with the reference length moved s levels towards the cutoff—together with (FLOW $\pm$) identifies, at every cutoff, the theory with data (m_0, y) with the theory with data $(m_0 + s, y_s)$, where $y_s - y \in [\lambda_{\sharp}s - c_5, B_{\sharp}s]$. For the limit family, every limit $S^{(\hat{w})}$ on $\mathbb{T}_{m_0}$ with $|\hat{w}| + 3\lambda_U s + c' \leq \varrho_2/2$ equals, after the rescaling $f_i \mapsto f_i(L^{-s} \cdot)$, a limit $S^{(\hat{w})}$ of the family on $\mathbb{T}_{m_0+s}$, with $\hat{w} - \hat{w} \in [\lambda_U s - c', 3\lambda_U s + c']$ (Section 7). Apart from (v-f), nothing relates distinct reference couplings on one torus.

- (v-e) (Continuity of one complete step at a fixed cutoff.) On one lattice, for two inputs in the class of the correlation theorem (complex sources allowed), the outputs of one complete step of Balaban's procedure, large-field operations included, differ by at most multiples, independent of the cutoff, of the difference of the inputs, measured in the weighted norms of the correlation theorem; K , k , the torus and the position enter no constant. Old large-field regions are forgotten geometrically from their first step on, by a factor $q_* < 1$ per step of age, paid from Balaban's entropy constant under a lower bound on the exponent p_0 . This is the step that (ii-T-a) and (v-b) iterate.

- (v-f) (Two blocking factors: convergent steps, one slope, injectivity.) Assume in addition

$$\varrho \geq \varrho_{\Lambda}, \tag{1.20}$$

a lower bound on the radius of (H7) by a number $\varrho_\Lambda = \varrho_\Lambda(L, \mathbf{p}, \gamma) \geq 6(B_\sharp + 1)$ chosen so that the window $|\hat{w}| \leq \varrho_2/2$ of offsets is at least four times as long as a bound, uniform in the coupling, for the steps $a_{n+1} - a_n$ between consecutive tuned bare inverse couplings (such a bound is given by (v-a)). There is a threshold $\gamma_\Lambda \in (0, \gamma_U]$, chosen last, such that the following holds for every $g \leq \gamma_\Lambda$ and every torus exponent at least the floor of Corollary 1.2; the subscript of $S^{(\hat{w})}$ indicates the torus. (1) The steps converge, $a_{n+1} - a_n \rightarrow \alpha_\infty \in [\lambda_U, 3\lambda_U]$, and the change of units of (v-d)(3) is exact with one shift: $S_{\mathbb{T}_{m_0}}^{(\hat{w})}(f_1, \dots, f_p) = S_{\mathbb{T}_{m_0+1}}^{(\hat{w}-\alpha_\infty)}(f_1(L^{-1}\cdot), \dots, f_p(L^{-1}\cdot))$ whenever $|\hat{w}|, |\hat{w} - \alpha_\infty| < \varrho_2/2$. (2) The number $b := \alpha_\infty / \log L$ depends neither on g nor on L : for two odd blocking factors $L \neq L'$ at least L_0 , each with its own choices in (H3) and with a common ϱ obeying (1.20) for both, the two families have the same b . Moreover, writing a bare inverse coupling x in each family as $x = a_n - \hat{w}$ with $\hat{w}$ in the window, and putting $\ell_L(x) := L^n e^{-\hat{w}/b}$ (well defined up to a factor tending to 1), the ratio $\ell_L(x)/\ell_{L'}(x)$ of the two length units converges as $x \rightarrow \infty$. (3) (Dilation at short distance.) For offsets u_1, u_2 in the window and torus exponents m, m' as above there are C and $r_0 > 0$ such that $|S_{\mathbb{T}_m}^{(u_1)}(\vec{f}) - S_{\mathbb{T}_{m'}}^{(u_2)}(\vec{f}(\cdot/\rho))| \leq C d^{4-p} \prod_i \|f_i\|_\infty$, with $\rho := e^{(u_1-u_2)/b}$, for $\mathbf{p} \in \{2, 3\}$, $d \leq r_0$ and every $\mathbf{p}$ -tuple $\vec{f}$ of C^2 functions of the class of Corollary 1.2 at scale d . (4) (Injectivity.) For $u_1 \neq u_2$ in the window, $S_{\mathbb{T}_m}^{(u_1)} \neq S_{\mathbb{T}_m}^{(u_2)}$ for every such m —their two-point functions at short distance already differ—and the sets of infinite-volume limit points of the tuned families with offsets u_1 and u_2 are disjoint. Not asserted: a dilation relating whole torus theories, or the sets of infinite-volume limits, at two offsets; a rate in (2); the value of b ; anything about long distances. This clause is proved in the companion account as an implication from clauses (iii), (iv) in every volume, (v) and Corollary 1.2 (Section 7).

(vi) Rotation invariance under a flux hypothesis. ($N = 2$; in the regime of (iii); for limits of (ii- $\mathbb{R}^4$) only.) There is a threshold $\gamma_{vi} \in (0, \gamma_J]$, chosen last, such that the following holds for every $g \leq \gamma_{vi}$. For a real antisymmetric 4×4 matrix $\omega \neq 0$ let $\xi(x) := \omega x$ be the Killing field of the rotations that it generates. Let $\chi_{\mathcal{R}}(x) := \chi_1(|x|/\mathcal{R})$ be the radial cut-off of Section 8, equal to 1 for $|x| \leq \frac{5}{4}\mathcal{R}$ and to 0 for $|x| \geq \frac{7}{4}\mathcal{R}$, and let $\mathfrak{B}_{\chi_{\mathcal{R}}}^{\text{sh}}$ be the shell functional (8.7) of the exact lattice rotational Ward identity of Section 8: an explicit gauge-invariant sum, over the plaquettes near the shell $\{\frac{5}{4}\mathcal{R} \leq |x| \leq \frac{7}{4}\mathcal{R}\}$ on which $\chi_{\mathcal{R}}$ varies, of plaquette and link variables weighted by ξ and by lattice derivatives of $\chi_{\mathcal{R}}$. It is the flux of the lattice rotation (angular-momentum) current of the rotations generated by ω through that shell. Let $S_{n;\infty}^T$ be a limit of (ii- $\mathbb{R}^4$) along a sequence $(g_0^{(j)}, K_j, m_j)$, $j \in J$, that satisfies the one-sided lower bound

$$K_j \geq Q(m_j)^j \quad (j \in J), \quad (1.21)$$

with $Q(m)$ the volume factor of (iii)(a); this lower bound on the cutoff is met along every iterated limit in which the cutoff is removed before the volume grows, and it is not needed for the conclusion (Remark 8.6). The hypothesis of this clause is that, for one such ω ,

$$\liminf_{\mathcal{R} \rightarrow \infty} \limsup_{j \rightarrow \infty, j \in J} \left| \langle \mathfrak{B}_{\chi_{\mathcal{R}}}^{\text{sh}}; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_{K_j, m_j}^{\text{conn}} \right| = 0 \quad (\text{H-IR})$$

for every $n \geq 2$ and every n -tuple $\vec{f}$ of real $C_c^\infty(\mathbb{R}^4)$ functions with pairwise disjoint supports: the flux through far spheres vanishes along one sequence of radii $\mathcal{R}_i \rightarrow \infty$, uniformly in the cutoff along the sequence considered. Given the ultraviolet half of the argument of Section 8 (part (A2))

there), (H-IR) is necessary as well as sufficient for the conclusion below (Remark 8.20). A sufficient condition is that the correlations with one insertion of the lattice stress tensor decay faster than $|y|^{-4}$, uniformly in the cutoff. Let $\Theta_{\nu\sigma}$, $\nu, \sigma \in \{0, 1, 2, 3\}$, be the stress-tensor density of the Wilson action, defined in Section 8 by (8.5): a gauge-invariant density built from the plaquette variables, of canonical dimension four, transforming as a symmetric tensor under W_4 , carrying the factor g_0^{-2} and traceless at tree level; write $\Theta_{\nu\sigma}(h)$ for its smearing with a function h . The sufficient condition is

$$\lim_{i \rightarrow \infty} \limsup_{j \rightarrow \infty, j \in J} \sum_{\Delta} \sup_h \left| \langle \Theta_{\nu\sigma}(h); \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_{K_j, m_j}^{\text{conn}} \right| = 0 \quad (\nu, \sigma \in \{0, 1, 2, 3\}) \quad (1.22)$$

for one sequence $\mathcal{R}_i \rightarrow \infty$, every $n \geq 2$ and every $\vec{f}$ as above, the sum running over the unit cubes Δ of the tiling of $\mathbb{R}^4$ by the translates of $[0, 1]^4$ by $\mathbb{Z}^4$ that meet $\{\mathcal{R}_i \leq |y| \leq 2\mathcal{R}_i\}$, and the supremum over the Lipschitz functions h with $\|h\|_{\sharp} \leq 1$ supported in the cube of side 2 concentric with Δ ; the same is required of the two further families of lattice densities of dimension four that occur beside Θ in the shell functional, as displayed in Hypothesis 8.2. In (1.22) the modulus is taken after smearing over a cube, that is, after summing over all plaquette orientations at a site. In particular it suffices that, for some $\delta > 0$, $\sup_{j \in J} |\langle \Theta_{\nu\sigma}(h); \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_{K_j, m_j}^{\text{conn}}| \leq C(\vec{f}) \|h\|_{\sharp} D^{-4-\delta}$ for every Lipschitz h supported in a cube of side 2 at distance $D \geq 1$ from $\bigcup_i \text{supp } f_i$, and likewise for the two further families. A decay slower than $|y|^{-4}$, uniform in the cutoff, would not suffice for the argument of Section 8. For orientation: at tree level this correlation decays like $|y|^{-8}$, and a mass gap uniform in the cutoff would give exponential decay; Theorem 0 proves no decay of it. Theorem 0 asserts nothing about the truth of (H-IR) or of (1.22); (H-IR) is a hypothesis of this clause and of no other.

Assume (H-IR) along this sequence. Then for every $n \geq 2$, every $\Lambda \in \text{O}(4)$ and all real $f_1, \dots, f_n \in C_c^1(\mathbb{R}^4)$ with pairwise disjoint supports,

$$S_{n;\infty}^T(f_1 \circ \Lambda^{-1}, \dots, f_n \circ \Lambda^{-1}) = S_{n;\infty}^T(f_1, \dots, f_n); \quad (1.23)$$

consequently the $S_{n;\infty}^T$ are invariant under $\mathbb{R}^4 \rtimes \text{O}(4)$, the centred Schwinger functions built from them are reflection positive at every hyperplane, and the elements of $\text{O}(4)$ that fix the time vector e_0 act on $\mathcal{H}^{\text{curv}}$ by unitary operators that fix Ω , commute with e^{-tH} , extend the action of W_3 , and depend strongly continuously on the group element.

Corollary 1.2 (Short-distance uniqueness on $\mathbb{R}^4$). ($N = 2$; in the regimes of clauses (iv) and (v).) For a fixed real offset $\hat{w}$, any two infinite-volume limit points of the tuned family of clause (v), along any sequences of cutoffs and volumes, and the torus limits $S^{(\hat{w})}$ on all tori above a floor depending on g , agree on the two-point function of test functions of scale $\mathbf{d}$ of the class of clause (iv) (non-negative, $\text{Lip } f_i \leq \|f_i\|_{\infty}/(\vartheta \mathbf{d})$, disjoint supports in one ball of radius $5\mathbf{d}$) up to $O(\mathbf{d}^2 \|f_1\|_{\infty} \|f_2\|_{\infty})$, and on the three-point function up to an error of order $\mathbf{d}$, for $\mathbf{d}$ below a fixed length depending on g . Since the two-point function is of order $g(\mathbf{d})^4 \|f_1\|_{\infty} \|f_2\|_{\infty}$ by clause (iv), below that length the two- and three-point functions are determined by the renormalised coupling up to these errors. This is proved in the companion account from the volume-transfer bound (1.19), the exact change of units with the flow bounds, and parts (b) and (c) of Theorem 7.2 (Section 7); in this paper it is used only as an input of clause (v-f).

Remark 1.3 (Not asserted). Theorem 0 asserts nothing beyond its displays. In particular it does not assert the following.

- (0) A sign of an individual increment $x_k - x_{k+1}$, or the one-loop value of an individual increment (the cumulative law is (0)(d)); that every integer is realised as a cutoff, or that $K(\cdot, g)$ or the

tuned bare coupling is monotone; regularity of β_{k+1} as a function of the couplings; convergence of g_{K-s} along first passage without passing to subsequences.

- (i) Any statement about the laws of Balaban's block variables; the pointwise upper bound [11, (0.1)].
- (iii) Anything for $n = 1$ or at coinciding points; decay in the separation $\mathbf{d}$; anything for $N \geq 3$.
- (ii) Convergence on $\mathbb{R}^4$ without passing to subsequences of volumes; uniqueness on $\mathbb{R}^4$ of the limit as the cutoff is removed; anything for $n = 1$ or at coinciding points, including a canonical extension of the Schwinger functions across the diagonals; the growth condition of [22] in the number of points, and hence field operators and Wightman functions; the spectral condition beyond $H \geq 0$; uniqueness of Ω ; the cluster property E4; a mass gap; invariance under all rotations (this is clause (vi), under its hypothesis); independence of the regularisation; on $\mathbb{R}^4$, limits for test functions that are only Lipschitz; any identification of $\mathcal{H}^{\text{curv}}$ with a space built from Balaban's block variables.
- (iv) Non-Gaussianity, or interaction in any sense beyond the displays; rotation invariance; a mass gap; anything about block observables or Wilson loops; the two-point law for coinciding, overlapping or signed test functions, for separations larger than $6\mathbf{d}$, for $\vartheta \rightarrow 0$, or for depths outside the range (F-depth), (F-UV); an approach of ζ'_{K-s} to 1 faster than the displayed bound; that the charge read off the two-point function has exactly the one-loop slope (only the running coupling of the construction has it, by (0)(d)); a bound on the large-field contributions uniform in the volume; convergence or uniqueness as $m \rightarrow \infty$; a transfer bound for four or more points; the power $\mathbf{d}^4$ in the transfer bound (1.19), which we expect to be the true one; anything for $N \geq 3$.
- (v) Anything for $\mathbf{p} = 1$, for $\log Z$ itself, or at coinciding points; Wilson loops; anything for $N \geq 3$; uniformity in the size of the torus; uniqueness in the volume (beyond Corollary 1.2); uniqueness of the state on Balaban's block variables; convergence of the effective actions; an exponential rate in any labelling by the bare coupling; a dilation relating whole torus theories, or the sets of infinite-volume limits, at two values of the parameter (at short distance see (v-f)(3)); the value of the number b of (v-f)(2), and a rate in (v-f)(2); the coupling-constant theorem [7, Theorem 2] and its inequality [7, (0.31)], which are not used.
- (vi) That (H-IR) or its sufficient condition holds; within Theorem 0, rotation invariance of limits taken along sequences that violate (1.21) (the floor is removed in the companion account; Remark 8.6); rotation covariance of the remaining tensor components of $F \otimes F$, of correlations of several stress tensors, of block observables or of Wilson loops; anything on a torus; clause (vi) along the tuned family of (v) on $\mathbb{R}^4$. Without (H-IR), Section 8 proves that the two- and three-point functions are rotation invariant at short distance to all orders in the running coupling, up to errors of order $\mathbf{d}^2$ and $\mathbf{d}$ at scale $\mathbf{d}$ (Proposition 8.16); exact invariance fails at every scale on every torus (Proposition 8.19, proved in the companion account).

Throughout, nothing is asserted for $\text{SU}(N)$, $N \geq 3$ (Remark 1.4).

Remark 1.4 (Other gauge groups). (1) Theorem 0 is stated for $\text{SU}(2)$. Clauses (0)(a)–(c) and (i) are written for $\text{SU}(N)$, every $N \geq 2$, in the companion account, except two group-specific estimates inside the bookkeeping of ultraviolet stability at $N \geq 3$ —the form, for general N , of the constants of the stability bounds of (i-b), and the verification that the arguments beneath clauses (0) and (i) that are written for $\text{SU}(2)$ carry over to $\text{SU}(N)$ —which are written in part (item (C15) of Section 2).

(2) Clause (0)(d) and clauses (ii)–(vi) are written for $SU(2)$. Beneath them, three further inputs are needed for $N \geq 3$: the two estimates of (1), completed; the correlation theorem of Section 3 for $SU(N)$, which also carries the two-point law of Sections 5 and 6; and the group-specific step of the uniqueness theorem behind clause (v). We do not expect any of them to present a new difficulty, but we have not carried them out, and nothing in Theorem 0 is asserted for $N \geq 3$.

(3) For a general compact simple gauge group G we expect the same statements to hold with the group-theoretic constants replaced accordingly. In our normalisation the constant $b_0 = (N^2 - 1)/2$ becomes $(\dim G)/2$, up to the normalisation of the trace, and the coefficient $11N/3$ of the running coupling in standard units becomes $\frac{11}{3}C_2(G)$, with $C_2(G)$ the quadratic Casimir of the adjoint representation. For a compact semisimple G , every invariant symmetric bilinear form on the Lie algebra is a sum of multiples of the Killing forms of its simple ideals, so the action carries one coupling for each simple factor, and we expect the same statements with one running coupling for each simple factor. This expectation rests on the observation that the group enters the construction only through its compactness and its Haar measure, through finitely many constants of its Lie algebra, through the definiteness of the invariant inner product on the Lie algebra, and through asymptotic freedom at one loop, that is, the positivity of the one-loop coefficient; all of these hold for every compact simple G . We expect this; we have not carried it out.

(4) Balaban’s papers are written for a general compact semisimple group. In [7, §0] he states: “We assume that G is semisimple and that it is a Lie subgroup of a group of complex unitary matrices”, adding that “a bigger part of our considerations does not depend on the semisimplicity assumption”; his small-field theorem “will be proved in this paper for an arbitrary semisimple compact Lie group $G \subset U(N)$, and for $d = 4$ ” (ibid.). The papers on the complete effective densities and on large fields [9, 10, 11] work with configurations taking values in the same group G . The papers on propagators [1, 2] concern the Gaussian theory of a real vector field, and [3] a Lie subgroup of a unitary group. In the papers [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11] the only statement formulated for $SU(2)$ alone is the announced coupling-constant theorem [7, Theorem 2], of which Balaban remarks that “the restriction to the group $SU(2)$ is superficial and is done only to simplify calculations”; that theorem is not used here.

(5) Abelian factors are excluded: a $U(1)$ factor is not asymptotically free, and the statements are not expected to hold for it in this form.

1.5 Plan of the proof

The clauses are proved in the order (0), (i), (iii), (ii), (iv), (v), (vi); each uses only the clauses before it, except that (0)(d) uses the bound on the coupling increments supplied by the correlation theorem of Section 3, and (ii-T-a) and the identification of limits in (ii-T-b) use clause (v).

Section 2: (0) and (i). Balaban’s ultraviolet stability theorem [11, Theorem 1] is cited by statement, with the representation [9, (2.18)] and the bounds it refers to; its one hypothesis is supplied by (0). The two clauses are proved together by one induction over the renormalisation steps: the increment β_{k+1} is a function of $g_0, \dots, g_k$ alone, independent of the torus and of K ; per-step bounds on it and the one-loop lemma for its Gaussian part β_{k+1}^0 give (FLOW $\pm$); stopping at first passage gives (0)(a)–(c) and the containment of all couplings in $(0, \gamma_H]$. This bookkeeping is proved in full; the inputs of the induction are taken from Balaban’s papers, as described in Subsection 1.1, or are proved in the companion account, except, for $N \geq 3$, the group-specific statements of Remark 1.4 (Section 9). The identity (i-d) holds because the densities are exact push-forwards of the Wilson measure. Clause (0)(d) is proved in full by telescoping the one-loop lemma against the bound on $|\beta_{k+1} - \beta_{k+1}^0|$ supplied by the correlation theorem; hence it holds in the regime of (iii).

Section 3: (iii). The correlation theorem runs Balaban’s induction with complex sources $w \in \mathcal{W}$: the sourced densities keep his format, the Wilson coefficient is shifted exactly by the running shift, the source-dependent vacuum energy is extracted as a sum of localised counterterms holomorphic in the source, and the remaining partition function is bounded above and below with constants free of K , g_0 and the position. We give its statement and the structure of its inductive proof; the induction is written out in the companion account (cited as [T-corr] from Section 3 on); it is proved there as an implication from the inputs beneath clauses (0) and (i) and from finitely many sub-statements, which are proved there as well (Section 9). Clauses (iii)(a) and (iii)(c) are read off from it by Cauchy estimates on the disc of sources. Clause (iii)(b) is proved completely: on a torus divided into cells of side $\ell(n, R, g)$, reflection positivity of the Wilson measure [23] and a chessboard estimate in all four directions bound the sourced partition function by periodised one-cell partition functions, and a re-centred counterterm (the cell average of the periodised bulk vacuum energies) cancels the bulk exactly, so that no covariance of the sourced vacuum energies is needed.

Section 4: (ii). Diagonal extraction from the bounds of (iii) gives the limits; invariance and reflection positivity pass from the lattice [23]; the per-ball bound gives the distribution property and, through an extension lemma for distributions with a separated-ball bound, the regularity condition E0; the reconstruction is that of [21]. Full convergence (ii-T-a) is taken from (v). What is not obtained (the growth condition in the number of points; clustering) is explained there.

Section 5: (iv) on one torus. On the smallest torus of the family the Gaussian part of the step at depth $s(\mathbf{d})$ is extracted exactly and gives the main term of (1.16); a room estimate bounds the remaining small-field terms, the large-field remainder is bounded separately, and the assembly gives $|R| \leq \frac{1}{2}$. Short steps are proved in full; for the long steps we give precise pointers to the companion account, where they are proved: the room estimate, deduced there from the correlation theorem of Section 3 and its inputs and therefore of the same status as that theorem, and the statements on the free kernel without block conditioning in the companion note on the free kernel. The two-sided comparison of the block-conditioned kernel rests on Balaban’s decay estimates for his propagators [1, Proposition 1.2], [2, Propositions 2.5–2.6], used by statement (Sections 5 and 9).

Section 6: (iv) in every volume. The volume-transfer theorem, with complete proofs: a trace inequality for mirror-periodic partition functions, from transfer operators and reflection positivity; a one-sided extensivity of Balaban’s vacuum normalisation constant, with the lemma from which it follows; the transfer theorem (a chessboard estimate on the large torus, the trace inequality and one common holomorphic counterterm reduce the difference of the two-point functions on two tori to a Cauchy estimate on a polydisc, with a constant free of K , g_0 , both volumes and the position); an averaging lemma over translated copies of the test pair with random signs, which gains $\mathbf{d}^4$ against the $\mathbf{d}^{-2}$ lost by the two norms $\|\cdot\|_\#$ and gives (1.19); and the assembly with Section 5, which gives clause (iv) in every volume and in every limit on $\mathbb{R}^4$. Of the large volume only reflection positivity of the bare Wilson measure and the lower stability bound at zero source are used. The new threshold $s_T(g)$ is a lower bound on the depth, of the same kind as those already in (F-depth).

Section 7: (v). The exact pin a_n , the depth-to-depth map $\mathbf{n}_n$, the convergence of the increments, the one-parameter family and its real-analyticity, the exhaustion, the renormalised coordinate and the change of units, with its form for the limit family: statements and proof structure, with exact pointers to the companion account. The clause is proved as an implication from the uniqueness theorem of the companion account; its supporting statements that we do not claim as proved—three written, with independent verification in progress, and two not yet written—are listed in Sections 7 and 9, together with one clause to be added to a definition. Clause (v-f), on two blocking factors, is proved in the companion account from clauses (iii)–(v) and Corollary 1.2; Section 7 outlines the argument.

Section 8: (vi). The exact lattice rotational Ward identity; the classification of the operators

of dimension six that can break $O(4)$ in the relevant channel; bounds, carried through Balaban's flow, on the insertions of these breaking terms; the deduction of (1.23) from (H-IR); and the proof that the sufficient condition (1.22) implies (H-IR). Clause (vi) is conditional on (H-IR). In the deduction, the Ward identity is proved in Section 8 and the classification in the companion account; both channels of the coefficient bound are proved in the companion account as implications of the same kind as clauses (iii)–(v); and the infrared half—the far flux vanishes if and only if the rotation defect does—is proved given the ultraviolet half. The lattice stress tensor, and the place at which an old large-field term is localised at the collar of a large-field region, are fixed by definitions. Without the hypothesis, the same identity, read on one torus with the cut-off at a fixed radius of order one, and the volume transfer of Section 6 give rotation invariance of the short-distance germ of the two- and three-point functions to all orders in the running coupling (Subsection 8.6).

Section 9 describes the parts of the companion account and lists, clause by clause, the statements used without proof—Balaban's, with references, among them those that we use as he asserts them, which are listed with their places in Subsection 9.1, and ours, with the place where their statements and the structure of their proofs are given—the statements whose proofs we defer, and the supporting statements of the uniqueness theorem behind clause (v); it ends with what is not claimed and with an outlook that is not part of Theorem 0.

2 Clauses (0) and (i): the coupling flow and ultraviolet stability

Clause (i) of Theorem 0 is, in the form of Proposition 2.4, Balaban's ultraviolet stability theorem [11, Theorem 1], whose single hypothesis is that every effective coupling lies in a fixed interval $(0, \gamma]$. Clause (0) supplies this hypothesis along the whole trajectory from the bare coupling g_0 to the first-passage cutoff $K = K(g_0, g)$, with two-sided bounds on the flow that are uniform in the cutoff, the volume and the position. The two clauses are the two halves of one induction on the scale index (Subsection 2.4). This section proves in full two things that we add to Balaban's theorems: the bookkeeping of one step, which turns per-step bounds on the coupling increment into the flow bounds, the first-passage arithmetic and the containment of all couplings in $(0, \gamma]$ (Subsection 2.3, complete proofs); and the exact one-loop law (0)(d) for the running coupling of the construction (Subsection 2.5, complete proof). By the *companion account* we mean the detailed exposition of the construction on which this paper rests, together with the complete proofs to which it refers (Section 9 describes its parts); when we use a statement proved there, we give its statement and the structure of its proof.

Hypotheses (H1)–(H6) of Section 1 are in force; (H7) is used only in Subsection 2.5. The statements of this section are written for $SU(N)$, $N \geq 2$, except (0)(d), stated for $N = 2$; at $N \geq 3$ their proofs use the group-specific statements (C15), written in part, and Theorem 0 states clauses (0) and (i) for $N = 2$ only (Remark 1.4). Logarithms are natural. We write $\mathbf{p}$ for the tuple of Balaban's auxiliary parameters, chosen after L in the order of (H3); γ and then $\gamma_H \leq \gamma$ are chosen after them.

2.1 Balaban's renormalisation transformations

For $0 \leq k \leq K$ let $\mathbb{T}^{(k)}$ be the lattice of spacing L^{k-K} on $\mathbb{T}_m$ (so $\mathbb{T}^{(0)}$ is the bare lattice, of spacing $\varepsilon = L^{-K}$), $\mathcal{V}_k$ the gauge configurations on its bonds, dV_k the product of normalised Haar measures and $|\mathbb{T}^{(k)}|$ its number of sites. Balaban's block average $\mathcal{M} : \mathcal{V}_k \rightarrow \mathcal{V}_{k+1}$ assigns to a bond of $\mathbb{T}^{(k+1)}$ a symmetrised average of parallel transports across the two blocks of L^4 sites centred at its end points [7, (0.3)–(0.12)], [3]; it commutes with gauge transformations and lattice symmetries. One

renormalisation transformation is the exact push-forward

$$(\mathbf{T}\rho)(V) := \int dU \delta(\mathcal{M}(U)V^{-1}) \rho(U), \quad V \in \mathcal{V}_{k+1} \quad (2.1)$$

[9, (0.1)]. Inside $\mathbf{T}\rho_k$ Balaban inserts exact decompositions of unity into small-field and large-field parts, with thresholds $\varepsilon_k = g_k p_0(g_k)$, where $p_0(g') = A_0(\log g'^{-2})^{p_0}$ with a constant A_0 and an integer exponent which, following Balaban, is also denoted p_0 , and an axial gauge fixing [7, (0.16)]. On the small-field part the block field V determines a background, the minimal configuration $U_{k+1}(V)$ of the Wilson action under the block constraint (the variational problem of [6]), and the integral over the fluctuation is controlled by the background-field propagators and their random-walk expansions [1, 2, 5] and by cluster expansions [8]. The effective action produced by one small-field step [7, Theorem 3] is $-g_{k+1}^{-2}A(U_{k+1})$ plus a sum of localised terms $E^{(k+1)}(X)$, indexed by the localisation domains $X \in \mathcal{D}_{k+1}$ (connected unions of cubes of side $M = L^{m'}$); each is gauge invariant, analytic on a complex neighbourhood of the configurations on X that are regular in Balaban's sense [7, (1.11)–(1.16)] (bounds on plaquette variables, on the current, and on the constrained minimal configurations built from block averages along towers of subdomains of X), and bounded there by $E_0 e^{-\kappa d_j(X)}$ [7, (1.18)], where $d_j(X)$ is the length, in units of M , of a shortest tree graph meeting every M -cube of X . (M , E_0 , the decay rate κ , A_0 , A_1 , p_0 are among the parameters $\mathbf{p}$.) By gauge invariance and lattice symmetry the only marginal local term is a multiple of the Wilson action [7, §§4–5]; it is absorbed by changing the coupling, $g_k^{-2} = g_{k+1}^{-2} + \beta_{k+1}$ [7, (0.20)], with β_{k+1} the coefficient [7, (1.22)] of Definition 2.6. Field-independent constants go into a vacuum energy.

Large-field regions are suppressed, by a factor of order $e^{-p_0(g_k)}$ per cube, but occur at every scale. Balaban keeps them in an inductive representation [9, (2.18)] of the effective densities $\rho_k := (\mathbf{RT})^k \rho_0$, $\rho_0 := \exp[-g_0^{-2}A(U) - E]$, the normalisation constant E cancelling from every expectation. It is a sum over *histories* — admissible sequences of small-field and large-field domains — of large-field integral operations [9, (2.19)–(2.22)] applied to characteristic functions times the exponential of an effective action of the form [9, (2.23)]

$$A_k(g_k^{-2}, U_k) = -A(g_k^{-2}(\cdot), U_k) + \mathbf{E}_k(U_k) + \mathbf{R}_k(U_k) + \mathbf{B}_k(U_k, \underline{A}) - E_k. \quad (2.2)$$

Here $A(c, U) = \sum_p c(x(p)) [1 - \text{Re tr } U(\partial p)]$ is the Wilson action with plaquette weights c at the barycentres $x(p)$; $g_k^{-2}(\cdot)$ equals g_k^{-2} away from large-field regions [9, (2.24)]; $\mathbf{E}_k$ is the regular part, renormalised and localised as in [9, (2.25)–(2.29)]; $\mathbf{R}_k$ collects the terms created by large-field operations, bounded by $g_j^{\kappa_0} e^{-\kappa d_j(X)}$ [9, (2.30)–(2.32)]; $\mathbf{B}_k$ collects boundary terms near large-field regions, which may depend on the fluctuation fields $\underline{A}$ stored there; E_k is a number. The operation $\mathbf{R}$ [10, 11] acts on a connected large-field component once it lies in a cube of side $100MR_k$, where R_k is the least power of L at least $(\log g_k^{-2})^r$ [9, (2.5)] for an integer $r \geq 2$ fixed first, and has waited a prescribed number of steps without a new large-field region being created in it. It integrates out the block field inside the component and replaces it by a normalised reference law [10, (1.100)–(1.102)]; on components of Balaban's first class, in the classification of [11, §1, p. 378], it removes the gauge-fixing δ -functions, on those of the second class they remain (we call such a component *live* while its block variables are variables of the density); and it expands the result into localised terms with the bounds [11, (1.98)–(1.102)]. The main result of the series is the following.

2.2 Ultraviolet stability

Theorem 2.1 (Ultraviolet stability; Balaban). *Let L be odd and large enough, and $\mathbf{p}$ chosen after L as in (H3). There is $\gamma > 0$, depending only on L , $\mathbf{p}$ and N , such that the following holds. Let*

$K \geq 0$ and $g_0 > 0$; let m be so large that the cubes of Balaban's large-field partitions of the scale- k torus, the largest of which have side $100MR_k$, fit in that torus for every $k \leq K$ (along the trajectory of clause (0) this is guaranteed by (H5), see the proof of Theorem 2.11(c)); let $\rho_k = (\mathbf{RT})^k \rho_0$ on $\mathbb{T}_m$ at spacing L^{-K} , and let $g_0, g_1, \dots$ be the effective couplings of the coupling-constant renormalisation. If $g_k \in (0, \gamma]$ for $0 \leq k \leq K$, then for every $k \leq K$ the density ρ_k has the form, and satisfies all the conditions and bounds, described in [9, §2]. In particular:

(a) $\rho_k dV_k$ has the representation [9, (2.18)]

$$\rho_k dV_k = \sum_h \mathbf{T}_k(h) \left[\chi_{k,h} \exp A_{k,h}(g_k^{-2}, U_k(V_k)) \right] dV_k, \quad (2.3)$$

summed over histories h , with large-field operations $\mathbf{T}_k(h)$ of the form [9, (2.19)–(2.22)], products $\chi_{k,h}$ of characteristic functions, and effective actions $A_{k,h}$ of the form (2.2);

(b) every localised term of $\mathbf{E}_k$ and $\mathbf{R}_k$ created at a level $j \leq k$ is localised in some $X \in \mathcal{D}_j$, depends on the background only through its restriction to X , is gauge invariant, analytic on Balaban's complex neighbourhood of the regular configurations on X [7, (1.11)–(1.16)], [9, (2.27)–(2.28)], and bounded there by a constant times $e^{-\kappa d_j(X)}$, the constant being $g_j^{\kappa_0}$ for the terms of $\mathbf{R}_k$ [9, (2.31)];

(c) the boundary terms and the sums over histories obey the bounds of [9, §2], among them the summed bounds of [9, Theorem 2].

This is [11, Theorem 1]. Its proof is the series [7, 8, 9, 10, 11], which rests on the averaging operations, propagators and minimal configurations of [1, 2, 3, 4, 5, 6]. Some intermediate assertions of that proof are printed without proof; Remark 2.2(2)–(3) names them and the form in which we use the theorem, and Subsection 2.4 says how each of them is treated.

Remark 2.2 (How the theorem is used). (1) Balaban notes that [7, Theorem 2] would remove the hypothesis on the couplings by second-order perturbation theory and that its proof is unpublished [11, p. 355]; we do not use it. The hypothesis is supplied by clause (0) (Proposition 2.10). Beyond this containment no asymptotic behaviour of the couplings is assumed; our arguments read in addition only the flow bounds over stretches of steps. (2) A few intermediate assertions of the series are made without proof, notably [9, pp. 276–278] and [11, pp. 379, 390]; the printed proof of the summed bounds [9, Theorem 2] uses two properties that fail on histories created by $\mathbf{R}$ (a separation granted only to domains determined by the renormalisation transformation, and a regularity of the block field on the shells between consecutive small-field domains); some printed displays must be read in the form that Balaban's definitions force (for instance the sign of a pair of boundary terms in [9, (3.15)], on p. 269, which we correct; this affects no estimate since the pair is bounded in modulus only); and the inclusion of regularity classes across cube sizes [8, pp. 9, 15], also asserted on [11, p. 375], cannot be used as printed (under Balaban's own convention of measuring a collar in the region's own cubes, the sentence of [8, p. 15] fails as a set inclusion whenever his enlargement constant is at most $1/4$); it is not used, the class of group (B) of Subsection 2.4 taking its place. Another sentence of [11, p. 375], that the old terms beside a new large-field region are analytic on the new class ("it is simple to see for all the old terms"), is used as he asserts it, read on that class (item (Pr2) of Subsection 9.1). None of these is used as a theorem; Subsection 2.4 says for each how it is treated. (3) The pointwise two-sided display [9, (2.50)] = [11, (0.1)] is not used: its upper member is false as printed at every scale at which a non-zero term of (2.3) carries a live second-class component (Proposition 2.5). Only the integrated bound (i-b) is asserted.

Remark 2.3 (The construction). The construction is Balaban's, with the choices he leaves open fixed once and for all (centring of partitions, cutoff functions, coupling profile, plaquette weights at barycentres) and three changes. (1) Wherever Balaban writes R_j , the construction reads $\check{R}_j$, the least power of L at least $(\log \check{x}_j)^r$, $\check{x}_j := \min_{i \leq j} g_i^{-2}$, because no per-step sign of $g_{j+1}^{-2} - g_j^{-2}$ is known. (2) $\mathbf{R}$ waits $\check{R}_k$ steps (the *waiting depth*), and a maturity condition on nested components is added. (3) At a step that creates a large-field region, Balaban's single-threshold decomposition is refined by two further thresholds, constant multiples of $\delta_k = (A_1/A_0)\varepsilon_k$ (one on the collar bonds of the new region, one cube-wise on the cubes of side $LM\check{R}_{k+1}$ of his domain Ω_{k+1} [9, (3.5)] less one layer of such cubes). This avoids two properties needed at a single threshold and not proved: the bound [9, (2.42)] on boxes at the rim of the integration domain, and a contractivity of the conditional-mean operator. The attachment at barycentres and the fixed coupling profile change no density, total or coupling; change (1) is Balaban's choice whenever the inverse couplings are non-increasing, and with it the sentences of [10] that use the decrease of cube sizes hold as printed at every level at which $\mathbf{R}$ acts (both proved in the companion account). That his one-step analysis applies after change (3), with constants of the same dependences and new conditions only of the kind of floors on parameters and ceilings on γ , is proved in the companion account from the inputs of Subsection 2.4.

Proposition 2.4 (Clause (i) of Theorem 0). *Assume (H1)–(H6) and let $K = K(g_0, g)$. For every $k \leq K$:*

(i-a) ρ_k has the representation (2.3) with the properties (a)–(c) of Theorem 2.1, analyticity being asserted on Balaban's domains read hereditarily (his regularity conditions on constrained minimal configurations imposed at every member of the towers of subdomains of X , under both of his rules for forming them); these domains lie in the class of configurations regular with one power of the scale ratio, which the induction carries. The exceptions to Theorem 2.1(b) are the boundary terms whose domain contains a live second-class region; let Z be the union of these regions. Those created by $\mathbf{R}$ beside no older live second-class region obey [11, (1.99)]

$$|\mathbf{B}'^{(j)}(X, (\mathbf{U}, \mathbf{J}))| \leq O(1) c_1 e^{-(1+\frac{1}{2}\beta_s)\kappa d_{j,Z}(X)}, \quad c_1 \in \{e^{-p_0(g_j)}, \alpha^{1/3}\}, \quad (2.4)$$

in the relative length $d_{j,Z}(X)$ of [11, (1.67)], with α a parameter of [11] and β_s Balaban's enlargement constant. For the others no bound is printed; they obey a bound of the same kind decaying in a tree length $d^\theta(X) \leq d_{j,Z}(X)$ (measured against enlargements of the live regions, defined in the companion account), and no lower bound of $d^\theta(X)$ by $d_{j,Z}(X)$ is asserted. The constants depend only on $L, \mathbf{p}, \gamma, N$, and on the levels only through the running couplings, with one exception: a ratio $\max\{1, (x_j/x_n)^2\}$, $j \leq n$, enters the summed bounds, and $g_j^{\kappa_0} \max\{1, (x_j/x_n)^2\} \leq 4g_j^{\kappa_0-4} g_n^4$ because $x_n \leq x_j + c_5 \leq 2x_j$ (flow bounds of clause (0), and $x_j > c_5$ by Proposition 2.10). No constant depends on K, m, g_0 or the position.

(i-b) There are $E_- = E_-(L, \mathbf{p}, N)$ and $E_+ \leq C(L, \mathbf{p}, \gamma, N)(1 + \log g_k^{-2})$ such that, with χ_k the indicator that no large-field region is present,

$$\chi_k e^{-g_k^{-2} A(U_k(V_k)) - E_- |\mathbb{T}^{(k)}|} dV_k \leq \rho_k dV_k \quad \text{on real fields,} \quad \int \rho_k dV_k \leq e^{E_+ |\mathbb{T}^{(k)}|}. \quad (2.5)$$

(i-c) $\int \rho_k dV_k = \int \rho_0 dU$.

(i-d) (Exact insertion.) Let $0 \leq s \leq K$, let F be a bounded Borel function of the block field V_{K-s} , and write $\rho_{K-s} dV_{K-s} = \sum_h \tau_h$ for the decomposition (2.3) into non-negative measures. Then,

exactly,

$$\langle F(\mathcal{M}^{K-s}U) \rangle_K = \frac{\sum_h \int F^{(h)} d\tau_h}{\sum_h \int d\tau_h}, \quad F^{(h)} = \mathbb{E}[F(\mathcal{M}^{K-s}U) | h, V_{K-s}], \quad |F^{(h)}| \leq \sup |F|, \quad (2.6)$$

where $\langle \cdot \rangle_K$ is the expectation in $\rho_0 dU / \int \rho_0 dU$ and the conditional expectation refers to Balaban's procedure run as a Markov chain from that measure, every outcome of every decomposition of unity being kept in the history. For curvature sources $w = \sum_i z_i f_i$ in the disc of Section 3, the sourced densities ρ_k^w keep this format, with the running shift described there, and $\int \rho_k^w dV_k = \int \rho_0^w dU$.

Status. (i-a) is Theorem 2.1, that is [11, Theorem 1], cited by statement and applied to the construction of Remark 2.3 with its hypothesis supplied by clause (0); that Balaban's argument applies to this construction in the hereditary reading, the summed bounds on every history, and the tree-length bound are proved in the companion account from the inputs of Subsection 2.4; among these inputs, assertions printed by Balaban without a proof covering our use are used as he asserts them, and the statements of ours are proved in the companion account, except, for $N \geq 3$ only, the group-specific statements of (C15), which we state there and whose proofs we defer. (i-b), (i-c) and the identity (i-d) are proved in the companion account from (i-a), the flow bounds of clause (0), the vacuum-energy computation below and the positivity dichotomy listed in group (B) of Subsection 2.4; we give the structure of the proofs.

Structure of the proofs. (i-b): the vacuum energy is made explicit, step by step (normalisation numbers built from $\log g_j$ and the gauge-fixing integral, whose logarithm is $(N^2 - 1) \log g_j + O(1)$, and unit-configuration values of local terms). For the history without large fields, $-E_k = \mathfrak{l}_k - \mathfrak{r}_k$ with $\mathfrak{l}_k = \frac{3}{2}(N^2 - 1)(L^4 - 1) \sum_{j=k}^{K-1} (\log g_j^{-2}) |\mathbb{T}^{(j+1)}| \geq 0$ and $|\mathfrak{r}_k| \leq C |\mathbb{T}^{(k)}|$, using $|x_{j+1} - x_j| \leq B_\sharp$ (Subsection 2.3). The lower member follows from that history, discarding $\mathfrak{l}_k$ and summing [9, (2.45)–(2.46)]; the upper member follows after integration over the live components from the majorants [11, (1.79)], [9, (2.43)–(2.44), (2.48)] and a summation over large-field histories of the type [9, (2.46)], the factors $e^{-p_0(g_j)}$ per cube dominating the logarithmic charges. (i-c): $\mathbf{T}$ preserves mass (Fubini, and every fibre of $\mathcal{M}$ has unit mass), as do decompositions of unity; $\mathbf{R}$ replaces a term τ by $m_\tau \mathfrak{s}$, with m_τ the integral of τ over the block field of the component and $\mathfrak{s}$ the normalised reference law; removing gauge-fixing δ -functions preserves integrals of gauge-invariant functions [11, p. 378], and so does dropping the characteristic function of [10, (1.89)] in the form used. The inputs are the non-negativity and finiteness of every term and of the reference normalisations. (i-d): every operation of the procedure is linear, positive, unital and mass-preserving on the pairs (measure with $F \circ \mathcal{M}^{K-s}$ inserted, measure without), and the dressing of a piece is the conditional expectation of $F(\mathcal{M}^{K-s}U)$ given the history and the current block field; at the close of step $K - s$ the pieces summed over histories are Balaban's terms τ_h .

Proposition 2.5 (The pointwise upper bound fails as printed). *Let $k \leq K$. Suppose every term τ_h of (2.3) is a non-negative measure on real block fields and some $\tau_{h_1} \neq 0$ carries a live second-class component, so that it is concentrated on the dV_k -null set $\mathcal{N}$ on which its gauge-fixing δ -functions fix certain block variables. Then $\rho_k dV_k$ is not absolutely continuous with respect to dV_k ; in particular $\rho_k dV_k \leq e^{E|\mathbb{T}^{(k)}|} dV_k$ fails for every $E \in \mathbb{R}$.*

Proof. $(\rho_k dV_k)(\mathcal{N}) \geq \tau_{h_1}(\mathcal{N}) > 0$, whereas $\int_{\mathcal{N}} f dV_k = 0$ for every measurable $f \geq 0$. $\square$

2.3 The coupling flow

Definition 2.6 (Coupling increment). Suppose steps $1, \dots, k$ have been performed along $g_0, \dots, g_k$. The complete regular part of step $k + 1$ is the sum of (a) the Gaussian normaliser $\log Z^{(k)}(U) -$

$\log Z^{(k)}(\mathbf{1})$ of the fluctuation measure of that step [7, (1.4)], [9, (3.28)], and (b) the logarithm of the fluctuation integral (the last logarithm in [9, (3.28)]), built from the effective action of scale k into which the aged large-field remainders of earlier steps are included [9, p. 270]; both on the whole lattice $\mathbb{Z}^4$ in scale- $(k+1)$ units. Let $\Pi_{\mu\nu}(u)$, $u \in \mathbb{Z}^4$, be the two-point tensor of its second variation at the trivial background, read through the linearisation at $V = \mathbf{1}$ of the map from block fields to minimal configurations. The *increment* is

$$\beta_{k+1} := \sum_{u \in \mathbb{Z}^4} \Pi_{\mu\nu}(u) u_\mu u_\nu \quad (\mu \neq \nu), \quad (2.7)$$

which is [7, (1.22)] and equals [9, (3.61)]. The same coefficient of piece (a) alone is β_{k+1}^0 ; it involves no coupling. A cluster of the expansion of piece (b) is *marked* if it contains an activity descending from a large-field remainder; the marked clusters contribute $\beta_{k+1}^{\mathfrak{R}}$, the rest together with piece (a) contribute $\beta_{k+1}^{\mathfrak{A}}$, and $\beta_{k+1} = \beta_{k+1}^{\mathfrak{A}} + \beta_{k+1}^{\mathfrak{R}}$.

The running inverse couplings are

$$x_0 := g_0^{-2}, \quad x_{k+1} := x_k - \beta_{k+1}, \quad g_k := x_k^{-1/2} \quad (x_k > 0), \quad (2.8)$$

Balaban's orientation [7, (0.18), (0.20)]; the recursion is run forward from g_0 , without declaring any lattice, as long as the induction of Subsection 2.4 defines the next increment.

Lemma 2.7 (Existence and independence of the increment). *Let $\kappa(4)$, K_0 depend only on the dimension, with $\sum_{X \ni z} e^{-\kappa(4)d_j(X)} \leq K_0$ on $\mathbb{Z}^4$ and on every torus [7, (0.26)]. Assume that, on the tori of every admissible side, the terms of the whole-torus fluctuation integral of step $k+1$, and each of their two sub-families, are analytic on a polydisc of radius $a > 0$ about the trivial background, real at real points, gauge invariant, and bounded there by $E^{\mathfrak{A}} e^{-\kappa d_{k+1}(X)}$ with $\kappa \geq 2\kappa(4)$, where a , $E^{\mathfrak{A}}$, κ do not depend on the side, and that their totals are covariant under the lattice symmetries [9, (3.58)]; and that the terms on domains inside a cube of side at most a quarter of the side of the torus coincide with the terms of one family on $\mathbb{Z}^4$. Then (2.7) converges absolutely, Π is the pointwise limit of the torus tensors and decays exponentially, β_{k+1} is real and independent of $\mu \neq \nu$, $\beta_{k+1} = \beta_{k+1}^{\mathfrak{A}} + \beta_{k+1}^{\mathfrak{R}}$, and β_{k+1} is a function of $g_0, \dots, g_k$ and $\mathfrak{p}$ alone; in particular it depends neither on K , nor on m , nor on the torus.*

Proved in the companion account (statement and structure given here). The Hessian at the trivial background of an analytic gauge-invariant function annihilates gauge directions [7, (4.13)–(4.15)]; the linearised minimal configuration decays exponentially by the random-walk expansion [5], uniformly in the torus; a Cauchy estimate and the polymer sum dominate the series; dominated convergence passes from tori to $\mathbb{Z}^4$, wrapping domains being long; and [7, §5] applies to the limit tensor. No rate of the torus limit is asserted or needed.

Per-step bounds. Three bounds on the increments enter.

(P1) (One-loop lemma.) With $c_0 := (11N^2/12\pi^2) \log L$ there is $c_2 = c_2(L, N) \geq 0$ such that for every $k \geq 0$

$$\left| \sum_{l=1}^k \beta_l^0 - k c_0 \right| \leq c_2, \quad \text{hence} \quad \left| \sum_{l=i+1}^j \beta_l^0 - (j-i) c_0 \right| \leq 2c_2 \quad (0 \leq i < j). \quad (2.9)$$

(P2) (Unmarked part.) There are C_{rem} , depending only on L, N and the parameters chosen before Balaban's small-field parameter ε_1^I , and C_β , depending only on L, N and the parameters chosen before γ , with $|\beta_l^{\mathfrak{A}} - \beta_l^0| \leq C_{\text{rem}}\varepsilon_1^I$ and $|\beta_l^{\mathfrak{A}}| \leq C_\beta$ for every $l \geq 1$ at which $g_0, \dots, g_{l-1} \in (0, \gamma]$, clause (i) holds below level l , the unmarked whole-lattice terms of step l satisfy the hypotheses of Lemma 2.7, and their non-Gaussian part obeys [8, (2.41)].

(P3) (Marked part.) Let $\kappa_0 \geq 17$ be Balaban's exponent in [9, (2.31)] (a lower bound only). There is C' , depending only on L, N, κ_0 and the parameters chosen before γ , with $|\beta_{k+1}^{\mathfrak{A}}| \leq C'g_k^{\kappa_0-7}$ whenever $g_0, \dots, g_k \in (0, \gamma]$, clause (i) holds at the levels at most k , and (2.13) below holds on $[0, k]$.

(P1) and (P2) are proved in the companion account; (P3) is proved there from the inputs of Subsection 2.4. Structure: (P1) The Gaussian integrals of steps $1, \dots, k$ compose, since the block averages compose to $\mathcal{M}^k$, and the coefficient (2.7) is linear and unchanged when a functional of a finer block field is read through the minimal configuration of a coarser one; so $\sum_{l \leq k} \beta_l^0 = B_k$, the coefficient of the composite Gaussian normaliser. With H the Wilson Hessian in background Feynman gauge, Δ_{FP} the Faddeev–Popov operator and $t := L^{-k}$, let B_k^{ref} be the coefficient of $-\frac{1}{2} \log \det(H + t^2) + \log \det(\Delta_{\text{FP}} + t^2)$ minus its value at the trivial background; then $|B_k - B_k^{\text{ref}}| \leq \frac{1}{2} C_{\text{cmp}}(L, N)$ and $|B_k^{\text{ref}} - (11C_{\mathfrak{g}}/24\pi^2) \log(1/t)| \leq C_{\text{det}}(N)$, with $C_{\mathfrak{g}} = 2N^2$ (Step 8 of the proof of Theorem 2.14) and constants free of k ; so $c_2 := C_{\text{det}} + \frac{1}{2} C_{\text{cmp}}$. No sign of an individual β_l^0 is asserted. (P2) Balaban's proof of [8, Lemma 3], at the cutoff radius $\delta_{l-1}/g_{l-1} = A_1(\log g_{l-1}^{-2})^{p_0}$ of [9], bounds the coefficient of the non-Gaussian part through [8, (2.41)]; in the norms of the analyticity domains the scale factors of the response and of the radius cancel; the second bound follows from the decay [7, (5.10)]. (P3) An induction on the level gives $|E^{(j), \mathfrak{A}}(X)| \leq C_{\mathfrak{A}} g_{j-1}^{\kappa_0-5} e^{-\kappa d_j(X)}$ for the marked whole-lattice terms of step j , with $C_{\mathfrak{A}}$ depending only on L and the parameters chosen before γ : freshly marked activities are bounded through [9, (2.31)], at a cost of five powers of the coupling; inherited ones contract by the renormalisation gain of Balaban's irrelevant terms, read at a Hölder exponent $\beta_H \in (0, 1)$ of [7, (3.27), (3.31)–(3.32), (4.18)], the sum over levels being $\sum_{n \geq 0} (1+n)^{5/2} L^{-\beta_H n} < \infty$. A Cauchy estimate in two directions passes to the coefficient.

Flow constants. The order of (H3) imposes the ceiling $\varepsilon_1^I \leq c_0/(2C_{\text{rem}})$, on a parameter chosen after C_{rem} , so that $|\beta_l^{\mathfrak{A}} - \beta_l^0| \leq \frac{1}{2}c_0$; and

$$\bar{\varsigma} := C'\gamma^{\kappa_0-8} \leq \min\{\frac{1}{4}c_0, 1\}, \quad \lambda_{\sharp} := \frac{1}{2}c_0 - \bar{\varsigma} \geq \frac{1}{4}c_0, \quad B_{\sharp} := C_\beta + \bar{\varsigma} \leq \bar{B} := C_\beta + 1, \quad c_5 := 2c_2, \quad (2.10)$$

the first condition, together with $\gamma \leq 1$, being a ceiling on γ . Since $g_k \leq \gamma \leq 1$, (P3) gives $|\beta_{k+1}^{\mathfrak{A}}| \leq \bar{\varsigma}$. The triple $(\frac{1}{4}c_0, c_5, \bar{B})$ is fixed before γ , and every later ceiling on γ involving a flow constant involves only this triple, so no circularity arises. For $0 < g' \leq 1$ put $g_-(g') := (g'^{-2} + B_{\sharp})^{-1/2}$, so that $g_-(g)$ is its value at g , and

$$\begin{aligned} K(g_0, g) &:= \min\{k \geq 0 : x_k \leq g^{-2} + B_{\sharp}\}, \\ R(g') &:= \min\{L^j : j \geq 0, L^j \geq (\log g'^{-2})^r\}, \\ m_{\mathbb{T}}(g') &:= 11 + m' + \log_L R(g'). \end{aligned} \quad (2.11)$$

Then (H6), $0 < g_0 \leq g_-(g)$, is equivalent to $x_0 \geq g^{-2} + B_{\sharp}$; (H5) is $m \geq m_{\mathbb{T}}(g_-(g))$; and the last threshold of (H3) obeys the terminal ceiling

$$\gamma_H \leq (\gamma^{-2} + c_5 + 1)^{-1/2}. \quad (2.12)$$

The bookkeeping of one step. The next three statements are proved here in full.

Lemma 2.8 (Two-sided flow bounds). *Let $n \geq 1$, $x_0 \in \mathbb{R}$, and $\beta_l^0, \beta_l^{\mathfrak{A}}, \beta_l^{\mathfrak{R}} \in \mathbb{R}$ for $1 \leq l \leq n$; put $\beta_l := \beta_l^{\mathfrak{A}} + \beta_l^{\mathfrak{R}}$ and $x_l := x_{l-1} - \beta_l$. Let $c_0 > 0$, $c_2, C_\beta \geq 0$, $0 \leq \bar{c} \leq \min\{\frac{1}{4}c_0, 1\}$, and assume, for $0 \leq i \leq j \leq n$ and $1 \leq l \leq n$,*

$$\sum_{l=i+1}^j \beta_l^0 \geq (j-i)c_0 - 2c_2, \quad |\beta_l^{\mathfrak{A}} - \beta_l^0| \leq \frac{1}{2}c_0, \quad |\beta_l^{\mathfrak{A}}| \leq C_\beta, \quad |\beta_l^{\mathfrak{R}}| \leq \bar{c}.$$

Then, with $\lambda_\sharp, B_\sharp, c_5$ as in (2.10),

$$\lambda_\sharp(j-i) - c_5 \leq x_i - x_j \leq B_\sharp(j-i), \quad 0 \leq i \leq j \leq n. \quad (2.13)$$

Proof. $x_i - x_j = \sum_{l=i+1}^j \beta_l$, and $\beta_l \leq |\beta_l^{\mathfrak{A}}| + |\beta_l^{\mathfrak{R}}| \leq B_\sharp$ gives the upper bound. Writing $\beta_l = \beta_l^0 + (\beta_l^{\mathfrak{A}} - \beta_l^0) + \beta_l^{\mathfrak{R}} \geq \beta_l^0 - \frac{1}{2}c_0 - \bar{c}$ and summing, $x_i - x_j \geq (j-i)c_0 - 2c_2 - (j-i)(\frac{1}{2}c_0 + \bar{c}) = \lambda_\sharp(j-i) - c_5$. Finally $\bar{c} \leq \frac{1}{4}c_0$ gives $\lambda_\sharp \geq \frac{1}{4}c_0$. $\square$

No per-step sign of $x_{l-1} - x_l$ is asserted, and none is used anywhere.

Lemma 2.9 (First passage). *Let $\lambda_\sharp, B_\sharp > 0$, $c_5 \geq 0$, $g > 0$, $t \geq g^{-2} + B_\sharp$, and let $(x_k)_{0 \leq k < n_{\max}}$, $n_{\max} \in \{1, 2, \dots\} \cup \{\infty\}$, be real with $x_0 = t$ and such that, for every $n \geq 1$ for which $x_0, \dots, x_{n-1}$ are defined and exceed g^{-2} , x_n is defined and (2.13) holds on $[0, n]$. Then*

$$K := \min\{k : x_k \text{ is defined and } x_k \leq g^{-2} + B_\sharp\}$$

is a well-defined non-negative integer, and: (a) $x_k > g^{-2} + B_\sharp$ for $k < K$ and $g^{-2} < x_K \leq g^{-2} + B_\sharp$; (b) (2.13) holds for $0 \leq i \leq j \leq K$; (c) $(t - g^{-2})/B_\sharp - 1 \leq K \leq 1 + (t - g^{-2} - B_\sharp + c_5)/\lambda_\sharp$.

Proof. Let $n_0 \geq 1$ be an integer with $n_0 > (t - g^{-2} - B_\sharp + c_5)/\lambda_\sharp$, and suppose no defined x_k , $k \leq n_0$, satisfies $x_k \leq g^{-2} + B_\sharp$. By induction on n , $x_0, \dots, x_{n_0}$ are then defined and exceed $g^{-2} + B_\sharp$ (if $x_0, \dots, x_{n-1}$ do, the hypothesis defines x_n and the supposition bounds it). But (2.13) with $(i, j) = (0, n_0)$ gives $x_{n_0} \leq t - \lambda_\sharp n_0 + c_5 < g^{-2} + B_\sharp$, a contradiction. So K exists, $K \leq n_0$, and $x_0, \dots, x_K$ are defined, the defined indices forming an initial segment. (a) Minimality gives the first claim. If $K = 0$, $x_K = t = g^{-2} + B_\sharp$; if $K \geq 1$, $x_0, \dots, x_{K-1}$ exceed g^{-2} and (2.13) with $(i, j) = (K-1, K)$ gives $x_K \geq x_{K-1} - B_\sharp > g^{-2}$. (b) For $K \geq 1$ this is the hypothesis with $n = K$; for $K = 0$ it reads $-c_5 \leq 0 \leq 0$. (c) By (b) with $(0, K)$, $t - g^{-2} - B_\sharp \leq t - x_K \leq B_\sharp K$. If $K \geq 1$, (a) and (b) with $(0, K-1)$ give $g^{-2} + B_\sharp < x_{K-1} \leq t - \lambda_\sharp(K-1) + c_5$, which rearranges to the upper bound; if $K = 0$ the right side is at least 1. $\square$

Proposition 2.10 (Containment: Balaban's hypothesis). *Assume (H4), (2.12), and $x_k > g^{-2}$ for all $k \leq K$. Then $x_k > g^{-2} \geq \gamma_H^{-2} \geq \gamma^{-2} + c_5 + 1$ for $k \leq K$. Hence $0 < g_k < \gamma$, the hypothesis of Theorem 2.1 holds along the whole trajectory, and every lower bound $x_k \geq x_*$ of the construction with $x_* \leq \gamma^{-2} + c_5 + 1$ holds at every level $k \leq K$.*

Proof. (H4) gives $g^{-2} \geq \gamma_H^{-2}$ and (2.12) gives $\gamma_H^{-2} \geq \gamma^{-2} + c_5 + 1 > \gamma^{-2}$. $\square$

Theorem 2.11 (Clause (0)(a)–(c) of Theorem 0). *Assume (H1)–(H6). Then:*

(a) $K = K(g_0, g)$ is a well-defined non-negative integer with

$$\frac{g_0^{-2} - g^{-2}}{B_\sharp} - 1 \leq K \leq 1 + \frac{g_0^{-2} - g^{-2} - B_\sharp + c_5}{\lambda_\sharp}; \quad (2.14)$$

in particular $K \rightarrow \infty$ as $g_0 \downarrow 0$, and the realised cutoffs are unbounded;

- (b) $g_k \in (0, g) \subset (0, \gamma_H]$ for every $k \leq K$, and $g_K \in [g_-(g), g)$;
- (c) $\lambda_\sharp(j-i) - c_5 \leq x_i - x_j \leq B_\sharp(j-i)$ for $0 \leq i \leq j \leq K$, and the K renormalisation steps on $\mathbb{T}_m$ at spacing L^{-K} produce exactly the couplings $g_0, \dots, g_K$.

Part (d) of clause (0) is Theorem 2.14.

Status. The deduction of (a)–(c) from (P1)–(P3) at every level up to K and from Lemma 2.7 is complete and given below; these inputs, and the identification of the torus steps with the whole-lattice increments in (c), are established jointly with clause (i) by the induction of Subsection 2.4, with the inputs and their status listed there.

Proof. The induction yields, for every $n \geq 1$ such that $x_i \geq \gamma^{-2}$ for all $i < n$: β_n is defined and is a function of $g_0, \dots, g_{n-1}$ and $\mathbf{p}$ (Lemma 2.7); and (P1)–(P3) hold at every $l \leq n$, so that (2.13) holds on $[0, n]$ by Lemma 2.8. By (H4) and (2.12), $g^{-2} > \gamma^{-2}$; by (H6), $t := g_0^{-2} \geq g^{-2} + B_\sharp$. So Lemma 2.9 applies: it gives (a), the flow bounds in (c), and $g^{-2} < x_k$ for $k \leq K$ with $x_K \leq g_-(g)^{-2}$, which with $g \leq \gamma_H$ is (b). The lower bound in (2.14) tends to infinity as $g_0 \downarrow 0$.

For the last assertion of (c): by Lemma 2.7 the sequence (x_k) is defined before any torus or cutoff, and K afterwards. On $\mathbb{T}_m$ at spacing L^{-K} the localised terms on non-wrapping domains coincide with those of $\mathbb{Z}^4$ (the torus-independence of Subsection 2.4), provided the torus at each scale $j \leq K$ holds the coarsest cubes of the construction. This is where (H5) enters: under a ceiling on γ the flow bounds give $\check{R}_j/\check{R}_{j+1} \in \{1, L\}$ (the cube-size lemma of the companion account, proved there), so $\check{R}_j \leq L^{K-j}\check{R}_K$; and $\check{x}_K \leq x_K \leq g_-(g)^{-2}$ gives $\check{R}_K \leq R(g_-(g))$. Hence $L^{11}M\check{R}_j \leq L^{m\mathbb{T}(g_-(g))+K-j} \leq L^{m+K-j}$, and the cubes of side $100M\check{R}_j$ fit in the scale- j torus, of side $2L^{m+K-j}$. So the K steps subtract exactly the increments of the sequence. No upper bound on K , m or L is imposed. $\square$

Corollary 2.12 (Coupling windows at fixed depth). *Let $g_0^{(i)} \downarrow 0$ and $K_i := K(g_0^{(i)}, g)$. Whenever $K_i \geq s$, $x_{K_i-s} \in (g^{-2} + \max\{0, \lambda_\sharp s - c_5\}, g^{-2} + B_\sharp(s+1)]$. Along a subsequence $g_{K_i-s} \rightarrow g_{\infty,s}$ for every $s \geq 0$, with $g_{\infty,s}^{-2} \in [g^{-2} + \max\{0, \lambda_\sharp s - c_5\}, g^{-2} + B_\sharp(s+1)]$; in particular $g_{\infty,0} \in [g_-(g), g]$.*

Proof. Theorem 2.11(c) with $(i, j) = (K_i - s, K_i)$, together with $g^{-2} < x_{K_i} \leq g^{-2} + B_\sharp$ and $x_{K_i-s} > g^{-2}$ from (b), gives the window; $K_i \rightarrow \infty$ by (a). The windows are compact subintervals of $(0, \infty)$; a diagonal extraction and the continuity of $x \mapsto x^{-1/2}$ give the rest. $\square$

What clause (0) does not assert. A per-step sign of $x_k - x_{k+1}$ (the lower flow bound holds only over stretches of steps, with the defect c_5); the one-loop value of an individual increment (Theorem 2.14 gives the cumulative law inside the regime of (H7)); that every integer is a cutoff, or that $K(\cdot, g)$ is monotone; continuity of β_{k+1} in $(g_0, \dots, g_k)$; convergence of g_{K-s} as $g_0 \downarrow 0$ without passing to subsequences. For a prescribed K the flow bounds place x_K only in an interval of length $(B_\sharp - \lambda_\sharp)K + c_5$, so they do not produce a bare coupling with $g_K = g$.

2.4 One induction, and the inputs of one step

Suppose $g_0, \dots, g_k$ are defined and lie in $(0, \gamma]$, and clause (i) holds at every level $j \leq k$. Step $k+1$ proceeds as follows. (1) The fluctuation integral at coupling g_k produces the whole-lattice effective action $E^{(k+1)}$. Its unmarked part has Bałaban's inductive properties by his one-step theorem [7, Theorem 3], [8] read in *family form* — for any family with his inductive properties and any coupling in $(0, \gamma]$ the integral produces a family with the same properties and constants, the couplings entering only through a threshold on g_k [8, (2.31)] — and its marked part obeys the bounds behind (P3). (2)

β_{k+1} is read off on $\mathbb{Z}^4$; it involves no object of index $k+1$, no torus and no K , so the induction is well founded. (3) (P1)–(P3) at $l = k+1$ give (2.13) on $[0, k+1]$; then x_{k+1} , $\check{R}_{k+1}$ and the thresholds of step $k+1$ are fixed. (4) If $x_{k+1} \geq \gamma^{-2}$, the operations **T** [9, §3] and **R** [10, 11] of step $k+1$ are performed and clause (i) at level $k+1$ follows as in Balaban’s inductive step, with the inputs below. Along the first-passage trajectory $x_k > g^{-2} > \gamma^{-2}$ (Proposition 2.10), so the containment needed in (1), (3), (4) holds at every step. The family form matters: [7, Theorem 3], read literally, concerns the couplings of Balaban’s recursion for the family without large-field remainders, and the couplings (2.8) agree with those up to index l if and only if $\beta_i^{\mathfrak{R}} = 0$ for all $i \leq l$ (induction on i). Of Balaban’s hypotheses on the couplings only containment is used.

The inputs fall into three groups.

(A) Balaban’s statements, used by statement. Theorem 2.1 and, within its proof: the one-step small-field analysis [7, §§1–5] (Theorem 3 there, the inductive properties (1.7), (1.18), (1.19), the identities (4.13)–(4.15), Lemma 4, the coefficient (1.22) and the analysis of §5); the cluster expansion of [8] with its Lemma 3 and bound (2.41); the step **T** on large-field histories [9, §3] and the identification [9, (3.57)–(3.64)]; the definition of the operation **R** [10, 11]; the averaging operations [3], in particular Propositions 2 and 6 there; the propagators and random-walk expansions [1, 2, 5], in particular [2, Propositions 2.3, 2.6, 2.7]; and the minimal configurations [6], in particular Proposition 9 there; its display (190), derived there from a display (189) that is not computed, is used as he asserts it and is listed in Subsection 9.1. Further displays of the series — the summed bounds [9, Theorem 2], the bounds [11, (1.98)–(1.102)], the normalisation [10, (0.4)] and the identity [10, (1.89)] — are used in forms proved in the companion account (group (B)).

(B) Statements proved in the companion account. Each is proved from the statements of (A) and from those of (C) named with it.

- (P1); the cube-size lemma; the vacuum-energy computation of (i-b); Lemma 2.7 and (P2) under their stated hypotheses, which the induction supplies: nothing from (C).
- The summed bounds [9, Theorem 2] on every history, beside the shells made by **R** included, with modified constants (in the bound [9, (2.44)] for a term created at level j and read at level n , a factor $\max\{1, (x_j/x_n)^2\}$): by a variant of the printed proof avoiding the two properties named in Remark 2.2(2), under hypotheses supplied by the induction; nothing from (C). The normalisation [10, (0.4)], derived in the proof of (i-c). The identity [10, (1.89)] in the form used, at the thresholds of the construction: from the statement that **R** acts only at levels exceeding the waiting depth $\check{R}_k$ (item (C4)).
- The fluctuation operator at a current not tied to the field (a *free current*; analyticity, complex symmetry, exponentially decaying kernel, positivity at real regular configurations with zero current), the reflection covariance of its localised expansion, and the bounds of [8] for five companion operators on one family of cells: from [2, Propositions 2.3, 2.6, 2.7].
- The operator bounds [7, (3.9), (3.14), (3.31)–(3.32)] at a current not tied to the field (the only printed argument for the Laplacian member uses an identity of [6] that fails there; the bound itself is not claimed to be false): from (A), the admissibility of the nested regions produced by **R** (proved there), and (C7).
- The class of configurations regular with one power of the scale ratio, carried along the maximal tower and the tower of [9, (2.13)], for which restriction to finer cubes is an identity, and its

reproduction at each step: from a lemma on uniform one-step increments (part of the previous item) and (C2), (C3), (C12).

- The independence, asserted without proof in [9, p. 278], of the localised terms of step $k + 1$ from the small-field region (their coincidence with the whole-lattice terms): from the items above, the format of the remainders below, the symmetry of the local Gaussian reference covariance at local data (proved there), and (C1), (C2).
- The format of the remainders created by $\mathbf{R}$, asserted in [11, p. 390]: the convergent series [8, (2.13)] with the changes of notation, obeying [11, (1.98)–(1.102)]. Bound half: from the inductive bounds at lower levels, the positivity dichotomy below, the admissibility of the nested regions produced by $\mathbf{R}$ (proved there), and (C5), (C6). Locality half: (C8).
- The positivity dichotomy, replacing the words “obviously positive” of [11]: at every regular real exterior field U , either $\mathbf{T}'_k(X, (U, 0))1 > 0$ for the top-integrated operation [11, (1.71)], or $\mathbf{T}'_k(X, \cdot) \equiv 0$. From clause (i) at level k and the next item.
- The operator theorem at free current, and the statement that the densities are decomposed over one fixed family of cells: from the fluctuation operator above, the statement [3, (139)] (proved in the companion account), the statement that $\mathbf{R}$ acts only at levels exceeding the waiting depth (item (C4)), and (C6), (C10), (C11).
- (P3): from the items above and (C2), (C6), (C9), (C12).
- The one-step theorem in family form, and the torus-independence of local terms (terms on non-wrapping domains are exactly the same functions on every admissible torus and on $\mathbb{Z}^4$): jointly with the induction, from the items above.
- (i-a)–(i-d) at level $k + 1$: from the items above, (C9), (C13), (C14), and, for $N \geq 3$, (C15).

(C) Further statements, by kind. Each statement named below is of one of three kinds: an assertion printed in Bałaban’s papers without a proof covering our use, which we use as he asserts it; a statement of ours that is settled, for the reason given with it; or a statement of ours that we state and whose proof we defer. We describe each precisely enough to locate it. Items (C1), (C3), (C4), and the statements of ours in (C2), (C5) and (C12)–(C14), are of the second kind. At $N = 2$ no statement is of the third kind; for $N \geq 3$ the group-specific statements of (C15) are.

- (C1) The regularity clause of the one-power class at source terms whose localisation domains meet two consecutive layers of the large-field regions. This needs no separate proof: it is a clause of the definition of the analyticity domains, read hereditarily as in (i-a), and under the cumulative reading of these domains used in the companion account, the condition read after the large-field operation is the source term’s own condition on the unitary factor that this operation leaves untouched; at real fields it holds automatically.
- (C2) The sentence of [9, p. 275] that on the difference of two consecutive small-field regions the configuration of the next step coincides with one of the earlier type and has its regularity properties and bounds, used as printed on the intermediate layers. Through it, the widened reading of [7, (1.12)] — on every cube of side at most a fixed multiple of LM in a domain, thin cubes included, a gauge in which the potential and its lattice gradient obey one bound proportional to $LM\alpha_0$ — and its reproduction by each step are proved in the companion account as an implication from [6, Theorem 1] and [4, Proposition 6]; as printed, [7, (1.12)] does not supply this reading.

- (C3) Analytic dependence of the quantities built along the towers at complex representatives of the one-power class. Proved in the companion account, and not an estimate: these finitely many quantities are defined through Bałaban’s minimal configurations, and holomorphy composes under their definition, with [6, Theorem 1, Proposition 9], [3, Proposition 2] and [7, (0.4)–(0.12), (1.8), (1.15)] used by statement.
- (C4) Every component of the large-field region meets the set on which a large-field region was created at that step or earlier, and keeps doing so under the later steps; this gives the strict inequality between the level and the waiting depth $\check{R}_k$ wherever $\mathbf{R}$ acts. This is proved in the companion account, as a combinatorial lemma from Bałaban’s definition of the large-field operation, both for the steps $\mathbf{T}$ and for the steps at which $\mathbf{R}$ replaces treated regions.
- (C5) The supports of the Wilson part of the densities at the regions prepared by the large-field operation, in the notation of (2.4). Proved in the companion account: what the induction reads here is the bookkeeping of supports in one localisation step, which follows from [11, pp. 360–366]—used by statement, except the estimate on p. 360 whose details are omitted there, which we use as he asserts it—under one further lower bound, on the waiting depth $\check{R}_k$, of the kind allowed by (H3).
- (C6) The statement of [11, p. 376] that the resummed boundary term $V(X^\sim, \cdot)$ “satisfies the same bound with a different constant”, used as he asserts it. A weaker bound, by the number of large-field levels present, is proved in the companion account; the printed one would follow from an analyticity radius, free of the coupling, the level and the domain, for the recently created boundary-type constituents, which [7] prints only for terms of energy type.
- (C7) Three sentences, of [3], of one of the papers [4, 5] and of [7] (at (3.26)), stating that certain bounds are obtained “in the same way”.
- (C8) The locality of the terms produced by $\mathbf{R}$ [11, p. 379], with the bookkeeping of new boundary terms and the resummation over hulls described in [11, p. 390].
- (C9) The assertion [9, pp. 276–277] that the analysis of [7, §§3–5] and [8] carries over, with the corresponding changes, to the cover by cubes of the scales of the large-field layers. This includes the two lemmas that [9, p. 276] cites as Lemmas 5 and 6 of the two-part paper [7, 8], whose two parts Bałaban numbers continuously; they are the lemmas on the fluctuation-field action in [8, §1], where the lemmas are numbered from 1.
- (C10) The inequalities (3.35), (3.37) of Bałaban’s paper on propagators in a background field [5] at the two lowest levels, read on a large-field region (they are asserted there for the global field spaces).
- (C11) The admissibility of the prepared large-field data [10, pp. 177–179] in the form used by the operator theorem at free current.
- (C12) The statements of [7] used as printed by the argument producing the one-power class, among them [7, (1.17)] at complex configurations of the class (not refuted; its printed derivation does not apply there) and the second inequality of the regularity condition verified before [7, Lemma 4], obtained there “in the same way” as the first [7, p. 279]; together with the uniqueness and locality of the minimiser that complete the proof of [7, Lemma 4], which are proved in the companion account.

- (C13) The statements of the companion account on the representation and on one step: three statements behind the tree-length decay of boundary terms beside live second-class regions (a resummation bound, its form inside the single induction, a statement on tubes about live regions), one of which rests on expansions of the sites of one step with inputs and outputs not depending on the large-field regions inside the strip; the inheritance statement for the representation; the containment statement used to sum boundary terms over finer cubes; the statements on terms with torus-wrapping supports, on terms created where a large-field region arises, on the resummations and on the bounds for $\mathbf{R}$; the treatment of applications of $\mathbf{R}$ that are stopped before completion, beside which the remark of [10, p. 191] on integrations interrupted by a new large-field region is used as he asserts it; and the statement fixing the single family of densities on which the induction runs, with the exactness of the two-threshold decomposition at the creating step. These are proved in the companion account. For an application of $\mathbf{R}$ stopped before completion this includes the statement that every later configuration lies in the domain of every term stored earlier, also at those collar bonds of the stored domains that have never been averaged; at these bonds the companion account imposes a condition of the same kind as at all other bonds, attached to the finest scale and shrinking at each later application of $\mathbf{R}$ by Balaban's schedule continued to that scale (Section 9).
- (C14) The steps, shared with the proof of the correlation theorem of Section 3, by which clause (i) sums boundary terms beside large-field regions: its resummation lemma and the weight lemma with the membership statement beneath it, which are proved in the companion account; and the statements of [11] taken there as printed, among them the sentence of [11, p. 375] that the old terms beside a new large-field region are analytic on the new class, read on the one-power class of group (B) (the inclusion of regularity classes asserted on the same page is not used; Remark 2.2(2)), and the lower bound $\exp(-O(1)|\Lambda|)$ for the normalising integrals of the large-field operation on a region Λ [11, (1.11)], which we use as he asserts them. The expansions of the fluctuation kernels that these steps use are those of group (A), used by statement.
- (C15) For $N \geq 3$: the reading for $\text{SU}(N)$ of the arguments for clause (i), and the constants of (i-b).

The independence of the cube size M of the constants of the random-walk expansion of the fluctuation kernels on the one family of cells, on which the consistency of the order (H3) rests, is proved in the companion account, for the kernel estimates on the whole lattice and, as an implication from the operator theorem at free current of group (B) and from (C4), on the one family of cells; with this, every choice in the order (H3) can be made (Subsection 1.3). The use of (C14) is not circular: the statements used are uniform in the level, their hypotheses at level k are supplied by (0) and (i) at level k , and the conclusion of the correlation theorem is an input of no step at a level at most K ; the companion account ranks all statements involved in a well-founded order (proved there). No input imposes an upper bound on L , M , K , m , a depth or a number of steps, or a lower bound on a coupling: the conditions on couplings are ceilings on γ , γ_H , chosen last, and every condition on a parameter is a floor or a ceiling attached to the latest parameter it reads, or one end of a window that earlier choices make non-empty. The lower bound on L is one existential threshold $L_0(N)$, never evaluated.

2.5 The exact one-loop law

Clause (0) places the average slope of x_k between $\lambda_{\sharp} \geq \frac{1}{4}c_0$ and $B_{\sharp}$. Inside the regime (H7) of the correlation theorem the average slope is exactly the one-loop value c_0 : over s steps the defect is

$o(s)$, uniformly in the cutoff and the volume. This is clause (0)(d), a statement about the coupling sequence of the construction. Besides clause (0)(a)–(c) its proof uses exactly (P1), in the two-sided form (2.9), and the following bound.

Proposition 2.13 (Closeness of the increment to its Gaussian part). *Let $N = 2$ and assume (H1)–(H6) and (H7), so that $g \leq \gamma_J$; among the ceilings defining γ_J is $\gamma_J \leq \gamma^\circ := (8c_2 + 4)^{-1/2}$. For every $g_0 \in (0, g_-(g)]$ and every $0 \leq k \leq K(g_0, g) - 1$,*

$$|\beta_{k+1} - \beta_{k+1}^0| \leq C_\beta e(g_k) + v_k \leq \lambda := \frac{1}{2}c_0, \quad (2.15)$$

where, for $0 < g' < 1$,

$$e(g') := \frac{4E_* A_1}{\varepsilon_1^1} g' (\log g'^{-2})^{p_0} + E_* \exp\left(-\frac{1}{4}\gamma_2 A_1^2 (\log g'^{-2})^{2p_0}\right), \quad v_k := 2C_\beta C_t C_m y_k^{-(\kappa_0-4)/2},$$

$y_k := \min_{i \leq k} x_i - 2c_2$, $C_m := \frac{5}{2} + \frac{5}{\lambda(\kappa_0-4)}$, A_1 , p_0 are Balaban's small-field constants ($\delta_k/g_k = A_1(\log g_k^{-2})^{p_0}$), ε_1^1 his small-field parameter, κ_0 his exponent in [9, (2.31)], subject to the floor $\kappa_0 \geq 17$ of (P3) (the bound (2.15) and the summations below need only $\kappa_0 \geq 7$), and E_* , γ_2 , C_t constants depending only on $(L, \mathbf{p}, N)$. No constant depends on K , k , m , g_0 or the position.

This bound is part (f) of the correlation theorem of Section 3 (Theorem 3.1), where what it rests on is stated. That theorem runs (2.8) from the same x_0 with the same increments (the coefficient (2.7) of the same complete regular part, with the same Gaussian parts) and stops at the largest K' with $x_j > g^{-2}$ for all $j \leq K'$; by Theorem 2.11(b), $K' \geq K$. Of e and v_k the proof uses only that e is continuous on $(0, 1)$ with $e(g') \rightarrow 0$ as $g' \downarrow 0$, and the explicit form of v_k .

Majorants. For $u \geq 1$ and $s \geq 1$ put

$$\bar{h}_u := (6c_2 + 4 + \lambda u)^{-1/2}, \quad \mathbf{e}_u := \sup_{0 < g' \leq \bar{h}_u} e(g'), \quad \mathbf{v}_u := 2C_\beta C_t C_m (4 + \lambda u)^{-(\kappa_0-4)/2}, \quad (2.16)$$

$$C_v := \sum_{u=1}^{\infty} \mathbf{v}_u, \quad \delta_e(s) := \frac{1}{s} \sum_{u=1}^s \mathbf{e}_u, \quad \Delta(s) := 2c_2 + C_v + C_\beta s \delta_e(s), \quad \Delta(0) := 2c_2. \quad (2.17)$$

$\bar{h}_u \leq \frac{1}{2}$ decreases to 0; $\mathbf{e}_u < \infty$ since e is continuous on $(0, 1)$ and tends to 0 at 0; $C_v < \infty$ since $(\kappa_0 - 4)/2 \geq \frac{3}{2}$. All depend only on $(L, \mathbf{p}, N)$.

Theorem 2.14 (Clause (0)(d): exact one-loop running). *Let $N = 2$ and assume (H1)–(H6) and (H7). For every $g_0 \in (0, g_-(g)]$, with $K = K(g_0, g)$, and all $0 \leq i < j \leq K$,*

$$|x_i - x_j - (j - i)c_0| \leq 2c_2 + \sum_{k=i}^{j-1} [C_\beta e(g_k) + v_k], \quad c_0 = \frac{11N^2}{12\pi^2} \log L; \quad (2.18)$$

in particular, for $0 \leq s \leq K$,

$$|x_{K-s} - x_K - sc_0| \leq 2c_2 + \sum_{u=1}^s [C_\beta e(g_{K-u}) + v_{K-u}] \leq \Delta(s), \quad (2.19)$$

and $\Delta(s)/s \rightarrow 0$. As Δ depends only on $(L, \mathbf{p}, N)$, for every $\eta > 0$ there is $S(\eta)$, depending only on η and $(L, \mathbf{p}, N)$, with $|x_{K-s} - x_K - sc_0| \leq \eta s$ whenever $S(\eta) \leq s \leq K$, for every $g \leq \gamma_J$ and $g_0 \in (0, g_-(g)]$. Consequently:

(c1) along the subsequences of Corollary 2.12, $g_{\infty,s}^{-2} \in [g^{-2} + s c_0 - \Delta(s), g^{-2} + B_{\sharp} + s c_0 + \Delta(s)]$, so $g_{\infty,s}^{-2}/(s c_0) \rightarrow 1$ as $s \rightarrow \infty$;

(c2) let $C : \{0, 1, 2, \dots\} \rightarrow [1, \infty)$ be non-decreasing with $\log C(s) = o(s)$, and for $0 < \mathbf{d} \leq 1$ let $s(\mathbf{d}) := \min\{s \geq 0 : C(s)L^{-s} \leq \mathbf{d}\}$. Then $s(\mathbf{d}) \log L = \log(1/\mathbf{d})(1 + o(1))$ as $\mathbf{d} \downarrow 0$, and

$$x(\mathbf{d}) := x_{K-s(\mathbf{d})} = \frac{11N^2}{12\pi^2} \log(1/\mathbf{d}) (1 + o(1)) \quad (\mathbf{d} \downarrow 0), \quad (2.20)$$

uniformly in $g_0 \in (0, g_-(g)]$ with $K(g_0, g) \geq s(\mathbf{d})$; the $o(1)$ depends only on g , C and $(L, \mathbf{p}, N)$, and the couplings depend neither on m nor on the position.

The depth attached to a separation $\mathbf{d}$ in Section 5 is of the form $s(\mathbf{d})$ in (c2), with $C(s)$ a constant times a power of $\log(g^{-2} + B_{\sharp}(s+1))$, so that $\log C(s) = O(\log \log s) = o(s)$ at fixed g .

Normalisation. The coupling is normalised so that its inverse square multiplies the Wilson action with the trace normalised to one, $g^{-2} = 2N/g_{\text{std}}^2$. In standard units the inverse square of the coupling grows by $11N/(24\pi^2) = (11N/3)/(8\pi^2)$ per e -fold of the scale, $11N/3$ being the universal first coefficient of the Yang–Mills β -function; the factor $2N$ between the two coefficients is this normalisation (Step 8 below). Thus (2.20) is the one-loop law of $\text{SU}(N)$ Yang–Mills theory, here at $N = 2$, obtained as a consequence of the control of the flow and not assumed.

Status. The proof is complete. Its inputs beyond clause (0)(a)–(c) are (P1), proved in the companion account (structure in Subsection 2.3), and Proposition 2.13, with the status of the correlation theorem stated in Section 3. No rarity of large fields beyond what these contain, no clustering, no gap and no uniqueness in the volume is used.

Proof. Preliminaries. By Theorem 2.11, $x_k > g^{-2}$ for $k \leq K$, and (2.15) holds for $0 \leq k \leq K-1$. Put $X := g^{-2}$; since $g \leq \gamma_J \leq \gamma^\circ$, $X \geq 8c_2 + 4$.

Step 1 (telescoping). For $0 \leq i < j \leq K$, by (2.8),

$$x_i - x_j = \sum_{l=i+1}^j \beta_l = \sum_{l=i+1}^j \beta_l^0 + \sum_{l=i+1}^j (\beta_l - \beta_l^0).$$

Step 2 (the one-loop lemma at both ends). By (P1) at $k = j$ and at $k = i$,

$$\left| \sum_{l=i+1}^j \beta_l^0 - (j-i)c_0 \right| = \left| \left(\sum_{l=1}^j \beta_l^0 - jc_0 \right) - \left(\sum_{l=1}^i \beta_l^0 - ic_0 \right) \right| \leq 2c_2.$$

Step 3 (the closeness bound, summed). For $i+1 \leq l \leq j$, $k := l-1 \in \{i, \dots, j-1\} \subset \{0, \dots, K-1\}$, so (2.15) gives $|\beta_l - \beta_l^0| \leq C_\beta e(g_{l-1}) + v_{l-1}$. With Steps 1–2 and the triangle inequality,

$$|x_i - x_j - (j-i)c_0| \leq 2c_2 + \sum_{l=i+1}^j |\beta_l - \beta_l^0| \leq 2c_2 + \sum_{k=i}^{j-1} [C_\beta e(g_k) + v_k],$$

which is (2.18). With $i = K-s$, $j = K$, $u := K-k$ it is the first inequality of (2.19): the defects summed are those of the s steps from level $K-s$ to level K , $u = 1, \dots, s$. For $s = 0$ both sides are $0 \leq 2c_2$.

Step 4 (the couplings u steps above level K). By (2.15), $|\beta_l - \beta_l^0| \leq \lambda$ for $1 \leq l \leq K$, so Steps 1–2 give $x_i - x_j \geq (j-i)c_0 - 2c_2 - (j-i)\lambda = \lambda(j-i) - 2c_2$ for $0 \leq i < j \leq K$. Let $1 \leq u \leq K$. With $(i, j) = (K-u, K)$ and $x_K > X$,

$$x_{K-u} > X + \lambda u - 2c_2 \geq 6c_2 + 4 + \lambda u, \quad \text{so} \quad g_{K-u} < \bar{h}_u.$$

Likewise $x_i \geq x_K + \lambda(K-i) - 2c_2 > X + \lambda u - 2c_2$ for every $i \leq K-u$, so $y_{K-u} = \min_{i \leq K-u} x_i - 2c_2 > X + \lambda u - 4c_2 \geq 4 + \lambda u > 0$.

Step 5 (the remnant defects are summable). v_{K-u} decreases in $y_{K-u} > 0$, so by Step 4 $v_{K-u} \leq \mathfrak{v}_u$ and $\sum_{u=1}^s v_{K-u} \leq C_v$ for $s \leq K$. Explicitly, as $4 + \lambda u \geq 1$ and $(\kappa_0 - 4)/2 \geq \frac{3}{2}$, $C_v \leq 2C_\beta C_t C_m \int_0^\infty (4 + \lambda t)^{-3/2} dt = 2C_\beta C_t C_m / \lambda$.

Step 6 (the small-field defects average to zero). By Step 4, $e(g_{K-u}) \leq \mathfrak{e}_u$ for $1 \leq u \leq K$. The sequence $(\mathfrak{e}_u)$ is non-increasing, as the intervals $(0, \bar{h}_u]$ shrink, and tends to 0: given $\eta > 0$ there is $g_\eta > 0$ with $e \leq \eta$ on $(0, g_\eta]$, then U with $\bar{h}_U \leq g_\eta$, and $\mathfrak{e}_u \leq \eta$ for $u \geq U$. Now fix $\eta > 0$ and $U = U(\eta) \geq 1$ with $\mathfrak{e}_u \leq \eta/2$ for $u \geq U$. For $s \geq S_e(\eta) := \max\{U, 2U\mathfrak{e}_1/\eta\}$,

$$\delta_e(s) = \frac{1}{s} \sum_{u=1}^U \mathfrak{e}_u + \frac{1}{s} \sum_{u=U+1}^s \mathfrak{e}_u \leq \frac{U\mathfrak{e}_1}{s} + \frac{\eta}{2} \leq \eta.$$

Hence $\delta_e(s) \rightarrow 0$, and $\sum_{u=1}^s C_\beta e(g_{K-u}) \leq C_\beta s \delta_e(s)$ for $1 \leq s \leq K$; $S_e(\eta)$ depends only on η and $(L, \mathfrak{p}, N)$.

Step 7 (assembly). Steps 3, 5 and 6 give, for $1 \leq s \leq K$,

$$|x_{K-s} - x_K - s c_0| \leq 2c_2 + C_v + C_\beta s \delta_e(s) = \Delta(s), \quad \frac{\Delta(s)}{s} = \frac{2c_2 + C_v}{s} + C_\beta \delta_e(s) \rightarrow 0.$$

Since Δ depends on none of K, g, g_0, m and the position, any $S(\eta)$ with $(2c_2 + C_v)/s \leq \eta/2$ and $C_\beta \delta_e(s) \leq \eta/2$ for $s \geq S(\eta)$ gives the uniform statement. This proves (2.18), (2.19) and the uniformity.

Step 8 (the coefficient and its normalisation). The constant of (P1) is $c_0 = (11C_g/24\pi^2) \log L$, where $\sum_a [t_a, [t_a, Y]] = C_g Y$ for an orthonormal basis $\{t_a\}$ of the Lie algebra, realised as traceless Hermitian matrices with $U = e^{iA}$, for $\langle Y, Y' \rangle = \text{tr}(YY')$, $\text{tr} \mathbf{1} = 1$. For $\text{SU}(N)$, with $\text{Tr} = N \text{tr}$ and $\text{Tr}(T_a T_b) = \frac{1}{2} \delta_{ab}$, one has $t_a = (2N)^{1/2} T_a$ and $\sum_a [T_a, [T_a, Y]] = NY$, so $C_g = 2N^2$ and $c_0 = (11N^2/12\pi^2) \log L$. In the lattice one-loop computation behind (P1) the gluon determinant contributes $5C_g/12\pi^2$ and the ghost determinant $C_g/24\pi^2$ per unit of $\log(1/t)$, $t = L^{-k}$. Units: with $\text{tr} \mathbf{1} = 1$ the weight $e^{-g_0^{-2} A(U)}$ is Wilson's with $g_0^{-2} = 2N/g_{\text{std}}^2$; the standard one-loop law makes g_{std}^{-2} grow by $11N/(24\pi^2)$ per e -fold; times $2N$ this is $11N^2/(12\pi^2)$ per e -fold here, and one step is $\log L$ e -folds.

Step 9 (consequence (c1)). Along the subsequence of Corollary 2.12, eventually $K_i \geq s$, $x_{K_i} \in (g^{-2}, g^{-2} + B_\sharp]$, and by (2.19) $x_{K_i-s} \in [g^{-2} + s c_0 - \Delta(s), g^{-2} + B_\sharp + s c_0 + \Delta(s)]$, an interval independent of i which therefore contains $g_{\infty, s}^{-2}$. Divided by $s c_0$, its end points are $1 + (g^{-2} - \Delta(s))/(s c_0)$ and $1 + (g^{-2} + B_\sharp + \Delta(s))/(s c_0)$, which tend to 1 by Step 7.

Step 10 (consequence (c2)). As $\log C(s) = o(s)$, $C(s)L^{-s} \rightarrow 0$ and $s(\mathfrak{d})$ is finite. From $C(s(\mathfrak{d}))L^{-s(\mathfrak{d})} \leq \mathfrak{d}$ and $C \geq 1$, $s(\mathfrak{d}) \log L \geq \log(1/\mathfrak{d})$, so $s(\mathfrak{d}) \rightarrow \infty$ as $\mathfrak{d} \downarrow 0$. If $s(\mathfrak{d}) \geq 1$, minimality gives $C(s(\mathfrak{d}) - 1)L^{-(s(\mathfrak{d})-1)} > \mathfrak{d}$, and $C(s(\mathfrak{d}) - 1) \leq C(s(\mathfrak{d}))$; hence

$$0 \leq s(\mathfrak{d}) \log L - \log(1/\mathfrak{d}) < \log L + \log C(s(\mathfrak{d})) = o(s(\mathfrak{d})),$$

that is, $s(\mathfrak{d}) \log L = \log(1/\mathfrak{d})(1 + o(1))$. Let $s := s(\mathfrak{d}) \leq K$. By (2.19) and Theorem 2.11, $x_{K-s} = x_K + s c_0 + \theta \Delta(s)$ with $|\theta| \leq 1$ and $x_K \in (g^{-2}, g^{-2} + B_\sharp]$, so

$$x_{K-s} = s c_0 \left[1 + \frac{x_K + \theta \Delta(s)}{s c_0} \right], \quad \left| \frac{x_K + \theta \Delta(s)}{s c_0} \right| \leq \frac{g^{-2} + B_\sharp}{s c_0} + \frac{\Delta(s)}{s c_0} \rightarrow 0$$

as $\mathfrak{d} \downarrow 0$, by Step 7, uniformly in $K \geq s$ and g_0 . Finally

$$\begin{aligned} s c_0 &= (c_0 / \log L) s \log L \\ &= (11N^2/12\pi^2) \log(1/\mathfrak{d})(1 + o(1)) \end{aligned}$$

by Step 8 and the tie just proved; multiplying the two factors $1 + o(1)$ gives (2.20). The base L and every auxiliary parameter have dropped out of the leading term. $\square$

Remark 2.15 (The charge read off the two-point function). Clause (iv) of Theorem 0 (Section 5) asserts that at separation $\mathbf{d}$, for test pairs of $\mathfrak{T}(\mathbf{d}; \vartheta)$, the connected two-point function of the smeared action density equals an explicit positive free kernel times $(N^2 - 1)/2$ times $g(\mathbf{d})^4$, $g(\mathbf{d}) := g_{K-s}(\mathbf{d})$, times a factor in $[\frac{2}{9}, \frac{8}{3}]$ (the square of a source factor in $[\frac{2}{3}, \frac{4}{3}]$ times $1 + R$, $|R| \leq \frac{1}{2}$). The charge $g_{\text{phys}}(\mathbf{d})$ obtained by dividing the two-point function by $(N^2 - 1)/2$ times that kernel and taking the fourth root therefore satisfies $g_{\text{phys}}(\mathbf{d})^{-2}/g(\mathbf{d})^{-2} \in [(\frac{3}{8})^{1/2}, (\frac{9}{2})^{1/2}]$, and by (2.20), with $b := 11N^2/(12\pi^2)$, this gives

$$\begin{aligned} \left(\frac{3}{8}\right)^{1/2} b &\leq \liminf_{\mathbf{d} \downarrow 0} \frac{g_{\text{phys}}(\mathbf{d})^{-2}}{\log(1/\mathbf{d})} \\ &\leq \limsup_{\mathbf{d} \downarrow 0} \frac{g_{\text{phys}}(\mathbf{d})^{-2}}{\log(1/\mathbf{d})} \leq \left(\frac{9}{2}\right)^{1/2} b, \end{aligned} \tag{2.21}$$

for any admissible choice of test pairs along the limit. These are bounds on logarithmic growth; they imply neither derivative bounds nor bounds on individual increments of the physical charge. Only the coupling of the construction is asserted to have the exact one-loop asymptotic coefficient.

What clause (0)(d) does not assert. The one-loop value or the sign of an individual increment β_{k+1} or β_{k+1}^0 ; any rate for the $o(s)$ in (2.19) or the $o(1)$ in (2.20); convergence of g_{K-s} along first passage without subsequences; anything about correlation functions; the case $N \geq 3$, which would follow by the same proof from Proposition 2.13 for $\text{SU}(N)$, which we have not carried out (Remark 1.4(2); (P1) holds for every $N \geq 2$).

Statements of this section used by citation. From Balaban's papers, by statement (on the printed proof see Remark 2.2(2) and Subsection 2.4): Theorem 2.1, which is [11, Theorem 1] with the representation of [9, §2], and the statements of group (A) of Subsection 2.4 from [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]. From the companion account (statements and structure given here): (P1)–(P3), Lemma 2.7, the one-step theorem in family form, the cube-size lemma, and the statements of group (B), among them the summed bounds of [9, Theorem 2] on every history, the format and bounds [11, (1.98)–(1.102)] of the remainders created by $\mathbf{R}$ (bound half), and (i-a)–(i-d). From Section 3: Proposition 2.13. Of group (C): the printed assertions in (C2) and (C5)–(C14) are used as Balaban asserts them, without a full proof in print, and are listed in Subsection 9.1; (C1), (C3), (C4) and the statements of ours in (C2), (C5) and (C12)–(C14) are settled, for the reasons given there; for $N \geq 3$ only, the statements of (C15) are stated, with their proofs deferred. Proved here in full: Lemmas 2.8 and 2.9, Propositions 2.10 and 2.5, Corollary 2.12, the deduction of Theorem 2.11 from its inputs, and Theorem 2.14 from its two inputs. Section 9 collects these statements clause by clause.

3 Clause (iii): correlation bounds

Throughout this section the gauge group is $\text{SU}(2)$, the hypotheses (H1)–(H6) are in force, and the radius $\varrho > 0$ of the source polydiscs is fixed after γ_H , as in (H7); every threshold γ_J below is chosen after ϱ . Clause (iii) of Theorem 0 has three parts:

- (a) a bound on the connected correlation functions S_n^T , $n \geq 2$, of the smeared action density on a fixed torus $\mathbb{T}_m$, uniform in the cutoff, with a constant that grows exponentially with the volume of $\mathbb{T}_m$;

- (b) the same bound with a constant independent of the volume and of the positions of the test functions, for test functions supported in balls of a fixed radius;
- (c) the analytic control, uniform in the cutoff, of the effect of a constant shift of the bare inverse coupling on the running inverse couplings (the *running shift*), which clauses (iv) and (v) use.

Parts (a) and (c) are consequences of the correlation theorem with sources, Theorem 3.1, which carries Balaban's renormalisation group through with a complex source placed in the coefficient of the Wilson action. Its proof is long and is given in the companion account, which we cite as [T-corr]; its status is stated at the end of §3.2, and Section 9 collects, under clause (iii), what the theorem rests on. In §3.2 we state the theorem in full, describe the structure of its proof, and say on what it rests; part (c) is deduced in §3.3 and part (a) in §3.4. Part (b) is deduced from Theorem 3.1(d) and the reflection positivity of the Wilson measure, by a chessboard estimate for complex observables and a re-centred counterterm; §§3.5–3.6 give this deduction with complete proofs.

3.1 Notation

Lattices and the sourced partition function. We realise $\mathbb{T}_m$ as $[-L^m, L^m]^4$ with opposite faces identified, and write $|x - y|$ for the Euclidean distance of the quotient. With $\varepsilon = L^{-K}$ the bare lattice is

$$\Lambda_0 := \left\{ x \in \varepsilon \mathbb{Z}^4 + \frac{\varepsilon}{2}(1, 1, 1, 1) : -L^m < x_\mu < L^m \right\}$$

as in [7, (0.1)]. For $0 \leq k \leq K$, Λ_k is the lattice of the centres of the cubes of side L^{k-K} with corners in $L^{k-K} \mathbb{Z}^4$, on which the block fields of step k live; $|\Lambda_k| = (2L^{m+K-k})^4$ is its number of sites, and Λ_K is the unit lattice. Bonds are oriented in the positive coordinate directions; $U = (U(b))_b$ is an $\text{SU}(2)$ -valued field on the bonds of Λ_0 and dU the product of the Haar measures. For a plaquette p of Λ_0 , $U(\partial p)$ is the ordered product of the bond variables around p , $x(p)$ is the barycentre of p , and $a_p(U) := 1 - \text{Re tr } U(\partial p) \in [0, 2]$. For a complex function v defined (at least) at the barycentres we put

$$A(v, U) := \sum_p v(x(p)) a_p(U), \quad (3.1)$$

which is linear in v ; the Wilson action is $A(U) = A(1, U)$, and $\mathcal{O}(f) = A(f, \cdot)$. With $x_0 := g_0^{-2}$ let

$$Z(v) := \int \exp[-A(x_0 - v, U)] dU = \int e^{-x_0 A(U) + A(v, U)} dU. \quad (3.2)$$

Then $Z(0) > 0$, and with $\langle \cdot \rangle$ the expectation in the Wilson measure $Z(0)^{-1} e^{-x_0 A(U)} dU$ we have $Z(v)/Z(0) = \langle e^{A(v, \cdot)} \rangle$. For real Lipschitz functions $f_1, \dots, f_n$ on $\mathbb{T}_m$ and $z \in \mathbb{C}^n$ put $w_z := \sum_i z_i f_i$. Then $Z(w_z)$ is the function $Z(z)$ of Section 1; it is entire in z (the integral over a compact space of an integrand entire in z), and

$$S_n^T(f_1, \dots, f_n) = \partial_{z_1} \cdots \partial_{z_n} \log Z(w_z) \big|_{z=0} = \langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle, \quad (3.3)$$

the joint cumulant of $\mathcal{O}(f_1), \dots, \mathcal{O}(f_n)$ under the Wilson measure.

Norms and polydiscs. $\|f\|_\infty$ is the supremum norm and $\text{Lip } f$ the Lipschitz constant with respect to the distance of $\mathbb{T}_m$ (in the units in which Λ_K has spacing one); $\|f\|_\# = \max\{\|f\|_\infty, 40M \text{Lip } f\}$, where M is Balaban's block size, one of the auxiliary parameters $\mathfrak{p}$ (a power of L). For a finite family $h = (h_1, \dots, h_{n_h})$ of real Lipschitz functions on $\mathbb{T}_m$ we write

$$\mathbb{D}_\varrho(h) := \left\{ \zeta \in \mathbb{C}^{n_h} : \sum_{l=1}^{n_h} |\zeta_l| \|h_l\|_\# < \varrho \right\},$$

and for $\vec{f} = (f_1, \dots, f_n)$ we abbreviate $\mathbb{D} := \mathbb{D}_\varrho(\vec{f})$ and $\frac{1}{2}\mathbb{D} := \mathbb{D}_{\varrho/2}(\vec{f})$.

Scales and localisation domains. We use Balaban's geometry [7, p. 257] in the units of $\mathbb{T}_m$. For $0 \leq j \leq K$, π_j is the partition of $\mathbb{T}_m$ into closed cubes of side

$$s_j := ML^{j-K}$$

(the cubes of side M in the units of Λ_j). A *localisation domain* at scale j is a finite union X of cubes of π_j that is connected through common walls; $\mathcal{D}_j$ is the set of these. $d_j(X)$ is the length of a shortest tree graph contained in X and meeting every cube of X , divided by s_j . X^+ is the union of the cubes of π_j that meet X , and $\text{plaq}(Y)$ is the set of plaquettes of Λ_0 that meet a set Y .

3.2 The correlation theorem with sources

Frozen auxiliary choices. Balaban's construction [7, 8, 9, 10, 11] makes auxiliary choices at every step: decompositions of unity into small-field and large-field regions with their thresholds, the large-field regions with their histories, localisation domains, background fields (minimisers of the Wilson action under the block constraints) and the operators built on them. In this section all these choices are those made for the source-free density at the real running couplings, once and for all; sources change the integrands, never the choices.

The coefficients $b_{k+1}(\zeta)$ and the running shift. For $\zeta \in \mathbb{C}$ with $|\zeta| < 2\varrho$, apply the construction, with the frozen choices, to the density $\exp[-A(x_0 - \zeta, U)]$ on the whole lattice, i.e. to the bare inverse coupling shifted by the constant ζ . The localised vacuum terms created at step $k+1$ form a family to which Balaban's coupling-constant renormalisation assigns a number: the number $\beta[\cdot]$ of [7, (1.20)–(1.22)], which is subtracted at step $k+1$, multiplied by $A(\varphi_{k+1}, U_{k+1})$, in [9, Sect. 3]. We denote it by $b_{k+1}(\zeta)$. At $\zeta = 0$ it is the coefficient of the coupling flow of clause (0): $x_{k+1} = x_k - b_{k+1}(0)$ (Section 2). The running shift is

$$\zeta_0(\zeta) := \zeta, \quad \zeta_{k+1}(\zeta) := \zeta_k(\zeta) + b_{k+1}(\zeta) - b_{k+1}(0), \quad (3.4)$$

so that $x_k - \zeta_k(\zeta)$ is the inverse coupling after k steps of the run started at $x_0 - \zeta$ with the frozen choices.

Sources and the class $\mathcal{W}$. For a complex weight w on the plaquettes of Λ_0 let $\rho_0^w := \exp[-A(x_0 - w, U)]$, and let ρ_k^w be the density of the block field V_k on Λ_k obtained from it by k of Balaban's steps with the frozen choices; dV_k is the product of the Haar measures of the bond variables of V_k , and U_k denotes Balaban's background field of step k , a field on the bonds of Λ_0 which is a function of V_k . Given an integer $K' \leq K$ (it is fixed in Theorem 3.1(c)), put for $j \leq K'$

$$r_j := \varrho \left(1 + \frac{1}{4} \sum_{i=j}^{K'} L^{-(K'-i)/2} \right) \in [\varrho, 2\varrho), \quad \mathfrak{l}_j := \frac{\varrho}{40M} L^{-(K'-j)/2},$$

let $\text{Lip}_j w := L^{j-K} \text{Lip } w$ be the Lipschitz constant of w (as a function of the barycentres) measured in units of Λ_j , and let $\mathcal{W}_j(X)$ be the set of complex functions w on $\text{plaq}(X^+)$ with $\sup |w| < r_j$ and $\text{Lip}_j w < \mathfrak{l}_j$; $\mathcal{W}_j := \mathcal{W}_j(\mathbb{T}_m)$. Then $\mathcal{W}_k \subset \mathcal{W}_j$ for $j \leq k$, and $\sum_l \zeta_l h_l \in \mathcal{W}_{K'}$ whenever h is a finite family of real Lipschitz functions and $\zeta \in \mathbb{D}_\varrho(h)$, because $\text{Lip}_{K'} \leq \text{Lip}$ and $40M \text{Lip } h_l \leq \|h_l\|_\#$.

Theorem 3.1 (Correlation theorem with sources). *Let the gauge group be $SU(2)$, assume (H1)–(H6), and let $\varrho > 0$. There is $\gamma_J = \gamma_J(L, \mathbf{p}, \gamma; \varrho) \in (0, \gamma_H]$, among whose defining conditions is $\gamma_J \leq \gamma^\circ := (8c_2 + 4)^{-1/2}$, such that for every $g \in (0, \gamma_J]$ there are an integer $b_* = b_*(g) \geq 0$, a constant $C_e \geq 0$ and a non-decreasing function $s \mapsto c_{\text{vol}}(s) \geq 0$ on the integers $s \geq b_*$, all depending only on $(L, \mathbf{p}, \gamma, g, \varrho)$ (and $c_{\text{vol}}(s)$ on s , through an upper bound in terms of $\log(g^{-2} + B_\#(s+1))$), for which the following holds for every $m \geq m_{\mathbb{T}}(g_-(g))$, every $g_0 \in (0, g_-(g)]$ and $K = K(g_0, g)$. We write $c_{\text{vol}} := c_{\text{vol}}(b_*)$.*

- (a) Coefficients. *For $k < K$, b_{k+1} is holomorphic on $\{|\zeta| < 2\varrho\}$ and real on the reals.*
- (b) Running shift. *For every $\varrho' \in [\varrho, 2\varrho]$ there are numbers $d_k(\varrho') \geq 0$, $k < K$, such that*

$$|b_{k+1}(\zeta) - b_{k+1}(0)| \leq d_k(\varrho') |\zeta| \quad (|\zeta| < \varrho'), \quad \sum_{k < K} d_k(\varrho') \leq \frac{C_{\mathfrak{D}}}{\lambda} \left[g (\log g^{-2})^{p_0} + \frac{g^{\kappa_0 - 6}}{\varrho'} \right] \leq \frac{1}{3}. \quad (3.5)$$

- (c) Run with sources. *Let s be an integer with $b_* \leq s \leq K$, and put $K' := K - s$. For every $k \leq K'$ and every $w \in \mathcal{W}_k$ the density ρ_k^w has the form of Balaban's representation [9, (2.18), (2.23)] — a sum over admissible sequences of large-field regions of large-field operations applied to characteristic functions times the exponential of an effective action — in which the Wilson term $-A(x_k(\cdot), U_k)$, with the position-dependent inverse coupling $x_k(\cdot)$ of [9, (2.24)], is replaced by $-A(x_k(\cdot) - \zeta_k^w, U_k)$, and the localised terms are supplemented by the differences between the local terms, computed with the source, and the frozen terms at the local value of the source. Here ζ_k^w is the complex function on plaquettes obtained from w by adding, at every level $j \leq k$ and every point y of Λ_j , the shift $b_j(w(p_y)) - b_j(0)$ at the value of the source at a plaquette p_y attached to y , spread by Balaban's partition of unity at scale j and the cut-off functions of his small-field regions; $|\zeta_k^w| \leq \frac{4}{3} \sup |w|$, and if $w \equiv \zeta$ is constant then $\zeta_k^w = \zeta_k(\zeta)$ wherever all these cut-off functions equal one. Every term of the representation is holomorphic in w , depends on w only through its values on $\text{plaq}(X^+)$, X the localisation domain of the term, and obeys Balaban's bounds; and $\int \rho_k^w dV_k = \int \rho_0^w dU$.*
- (d) Local vacuum-energy decomposition. *For s and K' as in (c), there are complex numbers $e_j(X; w)$, for $j \leq K'$, $X \in \mathcal{D}_j$ and $w \in \mathcal{W}_{K'}$ — the sourced vacuum energies: the values at the trivial field of the localised terms of level j with domain X , together with a logarithmic normalisation, minus the same quantity at $w = 0$ — such that:*

- (d1) $e_j(X; w)$ depends on w only through its values at the barycentres of the plaquettes in $\text{plaq}(X^+)$, and vanishes if these values all vanish;
- (d2) for every finite family h of real Lipschitz functions on $\mathbb{T}_m$, $\zeta \mapsto e_j(X; \sum_l \zeta_l h_l)$ is holomorphic on $\mathbb{D}_\varrho(h)$;
- (d3) $|e_j(X; w)| \leq C_e e^{-(1-\delta)\kappa d_j(X)}$ for every $w = \sum_l \zeta_l h_l$ with h as in (d2) and $\zeta \in \mathbb{D}_\varrho(h)$;
- (d4) with $\mathcal{E}(w) := \sum_{j \leq K'} \sum_{X \in \mathcal{D}_j} e_j(X; w)$, for every such w

$$|e^{-\mathcal{E}(w)} Z(w)| \leq e^{c_{\text{vol}}(s)|\Lambda_{K'}|} Z(0), \quad |\Lambda_{K'}| = L^{4s} |\Lambda_K|. \quad (3.6)$$

With $s = b_*$ this is an extensive bound in the volume $|\Lambda_K| = (2L^m)^4$ of the unit lattice, with the per-site constant $c_{\text{vol}} L^{4b_*}$, which depends on g only; this is the volume factor of (e), used in Sections 1, 4, 6 and 8.

(e) Torus bound. For $n \geq 2$ and real Lipschitz $f_1, \dots, f_n$ on $\mathbb{T}_m$ whose supports are pairwise at distance $\geq d > 0$,

$$|S_n^T(f_1, \dots, f_n)| \leq C_n(d) \prod_{i=1}^n \|f_i\|_{\sharp}, \quad (3.7)$$

where, with $Q := \exp(c_{\text{vol}}(2L^{m+b_*})^4)$, the constant C_{UV} of Lemma 3.5, and $\mathbf{c}_n := \sum_{\pi} (|\pi| - 1)!$ (the sum over the partitions π of $\{1, \dots, n\}$),

$$C_n(d) := \max \left\{ [C_{\text{UV}}(d; 2L^m) + \log 2 (2Q + 2)^n] \left(\frac{n}{Q}\right)^n, \mathbf{c}_n (12 (2L^{m+b_*})^4)^n \right\}.$$

$C_n(d)$ depends only on $(L, \mathbf{p}, \gamma, g, \varrho, m, n, d)$, not on K, g_0 or the positions of the supports. If $K \geq b_*$, then $\log Z(w_z)$ is holomorphic on $\{z : \sum_i |z_i| \|f_i\|_{\sharp} < \varrho/(Q + 1)\}$.

(f) Flow. For $0 \leq k \leq K - 1$,

$$|b_{k+1}(0) - \beta_{k+1}^0| \leq C_{\beta} e(g_k) + v_k \leq \lambda,$$

where β_{k+1}^0 is the Gaussian part of the coefficient (see (3.8) below) and $e(\cdot)$, v_k are the explicit functions of Proposition 2.13; and $\lambda(j - i) - 2c_2 \leq x_i - x_j \leq 3\lambda(j - i) + 2c_2$ for $0 \leq i < j \leq K$.

In (b), (d) and (f): $\lambda > 0$ is half the one-loop increment of the inverse coupling per block step (see the structure of the proof below); p_0 is the exponent of Balaban's small-field threshold $\delta_k = A_1 g_k (\log g_k^{-2})^{p_0}$ [9, Sect. 3]; $\kappa_0 \geq 7$ (a floor implied by the floor $\kappa_0 \geq 17$ of Section 2) is the exponent of his bound $|R^{(j)}(X)| \leq g_j^{\kappa_0} e^{-\kappa d_j(X)}$ on the remainder terms [9, (2.31)]; κ is his decay rate of localised terms and $\delta \in (0, \frac{1}{10})$ the slack of [8, (1.32)]; $C_{\mathfrak{D}}$ is a constant independent of g, K and m .

Theorem 3.1 is proved in [T-corr] (status at the end of this subsection); its part (e) is deduced from part (d) in §3.4, and its part (f) is obtained inside the inductive proof ("The coefficient and the flow" below). The companion account is written in a different normalisation, which we now translate.

Remark 3.2 (Normalisation). [T-corr] works on a torus of side one with bare spacing $L^{-K_J} n_T^{-1}$, where $n_T \geq 1$ is a free integer and K_J is the last step k such that $x_j > g^{-2}$ for all $j \leq k$ (the same flow, stopped at g^{-2} instead of the first passage of clause (0)). It performs the run with sources up to the level $K_J - m_J$, for any integer m_J above a floor $m_J \geq m^J(g)$ which ensures that all of Balaban's partitions at the couplings of the run divide the last lattice. Its constants do not depend on K_J, n_T or positions; the number of sites of its last lattice enters only the volume factor and the $\sharp$ -norm, and m_J enters only the per-site constant of the volume factor, through $\log x_{K_J - m_J}$ (with the choice $s = b_*$ below, this constant depends on g only). Parts (a) and (b) concern only the whole-lattice run with constant shifts, which [T-corr] performs on auxiliary tori whose sides are powers of L , at every depth and independently of the run with sources; their proof does not use the floor on m_J , and they hold for every K . For (c)–(e): $x_j > g^{-2}$ for $j \leq K$ by clause (0), so $K \leq K_J$; and the two-sided linear bounds of clause (0) on $x_i - x_j$, which persist as long as $x_j > g^{-2}$, bound $a_J := K_J - K$ by a constant a_* depending only on $(L, \mathbf{p}, \gamma)$: $x_K \leq g^{-2} + B_{\sharp}$ and $x_{K_J} > g^{-2}$ give $\lambda_{\sharp} a_J - c_5 < B_{\sharp}$, so one may take $a_* := \lceil (B_{\sharp} + c_5)/\lambda_{\sharp} \rceil$. Put $b_* := m^J(g)$ and add to the conditions on γ_J the requirement $m_T(g_-(g)) \geq a_*$, $b_* \geq a_*$ (both left sides tend to infinity as $g \downarrow 0$; this bounds γ_J from above and is imposed last). For an integer s with $b_* \leq s \leq K$ take $m_J := s + a_J$, which is admissible since $b_* \leq m_J \leq K_J$, and $n_T := 2L^{m - a_J}$: the bare lattice of [T-corr], dilated by the factor $2L^m$, is Λ_0 ; the run with sources stops at $K' := K - s$, whose lattice has $|\Lambda_{K'}| = L^{4s} (2L^m)^4$ sites; the per-site constant of the volume factor depends on s only

through $\log x_{K-s} \leq \log(g^{-2} + B_{\sharp}(s+1))$, and may be taken non-decreasing in s ; and the $\sharp$ -norm of [T-corr], which in our units is $\max\{\|f\|_{\infty}, 40M L^{-s} \text{Lip } f\}$, is at most ours, so that our polydiscs are contained in those of [T-corr]. With $s = b_*$ the per-site constant depends on g only; this is the choice made in (e), in §3.6 and in Section 6, while Section 5 stops the run at the depth $s(d)$ attached to the separation of the test functions. The case $K < b_*$ is elementary; see the proof of (e) in §3.4 and the paragraph preceding Lemma 3.11.

Structure of the proof

What is taken from Bałaban. The source-free run is Bałaban's: if the effective couplings stay in $(0, \gamma]$, the effective densities have the representation and satisfy all the conditions and bounds of [9, Sect. 2] (proved in [11, Theorem 1]); the hypothesis on the couplings is supplied by clause (0) (Section 2). From this run [T-corr] uses by statement only identities and source-free structural statements — the decompositions of unity and their thresholds, the large-field regions and their geometry, the background fields as functions of the block fields, and the operators built on them with their decay. The estimates on the terms of the expansion are derived again with the source present, except for the statements, listed under clause (iii) in Section 9, that are used as stated; among them is Bałaban's assertion that the old terms beside a newly created large-field region are analytic on the new class [11, p. 375], used as printed. For the measure [9, (3.29)], at the boundary of a new large-field region, [9, p. 276] states that the additional dependence of the action on the field tA_k “was already taken into account in the above considerations”; this is so for the potential terms, but not for the bilinear term coupling tA_k to the new fluctuation field, whose coefficient is of order one. [T-corr] supplies the missing argument, by an exact refinement of the decomposition of unity along the boundary of the new region.

Where the source sits. The source is carried in the coefficient of the Wilson action. In Bałaban's step [7, (2.1)–(2.12)], a complex coefficient $x_k - \xi$ in front of the Wilson action enters only through the factor $\mathfrak{s} := (x_k - \xi)g_k^2 = 1 - \xi g_k^2$ in front of the quadratic form of the fluctuation field and of the g_k^{-2} -part of the potential: the Jacobians and Haar densities do not come from the action, and the remaining operations of the step (translation to the background field, scaling by the real g_k , elimination of the constraint) do not see the coefficient. [T-corr] shows that the step, including the cluster expansion of [8, Sect. 2], tolerates $\mathfrak{s}$ in a fixed disc about 1 with Bałaban's bounds; a second proof keeps the Gaussian measure real and treats $\frac{1}{2}(1 - \mathfrak{s})$ times the quadratic form as one more localised potential.

The coefficient and the flow. At step $k+1$ the coefficient splits as

$$b_{k+1} = \beta_{k+1}^0 + f_{k+1}(\zeta_0, \dots, \zeta_k) + \mathfrak{q}_{k+1}. \quad (3.8)$$

Here β_{k+1}^0 , independent of ζ , is the renormalisation number of the localised logarithm of the Gaussian normalisation of the step at the background field; it obeys the one-loop law $\sum_{l=i+1}^j \beta_l^0 = (j-i)c_0 + \varsigma_i - \varsigma_j$ with $|\varsigma_l| \leq c_2$, where $c_0 > 0$ is the one-loop increment of the inverse coupling per block step and $c_2 \geq 0$ a constant (the leading coefficient of the coupling flow; Section 2); we put $\lambda := c_0/2$. The function f_{k+1} collects the dependence on the shifts of the earlier steps; it is holomorphic on the polydisc $\prod_{j \leq k} \{|\xi_j| < \sigma_0 x_j\}$, with radii proportional to x_j , and $|f_{k+1}| \leq C_{\beta} \mathfrak{e}_k$ with $\mathfrak{e}_k \asymp g_k (\log g_k^{-2})^{p_0}$ (σ_0, C_{β} constants). The term $\mathfrak{q}_{k+1}$ is the contribution of the large-field remnants of earlier steps; it obeys $|\mathfrak{q}_{k+1}| \leq \mathfrak{v}_k \asymp (\min_{i \leq k} x_i)^{-(\kappa_0-4)/2}$ on $\{|\zeta| < 2\varrho\}$, by Bałaban's bound [9, (2.31)] on the remainder terms. A Schwarz–Pick inequality in ℓ^1 — if $|f| \leq \omega$ on

$\prod_j D(0, \mathfrak{d}_j)$, then $\sum_j \mathfrak{d}_j |\partial_j f(\xi)| \leq \frac{4}{3}\omega$ whenever $|\xi_j| \leq \frac{1}{2}\mathfrak{d}_j$ for all j — and an induction on k on one fixed disc (the inductive hypothesis $|\zeta_j - \zeta| \leq \frac{1}{3}|\zeta|$ keeps $(\zeta_0, \dots, \zeta_k)$ in the half polydisc) give (3.5) with $d_k(\varrho') = \theta'_k + 2\mathfrak{v}_k/\varrho'$, $\theta'_k \asymp \mathfrak{e}_k g_k^2$; the sum over k is controlled by the linear growth of x_k . At $\zeta = 0$ one obtains $|b_{k+1}(0) - \beta_{k+1}^0| \leq C_\beta \mathfrak{e}_k + \mathfrak{v}_k \leq \lambda$, which with the one-loop law gives, inside the same induction, the two-sided bounds $\lambda(j-i) - 2c_2 \leq x_i - x_j \leq 3\lambda(j-i) + 2c_2$ for $0 \leq i < j \leq K$.

Two runs. The coefficients are produced by the *frozen whole-lattice run*: the construction with constant shifts ζ , $|\zeta| < 2\varrho$, performed on auxiliary tori whose sides are powers of L large enough for every partition at every level; this is legitimate because the terms that do not wrap around a torus do not depend on the torus. The *local run* on $\mathbb{T}_m$ carries a non-constant source $w \in \mathcal{W}_k$. Its new localised terms are holomorphic in w , read w on X^+ only, and coincide with the frozen ones when w is constant on X^+ . The renormalisation of step $k+1$ subtracts at each point y the frozen coefficient at the local value of the source, $b_{k+1}(w(p_y)) A(h_y \varphi_{k+1}, U_{k+1})$ (with Balaban's partition of unity h_y and the cut-off φ_{k+1} of the small-field region), and moves it into the Wilson coefficient. This add-and-subtract defines ζ_{k+1}^w , and it is the only place where the local run uses (3.5): $|\zeta_{k+1}^w| \leq \sup |w| (1 + \sum_j d_{j-1}(2\varrho)) \leq \frac{4}{3} \sup |w|$. The difference between a local term and the frozen term at the value of w at one plaquette is controlled by a freezing lemma: if $F(X; \cdot)$ is holomorphic and bounded by $\mathfrak{n}$ on $\mathcal{W}_j(X)$, $j < k$, $w \in \mathcal{W}_k(X)$ and $c \in \text{plaq}(X^+)$, then

$$|F(X; w) - F(X; w(c))| \leq 2\mathfrak{n} L^{-(k-j)/2} (1 + \text{diam}_j(X^+)/10M),$$

diam_j the diameter in units of Λ_j (Cauchy's estimate in a parameter interpolating between $w(c)$ and w ; the factor $L^{-(k-j)/2}$ is the Lipschitz condition of $\mathcal{W}_k$ seen at scale j). Each such difference is estimated from the a priori bound at its own level, so nothing compounds over the steps.

The inductive format and the step. The induction hypothesis (J_k) , $k \leq K$, collects: the two-sided bounds for levels $\leq k$; the holomorphy, reality and Lipschitz bounds of b_j and ζ_j ($j \leq k$) on $\{|\zeta| < 2\varrho\}$, with $|\zeta_j - \zeta| \leq \frac{1}{3}|\zeta|$; Balaban's bounds for the families of the frozen run (with his constant E_0), and the bound $\asymp (\min_{i < j} x_i)^{-(\kappa_0-2)/2}$ for the remnant families; Balaban's representation of the frozen densities for every constant shift $|\zeta| < 2\varrho$; and, if $k \leq K'$, the representation (c) of the sourced densities for every $w \in \mathcal{W}_k$. The step $(J_k) \Rightarrow (J_{k+1})$ has six parts: the Gaussian step at complex $\mathfrak{s}$, with the bound on f_{k+1} ; the frozen whole-lattice integral, giving $\mathfrak{q}_{k+1}$; the flow and the reference coupling x_{k+1} ; Balaban's step with domains and renormalisation [9, Sect. 3], first in the frozen run and then in the local run (with the add-and-subtract above); and Balaban's large-field renormalisation [10, 11] in both runs. Every move is an identity at fixed w , whence $\int \rho_k^w dV_k = \int \rho_0^w dU$; and b_{k+1} reads only data of levels $\leq k$, so the order of the steps has no circle. The bounds on terms are organised by kind — frozen renormalised groups, local-minus-frozen differences, remainder terms, boundary terms near large-field regions, and the source part of the Wilson term — per unit cell and summed over the scales. The marginal and irrelevant parts are controlled by Balaban's second proof of his Theorem 2 [9, pp. 283–284], run at general radii and complex kernels: after subtraction of $\beta[F] A(h_y, \cdot)$, a renormalised pointed group of scale j at an admissible background of scale $n' \geq j$ is bounded by a polynomial in $n' - j$ times $L^{-(4+\hat{\alpha})(n'-j)}$, for some $\hat{\alpha} > 0$, times ratios of the analyticity radii at the two scales, which are displayed.

Where the large-field bookkeeping enters. In three places. (i) In the large-field renormalisation step: the large-field operations are bounded by moduli, $|\mathbf{T}_k'^w(X)F| \leq e \cdot \mathfrak{M}_k(X) \sup |F|$ ($e = \exp 1$), with $\mathfrak{M}_k(X)$ the right side of [11, (1.79)] (adjusted to the refined decomposition of unity), and the Wilson factors with complex weight are controlled by $|e^{-A(x-\zeta, U)}| \leq e^{-(1-\vartheta_W)A(x, U)}$

for $SU(2)$ -valued U whenever $|\zeta| \leq \vartheta_W x$ (because $a_p \geq 0$), with $\vartheta_W = \frac{8}{3}\varrho g^2$. (ii) The boundary terms created next to large-field regions are carried with a decay in a length relative to the large-field region; this is the alternative form of the inductive assumption which [9, p. 262] states to be admissible (“All constructions and proofs of the procedure can be carried on with these inductive assumptions”). [T-corr] excises from the length the strips around the large-field regions still present, and shows that the large-field renormalisation returns its new boundary terms in the same format. (iii) In the final bound (3.6): Balaban’s proof of the upper bound in [9, (2.50), Corollary 3] is by moduli of the terms of the expansion; with sources, the real part of the last effective action is bounded as in [9, (2.49)] with $(1 - \vartheta_W)$ in front of the Wilson action and one more logarithmic term per large-field cell, and the gas of large-field components is summed by a polymer bound. Together with an extensive lower bound for the source-free partition function this gives (3.6), c_{vol} being the sum of the two extensive constants. The remnant part $\mathbf{q}_{k+1}$ of the coefficient is a fourth, smaller, entry point.

The output. The run with sources is stopped at K' . The numbers $e_j(X; w)$ of (d) are the vacuum values of the localised terms, $\mathcal{E} = \sum_{j,X} e_j(X; \cdot)$, and $Z = e^{\mathcal{E}} \tilde{Z}$, where $\tilde{Z}$ is the integral over the last block field of the sum over large-field histories of the remaining terms; (d1)–(d3) come from the representation (c), and (3.6) is (iii) above. Part (e) follows from (d) by two Cauchy estimates (§3.4).

The conditions on the parameters. The parameters are fixed in one order, as in (H3): L , with floors coming from Balaban’s own windows (for instance the window for his auxiliary integer of [10, (1.24)]); $\delta \in (0, (1 - 4/L)/10]$; $\kappa \geq \max\{10 \log L, \kappa_d, C_\delta/\delta\}$, where $\kappa \geq 10 \log L$ gives Balaban’s requirement $e^{-\kappa(n'-j)/2} < L^{-5(n'-j)}$ [9, p. 283] and κ_d, C_δ depend on the dimension only; κ_1 at least each of finitely many explicit affine functions of κ whose coefficients depend only on the dimension and on L , among them $16\kappa + 1$ and $4L\kappa + 1$, together with Balaban’s $\frac{1}{2}(\kappa_1 - 1) \geq 2L\kappa$ [8, p. 20]; the block sizes M_1 and M , a power of L with $M > L^2 M_1$, subject to floors such as $\delta_1 M \geq \kappa_1$ for a constant $\delta_1 > 0$ fixed after L ; Balaban’s exponents — $\kappa_0 \geq 7$, p_0 , and the integers $q_0 = q_1 =: q$ of his analyticity radii [9, (2.28)], with $q \geq p_0 + 7r_B + 1$ and, for the extraction estimate of Section 5, $q \geq 21p_0 + 1$ (1.9), r_B the exponent of his large-field scale R_k , the least power of L with $R_k \geq (\log g_k^{-2})^{r_B}$ — and his analysis constants; then γ ; and, after ϱ , the threshold γ_J . Every condition bounds a parameter from below in terms of those fixed before it, except the conditions on γ_J , which bound it from above and are imposed last. No condition bounds L , M , M_1 , κ , κ_1 , the exponents, K or m from above, and none bounds a coupling from below. [T-corr] closes by checking every sum over scales in the proof: each is dominated by a convergent series in the scale difference — geometric factors against polynomial ones, or the linear growth of x_k — so that no constant depends on K , k , m , the torus or the position.

Status. Theorem 3.1 is proved in the companion account [T-corr] as an implication (we give the statement and the structure of the proof). Besides Balaban’s theorems, used by statement as described, and the statements on which the source-free run of clause (i) rests, the proof in [T-corr] uses statements of Balaban as he asserts them, without full proof in print, and finitely many sub-statements of its own, all of which are proved there; the statements on which the source-free run rests are those of Subsection 2.4, as qualified there, and at $N = 2$, the case of this section, none of them is deferred; the statements of Balaban that the proof uses as he asserts them are listed in Subsection 9.1. Section 9 collects, under clause (iii), what the theorem rests on.

3.3 The running shift: clause (iii)(c)

Corollary 3.3 (Running shift). *Under the hypotheses of Theorem 3.1, for every $k \leq K$ the function $\zeta \mapsto \zeta_k(\zeta)$ of (3.4) is holomorphic on $\{|\zeta| < 2\varrho\}$ and real on the reals, and*

$$|\zeta_k(\zeta) - \zeta| \leq \frac{1}{3}|\zeta| \quad (|\zeta| < 2\varrho), \quad \sum_{k < K} \text{Lip}_{\{|\zeta| \leq \varrho\}} b_{k+1} \leq \frac{2}{3}, \quad \zeta'_k := \partial_\zeta \zeta_k(0) \in [\frac{2}{3}, \frac{4}{3}]. \quad (3.9)$$

Moreover $|\zeta'_k - 1| \leq C_\zeta g_k (\log g_k^{-2})^{p_0}$ for $k \leq K$, with C_ζ depending only on $(L, \mathbf{p}, \gamma, \varrho)$.

Proof. For $k < K$ put $h_k := b_{k+1} - b_{k+1}(0)$. By Theorem 3.1(a), h_k is holomorphic on $\{|\zeta| < 2\varrho\}$ and real on the reals, and $h_k(0) = 0$; by (b) with $\varrho' = 2\varrho$, $|h_k(\zeta)| \leq d_k(2\varrho)|\zeta|$ for $|\zeta| < 2\varrho$. Since $\zeta_k = \zeta + \sum_{j < k} h_j$, ζ_k is holomorphic and real on the reals, and $|\zeta_k(\zeta) - \zeta| \leq \sum_{j < k} d_j(2\varrho)|\zeta| \leq \frac{1}{3}|\zeta|$ by (3.5). Let $|\zeta| \leq \varrho$ and $0 < r' < \varrho$; the circle of radius r' about ζ lies in $\{|\zeta| < 2\varrho\}$, where $|h_k| \leq 2\varrho d_k(2\varrho)$, so Cauchy's estimate gives $|h'_k(\zeta)| \leq 2\varrho d_k(2\varrho)/r'$, and letting $r' \uparrow \varrho$, $|h'_k| \leq 2d_k(2\varrho)$ on the closed disc $\{|\zeta| \leq \varrho\}$. This disc is convex, so $\text{Lip}_{\{|\zeta| \leq \varrho\}} b_{k+1} \leq 2d_k(2\varrho)$, and the sum over $k < K$ is at most $\frac{2}{3}$. Finally $h'_k(0) = \lim_{\zeta \rightarrow 0} h_k(\zeta)/\zeta$ is real with $|h'_k(0)| \leq d_k(2\varrho)$, so $\zeta'_k(0) = 1 + \sum_{j < k} h'_j(0) \in [1 - \frac{1}{3}, 1 + \frac{1}{3}]$.

The bound on $|\zeta'_k - 1|$ is proved in the companion account; we give its structure. By the above, $|\zeta'_k - 1| \leq \sum_{j < k} d_j(2\varrho)$. The numbers $d_j(2\varrho) = \theta'_j + \mathbf{v}_j/\varrho$ are explicit non-increasing functions of $\min_{i \leq j} x_i$ (see “The coefficient and the flow” above), and by Theorem 3.1(f), $\min_{i \leq j} x_i \geq x_k + \lambda(k - j) - 2c_2$ for $j < k$. Summing over $j < k$ as for the total sum in (3.5), with x_k in place of g^{-2} , gives $\sum_{j < k} d_j(2\varrho) \leq (C_{\mathfrak{D}}/\lambda)[g_k(\log g_k^{-2})^{p_0} + g_k^{\kappa_0-6}/(2\varrho)]$, and $g_k^{\kappa_0-6} \leq g_k$ since $\kappa_0 \geq 7$. $\square$

In words: if the bare inverse coupling is changed from g_0^{-2} to $g_0^{-2} - \zeta$ while all auxiliary choices are kept, the inverse coupling after k steps changes from x_k to $x_k - \zeta_k(\zeta)$, where $\zeta \mapsto \zeta_k(\zeta)$ is a holomorphic near-identity map, uniformly in $k \leq K$ and in the cutoff. Clauses (iv) and (v) use (3.9) in this form (Sections 5 and 7).

3.4 Three elementary lemmas, and the torus bound: clause (iii)(a)

In this subsection and in §3.6 we assume the hypotheses of Theorem 3.1, and, where the numbers $e_j(X; w)$ occur, $K \geq b_*$ and K' as in Theorem 3.1(c).

Lemma 3.4 (Geometry and counting). *Let $j \leq K$ and $X \in \mathcal{D}_j$.*

- (i) *For $y_1, y_2 \in X^+$: $|y_1 - y_2| \leq s_j(d_j(X) + 8) \leq 4s_j(d_j(X) + 3)$.*
- (ii) *If $p_1, p_2 \in \text{plaq}(X^+)$ and $|x(p_1) - x(p_2)| \geq D$, then $d_j(X) \geq DL^{K-j}/(4M) - 4$.*
- (iii) *If X^+ meets a closed set $\Sigma \subset \mathbb{T}_m$ and some $p \in \text{plaq}(X^+)$ has $\text{dist}(x(p), \Sigma) \geq R + 2$, then $d_j(X) \geq RL^{K-j}/(4M) - 3$.*
- (iv) *For $x \in \mathbb{T}_m$ and $R > 0$, at most $\mathbf{a}(R, K - j)$ cubes of π_j meet a cube of π_j that meets the ball $\bar{B}(x, R + 2)$, where $\mathbf{a}(R, i) := (2(R + 2)L^i/M + 6)^4$.*
- (v) (Bálabán's tree sum.) $C_{\text{tree}} := \sup_{j \leq K} \sup_{\square \in \pi_j} \sum_{X \in \mathcal{D}_j, X \supset \square} e^{-(\frac{3}{4}-\delta)\kappa d_j(X)}$ *is finite and depends only on $(L, \mathbf{p})$.*

Proof. (i) Let $\square'_i \subset X^+$ be a cube of π_j containing y_i , and $\square_i \subset X$ a cube of X meeting $\square'_i$. A shortest tree graph $\mathcal{T} \subset X$ meeting every cube of X has length $s_j d_j(X)$; it meets $\square_1$ and $\square_2$ at points t_1, t_2 , which are joined in $\mathcal{T}$ by a path of length $\leq s_j d_j(X)$. Two closed cubes of side s_j

that meet lie in a cube of side $2s_j$, whose diameter is $4s_j$ in dimension four; so $|y_i - t_i| \leq 4s_j$ and $|y_1 - y_2| \leq s_j d_j(X) + 8s_j$. Finally $d + 8 \leq 4(d + 3)$ for $d \geq 0$. (ii) Pick $y_i \in p_i \cap X^+$. Every point of a plaquette is within ε of its barycentre, so $|y_1 - y_2| \geq D - 2\varepsilon$, and (i) gives $d_j(X) \geq (D - 2\varepsilon)/(4s_j) - 3 \geq D/(4s_j) - 4$, since $\varepsilon \leq s_j$; and $1/s_j = L^{K-j}/M$. (iii) Pick $y_1 \in X^+ \cap \Sigma$ and $y_2 \in p \cap X^+$. Then $|y_1 - y_2| \geq R + 2 - \varepsilon \geq R$, and (i) gives the claim. (iv) A cube of π_j meeting $\bar{B}(x, R + 2)$ lies in the cube of half-side $R + 2 + s_j$ about x , and a cube of π_j meeting it lies in the cube of half-side $R + 2 + 2s_j$ about x , which contains at most $(2(R + 2)/s_j + 5)^4$ cubes of the grid π_j (fewer if it wraps around $\mathbb{T}_m$). (v) This is [8, (1.26)]: $\sum_{X \in \mathcal{D}_j, X \supset \square'} e^{-\kappa d_j(X)} \leq O(1)$ for κ sufficiently large, applied at the rate $(\frac{3}{4} - \delta)\kappa \geq \frac{1}{2}\kappa$; that $\frac{1}{2}\kappa$ is large enough for it is one of the floors on κ . $\square$

Lemma 3.5 (Mixed derivatives of the local vacuum energies). *Let $n \geq 2$, $R > 0$, $d > 0$; let $h = (h_1, \dots, h_n)$ be real Lipschitz functions on $\mathbb{T}_m$, each supported in a closed ball of radius R , with supports pairwise at distance $\geq d$; and let $\mathcal{Y}_j \subset \mathcal{D}_j$, $j \leq K'$, be arbitrary subfamilies. Then $\Phi(\zeta) := \sum_{j \leq K'} \sum_{X \in \mathcal{Y}_j} e_j(X; \sum_l \zeta_l h_l)$ is holomorphic on $\mathbb{D}_\varrho(h)$, $\Phi(0) = 0$, and*

$$|\partial_{\zeta_1} \cdots \partial_{\zeta_n} \Phi(0)| \leq C_{UV}(d; R) \left(\frac{n}{\varrho}\right)^n \prod_{l=1}^n \|h_l\|_\sharp, \quad C_{UV}(d; R) := C_e C_{\text{tree}} \sum_{i \geq 0} a(R, i) e^{-\frac{\kappa}{4} [dL^i/(4M) - 4]_+}. \quad (3.10)$$

The series converges; $C_{UV}(d; R)$ depends only on $(L, p, \gamma, g, \varrho, d, R)$, not on K , g_0 , m or positions, and it is non-increasing in d . Moreover, with $\log_L^+ t := \max\{0, \log_L t\}$,

$$C_{UV}(d; R) \leq C'_{UV} \left(1 + \frac{R + 2}{d}\right)^4 (1 + \log_L^+(32M/d)), \quad (3.11)$$

with C'_{UV} depending only on C_e , C_{tree} and κ .

Proof. Φ is a finite sum; each term is holomorphic on $\mathbb{D}_\varrho(h)$ by (d2) and vanishes at 0 by (d1). By (d1), $e_j(X; \sum_l \zeta_l h_l)$ does not depend on ζ_l unless $h_l(x(p_l)) \neq 0$ for some $p_l \in \text{plaq}(X^+)$; hence its mixed derivative $\partial_{\zeta_1} \cdots \partial_{\zeta_n}$ at 0 vanishes unless such p_l exist for every l . For such X , $x(p_1)$ and $x(p_2)$ lie in the supports of h_1 and h_2 , so Lemma 3.4(ii) gives $d_j(X) \geq D_j := [dL^{K-j}/(4M) - 4]_+$. If some h_l vanishes there is nothing to prove. Otherwise, for $0 < v < 1$ the closed polydisc $\prod_l \{|\zeta_l| \leq (1 - v)\varrho/(n\|h_l\|_\sharp)\}$ lies in $\mathbb{D}_\varrho(h)$, and Cauchy's estimate with (d3) gives

$$|\partial_{\zeta_1} \cdots \partial_{\zeta_n} e_j(X; \cdot)(0)| \leq C_e e^{-(1-\delta)\kappa d_j(X)} \prod_{l=1}^n \frac{n\|h_l\|_\sharp}{(1-v)\varrho};$$

let $v \downarrow 0$. Let $\bar{B}(x_1, R) \supset \text{supp } h_1$. A point of p_1 lies in X^+ and within $\varepsilon \leq 1$ of $x(p_1) \in \bar{B}(x_1, R)$, so a cube of X^+ meets $\bar{B}(x_1, R + 2)$, and X contains a cube $\square$ meeting that cube; by Lemma 3.4(iv) there are at most $a(R, K - j)$ such $\square$. Writing $e^{-(1-\delta)\kappa d_j} \leq e^{-\frac{\kappa}{4} D_j} e^{-(\frac{3}{4}-\delta)\kappa d_j}$ and summing over $X \supset \square$ by Lemma 3.4(v), the contribution of scale j is at most $C_e C_{\text{tree}} a(R, K - j) e^{-\frac{\kappa}{4} D_j} (n/\varrho)^n \prod_l \|h_l\|_\sharp$. Summing over $j \leq K'$ with $i := K - j \geq 0$ gives (3.10). For the i with $dL^i/(4M) \geq 8$ the i -th term of the series is at most $C_e C_{\text{tree}} (2(R + 2)L^i/M + 6)^4 e^{-\kappa dL^i/(32M)}$, which is summable; there are finitely many other i ; and every term is non-increasing in d .

For (3.11) put $u_i := dL^i/M$; then $a(R, i) = (2(R + 2)u_i/d + 6)^4 \leq (1 + (R + 2)/d)^4 (2u_i + 6)^4$. At most $1 + \log_L^+(32M/d)$ indices have $u_i < 32$, and each of their terms is at most $70^4 C_e C_{\text{tree}} (1 + (R + 2)/d)^4$. If $u_i \geq 32$ then $[u_i/4 - 4]_+ \geq u_i/8$; the indices with $u_i \geq 32$, listed increasingly as $i_0 < i_0 + 1 < \cdots$, satisfy $u_{i_0+p} \geq 32L^p \geq 32(p + 1)$, and with $C_\kappa := \sup_{t \geq 0} (2t + 6)^4 e^{-\kappa t/64}$ their terms sum to at most $C_e C_{\text{tree}} C_\kappa (1 + (R + 2)/d)^4 \sum_{p \geq 1} e^{-\kappa p/2}$. So one may take $C'_{UV} := C_e C_{\text{tree}} (70^4 + C_\kappa \sum_{p \geq 1} e^{-\kappa p/2})$. $\square$

Lemma 3.6 (Schwarz and Cauchy). *Let $f_1, \dots, f_n$ be real Lipschitz functions with $\|f_i\|_{\sharp} > 0$, $|z|_{\sharp} := \varrho^{-1} \sum_i |z_i| \|f_i\|_{\sharp}$, $\mathfrak{r} \geq 1$ and $Q \geq 1$, and let h be holomorphic on $\{|z|_{\sharp} < 1/\mathfrak{r}\}$ with $h(0) = 1$ and $|h| \leq Q$ there. Then $|h(z) - 1| \leq (Q+1)\mathfrak{r}|z|_{\sharp}$ on $\{|z|_{\sharp} < 1/\mathfrak{r}\}$; the principal logarithm $\text{Log } h$ is holomorphic on $\{|z|_{\sharp} < 1/(\mathfrak{r}(Q+1))\}$; and*

$$|\partial_{z_1} \cdots \partial_{z_n} \text{Log } h(0)| \leq \log 2 (2Q+2)^n \mathfrak{r}^n \left(\frac{n}{\varrho}\right)^n \prod_{i=1}^n \|f_i\|_{\sharp}. \quad (3.12)$$

Proof. For $z \neq 0$ with $|z|_{\sharp} < 1/\mathfrak{r}$, the function $\phi(t) := h(tz/(\mathfrak{r}|z|_{\sharp})) - 1$ is holomorphic on $|t| < 1$ (where $|tz/(\mathfrak{r}|z|_{\sharp})|_{\sharp} = |t|/\mathfrak{r}$), $|\phi| \leq Q+1$ and $\phi(0) = 0$; by Schwarz's lemma $|\phi(t)| \leq (Q+1)|t|$, and $t = \mathfrak{r}|z|_{\sharp}$ gives the first claim. Hence $|h - 1| < 1$ on $\{|z|_{\sharp} < 1/(\mathfrak{r}(Q+1))\}$, where $\text{Log } h$ is therefore holomorphic. On the closed polydisc $\mathcal{P} := \prod_i \{|z_i| \leq \rho_i\}$, $\rho_i := \varrho/(2\mathfrak{r}(Q+1)n\|f_i\|_{\sharp})$, we have $|z|_{\sharp} \leq 1/(2\mathfrak{r}(Q+1))$, so $|h - 1| \leq \frac{1}{2}$ and $|\text{Log } h| \leq -\log(1 - \frac{1}{2}) = \log 2$. Cauchy's estimate for the coefficient of $z_1 \cdots z_n$ gives $|\partial_{z_1} \cdots \partial_{z_n} \text{Log } h(0)| \leq \log 2 / \prod_i \rho_i$, which is (3.12). $\square$

Proof of Theorem 3.1(e) from Theorem 3.1(d). If $K < b_*$, the bare lattice has

$$(2L^{m+K})^4 \leq (2L^{m+b_*})^4$$

sites and six times as many plaquettes, so $|\mathcal{O}(f_i)| \leq 12(2L^{m+b_*})^4 \|f_i\|_{\infty}$ because $0 \leq a_p \leq 2$. The joint cumulant of bounded random variables $Y_1, \dots, Y_n$ is $\sum_{\pi} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} \langle \prod_{i \in B} Y_i \rangle$, so $|S_n^T(f_1, \dots, f_n)| \leq \mathfrak{c}_n \prod_i 12(2L^{m+b_*})^4 \|f_i\|_{\infty}$, which is (3.7).

Let $K \geq b_*$; we may assume $\|f_i\|_{\sharp} > 0$ for all i , since otherwise $S_n^T = 0$. By (d2) for the family $\vec{f}$, $z \mapsto \mathcal{E}(w_z)$ is a finite sum of functions holomorphic on $\mathbb{D}$, and $\mathcal{E}(0) = 0$ by (d1). So $h(z) := e^{-\mathcal{E}(w_z)} Z(w_z)/Z(0)$ is holomorphic on $\mathbb{D} = \{|z|_{\sharp} < 1\}$ with $h(0) = 1$, and by (3.6) and $|\Lambda_{K'}| = (2L^{m+K-K'})^4 \leq (2L^{m+b_*})^4$, $|h| \leq Q$ on $\mathbb{D}$. Near $z = 0$, $\log Z(w_z) = \log Z(0) + \mathcal{E}(w_z) + \text{Log } h(z)$ (both sides are holomorphic, agree at 0 and have the same exponential), so

$$S_n^T(f_1, \dots, f_n) = \partial_{z_1} \cdots \partial_{z_n} \mathcal{E}(w_z)|_{z=0} + \partial_{z_1} \cdots \partial_{z_n} \text{Log } h(0).$$

Every subset of $\mathbb{T}_m$ lies in a closed ball of radius $2L^m$; so Lemma 3.5 with $h = \vec{f}$, $R = 2L^m$ and $\mathcal{Y}_j = \mathcal{D}_j$ bounds the first term by $C_{UV}(\mathbf{d}; 2L^m)(n/\varrho)^n \prod_i \|f_i\|_{\sharp}$, and Lemma 3.6 with $\mathfrak{r} = 1$, $Q = Q$ bounds the second by $\log 2 (2Q+2)^n (n/\varrho)^n \prod_i \|f_i\|_{\sharp}$. The function $\log Z(0) + \mathcal{E}(w_z) + \text{Log } h(z)$ is a logarithm of $Z(w_z)$ that is holomorphic on $\{|z|_{\sharp} < 1/(Q+1)\}$, by the second assertion of Lemma 3.6. $\square$

The constant $C_n(\mathbf{d})$ grows with the volume in two ways. The factor $Q = \exp(c_{\text{vol}}(2L^{m+b_*})^4)$ is the price of the extensive bound (3.6); it is the essential one, and it makes (3.7) useless in infinite volume. The growth of $C_{UV}(\mathbf{d}; 2L^m)$ with m only reflects the choice $R = 2L^m$: Lemma 3.5 counts the cubes near one support, and the support of f_1 was allowed to be the whole torus. Both are removed in §3.6 for test functions supported in balls of a fixed radius.

3.5 Reflection positivity and the chessboard estimate

Lattice symmetries. Let τ be an isometry of $\mathbb{T}_m$ that maps Λ_0 onto itself. For a bond b put $o_{\tau}(b) := +1$ if τ maps b onto a bond with the same orientation and $o_{\tau}(b) := -1$ otherwise; put $(U \circ \tau)(b) := U(\tau b)^{o_{\tau}(b)}$ and $T_{\tau}F(U) := F(U \circ \tau)$. Since $o_{\tau\tau'}(b) = o_{\tau}(\tau' b) o_{\tau'}(b)$, we have $(U \circ \tau) \circ \tau' = U \circ (\tau\tau')$ and $T_{\tau}T_{\tau'} = T_{\tau\tau'}$; T_{τ} is multiplicative and commutes with complex conjugation. The map $U \mapsto U \circ \tau$ preserves the product Haar measure (it relabels the bonds and inverts some variables) and the Wilson action ($a_p(U \circ \tau) = a_p(U)$, because Re tr of a holonomy is invariant under cyclic permutation and inversion); hence $\langle T_{\tau}F \rangle = \langle F \rangle$ for bounded measurable F .

Lemma 3.7 (Link reflection positivity [23]). *Let $\mu \in \{1, \dots, 4\}$, $\varpi \in \varepsilon\mathbb{Z}$, and let θ be the reflection $x_\mu \mapsto 2\varpi - x_\mu$ of $\mathbb{T}_m$. Then θ maps Λ_0 onto itself, its fixed set $\Pi := \{x_\mu = \varpi\} \cup \{x_\mu = \varpi + L^m\}$ contains no site of Λ_0 , and, with $H_\pm$ the two components of $\mathbb{T}_m \setminus \Pi$, $\mathfrak{A}(H_+)$ the bounded measurable functions of the variables of the bonds with both end points in H_+ , and $\Theta_\theta(F) := \overline{T_\theta F}$,*

$$\langle \Theta_\theta(F) F \rangle \geq 0 \quad (F \in \mathfrak{A}(H_+)).$$

Proof. Sites have $x_\mu \in \varepsilon(\mathbb{Z} + \frac{1}{2})$, whereas $\varpi, \varpi + L^m \in \varepsilon\mathbb{Z}$; and θ maps $\varepsilon(i + \frac{1}{2})$ to $\varepsilon((2\varpi/\varepsilon - i - 1) + \frac{1}{2})$. A bond either has both end points in H_+ , or both in H_- , or it is parallel to e_μ and crosses Π . For a crossing bond we write $U(b_y^\times)$ for the transporter from its end point in H_- to its end point $y \in H_+$. A plaquette containing no crossing bond lies in H_+ or in H_- ; a plaquette containing a crossing bond contains exactly two, $b_y^\times$ and $b_{y'}^\times$ with $y' = y + \varepsilon e_\nu$, $\nu \neq \mu$; we call it $p_{y,\nu}$ and let $b_{y,\nu}^\pm \subset H_\pm$ be its two other bonds, both oriented along $+e_\nu$. Then $\theta b_{y,\nu}^+ = b_{y,\nu}^-$ with the same orientation, so $(U \circ \theta)(b_{y,\nu}^+) = U(b_{y,\nu}^-)$, and $\text{Re tr } U(\partial p_{y,\nu}) = \text{Re tr } W_{y,\nu}$ with

$$W_{y,\nu} := U(b_y^\times) U(b_{y,\nu}^+) U(b_{y'}^\times)^{-1} U(b_{y,\nu}^-)^{-1}.$$

Let $A_\pm(U) := \sum_{p \subset H_\pm} a_p(U)$. Since θ maps the plaquettes in H_+ onto those in H_- , $A_- = T_\theta A_+$. Put $G_0 := F e^{-x_0 A_+} \in \mathfrak{A}(H_+)$ and $\varphi(W) := e^{x_0 \text{Re tr } W}$. With $n_\times$ the number of plaquettes $p_{y,\nu}$,

$$Z(0) \langle \Theta_\theta(F) F \rangle = e^{-x_0 n_\times} \int \overline{T_\theta G_0} G_0 \prod_{(y,\nu)} \varphi(W_{y,\nu}) dU. \quad (3.13)$$

Character expansion. $\text{Re tr } W$ is a positive multiple of $\chi(W) + \overline{\chi(W)}$, χ the character of the defining representation. Expanding the exponential, every power of $\chi + \bar{\chi}$ is the character of a representation, hence a combination of irreducible characters χ_ω with non-negative integer coefficients. So $\varphi = \sum_\omega a_\omega \chi_\omega$ with $a_\omega \geq 0$, and $\sum_\omega a_\omega \dim \omega = \varphi(\mathbf{1}) = e^{x_0}$. With unitary matrices D^ω of the irreducible representations, $\chi_\omega(PQ^{-1}) = \sum_{i,j} D_{ij}^\omega(P) \overline{D_{ij}^\omega(Q)}$, and $\sum_{i,j} |D_{ij}^\omega(P)| |D_{ij}^\omega(Q)| \leq \dim \omega$ by Cauchy–Schwarz, since $\sum_{i,j} |D_{ij}^\omega(P)|^2 = \dim \omega$.

Splitting the crossing bonds. If B and C are independent and Haar distributed, so is BC^{-1} . We may therefore write $U(b_y^\times) = B_y C_y^{-1}$ with independent Haar variables B_y , C_y , one pair for each crossing bond. By cyclicity of the trace, $\text{tr } W_{y,\nu} = \text{tr}(P_{y,\nu}^+ (P_{y,\nu}^-)^{-1})$, where

$$P_{y,\nu}^+ := C_y^{-1} U(b_{y,\nu}^+) C_{y'}, \quad P_{y,\nu}^- := B_y^{-1} U(b_{y,\nu}^-) B_{y'}.$$

Hence

$$\prod_{(y,\nu)} \varphi(W_{y,\nu}) = \sum_\alpha c_\alpha \prod_{(y,\nu)} D_{i_{y,\nu} j_{y,\nu}}^{\omega_{y,\nu}}(P_{y,\nu}^+) \overline{D_{i_{y,\nu} j_{y,\nu}}^{\omega_{y,\nu}}(P_{y,\nu}^-)}, \quad c_\alpha := \prod_{(y,\nu)} a_{\omega_{y,\nu}} \geq 0,$$

the sum running over $\alpha = (\omega_{y,\nu}, i_{y,\nu}, j_{y,\nu})_{(y,\nu)}$; the corresponding series of absolute values is bounded by $\prod_{(y,\nu)} \sum_\omega a_\omega \dim \omega = e^{x_0 n_\times}$, uniformly in all variables. Put

$$G_\alpha(U) := \int \prod_y dC_y \prod_{(y,\nu)} D_{i_{y,\nu} j_{y,\nu}}^{\omega_{y,\nu}}(C_y^{-1} U(b_{y,\nu}^+) C_{y'}),$$

a bounded function of the variables of the bonds in H_+ . Since $U(b_{y,\nu}^-) = (U \circ \theta)(b_{y,\nu}^+)$, the integral over the B_y of the second product is $\overline{T_\theta G_\alpha}$. By Fubini's theorem, justified by the uniform bound, (3.13) equals

$$e^{-x_0 n_\times} \sum_\alpha c_\alpha \int dU_+ dU_- \overline{T_\theta(G_0 G_\alpha)} (G_0 G_\alpha) = e^{-x_0 n_\times} \sum_\alpha c_\alpha \left| \int (G_0 G_\alpha) dU_+ \right|^2 \geq 0,$$

where $dU_{\pm}$ is the product of the Haar measures of the bonds in $H_{\pm}$; in the last step the dU_{-} -integral is transformed by the change of variables $U|_{H_{-}} \mapsto (U \circ \theta)|_{H_{+}}$, which preserves the product Haar measure. Divide by $Z(0) > 0$. $\square$

Consequently $(F, G)_{\theta} := \langle \Theta_{\theta}(F) G \rangle$ is a positive semidefinite Hermitian form on $\mathfrak{A}(H_{+})$ — Hermitian because $\overline{(F, G)_{\theta}} = \langle T_{\theta} F \cdot \bar{G} \rangle = \langle T_{\theta}(T_{\theta} F \cdot \bar{G}) \rangle = \langle F \cdot T_{\theta} \bar{G} \rangle = (G, F)_{\theta}$, using $\langle T_{\theta} \cdot \rangle = \langle \cdot \rangle$ and $T_{\theta}^2 = 1$ — and $|(F, G)_{\theta}|^2 \leq (F, F)_{\theta}(G, G)_{\theta}$. Lemma 3.7 holds for every compact gauge group with the Wilson action.

Cells and the cell-reflection group. Let ℓ be a power of L with $\ell \leq L^m$, so that $n_c := 2L^m/\ell$ is an even integer. For $\iota \in (\mathbb{Z}/n_c\mathbb{Z})^4$ let $c_{\iota} := \iota + [0, \ell]^4 \subset \mathbb{T}_m$; $\mathcal{C}$ is the set of these cells, $|\mathcal{C}| = n_c^4$; c° is the interior of c , and $\mathfrak{A}_c$ the bounded measurable functions of the variables of the bonds with both end points in c° . The faces of the cells lie on the planes $\{x_{\mu} \in \ell\mathbb{Z}\}$, and $\ell\mathbb{Z} \subset \varepsilon\mathbb{Z}$; so the reflections $\theta_{\mu,a} : x_{\mu} \mapsto 2a\ell - x_{\mu}$, $a \in \mathbb{Z}/n_c\mathbb{Z}$, are of the kind of Lemma 3.7, with fixed planes $\{x_{\mu} = a\ell\}$ and $\{x_{\mu} = (a + n_c/2)\ell\}$; on cell indices $\theta_{\mu,a}$ maps ι_{μ} to $2a - 1 - \iota_{\mu}$ and fixes ι_{ν} , $\nu \neq \mu$. Let $\mathfrak{G}$ be the group generated by the $\theta_{\mu,a}$.

Lemma 3.8 (The cell-reflection group). (i) $\mathfrak{G}$ acts simply transitively on $\mathcal{C}$; for $c, c' \in \mathcal{C}$ let $\mathfrak{g}_{c \rightarrow c'}$ be the element with $\mathfrak{g}_{c \rightarrow c'} c = c'$.

(ii) Every $\mathfrak{g} \in \mathfrak{G}$ has the form $x \mapsto (\sigma_{\mu} x_{\mu} + t_{\mu})_{\mu}$ with $\sigma_{\mu} = \pm 1$; $\text{sgn}(\mathfrak{g}) := \prod_{\mu} \sigma_{\mu}$ is a homomorphism, and $\text{sgn}(\mathfrak{g}_{c_{\iota} \rightarrow c_{\iota'}}) = (-1)^{\sum_{\mu} (\iota'_{\mu} - \iota_{\mu})}$.

(iii) Every $\mathfrak{g} \in \mathfrak{G}$ is an isometry mapping Λ_0 onto itself, cells onto cells, and c° onto $(\mathfrak{g}c)^{\circ}$.

(iv) For each c , exactly half of the cells c' have $\text{sgn}(\mathfrak{g}_{c \rightarrow c'}) = +1$.

Proof. For fixed μ , $\theta_{\mu,a}\theta_{\mu,a'}$ is the translation by $2(a - a')\ell e_{\mu}$, so the group generated by the $\theta_{\mu,a}$ consists of the $n_c/2$ translations by multiples of $2\ell e_{\mu}$ and the $n_c/2$ reflections $x_{\mu} \mapsto 2a\ell - x_{\mu}$; on indices they act as $\iota_{\mu} \mapsto \iota_{\mu} + 2t$ and $\iota_{\mu} \mapsto 2a - 1 - \iota_{\mu}$. Given ι_{μ} and ι'_{μ} , exactly one of these maps ι_{μ} to ι'_{μ} : a translation if $\iota'_{\mu} - \iota_{\mu}$ is even, a reflection if it is odd (parities are well defined because n_c is even). Elements for different μ commute; this gives (i) and the form in (ii), and sgn is multiplicative because the linear parts multiply. (iii) $2a\ell/\varepsilon \in \mathbb{Z}$, so $x_{\mu} \mapsto 2a\ell - x_{\mu}$ maps $\varepsilon(\mathbb{Z} + \frac{1}{2})$ onto itself; and cells go to cells. (iv) $c_{\iota} \mapsto c_{\iota + e_1}$ is a bijection of $\mathcal{C}$ that changes the parity of $\sum_{\mu} \iota_{\mu}$. $\square$

Chessboard copies. For $\mathfrak{g} \in \mathfrak{G}$, $T_{\mathfrak{g}}$ maps $\mathfrak{A}_c$ to $\mathfrak{A}_{\mathfrak{g}c}$. For $F \in \mathfrak{A}_{c_0}$ and $c' \in \mathcal{C}$ put, with $\mathfrak{g} := \mathfrak{g}_{c_0 \rightarrow c'}$,

$$F^{[c']} := T_{\mathfrak{g}} F \quad \text{if } \text{sgn}(\mathfrak{g}) = +1, \quad F^{[c']} := \overline{T_{\mathfrak{g}} F} \quad \text{if } \text{sgn}(\mathfrak{g}) = -1$$

(the copies of F based at c_0 ; $F^{[c_0]} = F$). For every face reflection $\theta = \theta_{\mu,a}$ and all $c', c'' \in \mathcal{C}$:

- (k1) $\Theta_{\theta}(F^{[c']}) = F^{[\theta c']}$;
- (k2) $(F^{[c']})^{[c'']} = F^{[c'']}$, the copies of $F^{[c']} \in \mathfrak{A}_{c'}$ being based at c' ;
- (k3) $(FG)^{[c']} = F^{[c']}G^{[c']}$ and $(\bar{F})^{[c']} = \overline{F^{[c]}}$.

Indeed, $\theta \mathfrak{g}_{c_0 \rightarrow c'} = \mathfrak{g}_{c_0 \rightarrow \theta c'}$ by uniqueness and $\text{sgn}(\theta \mathfrak{g}) = -\text{sgn}(\mathfrak{g})$, which with $T_{\theta} T_{\mathfrak{g}} = T_{\theta \mathfrak{g}}$ and the definition of Θ_{θ} gives (k1); $\mathfrak{g}_{c' \rightarrow c''} \mathfrak{g}_{c_0 \rightarrow c'} = \mathfrak{g}_{c_0 \rightarrow c''}$ and the multiplicativity of sgn give (k2); (k3) holds because $T_{\mathfrak{g}}$ is multiplicative and commutes with conjugation. The *chessboard value* of $F \in \mathfrak{A}_{c_0}$ is $\mathbf{c}(F) := \langle \prod_{c' \in \mathcal{C}} F^{[c']} \rangle$. It is real and non-negative: for a face reflection θ , the interior of every

cell lies in exactly one of the open halves $H_{\pm}(\theta)$ of $\mathbb{T}_m \setminus \Pi$; the products $G_{\pm}$ of the $F^{[c']}$ over the cells with $(c')^{\circ} \subset H_{\pm}(\theta)$ lie in $\mathfrak{A}(H_{\pm})$, and $\Theta_{\theta}(G_{+}) = G_{-}$ by (k1); so $\mathfrak{c}(F) = \langle \Theta_{\theta}(G_{+}) G_{+} \rangle \geq 0$ by Lemma 3.7. The *chessboard norm* is $\|F\|_{\text{ch}} := \mathfrak{c}(F)^{1/|C|}$. By (k2) the copies of $F^{[c']}$ based at c' are the copies of F , so $\|F^{[c']}\|_{\text{ch}} = \|F\|_{\text{ch}}$; and $\|1\|_{\text{ch}} = 1$.

Lemma 3.9 (Chessboard estimate). *For $F_c \in \mathfrak{A}_c$, $c \in \mathcal{C}$,*

$$\left| \left\langle \prod_{c \in \mathcal{C}} F_c \right\rangle \right| \leq \prod_{c \in \mathcal{C}} \|F_c\|_{\text{ch}}.$$

This is the chessboard estimate of Fröhlich, Israel, Lieb and Simon [17], here for complex observables, the antilinear reflection Θ_{θ} , link reflections of a lattice gauge theory and observables of the open cells; we give the proof.

Proof. For a configuration $Y = (Y_c)_{c \in \mathcal{C}}$, $Y_c \in \mathfrak{A}_c$, write $\mathbf{A}(Y) := \langle \prod_c Y_c \rangle$. For a face reflection θ let $\mathcal{C}_{\pm}(\theta)$ be the set of cells with interior in $H_{\pm}(\theta)$; let $Y^{\theta,+}$ be the configuration equal to Y_c for $c \in \mathcal{C}_{+}(\theta)$ and to $\Theta_{\theta}(Y_{\theta c})$ for $c \in \mathcal{C}_{-}(\theta)$, and $Y^{\theta,-}$ the one equal to $\Theta_{\theta}(Y_{\theta c})$ on $\mathcal{C}_{+}(\theta)$ and to Y_c on $\mathcal{C}_{-}(\theta)$.

Step 0 (Schwarz). $|\mathbf{A}(Y)|^2 \leq \mathbf{A}(Y^{\theta,+}) \mathbf{A}(Y^{\theta,-})$, and both factors are non-negative. Indeed, with $G_{\pm} := \prod_{c \in \mathcal{C}_{\pm}(\theta)} Y_c \in \mathfrak{A}(H_{\pm})$ and Θ_{θ} an involution, $\mathbf{A}(Y) = \langle \Theta_{\theta}(\Theta_{\theta} G_{-}) G_{+} \rangle = (\Theta_{\theta} G_{-}, G_{+})_{\theta}$, so

$$\begin{aligned} |\mathbf{A}(Y)|^2 &\leq (\Theta_{\theta} G_{-}, \Theta_{\theta} G_{-})_{\theta} (G_{+}, G_{+})_{\theta} \\ &= \langle G_{-} \Theta_{\theta}(G_{-}) \rangle \langle \Theta_{\theta}(G_{+}) G_{+} \rangle. \end{aligned}$$

Since $\Theta_{\theta}(G_{+}) = \prod_{c \in \mathcal{C}_{-}(\theta)} \Theta_{\theta}(Y_{\theta c})$, $\langle \Theta_{\theta}(G_{+}) G_{+} \rangle = \mathbf{A}(Y^{\theta,+})$, and likewise $\langle G_{-} \Theta_{\theta}(G_{-}) \rangle = \mathbf{A}(Y^{\theta,-})$; both are ≥ 0 by Lemma 3.7, applied to G_{+} and to $G' := \Theta_{\theta} G_{-} \in \mathfrak{A}(H_{+})$ respectively, since $\langle G_{-} \Theta_{\theta}(G_{-}) \rangle = \langle \Theta_{\theta}(G') G' \rangle$.

Step 1 (spreading). Let Y be a configuration, $c_0 \in \mathcal{C}$ and $F := Y_{c_0}$. There are face reflections $\theta_1, \dots, \theta_T$ and signs such that $Y^0 := Y$, $Y^i := (Y^{i-1})^{\theta_i, \pm}$ ends with the chessboard configuration $Y^T = (F^{[c]})_{c \in \mathcal{C}}$. To see this, call a set of cells a *box* if it is $\{c_i : \iota_{\mu} \in S_{\mu} \ \forall \mu\}$ with each S_{μ} a set of consecutive residues mod n_c , and keep track of a box $\mathcal{B}$ such that the current configuration equals $F^{[c]}$ at every $c \in \mathcal{B}$; initially $\mathcal{B} = \{c_0\}$. Treat the directions $\mu = 1, 2, 3, 4$ in turn, with $S_{\nu} = \mathbb{Z}/n_c\mathbb{Z}$ for the directions already treated and S_{ν} a single residue for the others. Let $s := |S_{\mu}| < n_c$. If $s \leq n_c/2$ and $S_{\mu} = \{a_0, \dots, a_0 + s - 1\}$, take $\theta := \theta_{\mu, a_0+s}$: one of its halves consists of the cells with $\iota_{\mu} \in \{a_0 + s - n_c/2, \dots, a_0 + s - 1\}$ and contains $\mathcal{B}$; keeping that half, the new configuration carries $\Theta_{\theta}(F^{[c]}) = F^{[\theta c]}$ at θc for $c \in \mathcal{B}$ by (k1), so $\mathcal{B} \cup \theta \mathcal{B}$ is a box whose μ -arc has length $2s$. If $n_c/2 < s < n_c$, S_{μ} contains $n_c/2$ consecutive residues $a_0, \dots, a_0 + n_c/2 - 1$, which form a half of $\theta := \theta_{\mu, a_0}$; keeping that half fills the μ -arc, again by (k1). After finitely many steps $S_{\mu} = \mathbb{Z}/n_c\mathbb{Z}$ for all μ .

Step 2 (zero norms). Let $\mathcal{F} := \{F_c^{[c']} : c, c' \in \mathcal{C}\}$, a finite set closed under every Θ_{θ} by (k1); every configuration met below has its entries in $\mathcal{F}$, so $|\mathbf{A}| \leq \Xi := \max\{1, \max_c \|F_c\|_{\infty}\}^{|C|}$ for all of them. Suppose $\|F_{c_0}\|_{\text{ch}} = 0$ for some c_0 , and run Step 1 from $X := (F_c)_c$ and c_0 . By Step 0, $|\mathbf{A}(Y^{i-1})|^2 \leq \Xi \mathbf{A}(Y^i)$ for each i , hence $|\mathbf{A}(X)|^{2^T} \leq \Xi^{2^T-1} \mathbf{A}(Y^T) = \Xi^{2^T-1} \mathfrak{c}(F_{c_0}) = 0$, and the lemma holds.

Step 3 (maximiser). Assume now $\|F_c\|_{\text{ch}} > 0$ for all c . Call Y admissible if $Y_{c'} \in \mathcal{F} \cap \mathfrak{A}_{c'}$ for every c' ; there are finitely many, $X := (F_c)_c$ is one ($F_c = F_c^{[c]}$), and by (k2) every entry of an admissible configuration has chessboard norm in $\{\|F_c\|_{\text{ch}}\} \subset (0, \infty)$. Put $\Gamma(Y) := |\mathbf{A}(Y)| / \prod_{c'} \|Y_{c'}\|_{\text{ch}}$ and $\Gamma_* := \max_Y \Gamma(Y)$ over admissible Y ; $\Gamma_* \geq 1$, since the chessboard configuration of F_{c_0} is admissible with $\Gamma = \mathfrak{c}(F_{c_0}) / \|F_{c_0}\|_{\text{ch}}^{|C|} = 1$. For admissible Y and a face reflection θ , $Y^{\theta, \pm}$ are admissible, and

since $\|\Theta_\theta Z\|_{\text{ch}} = \|Z\|_{\text{ch}}$ for $Z \in \mathcal{F}$ (by (k1), (k2)), $\prod_{c'} \|Y_{c'}^{\theta,+}\|_{\text{ch}} \prod_{c'} \|Y_{c'}^{\theta,-}\|_{\text{ch}} = (\prod_{c'} \|Y_{c'}\|_{\text{ch}})^2$; Step 0 gives $\Gamma(Y)^2 \leq \Gamma(Y^{\theta,+})\Gamma(Y^{\theta,-})$. Hence if $\Gamma(Y) = \Gamma_*$ then $\Gamma(Y^{\theta,\pm}) = \Gamma_*$: the images of a maximiser under Step 0 are maximisers.

Step 4 (conclusion). Let Y be a maximiser, fix c_0 and put $F := Y_{c_0}$. Then $F = F_{c_1}^{[c_0]}$ for some c_1 , so by (k2) $F^{[c]} = F_{c_1}^{[c]}$ for every c , and $\|F\|_{\text{ch}} = \|F_{c_1}\|_{\text{ch}} > 0$. Running Step 1 from Y and c_0 , every configuration of the sequence is a maximiser by Step 3; so the chessboard configuration $(F^{[c]})_c$ is a maximiser, and its value of Γ is $\mathfrak{c}(F)/\|F\|_{\text{ch}}^{|C|} = 1$. Hence $\Gamma_* = 1$, and $|\mathbf{A}(X)| = \Gamma(X) \prod_c \|F_c\|_{\text{ch}} \leq \prod_c \|F_c\|_{\text{ch}}$. $\square$

3.6 The per-ball volume-free bound: clause (iii)(b)

The idea is as follows. By the chessboard estimate, $|Z(w_z)/Z(0)|$ is bounded by a product, over the cells that contain sources, of $|C|$ -th roots of partition functions with a *periodised* source, i.e. with copies of the source in every cell. These are controlled by (3.6), at the price $e^{c_{\text{vol}}|\Lambda_{K'}|}$, whose $|C|$ -th root is bounded in terms of the cell volume only; but only up to the factor $e^{\text{Re } \mathcal{E}}$ of the periodised source. The vacuum energy $\mathcal{E}(w_z)$ is not bounded uniformly in the cutoff: its first derivatives at $z = 0$ are the one-point functions $\langle \mathcal{O}(f_i) \rangle$, which grow like the number of plaquettes in the supports; only its mixed derivatives are controlled (Lemma 3.5). So no modulus of $\mathcal{E}$ may be taken. We therefore change the counterterm: we replace $\mathcal{E}(w_z)$ by a holomorphic function $\mathcal{E}''(z)$, built from the vacuum energies of transported copies of the sources, whose real part *equals* the cell average of the real part of the periodised vacuum energy. The divergent bulk parts then cancel identically, while the mixed derivative of $\mathcal{E}''$ is controlled by Lemma 3.5, because each transported family of sources has the same support geometry as the original one.

Theorem 3.10 (Per-ball volume-free bound). *Assume the hypotheses of Theorem 3.1. Let $n \geq 2$, $R > 0$, $\mathbf{d} > 0$, and let $\ell = \ell(n, R)$ be the least power of L with $\ell \geq 8n(R + 1)$. There is $C_n^\infty(\mathbf{d}; R)$, depending only on $(L, \mathbf{p}, \gamma, g, \varrho, n, \mathbf{d}, R)$ and non-increasing in $\mathbf{d}$, such that for every $m \geq m_{\mathbb{T}}(g_-(g))$ with $L^m \geq \ell$, every $g_0 \in (0, g_-(g)]$ (and $K = K(g_0, g)$), and all real Lipschitz functions $f_1, \dots, f_n$ on $\mathbb{T}_m$, each supported in a closed ball of radius R , with supports pairwise at distance $\geq \mathbf{d}$,*

$$|S_n^T(f_1, \dots, f_n)| \leq C_n^\infty(\mathbf{d}; R) \prod_{i=1}^n \|f_i\|_{\sharp}. \quad (3.14)$$

One may take

$$C_n^\infty(\mathbf{d}; R) := \max \left\{ \left[2C_{\text{UV}}(\mathbf{d}; R) + 2^n \log 2 (2Q_\infty^n + 2)^n \left(\frac{n}{\varrho} \right)^n \right], \right. \\ \left. \mathfrak{c}_n (12(2RL^{b_*} + 1)^4)^n \right\}, \quad (3.15)$$

$$Q_\infty := \exp(c_{\text{vol}} L^{4b_*} \ell^4 + 2C_{\text{face}}(R)),$$

with C_{UV} from Lemma 3.5, C_{face} from Lemma 3.13, and $\mathfrak{c}_n$ as in Theorem 3.1(e).

The conditions $m \geq m_{\mathbb{T}}(g_-(g))$ and $L^m \geq \ell$ are floors on the volume; nothing bounds m or K from above. The constant depends on the common radius R of the balls and on $\mathbf{d}$, not on the distances between the balls or on the diameter of their union. No decay in $\mathbf{d}$ is asserted. Section 4 uses the theorem in this per-ball form: it makes the bound uniform under translations of some of the test functions away from the others, which is what the Osterwalder–Schrader reconstruction requires; a bound whose constant depends on the diameter of the union of the supports would not be uniform under such translations.

The case $K < b_*$. Then $\varepsilon = L^{-K} > L^{-b_*}$. For each of the six orientations, the barycentres of the plaquettes form a translate of $\varepsilon\mathbb{Z}^4$, of which at most $(2R/\varepsilon + 1)^4$ points lie in a ball of radius R ; so $|\mathcal{O}(f_i)| \leq 2\|f_i\|_\infty \cdot 6(2RL^{b_*} + 1)^4$, and the cumulant formula used in the proof of Theorem 3.1(e) gives (3.14) with the second member of (3.15). From now on $K \geq b_*$, $K' := K - b_*$ is the stopping level of Theorem 3.1(c) with $s = b_*$, $\|f_i\|_\# > 0$ for all i (otherwise $S_n^T = 0$), $\bar{B}_i \supset \text{supp } f_i$ is a closed ball of radius R , and $\mathcal{C}, \mathfrak{G}$ are the cells and the cell-reflection group of §3.5 with $\ell = \ell(n, R) \leq L^m$.

Lemma 3.11 (Placement). *There is $\mathbf{a} \in \varepsilon\mathbb{Z}^4$ such that every translated ball $\bar{B}_i + \mathbf{a}$ lies at distance $\geq R + 2$ from all the planes $\{x_\mu \in \ell\mathbb{Z}\}$, hence in the interior of one cell; and $S_n^T(f_1(\cdot - \mathbf{a}), \dots, f_n(\cdot - \mathbf{a})) = S_n^T(f_1, \dots, f_n)$.*

Proof. Fix μ , and let $x_{i,\mu}$ be the μ -th coordinate of the centre of $\bar{B}_i$. The ball $\bar{B}_i + te_\mu$ comes within distance $R + 2$ of $\{x_\mu \in \ell\mathbb{Z}\}$ if and only if $\text{dist}(x_{i,\mu} + t, \ell\mathbb{Z}) < 2R + 2$, i.e. if and only if t lies in a certain open arc of length $4R + 4$ of the circle $\mathbb{R}/\ell\mathbb{Z}$. The complement of the union of these n arcs has measure $\geq \ell - 4n(R + 1) \geq 4n(R + 1)$ and consists of at most n closed arcs; one of them has length $\geq 4(R + 1) > \varepsilon$ and therefore contains a point $\mathbf{a}_\mu$ of $\varepsilon\mathbb{Z}/\ell\mathbb{Z}$ (recall $\ell \in \varepsilon\mathbb{Z}$). Put $\mathbf{a} := (\mathbf{a}_\mu)_\mu$. The translation $\tau : x \mapsto x + \mathbf{a}$ maps Λ_0 onto itself and $x(\tau p) = x(p) + \mathbf{a}$, so $A(\sum_i z_i f_i(\cdot - \mathbf{a}), U) = A(w_z, U \circ \tau)$, and $Z(\sum_i z_i f_i(\cdot - \mathbf{a})) = Z(w_z)$ by the invariance of the Wilson measure. $\square$

Henceforth the sources are placed: we replace f_i by $f_i(\cdot - \mathbf{a})$ and $\bar{B}_i$ by $\bar{B}_i + \mathbf{a}$, which changes neither S_n^T , nor the radii, nor the distances, nor the norms. For a cell c let $I_c := \{i : \bar{B}_i \subset c^\circ\}$, and let $\mathcal{C}_{\text{occ}} := \{c : I_c \neq \emptyset\}$; then $\sum_c |I_c| = n$ and $|\mathcal{C}_{\text{occ}}| \leq n$. For $z \in \mathbb{C}^n$, $c \in \mathcal{C}_{\text{occ}}$ and $\mathbf{g} \in \mathfrak{G}$ put

$$w_{c,z} := \sum_{i \in I_c} z_i f_i, \quad F_{c,z} := e^{A(w_{c,z}, \cdot)}, \quad u_{c,\mathbf{g},z} := w_{c,z} \circ \mathbf{g}^{-1} = \sum_{i \in I_c} z_i f_i \circ \mathbf{g}^{-1},$$

$F_{c,z} := 1$ for $c \notin \mathcal{C}_{\text{occ}}$, and $\xi^{(+)} := \xi$, $\xi^{(-)} := \bar{\xi}$ for numbers and functions.

Lemma 3.12 (Periodised weights). *Let $c \in \mathcal{C}_{\text{occ}}$ and $z \in \mathbb{C}^n$.*

- (i) $F_{c,z} \in \mathfrak{A}_c$, and $\prod_{c' \in \mathcal{C}} F_{c',z} = e^{A(w_z, \cdot)}$; hence $Z(w_z)/Z(0) = \langle \prod_{c'} F_{c',z} \rangle$.
- (ii) For $\mathbf{g} \in \mathfrak{G}$, the copy of $F_{c,z}$ based at c in the cell $\mathbf{g}c$ is $F_{c,z}^{[\mathbf{g}c]} = e^{A(u_{c,\mathbf{g},z}^{(\text{sgn } \mathbf{g})}, \cdot)}$. Hence $\prod_{c'} F_{c',z}^{[c']} = e^{A(w_{c,z}^{\text{ch}}, \cdot)}$ with the periodised weight

$$w_{c,z}^{\text{ch}} := \sum_{\mathbf{g} \in \mathfrak{G}} u_{c,\mathbf{g},z}^{(\text{sgn } \mathbf{g})} = \sum_{i \in I_c} (z_i h_{c,i}^+ + \bar{z}_i h_{c,i}^-), \quad h_{c,i}^\pm := \sum_{\mathbf{g} : \text{sgn } \mathbf{g} = \pm 1} f_i \circ \mathbf{g}^{-1},$$

and $\|F_{c,z}\|_{\text{ch}}^{|C|} = Z(w_{c,z}^{\text{ch}})/Z(0) \in [0, \infty)$.

- (iii) The summand of $w_{c,z}^{\text{ch}}$ of index $\mathbf{g}$ is supported in the balls $\mathbf{g}(\bar{B}_i)$, $i \in I_c$, which lie in $(\mathbf{g}c)^\circ$ at distance $\geq R + 2$ from all the planes $\{x_\mu \in \ell\mathbb{Z}\}$. The $h_{c,i}^\pm$ are real Lipschitz functions with $\|h_{c,i}^\pm\|_\# = \|f_i\|_\#$. If $z \in \frac{1}{2}\mathbb{D}$, then $(z_i, \bar{z}_i)_{i \in I_c} \in \mathbb{D}_\ell((h_{c,i}^+, h_{c,i}^-)_{i \in I_c})$; and for every $z \in \mathbb{D}$ and $\mathbf{g} \in \mathfrak{G}$, both $(z_i)_{i \in I_c}$ and $(\bar{z}_i)_{i \in I_c}$ lie in $\mathbb{D}_\ell((f_i \circ \mathbf{g}^{-1})_{i \in I_c})$.

Proof. (i) If $w_{c,z}(x(p)) \neq 0$ then $x(p) \in \bar{B}_i$ for some $i \in I_c$; the bonds of p are within $\varepsilon \leq 1$ of $x(p)$, and $\bar{B}_i$ is at distance $\geq R + 2$ from ∂c , so they have both end points in c° . The product formula holds because A is linear in the weight and $w_z = \sum_{c'} w_{c',z}$. (ii) For a weight v and $\mathbf{g} \in \mathfrak{G}$, $T_{\mathbf{g}}(e^{A(v, \cdot)})(U) = \exp \sum_p v(x(p)) a_{\mathbf{g}p}(U) = e^{A(v \circ \mathbf{g}^{-1}, U)}$, because $x(\mathbf{g}p) = \mathbf{g}x(p)$; and $\overline{e^{A(v, U)}} = e^{A(\bar{v}, U)}$

because a_p is real. Since $\mathfrak{g}_{c \rightarrow \mathfrak{g}c} = \mathfrak{g}$, this gives $F_{c,z}^{[\mathfrak{g}c]}$; multiplying over $\mathfrak{g} \in \mathfrak{G}$, i.e. over all cells (Lemma 3.8(i)), gives the product, and the second expression for $w_{c,z}^{\text{ch}}$ uses that the f_i are real. Then $\mathfrak{c}(F_{c,z}) = \langle e^{A(w_{c,z}^{\text{ch}})} \rangle = Z(w_{c,z}^{\text{ch}})/Z(0)$. (iii) $\mathfrak{g}$ is an isometry mapping cells to cells and the planes $\{x_\mu \in \ell\mathbb{Z}\}$ to themselves, which gives the first assertion; in particular the summands have disjoint supports, $\|h_{c,i}^\pm\|_\infty = \|f_i\|_\infty$, and $h_{c,i}^\pm$ vanishes on the faces of all cells. If x, y lie in one closed cell, the segment between them inside the cell realises their distance (because $\ell \leq L^m$), so $|h_{c,i}^\pm(x) - h_{c,i}^\pm(y)| \leq \text{Lip}(f_i)|x - y|$, $\mathfrak{g}^{-1}$ being an isometry. If they lie in different cells, a shortest segment from x to y leaves the cell of x at a point x^* and enters the cell of y at a point y^* ; since $h_{c,i}^\pm(x^*) = h_{c,i}^\pm(y^*) = 0$, $|h_{c,i}^\pm(x) - h_{c,i}^\pm(y)| \leq \text{Lip}(f_i)(|x - x^*| + |y^* - y|) \leq \text{Lip}(f_i)|x - y|$. So $\|h_{c,i}^\pm\|_\# \leq \|f_i\|_\#$, with equality by restriction to one cell. Finally $\sum_{i \in I_c} (|z_i| + |\bar{z}_i|)\|f_i\|_\# < \varrho$ if $z \in \frac{1}{2}\mathbb{D}$, and $\|f_i \circ \mathfrak{g}^{-1}\|_\# = \|f_i\|_\#$. $\square$

Bulk and face parts. For $j \leq K'$ and $c' \in \mathcal{C}$ let $\mathcal{X}_{c',j} := \{X \in \mathcal{D}_j : X^+ \subset (c')^\circ\}$ and $\mathcal{X}_{\partial,j} := \mathcal{D}_j \setminus \bigcup_{c'} \mathcal{X}_{c',j}$ (a disjoint decomposition). For $w \in \mathcal{W}_{K'}$ put

$$\begin{aligned}\mathcal{E}_{c'}^{\text{bulk}}(w) &:= \sum_{j \leq K'} \sum_{X \in \mathcal{X}_{c',j}} e_j(X; w), \\ \mathcal{E}^{\text{bulk}}(w) &:= \sum_{c' \in \mathcal{C}} \mathcal{E}_{c'}^{\text{bulk}}(w), \\ \mathcal{E}^\partial(w) &:= \sum_{j \leq K'} \sum_{X \in \mathcal{X}_{\partial,j}} e_j(X; w),\end{aligned}$$

so that $\mathcal{E} = \mathcal{E}^{\text{bulk}} + \mathcal{E}^\partial$.

Lemma 3.13 (Locality across cells, and the face terms). *(i) If $X \in \mathcal{X}_{\partial,j}$, then X^+ meets a plane $\{x_\mu \in \ell\mathbb{Z}\}$.*

(ii) Let $X \in \mathcal{X}_{c',j}$, and let $w, w' \in \mathcal{W}_{K'}$ coincide on c' and vanish outside c' except on balls of radius R contained in other cells at distance $\geq R + 2$ from the planes $\{x_\mu \in \ell\mathbb{Z}\}$. Then $e_j(X; w) = e_j(X; w')$.

(iii) Let $v = \sum_{\eta \in \mathbf{H}} v_\eta$, where each v_η vanishes outside a closed ball $\bar{B}'_\eta$ of radius R at distance $\geq R + 2$ from all the planes $\{x_\mu \in \ell\mathbb{Z}\}$, and suppose that $v = \sum_l \zeta_l h_l$ for a finite family h of real Lipschitz functions and some $\zeta \in \mathbb{D}_\varrho(h)$. Then

$$\sum_{j \leq K'} \sum_{X \in \mathcal{X}_{\partial,j}} |e_j(X; v)| \leq |\mathbf{H}| C_{\text{face}}(R), \quad C_{\text{face}}(R) := C_e C_{\text{tree}} \sum_{i \geq 0} \mathbf{a}(R, i) e^{-\frac{\kappa}{4} [RL^i / (4M) - 3]_+},$$

a convergent series, independent of K, g_0, m, ℓ and positions.

Proof. (i) X is connected and every cube of X^+ meets X , so X^+ is a connected union of closed cubes. The complement of the planes $\{x_\mu \in \ell\mathbb{Z}\}$ is the disjoint union of the open cells. If X^+ meets two different open cells, a path in X^+ joining them meets a plane; otherwise X^+ meets at most one open cell and is not contained in it, so it contains a point outside all open cells, i.e. on a plane. (ii) Let $p \in \text{plaq}(X^+)$. Then $x(p)$ is within ε of a point of $(c')^\circ$. A point of $(c')^\circ$ lies outside every other open cell c'' , hence at distance $\geq R + 2$ from a ball in c'' at distance $\geq R + 2$ from $\partial c''$; so $x(p)$ is at distance $> R$ from such balls. Thus either $x(p) \in c'$, where $w = w'$, or $x(p)$ lies in another cell but on none of these balls, where $w = w' = 0$. By (d1), $e_j(X; w) = e_j(X; w')$. (iii) By (d1), $e_j(X; v) \neq 0$ requires some $p \in \text{plaq}(X^+)$ with $x(p) \in \bar{B}'_\eta$ for some η ; for $X \in \mathcal{X}_{\partial,j}$, X^+ meets a plane by (i), and

Lemma 3.4(iii), with Σ the union of the planes, gives $d_j(X) \geq [RL^{K-j}/(4M)-3]_+$. Assign each such X to one such η ; as in the proof of Lemma 3.5, X then contains one of at most $\mathbf{a}(R, K-j)$ cubes (Lemma 3.4(iv) about the centre of $\bar{B}'_\eta$). With (d3), $e^{-(1-\delta)\kappa d_j} \leq e^{-\frac{\kappa}{4}d_j} e^{-(\frac{3}{4}-\delta)\kappa d_j}$ and Lemma 3.4(v), the terms assigned to η at scale j contribute at most $C_e C_{\text{tree}} \mathbf{a}(R, K-j) e^{-\frac{\kappa}{4}[RL^{K-j}/(4M)-3]_+}$. Sum over η and over $j \leq K'$, with $i := K-j \geq 0$; the series converges as in Lemma 3.5. $\square$

The re-centred counterterm. For $z \in \mathbb{D}$ we define

$$\mathcal{E}''(z) := \sum_{c \in \mathcal{C}_{\text{occ}}} \frac{1}{|\mathcal{C}|} \sum_{\mathbf{g} \in \mathfrak{G}} \left[\mathcal{E}_{\mathbf{gc}}^{\text{bulk}}(u_{c,\mathbf{g},z}^{(\text{sgn } \mathbf{g})}) \right]^{(\text{sgn } \mathbf{g})} + \mathcal{E}^\partial(w_z), \quad \mathbf{h}''(z) := e^{-\mathcal{E}''(z)} \frac{Z(w_z)}{Z(0)}. \quad (3.16)$$

The weights inserted lie in $\mathcal{W}_{K'}$ by Lemma 3.12(iii). The bulk vacuum energy of each occupied cell is replaced by the average over all cells of the bulk vacuum energies of its transported copies, the values on cells of odd parity being conjugated; the face part is that of Theorem 3.1.

Proof of Theorem 3.10. Step 1: holomorphy and the mixed derivative of $\mathcal{E}''$. For $\text{sgn } \mathbf{g} = +1$ the summand of index $(c, \mathbf{g})$ is $\Phi_{c,\mathbf{g}}(z) := \mathcal{E}_{\mathbf{gc}}^{\text{bulk}}(\sum_{i \in I_c} z_i f_i \circ \mathbf{g}^{-1})$, holomorphic on $\mathbb{D}$ by (d2) for the family $(f_i \circ \mathbf{g}^{-1})_{i \in I_c}$ (Lemma 3.12(iii)). For $\text{sgn } \mathbf{g} = -1$, since the f_i are real, $u_{c,\mathbf{g},z}^{(-)} = \sum_{i \in I_c} \bar{z}_i f_i \circ \mathbf{g}^{-1}$, and the summand is $\overline{\Phi_{c,\mathbf{g}}(\bar{z})}$; as $\mathbb{D}$ is invariant under conjugation, this is holomorphic on $\mathbb{D}$, and its derivatives at 0 are the conjugates of those of $\Phi_{c,\mathbf{g}}$. $\mathcal{E}^\partial(w_z)$ is holomorphic on $\mathbb{D}$ by (d2), and $\mathcal{E}''(0) = 0$ by (d1). If $|\mathcal{C}_{\text{occ}}| \geq 2$, each bulk summand depends only on $(z_i)_{i \in I_c}$ with $I_c \neq \{1, \dots, n\}$, so $\partial_{z_1} \cdots \partial_{z_n}$ annihilates it. If $|\mathcal{C}_{\text{occ}}| = 1$, so that $I_c = \{1, \dots, n\}$, Lemma 3.5 applied to the family $(f_i \circ \mathbf{g}^{-1})_{i \leq n}$ — supported in the balls $\mathbf{g}(\bar{B}_i)$ of radius R , pairwise at distance $\geq \mathbf{d}$ because $\mathbf{g}$ is an isometry — with $\mathcal{Y}_j := \mathcal{X}_{\mathbf{gc},j}$ gives $|\partial_{z_1} \cdots \partial_{z_n} \Phi_{c,\mathbf{g}}(0)| \leq C_{\text{UV}}(\mathbf{d}; R)(n/\varrho)^n \prod_i \|f_i\|_\#$ for every $\mathbf{g}$, and the same bound for the conjugated summands; their average over the $|\mathcal{C}| = |\mathfrak{G}|$ elements obeys the same bound. Lemma 3.5 with $h = \bar{f}$ and $\mathcal{Y}_j := \mathcal{X}_{\partial,j}$ bounds the face part. Hence

$$|\partial_{z_1} \cdots \partial_{z_n} \mathcal{E}''(0)| \leq 2C_{\text{UV}}(\mathbf{d}; R) \left(\frac{n}{\varrho}\right)^n \prod_{i=1}^n \|f_i\|_\#, \quad (3.17)$$

with no factor depending on the volume.

Step 2: the real part of $\mathcal{E}''$. Let $z \in \frac{1}{2}\mathbb{D}$. Since $\text{Re } \bar{\xi} = \text{Re } \xi$,

$$\text{Re } \mathcal{E}''(z) = \sum_{c \in \mathcal{C}_{\text{occ}}} \frac{1}{|\mathcal{C}|} \sum_{\mathbf{g} \in \mathfrak{G}} \text{Re } \mathcal{E}_{\mathbf{gc}}^{\text{bulk}}(u_{c,\mathbf{g},z}^{(\text{sgn } \mathbf{g})}) + \text{Re } \mathcal{E}^\partial(w_z).$$

For $X \in \mathcal{X}_{\mathbf{gc},j}$, the weights $w_{c,z}^{\text{ch}}$ and $u_{c,\mathbf{g},z}^{(\text{sgn } \mathbf{g})}$ coincide on $\mathbf{gc}$ (the summand of index $\mathbf{g}$ is the only one supported there), and outside $\mathbf{gc}$ both vanish except on the balls of Lemma 3.12(iii); by Lemma 3.13(ii), $e_j(X; u_{c,\mathbf{g},z}^{(\text{sgn } \mathbf{g})}) = e_j(X; w_{c,z}^{\text{ch}})$. Hence $\mathcal{E}_{\mathbf{gc}}^{\text{bulk}}(u_{c,\mathbf{g},z}^{(\text{sgn } \mathbf{g})}) = \mathcal{E}_{\mathbf{gc}}^{\text{bulk}}(w_{c,z}^{\text{ch}})$, and since $\mathbf{g} \mapsto \mathbf{gc}$ is a bijection of $\mathfrak{G}$ onto $\mathcal{C}$,

$$\text{Re } \mathcal{E}''(z) = \sum_{c \in \mathcal{C}_{\text{occ}}} \frac{1}{|\mathcal{C}|} \text{Re } \mathcal{E}^{\text{bulk}}(w_{c,z}^{\text{ch}}) + \text{Re } \mathcal{E}^\partial(w_z) \quad (z \in \tfrac{1}{2}\mathbb{D}). \quad (3.18)$$

Step 3: the chessboard side. Let $z \in \frac{1}{2}\mathbb{D}$. By Lemma 3.12(i) and Lemma 3.9, $|Z(w_z)/Z(0)| \leq \prod_{c \in \mathcal{C}_{\text{occ}}} \|F_{c,z}\|_{\text{ch}}$. By Lemma 3.12(ii), (iii) and (3.6) applied to the family $(h_{c,i}^+, h_{c,i}^-)_{i \in I_c}$ at the point $(z_i, \bar{z}_i)_{i \in I_c}$ of its polydisc,

$$\|F_{c,z}\|_{\text{ch}}^{|\mathcal{C}|} = \frac{Z(w_{c,z}^{\text{ch}})}{Z(0)} = |e^{\mathcal{E}(w_{c,z}^{\text{ch}})}| \frac{|e^{-\mathcal{E}(w_{c,z}^{\text{ch}})} Z(w_{c,z}^{\text{ch}})|}{Z(0)} \leq e^{\text{Re } \mathcal{E}(w_{c,z}^{\text{ch}})} e^{C_{\text{vol}} |\Lambda_{K'}|}.$$

Since $|\Lambda_{K'}|/|\mathcal{C}| = (2L^{m+K-K'})^4/(2L^m/\ell)^4 = \ell^4 L^{4(K-K')} \leq \ell^4 L^{4b_*}$,

$$\|F_{c,z}\|_{\text{ch}} \leq \exp\left\{\frac{1}{|\mathcal{C}|} \text{Re } \mathcal{E}^{\text{bulk}}(w_{c,z}^{\text{ch}}) + \frac{1}{|\mathcal{C}|} \text{Re } \mathcal{E}^{\partial}(w_{c,z}^{\text{ch}}) + c_{\text{vol}} \ell^4 L^{4b_*}\right\}.$$

Step 4: the bulk cancels. For $z \in \frac{1}{2}\mathbb{D}$, Steps 2 and 3 give

$$|\mathbf{h}''(z)| = e^{-\text{Re } \mathcal{E}''(z)} \left| \frac{Z(w_z)}{Z(0)} \right| \leq \exp\left\{ \sum_{c \in \mathcal{C}_{\text{occ}}} \frac{1}{|\mathcal{C}|} \text{Re } \mathcal{E}^{\partial}(w_{c,z}^{\text{ch}}) - \text{Re } \mathcal{E}^{\partial}(w_z) + |\mathcal{C}_{\text{occ}}| c_{\text{vol}} \ell^4 L^{4b_*} \right\} :$$

the bulk vacuum energies, which are not bounded uniformly in the cutoff, cancel identically. Lemma 3.13(iii) applies to $w_{c,z}^{\text{ch}}$, a sum of pieces supported in the $|\mathcal{C}| |I_c|$ balls $\mathfrak{g}(\bar{B}_i)$ (Lemma 3.12(iii)), and to w_z , a sum of n pieces supported in the $\bar{B}_i$; so $|\mathcal{C}|^{-1} |\mathcal{E}^{\partial}(w_{c,z}^{\text{ch}})| \leq |I_c| C_{\text{face}}(R)$ and $|\mathcal{E}^{\partial}(w_z)| \leq n C_{\text{face}}(R)$. With $\sum_c |I_c| = n$, $|\mathcal{C}_{\text{occ}}| \leq n$ and $c_{\text{vol}} \geq 0$,

$$|\mathbf{h}''(z)| \leq \exp\{2n C_{\text{face}}(R) + n c_{\text{vol}} \ell^4 L^{4b_*}\} = Q_{\infty}^n \quad (z \in \tfrac{1}{2}\mathbb{D}). \quad (3.19)$$

Step 5: conclusion. $\mathbf{h}''$ is holomorphic on $\mathbb{D}$ (by Step 1, since $Z(w_z)$ is entire and $Z(0) > 0$), $\mathbf{h}''(0) = 1$, and $|\mathbf{h}''| \leq Q_{\infty}^n$ on $\frac{1}{2}\mathbb{D} = \{|z|_{\#} < \frac{1}{2}\}$. Lemma 3.6 with $\mathfrak{r} = 2$ and $Q = Q_{\infty}^n$ gives $|\partial_{z_1} \cdots \partial_{z_n} \text{Log } \mathbf{h}''(0)| \leq 2^n \log 2 (2Q_{\infty}^n + 2)^n (n/\varrho)^n \prod_i \|f_i\|_{\#}$. Near $z = 0$, $\log Z(w_z) = \log Z(0) + \mathcal{E}''(z) + \text{Log } \mathbf{h}''(z)$, so

$$S_n^T(f_1, \dots, f_n) = \partial_{z_1} \cdots \partial_{z_n} \mathcal{E}''(0) + \partial_{z_1} \cdots \partial_{z_n} \text{Log } \mathbf{h}''(0),$$

and (3.17) gives (3.14) with the first member of (3.15). By Lemma 3.11 the bound holds for the sources before placement. Only C_{UV} depends on $\mathbf{d}$, and it is non-increasing. The constants C_e , c_{vol} , b_* , C_{tree} , κ , δ , M depend only on $(L, \mathfrak{p}, \gamma, g, \varrho)$ (Theorem 3.1, Lemma 3.4(v)), ℓ only on (n, R, L) , and $\mathbf{a}$, C_{UV} , C_{face} are series that do not involve K , g_0 , m or the positions; so $C_n^{\infty}(\mathbf{d}; R)$ has the stated dependence. $\square$

Remark 3.14 (What the proof uses). From Theorem 3.1 the proof uses only (d1)–(d4), at the weights w_z , $u_{c,\mathfrak{g},z}$ and $u_{c,\mathfrak{g},z}^{(-)}$ ($z \in \mathbb{D}$) and $w_{c,z}^{\text{ch}}$ ($z \in \frac{1}{2}\mathbb{D}$), each of the form $\sum_l \zeta_l h_l$ with ζ in the polydisc of a finite family of real Lipschitz functions; besides, it uses Lemma 3.7, Bałaban's tree sum [8, (1.26)] and elementary complex analysis. It uses no covariance or reality property of the sourced vacuum energies under the symmetries of the cells, and no compatibility of the cells with Bałaban's partitions (ℓ depends on (n, R, L) only). Keeping the counterterm $\mathcal{E}(w_z)$ of Theorem 3.1 would require, in Step 4, the identity $\text{Re } \mathcal{E}^{\text{bulk}}(w_{c,z}^{\text{ch}}) = |\mathcal{C}| \text{Re } \mathcal{E}_c^{\text{bulk}}(w_{c,z})$, a covariance statement that Theorem 3.1 does not make; the re-centring makes it unnecessary. The only moduli taken are those of $Z(w_z)/Z(0)$, bounded by chessboard values which are non-negative by reflection positivity, and of $e^{-\mathcal{E}} Z$ in (3.6). The argument does not depend on the gauge group, which enters only through Theorem 3.1.

Status of clause (iii). Clause (iii)(a) is Theorem 3.1(e), deduced above from Theorem 3.1(d); clause (iii)(b) is Theorem 3.10, proved above in full from Theorem 3.1(d), Lemma 3.7 and [8, (1.26)]; clause (iii)(c) is Corollary 3.3, deduced above from Theorem 3.1(a), (b). Theorem 3.1 itself: see the status paragraph at the end of §3.2; Section 9 collects what it rests on.

4 Clause (ii): continuum limits and the Osterwalder–Schrader axioms

Throughout this section the gauge group is $SU(2)$, hypotheses (H1)–(H7) are in force, $\varrho > 0$ is fixed and $0 < g \leq \gamma_J$ as in (H7). We remove the cutoff from the connected Schwinger functions of the smeared action density on a fixed torus, and the cutoff and the volume together on $\mathbb{R}^4$ (Theorems 4.8 and 4.10); we prove that every limit on $\mathbb{R}^4$ has the regularity property (E0), invariance under translations and under the hypercubic group, reflection positivity (E2) and symmetry (E3) of Osterwalder and Schrader, in the form stated in Theorem 4.12 (invariance under the full rotation group is clause (vi), Section 8, and is not proved in this section); and we carry out their reconstruction of a Hilbert space with a vacuum vector, a positive Hamiltonian and momenta (Theorem 4.17). The inputs from the construction are clause (0), which gives $K \rightarrow \infty$ as $g_0 \downarrow 0$, and clause (iii), in the two forms (4.3) and (4.4) recalled below from Section 3; clauses (iv) and (v) enter only Remarks 4.18 and 4.9. Everything else is proved in this section, from exact symmetries of the lattice measure and lattice reflection positivity [23], by standard functional analysis; in particular the passage to infinite volume and the regularity property (E0) are proved here in full. Every statement of this section is therefore an implication from clauses (0) and (iii); what they rest on is stated in Sections 2 and 3 and collected in Section 9.

For a non-abelian gauge group the curvature $F_{\mu\nu}$ is not gauge invariant (it transforms by conjugation), so an Osterwalder–Schrader statement for the curvature can only concern gauge-invariant local functions of F . We treat the action density: the field is the smeared action density $\mathcal{O}(f)$ of Section 1, a lattice version of the smeared density $\int f \operatorname{tr} F_{\mu\nu} F_{\mu\nu}$. Its Schwinger functions are considered at pairwise disjoint supports only (Remark 4.19).

4.1 Setting, the input from clause (iii), and lattice lemmas

Lattices. For $m \geq 0$ let $\mathbb{T}_m = \mathbb{R}^4 / 2L^m \mathbb{Z}^4$, with its flat metric $d_{\mathbb{T}}$ and the quotient map $q_m: \mathbb{R}^4 \rightarrow \mathbb{T}_m$. At cutoff K the bare lattice in $\mathbb{T}_m$ has spacing $\varepsilon = L^{-K}$; as in [7], its sites are the points of $\varepsilon(\mathbb{Z} + \frac{1}{2})^4$ modulo $2L^m \mathbb{Z}^4$, bonds join nearest neighbours, plaquettes are the elementary squares, and $x(p)$ denotes the barycentre of the plaquette p . Thus a hyperplane $\{x^\mu = c\}$ with $c \in \varepsilon\mathbb{Z}$ contains no site. Points are written $x = (x^0, \vec{x})$, $\vec{x} = (x^1, x^2, x^3)$, with unit vectors $e_0, \dots, e_3$. W_4 is the group of signed permutations of the four coordinates and $W_3 \subset W_4$ the subgroup fixing x^0 ; both act on $\mathbb{R}^4$ and on $\mathbb{T}_m$ fixing the origin. We write $\theta(x^0, \vec{x}) = (-x^0, \vec{x})$, $(w \cdot f)(x) = f(w^{-1}x)$ and $(\tau_a f)(x) = f(x - a)$.

Schwinger functions. Let $a_p(U) := 1 - \operatorname{Re} \operatorname{tr} U(\partial p) \in [0, 2]$, so that $A(U) = \sum_p a_p(U)$, and let $\langle \cdot \rangle_{K,m}$ be the expectation in the probability measure $\mu_{K,m}$ proportional to $e^{-g_0^{-2} A(U)} dU$, where $g_0 \in (0, g_-(g)]$ and $K = K(g_0, g)$ is the cutoff fixed by first passage (Section 1); by clause (0), $K \rightarrow \infty$ as $g_0 \downarrow 0$. For a real function h on $\mathbb{T}_m$, $\mathcal{O}(h) = \sum_p h(x(p)) a_p(U)$ is a bounded random variable, and for $n \geq 2$

$$S_{n;K,m}^T(h_1, \dots, h_n) = \partial_{z_1} \cdots \partial_{z_n} \log Z(z) \big|_{z=0} = \kappa_n(\mathcal{O}(h_1), \dots, \mathcal{O}(h_n)) \quad (4.1)$$

is the joint cumulant under $\mu_{K,m}$. Write $:X: = X - \langle X \rangle_{K,m}$. For $N \geq 1$ and real $h_1, \dots, h_N$ with pairwise disjoint supports, the moment–cumulant formula gives

$$S_{N;K,m}(h_1, \dots, h_N) := \left\langle \prod_{a=1}^N : \mathcal{O}(h_a) : \right\rangle_{K,m} = \sum_{\pi \in \Pi_{\geq 2}[N]} \prod_{B \in \pi} S_{|B|;K,m}^T((h_a)_{a \in B}), \quad (4.2)$$

where $\Pi_{\geq 2}[N]$ is the set of partitions of $\{1, \dots, N\}$ into blocks of size at least two: blocks of size one drop out because the variables are centred, and cumulants of order at least two are unchanged by centring. Thus $S_{1;K,m} = 0$, and we put $S_{0;K,m} := 1$. We call the S_N the *centred Schwinger functions*.

Test functions. As in Section 1, $\|f\|_{\sharp} = \max\{\|f\|_{\infty}, 40M \text{Lip } f\}$. For C^1 functions on $\mathbb{R}^4$ or on $\mathbb{T}_m$ one has $\text{Lip } f = \|\nabla f\|_{\infty}$, hence $\|f\|_{\sharp} \leq c_{\sharp}\|f\|_{C^1}$, where $\|f\|_{C^1} := \max\{\|f\|_{\infty}, \|\nabla f\|_{\infty}\}$ and $c_{\sharp} := 40M \geq 1$. Let X denote $\mathbb{R}^4$ or a torus $\mathbb{T}_m$. A *separated n -tuple* on X is an n -tuple $\vec{f} = (f_1, \dots, f_n)$ of real C^1 functions on X with compact, pairwise disjoint supports; $\mathfrak{S}_n(X)$ denotes the set of these, and $d(\vec{f}) := \min_{i \neq k} \text{dist}(\text{supp } f_i, \text{supp } f_k) > 0$. We write $X_{\neq}^n := \{x \in X^n : x_i \neq x_k \text{ for } i \neq k\}$ and $\Delta_n := X^n \setminus X_{\neq}^n$.

The input from clause (iii). Let $\ell(n, R, g)$ be the least power of L which is at least $\max\{8n(R+1), L^{m_{\mathbb{T}}(g-(g))}\}$. We use clause (iii) in the following two forms (Theorems 3.1(e) and 3.10).

- (iii)(a) For every $m \geq m_{\mathbb{T}}(g-(g))$, $n \geq 2$ and $\mathbf{d} > 0$ there is $C_n(\mathbf{d}) < \infty$, depending on m but not on K, g_0 or the positions of the supports, such that for all $g_0 \in (0, g-(g)]$ and all real Lipschitz functions $h_1, \dots, h_n$ on $\mathbb{T}_m$ with supports pairwise at distance at least $\mathbf{d}$,

$$|S_{n;K,m}^T(h_1, \dots, h_n)| \leq C_n(\mathbf{d}) \prod_{i=1}^n \|h_i\|_{\sharp}. \quad (4.3)$$

- (iii)(b) For every $n \geq 2$, $R > 0$ and $\mathbf{d} > 0$ there is $C_n^{\infty}(\mathbf{d}; R) < \infty$, independent of K, g_0, m and of the positions of the supports, such that for all m with $L^m \geq \ell(n, R, g)$, all $g_0 \in (0, g-(g)]$, and all real Lipschitz functions $h_1, \dots, h_n$ on $\mathbb{T}_m$, each supported in a ball of radius R , with supports pairwise at distance at least $\mathbf{d}$,

$$|S_{n;K,m}^T(h_1, \dots, h_n)| \leq C_n^{\infty}(\mathbf{d}; R) \prod_{i=1}^n \|h_i\|_{\sharp}. \quad (4.4)$$

Of the constant in (iii)(b) we use one further property, which follows from its explicit form (3.15): it depends on $\mathbf{d}$ only through the summand containing the series $C_{UV}(\mathbf{d}; R)$ of Lemma 3.5, and by (3.11) that series is at most $C'_{UV}(1 + (R+2)/\mathbf{d})^4 (1 + \log_L^+(32M/\mathbf{d}))$, with C'_{UV} independent of $\mathbf{d}$. The other summands of $C_n^{\infty}(\mathbf{d}; R)$ do not depend on $\mathbf{d}$. Since, for $R = 1$ and $\mathbf{d} \leq 1$, $(1 + 3/\mathbf{d})^4 \leq 4^4 \mathbf{d}^{-4}$ and $\log_L^+(32M/\mathbf{d}) \leq \log_L(32M) + 1/\mathbf{d}$, while for $\mathbf{d} \geq 1$ both factors are bounded, it follows that

$$G_n := \sup_{\mathbf{d} > 0} \min\{1, \mathbf{d}^5\} C_n^{\infty}(\mathbf{d}; 1) < \infty, \quad (4.5)$$

and G_n is independent of K, g_0, m and positions.

Lattice lemmas.

Lemma 4.1 (exact symmetries). *Let γ be a translation of $\mathbb{T}_m$ by a vector of $\varepsilon\mathbb{Z}^4$, or an element of W_4 . For real functions $h_1, \dots, h_n$ on $\mathbb{T}_m$ the joint laws under $\mu_{K,m}$ of $(\mathcal{O}(\gamma \cdot h_i))_{i \leq n}$ and of $(\mathcal{O}(h_i))_{i \leq n}$ coincide. In particular $S_{n;K,m}^T(\gamma \cdot h_1, \dots, \gamma \cdot h_n) = S_{n;K,m}^T(h_1, \dots, h_n)$ and $\langle \mathcal{O}(\gamma \cdot h) \rangle_{K,m} = \langle \mathcal{O}(h) \rangle_{K,m}$.*

Proof. γ maps $\varepsilon(\mathbb{Z} + \frac{1}{2})^4$ and $2L^m\mathbb{Z}^4$ onto themselves (for sign changes because $-(k + \frac{1}{2}) = (-k-1) + \frac{1}{2}$), so it permutes sites, bonds and plaquettes, with $x(\gamma p) = \gamma x(p)$. Put $(U \circ \gamma)(b) := U(\gamma b)$,

replaced by its inverse when γ reverses the orientation of b . Since the real part of the trace of a holonomy is unchanged under cyclic relabelling and inversion, $a_{\gamma p}(U) = a_p(U \circ \gamma)$, hence $\mathcal{O}(\gamma \cdot h)(U) = \sum_p h(\gamma^{-1}x(p)) a_p(U) = \sum_{p'} h(x(p')) a_{\gamma p'}(U) = \mathcal{O}(h)(U \circ \gamma)$. The map $U \mapsto U \circ \gamma$ preserves the product Haar measure (Haar measure is invariant under inversion) and the action A . $\square$

Lemma 4.2 (transport to tori). *Let $\rho > 0$, $B_\rho := \{x \in \mathbb{R}^4 : |x| < \rho\}$ and $L^m \geq 2\rho + 2$. Then q_m is isometric on B_ρ . For $f \in C_c^1(\mathbb{R}^4)$ with $\text{supp } f \subset B_\rho$ let $f^{(m)} := f \circ (q_m|_{B_\rho})^{-1}$ on $q_m(B_\rho)$ and $f^{(m)} := 0$ elsewhere. Then $f^{(m)} \in C^1(\mathbb{T}_m)$ and $\|f^{(m)}\|_\# = \|f\|_\#$; distances between supports are preserved; if $\text{supp } f$ lies in a ball of radius R , so does $\text{supp } f^{(m)}$; $(w \cdot f)^{(m)} = w \cdot f^{(m)}$ for $w \in W_4$; and $(\tau_a f)^{(m)} = \tau_{q_m(a)} f^{(m)}$ whenever also $\text{supp } \tau_a f \subset B_\rho$.*

Proof. For $x, y \in B_\rho$ and $k \in \mathbb{Z}^4 \setminus \{0\}$, $|x - y + 2L^m k| \geq 2L^m - 2\rho > |x - y|$, so $d_{\mathbb{T}}(q_m x, q_m y) = |x - y|$. Since $\text{supp } f$ is a compact subset of B_ρ and q_m is a local isometry, $f^{(m)}$ is C^1 with $\|f^{(m)}\|_\infty = \|f\|_\infty$ and $\|\nabla f^{(m)}\|_\infty = \|\nabla f\|_\infty$, and on the flat torus the Lipschitz constant of a C^1 function is the supremum of its gradient. Distances between supports are preserved by the isometry; q_m is 1-Lipschitz, which gives the statement on balls; the last two identities follow from $q_m \circ w = w \circ q_m$ and $q_m \circ \tau_a = \tau_{q_m(a)} \circ q_m$. $\square$

For $\vec{f} \in \mathfrak{S}_n(\mathbb{R}^4)$ with supports in B_ρ and $L^m \geq 2\rho + 2$ we write

$$S_{n;K,m}^T(\vec{f}) := S_{n;K,m}^T(f_1^{(m)}, \dots, f_n^{(m)}),$$

and similarly for $S_{N;K,m}$.

Lemma 4.3 (half-spaces and reflection). *Let $H_+ := \{x \in \mathbb{T}_m : 0 < x^0 < L^m\}$, the coordinate x^0 being taken in $(-L^m, L^m]$. If $\text{supp } h \subset H_+$, then every plaquette p with $h(x(p)) \neq 0$ has all its bonds in H_+ , so $\mathcal{O}(h)$ is a function of the variables of the bonds contained in H_+ . Moreover $\mathcal{O}(h)(\theta \cdot U) = \mathcal{O}(h \circ \theta)(U)$, where $(\theta \cdot U)(b) := U(\theta b)$, replaced by its inverse for bonds parallel to e_0 , and $\langle \mathcal{O}(h \circ \theta) \rangle_{K,m} = \langle \mathcal{O}(h) \rangle_{K,m}$.*

Proof. A plaquette in a $(0, \nu)$ -plane has corners at times $(k + \frac{1}{2})\varepsilon$ and $(k + \frac{3}{2})\varepsilon$ and its barycentre at time $(k + 1)\varepsilon$. If $(k + 1)\varepsilon \in (0, L^m)$ then, since 0 and L^m are multiples of ε , $\varepsilon \leq (k + 1)\varepsilon \leq L^m - \varepsilon$, and all corner times lie in $[\varepsilon/2, L^m - \varepsilon/2]$. A plaquette in a (ν, λ) -plane with $\nu, \lambda \neq 0$ lies at the time of its barycentre. The remaining assertions are Lemma 4.1 with $\gamma = \theta$. $\square$

If $f \in C_c^1(\mathbb{R}^4)$, $\text{supp } f \subset B_\rho \cap \{x^0 > 0\}$ and $L^m \geq 2\rho + 2$, then $\text{supp } f^{(m)} \subset H_+$.

Lemma 4.4 (lattice reflection positivity). *Let $m \geq 1$ (so that $\varepsilon \leq 1 < L^m$), let H_+ be as in Lemma 4.3 and $H_- := \theta H_+$, and let F, G be bounded measurable gauge-invariant functions of the variables of the bonds contained in H_+ . Then*

$$\langle \overline{F(\theta \cdot U)} F(U) \rangle_{K,m} \geq 0, \quad \langle \overline{F(\theta \cdot U)} G(U) \rangle_{K,m} = \overline{\langle G(\theta \cdot U) F(U) \rangle_{K,m}}. \quad (4.6)$$

This is the reflection positivity of the Wilson action in hyperplanes containing no sites, due to Osterwalder and Seiler [23]; we recall the argument in the present setting.

Proof. The reflection θ fixes the hyperplanes $\Pi_0 = \{x^0 = 0\}$ and $\Pi_1 = \{x^0 = L^m\}$, which contain no sites, and $\mathbb{T}_m \setminus (\Pi_0 \cup \Pi_1) = H_+ \cup H_-$. A bond lies in H_+ , lies in H_- , or crosses $\Pi_0 \cup \Pi_1$; a crossing bond joins a site $y \in H_+$ at time $\varepsilon/2$ or $L^m - \varepsilon/2$ to θy , and since $\varepsilon < L^m$ distinct crossing bonds have no common endpoint. Put $\beta := g_0^{-2}$. The integrand $\overline{F(\theta \cdot U)} F(U) e^{-\beta A(U)}$ is gauge invariant;

by a gauge transformation at the endpoints in H_+ of the crossing bonds (depending only on the crossing variables) and the translation invariance of Haar measure, the crossing variables may be set equal to $\mathbf{1}$ in the integral. A plaquette containing no crossing bond lies in H_+ or in H_- ; a plaquette containing one contains exactly two, and after the gauge fixing its holonomy is conjugate to $U(b)U(\theta b)^{-1}$ or to its inverse, where $b \subset H_+$ is its bond parallel to Π_0 and $\theta b \subset H_-$ is oriented like b . Now $e^{\beta \operatorname{Re} \operatorname{tr} W} = \sum_{\lambda} a_{\lambda} \chi_{\lambda}(W)$, summed over the irreducible unitary representations D^{λ} of the gauge group with characters χ_{λ} , with $a_{\lambda} \geq 0$ and $\sum_{\lambda} a_{\lambda} \dim D^{\lambda} = e^{\beta}$ (the function $\operatorname{Re} \operatorname{tr}$ is a non-negative combination of characters, and a product of characters is a non-negative integer combination of characters); and $\chi_{\lambda}(VW^{-1}) = \sum_{i,k} D_{ik}^{\lambda}(V) \overline{D_{ik}^{\lambda}(W)}$, where $\sum_{i,k} |D_{ik}^{\lambda}(V)| |D_{ik}^{\lambda}(W)| \leq \dim D^{\lambda}$ by the Cauchy–Schwarz inequality and the unitarity of D^{λ} . So the expansion of each crossing Boltzmann factor converges absolutely and uniformly, the sum of the absolute values of its terms being at most e^{β} . Hence the product of the Boltzmann factors of the crossing plaquettes equals $e^{-\beta n_{\times}} \sum_{\alpha} c_{\alpha} G_{\alpha}(U_+) \overline{G_{\alpha}(\theta \cdot U_-)}$, an absolutely convergent sum with $c_{\alpha} \geq 0$, where $n_{\times}$ is the number of crossing plaquettes, $U_{\pm}$ are the variables of the bonds in $H_{\pm}$, the G_{α} are bounded functions of the variables in H_+ , and $\theta \cdot U_-$ denotes the configuration $b \mapsto (\theta \cdot U)(b)$ on the bonds $b \subset H_+$. The plaquettes in H_- are the images under θ of those in H_+ , so their total action is $A_+(\theta \cdot U_-)$, where A_+ is the action of the plaquettes in H_+ . As $U_- \mapsto \theta \cdot U_-$ preserves Haar measure,

$$\int \overline{F(\theta \cdot U)} F(U) e^{-\beta A(U)} dU = e^{-\beta n_{\times}} \sum_{\alpha} c_{\alpha} \left| \int F(U_+) G_{\alpha}(U_+) e^{-\beta A_+(U_+)} dU_+ \right|^2 \geq 0.$$

The identity in (4.6) follows from the invariance of $\mu_{K,m}$ under $U \mapsto \theta \cdot U$ (Lemma 4.1) and from $\theta \cdot (\theta \cdot U) = U$. $\square$

Lemma 4.5 (translation modulus). *For $f \in C_c^1(\mathbb{R}^4)$ and $a, b \in \mathbb{R}^4$,*

$$\|\tau_a f - \tau_b f\|_{\#} \leq \varpi_f(|a - b|) := \max \{ \|\nabla f\|_{\infty} |a - b|, 40M \omega_{\nabla f}(|a - b|) \},$$

where $\omega_{\nabla f}(\delta) := \sup_{|x-y| \leq \delta} |\nabla f(x) - \nabla f(y)|$ tends to 0 as $\delta \downarrow 0$. The same holds on $\mathbb{T}_m$.

Proof. The mean value theorem bounds $\|\tau_a f - \tau_b f\|_{\infty}$; $\operatorname{Lip}(\tau_a f - \tau_b f) = \|\nabla f(\cdot - a) - \nabla f(\cdot - b)\|_{\infty} \leq \omega_{\nabla f}(|a - b|)$; and ∇f , being continuous with compact support, is uniformly continuous. $\square$

For merely Lipschitz f the left side need not tend to zero: for $f(x) = |x|^1 |\chi(x)|$, with $\chi \in C_c^1(\mathbb{R}^4)$ equal to 1 near the origin, $\operatorname{Lip}(\tau_{se_1} f - f) \geq 2$ for all small $s \neq 0$. This is why the continuum limits below are taken on C^1 test functions; since $C_c^1 \supset C_c^{\infty}$, nothing is lost for distributions.

Lemma 4.6 (linearity in the test function). *Fix K, m and $N \geq 1$, and let V be either $\mathbb{T}_m$, or B_{ρ} with $L^m \geq 2\rho + 2$, identified with $q_m(B_{\rho})$ by Lemma 4.2. For a function F on V^N put*

$$\begin{aligned} \Phi_{K,m}(F) &:= \sum_{p_1, \dots, p_N} F(x(p_1), \dots, x(p_N)) \\ &\quad \prod_{a=1}^N (a_{p_a}(U) - \langle a_{p_a} \rangle_{K,m}), \\ \Psi_{K,m}(F) &:= \sum_{p_1, \dots, p_N} F(x(p_1), \dots, x(p_N)) \\ &\quad \kappa_N(a_{p_1}, \dots, a_{p_N}), \end{aligned}$$

the sums running over N -tuples of plaquettes with barycentres in V . Both are linear in F and depend only on the values of F ; $\Phi_{K,m}(f_1 \otimes \dots \otimes f_N) = \prod_a : \mathcal{O}(f_a) :$ and $\Psi_{K,m}(f_1 \otimes \dots \otimes f_N) = \kappa_N(\mathcal{O}(f_1), \dots, \mathcal{O}(f_N))$; $\Phi_{K,m}(F) \Phi_{K,m}(G) = \Phi_{K,m}(F \otimes G)$; $\overline{\Phi_{K,m}(F)} = \Phi_{K,m}(\bar{F})$; and, as $\theta V = V$, $\Phi_{K,m}(F)(\theta \cdot U) = \Phi_{K,m}(\Theta F)(U)$, where $(\Theta F)(x_1, \dots, x_N) := F(\theta x_1, \dots, \theta x_N)$.

Proof. Linearity is visible, and the tensor identities follow by expanding the products, since $:\mathcal{O}(h): = \sum_p h(x(p))(a_p - \langle a_p \rangle_{K,m})$ and κ_N is multilinear. The double sum defining $\Phi_{K,m}(F)\Phi_{K,m}(G)$ is the sum defining $\Phi_{K,m}(F \otimes G)$. The variables a_p are real. For the reflection use $a_p(\theta \cdot U) = a_{\theta p}(U)$, $\langle a_p \rangle_{K,m} = \langle a_{\theta p} \rangle_{K,m}$ and $x(\theta p) = \theta x(p)$ (Lemmas 4.1 and 4.3), and relabel $p \mapsto \theta p$. $\square$

For $V \subset X$ let $\mathcal{T}_N(V)$ be the complex span of the tensors $f_1 \otimes \cdots \otimes f_N$ of real C^1 functions on X with compact, pairwise disjoint supports contained in V , and $\mathcal{T}_0(V) := \mathbb{C}$. By Lemma 4.6, $S_{N;K,m}(F) := \langle \Phi_{K,m}(F) \rangle_{K,m}$ and $S_{N;K,m}^T(F) := \Psi_{K,m}(F)$ are well-defined linear functionals of the function $F \in \mathcal{T}_N(X)$ (for $X = \mathbb{R}^4$: of $F \in \mathcal{T}_N(B_\rho)$, $L^m \geq 2\rho + 2$), independent of its representation as a sum of tensors, which on separated tensors are (4.2) and (4.1).

Lemma 4.7 (local estimate). *Let $n \geq 2$, $c \geq 1$, $0 < \sigma \leq \frac{1}{32}$, $y = (y_1, \dots, y_n) \in (\mathbb{R}^4)^n$, $q_i := y_i + [-\frac{\sigma}{2}, \frac{\sigma}{2}]^4$ and $q'_i := y_i + [-\sigma, \sigma]^4$, with $q'_1, \dots, q'_n$ pairwise disjoint. Let T be a distribution on an open subset of $(\mathbb{R}^4)^n_{\neq}$ containing $\prod_i q'_i$ such that*

$$|T(\varphi_1 \otimes \cdots \otimes \varphi_n)| \leq C_T \prod_{i=1}^n \max\{\|\varphi_i\|_\infty, c\|\nabla \varphi_i\|_\infty\} \quad (4.7)$$

for all real $\varphi_i \in C_c^\infty(\text{int } q'_i)$. Then for every smooth Φ with $\text{supp } \Phi \subset \prod_i q_i$,

$$|T(\Phi)| \leq K_n C_T \sigma^{-n} |\Phi|_{5n+1, \sigma}, \quad |\Phi|_{p, \sigma} := \max_{|\alpha| \leq p} \sigma^{|\alpha|} \|\partial^\alpha \Phi\|_\infty,$$

with K_n depending only on n and c . Moreover Φ is the limit in $C_c^\infty(\prod_i \text{int } q'_i)$ of finite sums of tensors $\varphi_1 \otimes \cdots \otimes \varphi_n$ with $\varphi_i \in C_c^\infty(\text{int } q'_i)$. The same holds with $\mathbb{T}_m$ in place of $\mathbb{R}^4$.

Proof. Let $Q' := \prod_i q'_i = y + [-\sigma, \sigma]^{4n}$ and, for ξ in $\Lambda_\sigma := (\pi/\sigma)\mathbb{Z}^{4n}$,

$$\hat{\Phi}(\xi) := (2\sigma)^{-4n} \int_{Q'} \Phi(x) e^{-i\xi \cdot (x-y)} dx.$$

Since Φ vanishes near $\partial Q'$ and its support has volume at most σ^{4n} , p -fold integration by parts in one coordinate gives

$$(1 + \sigma|\xi|_\infty)^p |\hat{\Phi}(\xi)| \leq 2^{p-4n} |\Phi|_{p, \sigma} \quad (p \geq 0). \quad (4.8)$$

Fix $\chi \in C_c^\infty((-1, 1)^4)$ with $0 \leq \chi \leq 1$ and $\chi = 1$ on $[-\frac{3}{4}, \frac{3}{4}]^4$, let $C_\chi := \max\{1, \|\nabla \chi\|_\infty\}$, $\chi_i(u) := \chi((u - y_i)/\sigma)$ and $g_{i, \xi}(u) := \chi_i(u) e^{i\xi_i \cdot (u - y_i)}$. The 2σ -periodic extension of $\Phi|_{Q'}$ is smooth and is the sum of its Fourier series; multiplying by $\prod_i \chi_i(x_i)$, which equals 1 on $\text{supp } \Phi$ and vanishes off Q' , we get $\Phi = \sum_{\xi \in \Lambda_\sigma} \hat{\Phi}(\xi) g_{1, \xi} \otimes \cdots \otimes g_{n, \xi}$. By (4.8) the partial sums converge in every C^k -norm, and their supports lie in the fixed compact set $\prod_i \text{supp } \chi_i \subset \prod_i \text{int } q'_i$; this proves the density statement, and $T(\Phi) = \sum_{\xi} \hat{\Phi}(\xi) T(\otimes_i g_{i, \xi})$. The real and imaginary parts of $g_{i, \xi}$ are real smooth functions supported in $\text{int } q'_i$, of sup norm at most 1 and gradient at most $C_\chi/\sigma + 2|\xi|_\infty$; expanding $\otimes_i g_{i, \xi}$ into 2^n real tensors and using (4.7) and $cC_\chi/\sigma \geq 1$,

$$|T(\otimes_i g_{i, \xi})| \leq 2^n (2cC_\chi \sigma^{-1} (1 + \sigma|\xi|_\infty))^n C_T.$$

With (4.8) for $p = 5n + 1$, $|T(\Phi)| \leq 2^{n+1} (4cC_\chi)^n \sigma^{-n} C_T |\Phi|_{5n+1, \sigma} \sum_{\zeta \in \mathbb{Z}^{4n}} (1 + \pi|\zeta|_\infty)^{-4n-1}$, and the last sum is finite because $\#\{\zeta : |\zeta|_\infty = r\} \leq 8n(2r + 1)^{4n-1}$. $\square$

The reflection form. Let X be $\mathbb{R}^4$ or $\mathbb{T}_m$, and $H := \{x \in \mathbb{R}^4 : x^0 > 0\}$, resp. $H := H_+$; thus $\theta H \cap H = \emptyset$. Let $\mathcal{P}(H)$ be the space of finite sequences $F = (F_N)_{N \geq 0}$ with $F_N \in \mathcal{T}_N(H)$, and put $(\Theta \bar{F})_N(x_1, \dots, x_N) := \overline{F_N(\theta x_1, \dots, \theta x_N)}$. In each tensor term of $(\Theta \bar{F})_N \otimes G_{N'}$ the first N factors have pairwise disjoint supports in θH and the last N' factors pairwise disjoint supports in H , so $(\Theta \bar{F})_N \otimes G_{N'} \in \mathcal{T}_{N+N'}(X)$, whatever the overlaps between different tensor terms. Given linear functionals S_N on $\mathcal{T}_N(X)$ with $S_0 = 1$ and $S_1 = 0$, we put

$$(F, G)_{\text{OS}} := \sum_{N, N' \geq 0} S_{N+N'}((\Theta \bar{F})_N \otimes G_{N'}), \quad F, G \in \mathcal{P}(H), \quad (4.9)$$

and call $(S_N)_N$ *reflection positive* if $(F, F)_{\text{OS}} \geq 0$ for all $F \in \mathcal{P}(H)$. For $\mu \in \{0, 1, 2, 3\}$ and $c \in \mathbb{R}$, the same definitions with the reflection $\theta_{\mu, c}: x^\mu \mapsto 2c - x^\mu$ in place of θ and with $H_{\mu, c} := \{x^\mu > c\}$ (on $\mathbb{T}_m$: $\{c < x^\mu < c + L^m\}$; then $\theta_{\mu, c}$ fixes the two hyperplanes $\{x^\mu = c\}$ and $\{x^\mu = c + L^m\}$) define reflection positivity at the hyperplane $\{x^\mu = c\}$.

4.2 Continuum limits on a fixed torus and on $\mathbb{R}^4$

Theorem 4.8 (limits on a fixed torus, first passage). *Fix $m \geq m_{\mathbb{T}}(g_-(g))$ and $g_0^{(i)} \in (0, g_-(g)]$ with $K_i := K(g_0^{(i)}, g) \rightarrow \infty$. There is an infinite set $I \subset \mathbb{N}$ such that for every $n \geq 2$ and every $\vec{f} \in \mathfrak{S}_n(\mathbb{T}_m)$ the limit*

$$S_{n;m}^T(\vec{f}) := \lim_{i \in I} S_{n;K_i,m}^T(\vec{f})$$

exists. For every infinite I with this property:

- (a) $S_{n;m}^T$ is real, symmetric, and multilinear on separated tuples (in each argument, among functions keeping the tuple separated), and $|S_{n;m}^T(\vec{f})| \leq C_n(d) \prod_i \|f_i\|_{\sharp}$ whenever $d(\vec{f}) \geq d$;
- (b) $S_{n;m}^T$ extends uniquely to a distribution on $(\mathbb{T}_m)^n_{\neq}$, of order at most $5n + 1$ on every compact subset;
- (c) $S_{n;m}^T$ is invariant under all translations of $\mathbb{T}_m$ and under W_4 ;
- (d) the centred Schwinger functions $S_{N;m} := \lim_{i \in I} S_{N;K_i,m}$ on $\mathcal{T}_N(\mathbb{T}_m)$, which on separated tensors are given by (4.2) in terms of the $S_{n;m}^T$ ($S_{0;m} = 1$, $S_{1;m} = 0$), are reflection positive at every hyperplane $\{x^\mu = c\}$, $\mu \in \{0, 1, 2, 3\}$, $c \in \mathbb{R}$.

On a torus we assert only (a)–(d); the Hilbert space obtained from (d) at $\{x^0 = 0\}$ by the construction of (R1) below is used once, in Corollary 6.25(a), and we construct no semigroup or Hamiltonian there.

Remark 4.9 (the tuned family). Clause (ii) also contains a statement without subsequences on a fixed torus, for the family of bare couplings tuned at the reference length that is introduced in clause (v): along that family the connected functions converge as the cutoff is removed, for all separated tuples of real Lipschitz functions, at an explicit rate, and the limits obey the bound of clause (iii)(a). It is part of clause (v) and holds under the hypotheses of that clause, including its own ceiling γ_U on the coupling; we refer to Section 7 for the statement and for the structure of the proof. Given this convergence and this bound, properties (a)–(d) of Theorem 4.8 hold for the tuned limits (with (c) on C^1 tuples): Steps 4–7 of the proof below use only convergence on separated C^1 tuples, the bound for the limit functional, the exact lattice symmetries and Lemma 4.4. The first-passage limits of Theorem 4.8 are in general obtained only along subsequences; clause (v) describes them further.

Theorem 4.10 (limits on $\mathbb{R}^4$ along a joint subsequence of cutoffs and volumes). *Let $g_0^{(j)} \in (0, g_-(g)]$ with $K_j := K(g_0^{(j)}, g) \rightarrow \infty$, and let $m_j \geq m_{\mathbb{T}}(g_-(g))$ be integers with $m_j \rightarrow \infty$. There is an infinite set $J \subset \mathbb{N}$ such that for every $n \geq 2$ and every $\vec{f} \in \mathfrak{S}_n(\mathbb{R}^4)$ the limit*

$$S_{n;\infty}^T(\vec{f}) := \lim_{j \in J} S_{n;K_j,m_j}^T(\vec{f})$$

exists, the terms being defined for all large j (Lemma 4.2). For every infinite J with this property:

(a) $S_{n;\infty}^T$ is real, symmetric, and multilinear on separated tuples, and

$$|S_{n;\infty}^T(\vec{f})| \leq C_n^\infty(\mathbf{d}; R) \prod_{i=1}^n \|f_i\|_{\#} \quad (4.10)$$

whenever each f_i is supported in a ball of radius R and $d(\vec{f}) \geq \mathbf{d}$; the constant is that of (4.4), independent of positions;

(b) $S_{n;\infty}^T$ extends uniquely to a distribution on $(\mathbb{R}^4)_{\neq}^n$, of order at most $5n + 1$ on every compact subset;

(c) $S_{n;\infty}^T(\gamma \cdot f_1, \dots, \gamma \cdot f_n) = S_{n;\infty}^T(\vec{f})$ for every γ in the group $\mathbb{R}^4 \rtimes W_4$ generated by the translations and W_4 , and the distributions of (b) are invariant under the diagonal action of this group.

Not asserted: convergence for merely Lipschitz tuples, independence of the limit from J , and anything for $n = 1$ or at coinciding points.

Proof of Theorems 4.8 and 4.10. We treat both cases together. X denotes $\mathbb{T}_m$ or $\mathbb{R}^4$; we index the sequence by j in both cases (with $m_j = m$ for the torus), write $S_{n;j}^T := S_{n;K_j,m_j}^T$, and in the case $X = \mathbb{R}^4$ regard $S_{n;j}^T(\vec{f})$ as defined for those j with $L^{m_j} \geq 2\rho + 2$, where B_ρ contains the supports. The limits are denoted $S_{n;\infty}^T$ in both cases.

Step 0: the uniform bound. For all $n \geq 2$, $R > 0$, $\mathbf{d} > 0$ and $\rho > 0$ there are $C = C(n, \mathbf{d}, R) < \infty$ and $j_0 = j_0(n, R, \rho)$ such that for all $j \geq j_0$ and all real C^1 functions $f_1, \dots, f_n$ on X , each supported in a ball of radius R (and, if $X = \mathbb{R}^4$, in B_ρ), with supports pairwise at distance at least $\mathbf{d}$,

$$|S_{n;j}^T(\vec{f})| \leq C \prod_{i=1}^n \|f_i\|_{\#}. \quad (4.11)$$

For $X = \mathbb{T}_m$ this is (4.3), with $C = C_n(\mathbf{d})$ and $j_0 = 1$. For $X = \mathbb{R}^4$ take $C = C_n^\infty(\mathbf{d}; R)$ and j_0 such that $L^{m_j} \geq \max\{2\rho + 2, \ell(n, R, g)\}$ for $j \geq j_0$; then (4.11) is (4.4) applied to the $f_i^{(m_j)}$ (Lemma 4.2).

Step 1: equicontinuity. Let $Q_1, \dots, Q_n \subset X$ be compact, each contained in a ball of radius R' (and in B_ρ if $X = \mathbb{R}^4$), pairwise at distance at least $\mathbf{d}'$. For real C^1 functions f_i, g_i supported in Q_i and $j \geq j_0(n, R', \rho)$,

$$|S_{n;j}^T(\vec{f}) - S_{n;j}^T(\vec{g})| \leq C(n, \mathbf{d}', R') \sum_{i=1}^n \|f_i - g_i\|_{\#} \prod_{l < i} \|g_l\|_{\#} \prod_{l > i} \|f_l\|_{\#}. \quad (4.12)$$

Indeed, by multilinearity of κ_n and linearity of $f \mapsto \mathcal{O}(f)$ the difference is the sum over i of $S_{n;j}^T(g_1, \dots, g_{i-1}, f_i - g_i, f_{i+1}, \dots, f_n)$, and each of these tuples satisfies the hypotheses of (4.11).

Step 2: a countable skeleton and one diagonal sequence. Let $\mathcal{Q}$ be the countable family of finite unions of closed boxes with rational corners in $\mathbb{R}^4$ (for $X = \mathbb{T}_m$: of images under q_m of such boxes of

side less than L^m). For $Q \in \mathcal{Q}$ the space $C_Q^1 := \{f \in C^1(X; \mathbb{R}) : f = 0 \text{ off } Q\}$ is separable for $\|\cdot\|_{C^1}$, since $f \mapsto (f, \partial_1 f, \dots, \partial_4 f)$ embeds it isometrically into $C(Q'')^5$ for a closed box $Q'' \supset Q$ (resp. into $C(\mathbb{T}_m)^5$); fix a countable $\|\cdot\|_{C^1}$ -dense subset $D_Q \subset C_Q^1$. Two facts will be used: $\|f\|_{\sharp} \leq c_{\sharp} \|f\|_{C^1}$; and for every real $f \in C_c^1(X)$ and every open $W \supset \text{supp } f$ there is $Q \in \mathcal{Q}$ with $\text{supp } f \subset Q \subset W$ (cover the compact support by finitely many open rational boxes with closures in W). Let

$$\mathfrak{F} := \bigcup_{n \geq 2} \{(g_1, \dots, g_n) : Q_1, \dots, Q_n \in \mathcal{Q} \text{ pairwise disjoint, } g_i \in D_{Q_i}\},$$

a countable subset of $\bigcup_n \mathfrak{S}_n(X)$. For $\vec{g} \in \mathfrak{F}$ the sequence $(S_{n,j}^T(\vec{g}))_j$ is defined for all large j and bounded, by (4.11): the Q_i are compact and pairwise at positive distance. By the Cantor diagonal procedure there is an infinite J (for the torus: I) along which $S_{n,j}^T(\vec{g})$ converges for every $\vec{g} \in \mathfrak{F}$.

Step 3: convergence on all separated tuples. Let $\vec{f} \in \mathfrak{S}_n(X)$, $d := d(\vec{f})$ and, if $X = \mathbb{R}^4$, $\bigcup_i \text{supp } f_i \subset B_\rho$. Choose $Q_i \in \mathcal{Q}$ with $\text{supp } f_i \subset Q_i \subset \{x : \text{dist}(x, \text{supp } f_i) < d/3\}$. The Q_i are pairwise at distance at least $d/3$, and they lie in a ball of radius R' , where $R' := \rho + d/3$ if $X = \mathbb{R}^4$ (the ball $B_{R'}$) and R' is the diameter of $\mathbb{T}_m$ otherwise. For $\eta > 0$ pick $g_i \in D_{Q_i}$ with $\|f_i - g_i\|_{C^1} \leq \eta$. Then $\vec{g} \in \mathfrak{F}$ and, by (4.12), for all large $j \in J$,

$$|S_{n,j}^T(\vec{f}) - S_{n,j}^T(\vec{g})| \leq r(\eta) := C(n, d/3, R') c_{\sharp} \eta \sum_{i=1}^n \prod_{l \neq i} (\|f_l\|_{\sharp} + c_{\sharp} \eta).$$

Since $(S_{n,j}^T(\vec{g}))_{j \in J}$ converges, $\limsup_{j \in J} S_{n,j}^T(\vec{f}) - \liminf_{j \in J} S_{n,j}^T(\vec{f}) \leq 2r(\eta)$ for every $\eta > 0$, and $r(\eta) \rightarrow 0$ as $\eta \downarrow 0$. Hence the limit exists. This proves the existence statements of both theorems.

Step 4: properties (a). From now on J is any infinite set along which the limits exist for all separated tuples. Reality, symmetry and multilinearity hold for every j (joint cumulants of real bounded random variables, Lemma 4.6) and pass to the limit; so does (4.11), which gives the bounds in (a), since the lower bounds on j required in Step 0 hold for all large j . By Lemma 4.6 the limit extends to a linear functional $S_{n,\infty}^T(F) = \lim_{j \in J} \Psi_{K_j, m_j}(F)$ of the function $F \in \mathcal{T}_n(X)$; likewise $S_{N,\infty}(F) := \lim_{j \in J} S_{N; K_j, m_j}(F)$ exists for every $F \in \mathcal{T}_N(X)$, is linear in F , and equals $\sum_{\pi \in \Pi_{\geq 2}[N]} \prod_{B \in \pi} S_{|B|,\infty}^T((f_a)_{a \in B})$ on a separated tensor $f_1 \otimes \dots \otimes f_N$.

Step 5: distributions, (b). Let $U_1, \dots, U_n \subset X$ be open balls of radius at most R_U (less than $L^m/4$ if $X = \mathbb{T}_m$), pairwise at distance at least $d_U > 0$. For real $\varphi_i \in C_c^\infty(U_i)$, extended complex-multilinearly, $B(\varphi_1, \dots, \varphi_n) := S_{n,\infty}^T(\vec{\varphi})$ satisfies $|B(\vec{\varphi})| \leq 2^n C(n, d_U, R_U) \prod_i c_{\sharp} \|\varphi_i\|_{C^1}$ by (a), so it is separately continuous. By the Schwartz kernel theorem there is a unique $T_U \in \mathcal{D}'(\prod_i U_i)$ with $T_U(\varphi_1 \otimes \dots \otimes \varphi_n) = B(\vec{\varphi})$; uniqueness holds on every product of open sets, because finite sums of tensors are dense in the test functions of a product (Lemma 4.7 and a partition of unity). Products of such balls cover $X_{\neq}^n$ (around $x \in X_{\neq}^n$ take balls of radius less than $\frac{1}{3} \min_{i \neq k} d(x_i, x_k)$, and less than $L^m/4$ on $\mathbb{T}_m$). On an overlap $\prod_i (U_i \cap U'_i)$ the distributions T_U and $T_{U'}$ take the same values on tensors, hence coincide. By the gluing property of distributions there is a unique $S_{n,\infty}^T \in \mathcal{D}'(X_{\neq}^n)$ restricting to every T_U , and every distribution on $X_{\neq}^n$ taking the values $S_{n,\infty}^T(\vec{\varphi})$ on separated smooth tensors coincides with it. For the order, let $\mathcal{K} \subset \prod_i U_i$ be compact, choose $\sigma \leq \frac{1}{32}$ so small that every product of closed cubes of side 2σ meeting $\mathcal{K}$ lies in $\prod_i U_i$, and a finite smooth partition of unity on $\mathcal{K}$ by functions supported in products of cubes of side σ , with k -th derivatives bounded by $C_k \sigma^{-k}$. The hypothesis (4.7) holds for T_U with $C_T = C(n, d_U, R_U)$ and $c = c_{\sharp}$, by (a); Lemma 4.7 bounds the value of T_U on each piece of $\Phi \in C_c^\infty$, $\text{supp } \Phi \subset \mathcal{K}$, by a constant times $\|\Phi\|_{C^{5n+1}}$. Hence $|S_{n,\infty}^T(\Phi)| \leq C_{\mathcal{K}} \|\Phi\|_{C^{5n+1}}$, and by a finite partition of unity the same holds on every compact subset of $X_{\neq}^n$.

Step 6: invariance, (c). Let $w \in W_4$ and $\vec{f} \in \mathfrak{S}_n(X)$. Then $w \cdot \vec{f} \in \mathfrak{S}_n(X)$ and $S_{n;j}^T(w \cdot \vec{f}) = S_{n;j}^T(\vec{f})$ for all large j (Lemmas 4.1 and 4.2); pass to the limit. If $a \in L^{-t}\mathbb{Z}^4$, then a (resp. $q_{m_j}(a)$) is a lattice vector at every cutoff $K_j \geq t$, so $S_{n;j}^T(\tau_a \vec{f}) = S_{n;j}^T(\vec{f})$ for all large j ; hence the limit is invariant under the dense subgroup $\bigcup_t L^{-t}\mathbb{Z}^4$. For general a , let $a_k \rightarrow a$ with a_k in this subgroup and $|a_k - a| \leq \min\{1, d/4\}$, $d := d(\vec{f})$, and let R be such that each f_i is supported in a ball of radius R . Telescoping as in Step 1, now for the limit functional, every intermediate tuple consists of functions supported in balls of radius $R+1$ at pairwise distance at least $d/2$; by (a) and Lemma 4.5,

$$|S_{n;\infty}^T(\tau_{a_k} \vec{f}) - S_{n;\infty}^T(\tau_a \vec{f})| \leq C(n, d/2, R+1) \sum_{i=1}^n \varpi_{f_i}(|a_k - a|) \prod_{l \neq i} \|f_l\|_{\#} \rightarrow 0.$$

Hence $S_{n;\infty}^T(\tau_a \vec{f}) = \lim_k S_{n;\infty}^T(\tau_{a_k} \vec{f}) = S_{n;\infty}^T(\vec{f})$. Invariance of the distributions follows from invariance on tensors and the uniqueness in Step 5. By linearity, $S_{N;\infty}$ on $\mathcal{T}_N(X)$ is invariant as well.

Step 7: reflection positivity on tori, Theorem 4.8(d). Let $X = \mathbb{T}_m$ and $F \in \mathcal{P}(H_+)$, and put $\mathbb{F}_j := \sum_N \Phi_{K_j,m}(F_N)$, with $\Phi_{K_j,m}(F_0) := F_0$. By Lemma 4.3, $\mathbb{F}_j$ is a bounded gauge-invariant function of the variables of the bonds in H_+ . By Lemma 4.6, $\overline{\mathbb{F}_j(\theta \cdot U)} = \sum_N \Phi_{K_j,m}((\Theta \bar{F})_N)(U)$ and

$$\langle \overline{\mathbb{F}_j(\theta \cdot U)} \mathbb{F}_j(U) \rangle_{K_j,m} = \sum_{N,N'} S_{N+N';K_j,m}((\Theta \bar{F})_N \otimes F_{N'}),$$

which is non-negative by Lemma 4.4 and converges to $(F, F)_{\text{OS}}$ along J by Step 4 (finitely many terms). This is positivity at $\{x^0 = 0\}$. For $\mu \in \{0, 1, 2, 3\}$ and $c \in \mathbb{R}$ let $\gamma := \tau_{ce_\mu} \circ w_{0\mu}$, with $w_{0\mu} \in W_4$ exchanging the coordinates x^0 and x^μ (the identity if $\mu = 0$). Then $\gamma H_+ = H_{\mu,c}$ and $\gamma \theta \gamma^{-1} = \theta_{\mu,c}$, so $F \mapsto \gamma \cdot F$ maps $\mathcal{P}(H_+)$ onto $\mathcal{P}(H_{\mu,c})$ and, by the invariance of the $S_{N;\infty}$ under γ (Step 6), carries the form at $\{x^0 = 0\}$ to the form at $\{x^\mu = c\}$. $\square$

The proof of Step 7 applies verbatim on $\mathbb{R}^4$, with Φ_{K_j,m_j} taken on $V = B_\rho$, where B_ρ contains all the supports occurring in F , and $L^{m_j} \geq 2\rho + 2$: by Lemma 4.2 and the remark following Lemma 4.3, $\mathbb{F}_j$ is then a function of the bonds in $H_+ \subset \mathbb{T}_{m_j}$. This gives (E2) in Theorem 4.12 below, and also, for $F, G \in \mathcal{P}(\{x^0 > 0\})$,

$$(F, G)_{\text{OS}} = \lim_{j \in J} \langle \overline{\mathbb{F}_j(\theta \cdot U)} \mathbb{G}_j(U) \rangle_{K_j,m_j}, \quad (F, G)_{\text{OS}} = \overline{(G, F)_{\text{OS}}}, \quad (4.13)$$

the second identity by the second part of (4.6).

4.3 The Osterwalder–Schrader properties on $\mathbb{R}^4$

Remark 4.11 (numbering). We label properties by the numbering of Osterwalder and Schrader [21]: (E0) the distribution property (the Schwinger functions are continuous linear functionals on Schwartz test functions vanishing at coinciding arguments), (E1) Euclidean invariance, (E2) reflection positivity, (E3) symmetry, (E4) the cluster property; and (E0') for the linear growth condition of [22], a bound on the n -point function by one Schwartz norm of degree proportional to n with constants growing at most like a power of $n!$. Each property is stated below together with its test-function space, so that nothing depends on matching our formulation with the precise wording of those papers. The formulation through a generating functional, which presupposes moments at coinciding smearings, is not the appropriate frame for a composite field of dimension four, and we do not use it.

For $n \geq 1$ let ${}^0\mathcal{S}(\mathbb{R}^{4n}) := \{\varphi \in \mathcal{S}(\mathbb{R}^{4n}) : \partial^\alpha \varphi = 0 \text{ on } \Delta_n \text{ for every multi-index } \alpha\}$, with the topology induced by the seminorms

$$\|\varphi\|_{k,p} := \max_{|\alpha| \leq p} \sup_{x \in \mathbb{R}^{4n}} (1 + |x|)^k |\partial^\alpha \varphi(x)| \quad (k, p \geq 0)$$

of $\mathcal{S}(\mathbb{R}^{4n})$. It is the intersection of the kernels of the continuous functionals $\varphi \mapsto \partial^\alpha \varphi(x)$, $x \in \Delta_n$, hence a closed subspace and a Fréchet space; $C_c^\infty((\mathbb{R}^4)_{\neq}^n) \subset {}^0\mathcal{S}(\mathbb{R}^{4n})$ continuously; and ${}^0\mathcal{S}(\mathbb{R}^4) = \mathcal{S}(\mathbb{R}^4)$. A linear functional on ${}^0\mathcal{S}(\mathbb{R}^{4n})$ is continuous if and only if it is bounded by a constant times one seminorm $\|\cdot\|_{k,p}$.

Theorem 4.12 (Osterwalder–Schrader properties). *Let $(S_{n;\infty}^T)_{n \geq 2}$ be the connected Schwinger functions of a limit as in Theorem 4.10, and for $N \geq 2$ let $S_{N;\infty} := \sum_{\pi \in \Pi_{\geq 2}[N]} \bigotimes_{B \in \pi} S_{|B|;\infty}^T$, a distribution on $(\mathbb{R}^4)_{\neq}^N$ which on separated smooth tensors coincides with the limit of the centred Schwinger functions (Step 4); $S_{0;\infty} := 1$, $S_{1;\infty} := 0$.*

(E0) *For every $n \geq 2$, $S_{n;\infty}^T$ extends uniquely from $C_c^\infty((\mathbb{R}^4)_{\neq}^n)$ to a continuous linear functional on ${}^0\mathcal{S}(\mathbb{R}^{4n})$, and*

$$|S_{n;\infty}^T(\varphi)| \leq \mathbf{a}_n G_n \|\varphi\|_{4n+2, 10n+3} \quad (\varphi \in {}^0\mathcal{S}(\mathbb{R}^{4n})), \quad (4.14)$$

with G_n from (4.5) and $\mathbf{a}_n$ depending only on n and M . Likewise, for every $N \geq 2$, $S_{N;\infty}$ extends uniquely to a continuous linear functional on ${}^0\mathcal{S}(\mathbb{R}^{4N})$, bounded by a constant times $\|\cdot\|_{4N+2, 10N+5[N/2]-2}$. Thus $S_{0;\infty} = 1$ and $S_{N;\infty} \in {}^0\mathcal{S}'(\mathbb{R}^{4N})$ for every $N \geq 1$. All these functionals are real: $S(\bar{\varphi}) = \overline{S(\varphi)}$.

(E1) *The functionals of (E0) are invariant under the diagonal action of $\mathbb{R}^4 \rtimes W_4$. (Invariance under all of $O(4)$ is clause (vi), under the infrared hypothesis stated there; Section 8.)*

(E2) *For every $\mu \in \{0, 1, 2, 3\}$ and $c \in \mathbb{R}$ the centred Schwinger functions are reflection positive at the hyperplane $\{x^\mu = c\}$:*

$$\sum_{N, N' \geq 0} S_{N+N';\infty}((\Theta_{\mu,c}\bar{F})_N \otimes F_{N'}) \geq 0 \quad (4.15)$$

for every finite sequence $(F_N)_{N \geq 0}$, $F_0 \in \mathbb{C}$, in each of the following three cases: (i) $F_N \in \mathcal{T}_N(H_{\mu,c})$; (ii) $F_N \in C_c^\infty(\Omega_{\mu,c}^{(N)})$, where $\Omega_{\mu,c}^{(N)} := \{x \in (\mathbb{R}^4)_{\neq}^N : x_a^\mu > c \text{ for all } a\}$; (iii) F_N in the space $\mathcal{S}_{\Omega_{\mu,c}^{(N)}}$ of Schwartz functions on $\mathbb{R}^{4N}$ vanishing together with all their derivatives outside $\Omega_{\mu,c}^{(N)}$ (a space which contains the Schwartz functions supported in the closed time-ordered region $\{c \leq x_1^\mu \leq \dots \leq x_N^\mu\}$), with $S_{N+N'}$ the extension of (E0). Here $(\Theta_{\mu,c}\bar{F})_N(x) := F_N(\theta_{\mu,c}x_1, \dots, \theta_{\mu,c}x_N)$.

(E3) *The functionals of (E0) are symmetric under permutations of their n arguments in $\mathbb{R}^4$.*

The proof of (E0) rests on an extension lemma for distributions with a separated-ball bound, Proposition 4.16. We first fix auxiliary functions. Let $\omega \in C_c^\infty((\frac{1}{2}, 2))$ with $\omega \geq 0$ and $\sum_{j \in \mathbb{Z}} \omega(2^j t) = 1$ for $t > 0$ (take $\omega_0 \geq 0$ in $C_c^\infty((\frac{1}{2}, 2))$, positive on $[2^{-1/2}, 2^{1/2}]$, and $\omega := \omega_0 / \sum_j \omega_0(2^j \cdot)$), and $\bar{\omega}(t) := \sum_{j \leq 0} \omega(2^j t)$; then $\bar{\omega}$ is smooth, $0 \leq \bar{\omega} \leq 1$, $\bar{\omega} = 0$ on $(0, \frac{1}{2}]$ and $\bar{\omega} = 1$ on $[2, \infty)$. Let $\vartheta \in C_c^\infty((-1, 1))$ with $0 \leq \vartheta \leq 1$ and $\sum_{l \in \mathbb{Z}} \vartheta(t-l) = 1$ (for instance the convolution of the indicator of $[-\frac{1}{2}, \frac{1}{2}]$ with a non-negative mollifier supported in $(-\frac{1}{2}, \frac{1}{2})$).

Lemma 4.13 (regularised distance). *Let $\lambda_1, \dots, \lambda_r$ be non-zero linear forms and $V_1, \dots, V_t$ proper linear subspaces of $\mathbb{R}^D$, $r + t \geq 1$, let $\Omega := \{x : \lambda_b(x) > 0 \text{ for all } b\} \setminus \bigcup_e V_e$, assumed non-empty, and $d_\Omega(x) := \text{dist}(x, \mathbb{R}^D \setminus \Omega)$. On Ω put*

$$r_\Omega(x) := \left(\sum_b (\lambda_b(x)/|\lambda_b|)^{-2} + \sum_e |P_e^\perp x|^{-2} \right)^{-1/2},$$

$P_e^\perp$ the orthogonal projection onto $V_e^\perp$. Then r_Ω is smooth on Ω , $(r+t)^{-1/2}d_\Omega \leq r_\Omega \leq d_\Omega$, and $|\partial^\alpha r_\Omega| \leq C_\alpha d_\Omega^{1-|\alpha|}$ on Ω , with C_α depending only on $D, r+t$ and α .

Proof. For $x \in \Omega$, $d_\Omega(x)$ is the minimum of the distances from x to the closed half-spaces $\{\lambda_b \leq 0\}$ and to the subspaces V_e , that is, the minimum of the functions $h = \lambda_b/|\lambda_b|$ and $h = |P_e^\perp \cdot|$, all of which are $\geq d_\Omega$. Hence $u := \sum h^{-2}$ satisfies $d_\Omega^{-2} \leq u \leq (r+t)d_\Omega^{-2}$, which gives the two-sided bound for $r_\Omega = u^{-1/2}$. Each h^2 is a polynomial of degree two with $|\partial^\beta h^2| \leq 2h^{2-|\beta|}$ for $|\beta| \leq 2$; differentiating $h^2 \cdot h^{-2} = 1$ and inducting on $|\alpha|$ gives $|\partial^\alpha h^{-2}| \leq C_\alpha h^{-2-|\alpha|} \leq C_\alpha d_\Omega^{-2-|\alpha|}$, so $|\partial^\alpha u| \leq C_\alpha d_\Omega^{-2-|\alpha|}$. Finally $\partial_\iota r_\Omega = -\frac{1}{2}r_\Omega^3 \partial_\iota u$ for each coordinate ι , and induction on $|\alpha|$ with the Leibniz rule gives $|\partial^\alpha r_\Omega| \leq C_\alpha d_\Omega^{1-|\alpha|}$. $\square$

Lemma 4.14 (density). *In the setting of Lemma 4.13 let $\mathcal{S}_\Omega := \{f \in \mathcal{S}(\mathbb{R}^D) : \partial^\alpha f = 0 \text{ on } \mathbb{R}^D \setminus \Omega \text{ for all } \alpha\}$, a closed subspace of $\mathcal{S}(\mathbb{R}^D)$. Then $C_c^\infty(\Omega)$ is dense in $\mathcal{S}_\Omega$ for the topology of $\mathcal{S}(\mathbb{R}^D)$. In particular $C_c^\infty((\mathbb{R}^4)_{\neq}^n)$ is dense in ${}^0\mathcal{S}(\mathbb{R}^{4n}) = \mathcal{S}(\mathbb{R}^{4n})_{\neq}$.*

Proof. For $0 < \lambda \leq 1$ let $A_\lambda := \bar{\omega}(r_\Omega/\lambda)$ on Ω and $A_\lambda := 0$ off Ω . It vanishes where $d_\Omega < \lambda/2$, so it is smooth; it equals 1 where $r_\Omega \geq 2\lambda$; and by Lemma 4.13 and the chain rule $|\partial^\gamma A_\lambda| \leq C_\gamma \lambda^{-|\gamma|}$, the derivatives of positive order being supported where $\lambda/2 \leq d_\Omega \leq C_1\lambda$, $C_1 := 2(r+t)^{1/2}$. Let $\beta \in C_c^\infty(\mathbb{R}^D)$ with $0 \leq \beta \leq 1$ and $\beta = 1$ on the unit ball, and $\beta_R(x) := \beta(x/R)$. For $f \in \mathcal{S}_\Omega$ the function $A_\lambda \beta_R f$ lies in $C_c^\infty(\Omega)$. Fix k, p . (i) The function $1 - A_\lambda$ is supported in $\{r_\Omega \leq 2\lambda\} \cup (\mathbb{R}^D \setminus \Omega)$, where $d_\Omega \leq C_1\lambda$. For such x and $|\beta'| \leq p$, let $y \notin \Omega$ with $|x-y| = d_\Omega(x)$; all derivatives of f vanish at y , and Taylor's formula of order $p+1$ at y gives $|\partial^{\beta'} f(x)| \leq C(C_1\lambda)^{p+1}(1+C_1)^k(1+|x|)^{-k}\|f\|_{k,2p+1}$, because $1+|x| \leq (1+C_1)(1+|w|)$ for w on the segment $[y, x]$. With the Leibniz rule, $\|(1-A_\lambda)f\|_{k,p} \leq C\lambda\|f\|_{k,2p+1}$. (ii) $A_\lambda(1-\beta_R)f$ is supported in $\{|x| \geq R\}$ and its derivatives of order $\leq p$ are bounded by $C\lambda^{-p}$ times those of f , so $\|A_\lambda(1-\beta_R)f\|_{k,p} \leq C\lambda^{-p}(1+R)^{-1}\|f\|_{k+1,p}$. Taking $\lambda_l = 2^{-l}$ and $R_l = 2^{l^2}$, $A_{\lambda_l}\beta_{R_l}f \rightarrow f$ in every seminorm $\|\cdot\|_{k,p}$. For the last statement take $r = 0$ and $\{V_e\} = \{\{x_i = x_k\} : i < k\}$. $\square$

From now on $n \geq 2$, $r := r_{(\mathbb{R}^4)_{\neq}^n}$ and $d_\Delta := \text{dist}(\cdot, \Delta_n)$. Since $\text{dist}(x, \{x_i = x_k\}) = |x_i - x_k|/\sqrt{2}$, Lemma 4.13 gives $t_n^{-1/2}d_\Delta \leq r \leq d_\Delta$ with $t_n := n(n-1)/2$, and $|x_i - x_k| \geq \sqrt{2}d_\Delta(x)$ for $i \neq k$. For $j \geq 0$ let $s_j := 2^{-j}$, $\tau_j := s_j/64$, $\omega_0^\Delta := \bar{\omega} \circ r$, $\omega_j^\Delta := \omega(2^j r)$ for $j \geq 1$ (extended by 0 on Δ_n), and for $z \in \mathbb{Z}^{4n}$

$$\chi_{j,z}(x) := \omega_j^\Delta(x) \prod_{\iota=1}^{4n} \vartheta(u_\iota/\tau_j - z_\iota),$$

where $(u_1, \dots, u_{4n})$ are the real coordinates of $x \in \mathbb{R}^{4n}$. Let $\mathcal{P} := \{(j, z) : \chi_{j,z} \not\equiv 0\}$; write $c_{j,z} := \tau_j z$, with components $c_1, \dots, c_n \in \mathbb{R}^4$, and let q'_i be the closed cube of side $s_j/16$ centred at c_i .

Lemma 4.15 (a Whitney-type partition of unity). *With constants depending only on n (and on ω, ϑ):*

(P1) $\chi_{j,z} \in C_c^\infty((\mathbb{R}^4)_{\neq}^n)$, $\text{supp } \chi_{j,z} \subset \prod_i (c_i + [-\frac{s_j}{64}, \frac{s_j}{64}]^4)$, $\sum_{(j,z) \in \mathcal{P}} \chi_{j,z} = 1$ on $(\mathbb{R}^4)_{\neq}^n$ with at most $3 \cdot 2^{4n}$ non-zero terms at each point, and $|\partial^\gamma \chi_{j,z}| \leq C_\gamma 2^{j|\gamma|}$.

- (P2) For $(j, z) \in \mathcal{P}$: q'_i lies in the closed ball of radius $s_j/16$ about c_i ; $\text{dist}(q'_i, q'_k) \geq s_j/4$ for $i \neq k$; if $j \geq 1$ then $d_\Delta \leq 2ns_j$ on $\text{supp } \chi_{j,z}$; and $1 + |c_{j,z}| \leq (1 + \sqrt{n})(1 + |x|)$ for $x \in \text{supp } \chi_{j,z}$.
- (P3) For every integer $h \geq 0$, the number of z with $(j, z) \in \mathcal{P}$ and $h \leq |c_{j,z}| < h + 1$ is at most $C(1 + h)^{4n-4} s_j^{4-4n}$ if $j \geq 1$, and at most $C(1 + h)^{4n}$ if $j = 0$.

Proof. (P1) ω_j^Δ vanishes on the neighbourhood $\{d_\Delta < 2^{-j-1}\}$ of Δ_n (as $r \leq d_\Delta$), so it is smooth on $\mathbb{R}^{4n}$; for $j \geq 1$ its support lies in $\{2^{-j-1} \leq r \leq 2^{1-j}\}$, and that of ω_0^Δ in $\{r \geq \frac{1}{2}\}$. The support statement follows from $\text{supp } \vartheta \subset (-1, 1)$. On $(\mathbb{R}^4)_{\neq}^n$, $\sum_{j \geq 0} \omega_j^\Delta = \sum_{j \in \mathbb{Z}} \omega(2^j r) = 1$ and $\sum_z \prod_l \vartheta(u_l/\tau_j - z_l) = 1$; at most three values of j contribute (the condition $2^{-j-1} \leq r \leq 2^{1-j}$ confines j to an interval of length two), and for each at most 2^{4n} values of z . On the support of the derivatives of ω_j^Δ we have $d_\Delta \geq r \geq 2^{-j-1}$, so the chain rule and Lemma 4.13 give $|\partial^\gamma \omega_j^\Delta| \leq C_\gamma \sum_{q \leq |\gamma|} 2^{jq} d_\Delta^{q-|\gamma|} \leq C_\gamma 2^{j|\gamma|}$; the lattice factor contributes $\tau_j^{-|\gamma|}$. (P2) A cube of side ℓ in $\mathbb{R}^4$ lies in the closed ball of radius ℓ about its centre. Pick $x^* \in \text{supp } \chi_{j,z}$; then $d_\Delta(x^*) \geq r(x^*) \geq s_j/2$, so $|x_i^* - x_k^*| \geq s_j/\sqrt{2}$, while $x_i^* \in q'_i$ and $\text{diam } q'_i = s_j/8$; hence $\text{dist}(q'_i, q'_k) \geq s_j/\sqrt{2} - s_j/4 \geq s_j/4$. If $j \geq 1$, $r(x^*) \leq 2s_j$, so $d_\Delta(x^*) \leq 2t_n^{1/2} s_j \leq \sqrt{2} ns_j$, and d_Δ is 1-Lipschitz while $\text{supp } \chi_{j,z}$ has diameter at most $\sqrt{n} s_j/16$; so $d_\Delta \leq 2ns_j$ there. Finally $|x - c_{j,z}| \leq \sqrt{n} s_j/32 \leq \sqrt{n}$ on the support. (P3) The cubes $\tau_j(z + [-\frac{1}{2}, \frac{1}{2}]^{4n})$ have disjoint interiors and volume τ_j^{4n} , and each lies within distance $\sqrt{n} s_j/32$ of $c_{j,z}$. For $j \geq 1$, by (P2) and these distances, the relevant ones lie in $E := \{d_\Delta \leq 3ns_j, |x| \leq (1 + \sqrt{n})(h + 1)\} \subset \bigcup_{i < k} \{P_{ik}^\perp x \leq 3ns_j, |x| \leq (1 + \sqrt{n})(h + 1)\}$, $P_{ik}^\perp$ the orthogonal projection onto $\{x_i = x_k\}^\perp \cong \mathbb{R}^4$; by Fubini each set of the union has volume at most $C(1 + h)^{4n-4} s_j^4$, and dividing by τ_j^{4n} gives the bound. For $j = 0$ use $|x| \leq (1 + \sqrt{n})(h + 1)$ only. $\square$

Proposition 4.16 (extension to test functions vanishing on the diagonals). *Let $n \geq 2$, $T \in \mathcal{D}'((\mathbb{R}^4)_{\neq}^n)$, $a \geq 0$, $c \geq 1$ and $G < \infty$, and suppose that for every $\mathbf{d} > 0$ and all real $\varphi_1, \dots, \varphi_n \in C_c^\infty(\mathbb{R}^4)$, each supported in a closed ball of radius at most $\frac{1}{16}$, with supports pairwise at distance at least $\mathbf{d}$,*

$$|T(\varphi_1 \otimes \dots \otimes \varphi_n)| \leq G \max\{1, \mathbf{d}^{-a}\} \prod_{i=1}^n \max\{\|\varphi_i\|_\infty, c\|\nabla \varphi_i\|_\infty\}. \quad (4.16)$$

Then T extends uniquely to a continuous linear functional $\tilde{T}$ on ${}^0\mathcal{S}(\mathbb{R}^{4n})$; it is given by the absolutely convergent series $\tilde{T}(\varphi) = \sum_{(j,z) \in \mathcal{P}} T(\chi_{j,z}\varphi)$, and $|\tilde{T}(\varphi)| \leq \mathbf{a} G \|\varphi\|_{4n+2, 10n+[a]-2}$ with $\mathbf{a}$ depending only on n , a and c .

Proof. Let $\varphi \in {}^0\mathcal{S}(\mathbb{R}^{4n})$, $(j, z) \in \mathcal{P}$, $\Phi_{j,z} := \chi_{j,z}\varphi$, $\sigma_j := s_j/32$ and $p := 5n + 1$.

Local bound. By (P1)–(P2), Lemma 4.7 applies to $\Phi_{j,z}$ with $\sigma = \sigma_j \leq \frac{1}{32}$ and the cubes centred at $c_1, \dots, c_n$: its real test functions are supported in the cubes q'_i , which lie in closed balls of radius $s_j/16 \leq \frac{1}{16}$ and are pairwise at distance $\geq s_j/4$, so (4.16) gives (4.7) with $C_T = G \max\{1, (s_j/4)^{-a}\} = 4^a G s_j^{-a}$. Hence

$$|T(\Phi_{j,z})| \leq K_n 4^a 32^n G s_j^{-n-a} |\Phi_{j,z}|_{p, \sigma_j}.$$

Flatness and decay. By (P1), $\sigma_j^{|\gamma|} |\partial^\gamma \chi_{j,z}| \leq C_\gamma 32^{-|\gamma|}$, so by the Leibniz rule $|\Phi_{j,z}|_{p, \sigma_j} \leq C \sup\{|\partial^\beta \varphi(x)| : x \in \text{supp } \chi_{j,z}, |\beta| \leq p\}$. Let $k := 4n + 2$ and $m := 5n + [a] - 3$. If $j = 0$, (P2) gives $|\partial^\beta \varphi(x)| \leq (1 + \sqrt{n})^k (1 + |c_{0,z}|)^{-k} \|\varphi\|_{k,p}$. If $j \geq 1$, let $y \in \Delta_n$ with $|x - y| = d_\Delta(x) \leq 2ns_j \leq n$. All derivatives of $\partial^\beta \varphi$ vanish at y , and Taylor's formula of order m at y gives

$$\begin{aligned} |\partial^\beta \varphi(x)| &\leq \frac{(4n)^m}{m!} (2ns_j)^m \sup_{w \in [y, x], |\gamma| = m} |\partial^{\beta+\gamma} \varphi(w)| \\ &\leq C s_j^m (1 + |c_{j,z}|)^{-k} \|\varphi\|_{k, p+m}, \end{aligned}$$

since $1 + |x| \leq (1 + n)(1 + |w|)$ on $[y, x]$ and by (P2).

Summation. Note $p + m = 10n + \lceil a \rceil - 2$. Group the z according to $h \leq |c_{j,z}| < h + 1$ and use (P3). For $j \geq 1$ the power of s_j is $-n - a + m + 4 - 4n = 1 + \lceil a \rceil - a \geq 1$, so

$$\sum_{j \geq 1} \sum_z |T(\Phi_{j,z})| \leq C G \|\varphi\|_{k,p+m} \sum_{j \geq 1} s_j^{1+\lceil a \rceil - a} \sum_{h \geq 0} (1+h)^{4n-4-k} < \infty,$$

the inner sum being $\sum_h (1+h)^{-6}$; and $\sum_z |T(\Phi_{0,z})| \leq C G \|\varphi\|_{k,p} \sum_{h \geq 0} (1+h)^{4n-k} < \infty$. Hence the series defining $\tilde{T}$ converges absolutely, and $\tilde{T}$ is linear and bounded by $\mathfrak{a} G \|\varphi\|_{4n+2,10n+\lceil a \rceil - 2}$; thus it is continuous on ${}^0\mathcal{S}(\mathbb{R}^{4n})$.

Consistency and uniqueness. If $\varphi \in C_c^\infty((\mathbb{R}^4)_{\neq}^n)$, then $\mathfrak{r}$ has a positive minimum on $\text{supp } \varphi$, so only finitely many (j, z) have $\chi_{j,z} \varphi \neq 0$, $\varphi = \sum \chi_{j,z} \varphi$ is a finite sum, and $\tilde{T}(\varphi) = T(\varphi)$. Two continuous extensions agree on the dense subspace $C_c^\infty((\mathbb{R}^4)_{\neq}^n)$ (Lemma 4.14), hence everywhere. $\square$

Proof of Theorem 4.12. (E0) Let $\varphi_1, \dots, \varphi_n$ be real, smooth, supported in closed balls of radius at most $\frac{1}{16}$, pairwise at distance at least $\mathfrak{d}$. Each lies in a ball of radius 1, so by Theorem 4.10(a),(b) and (4.5),

$$|S_{n;\infty}^T(\varphi_1 \otimes \dots \otimes \varphi_n)| \leq C_n^\infty(\mathfrak{d}; 1) \prod_i \|\varphi_i\|_{\#} \leq G_n \max\{1, \mathfrak{d}^{-5}\} \prod_i \max\{\|\varphi_i\|_\infty, c_{\#} \|\nabla \varphi_i\|_\infty\}.$$

Proposition 4.16 with $a = 5$, $c = c_{\#}$ gives the extension and (4.14). For $S_{N;\infty}$: the tensor product of the distributions $S_{|B|;\infty}^T$, $B \in \pi$, is a distribution on $\prod_B (\mathbb{R}^4)_{\neq}^B \supset (\mathbb{R}^4)_{\neq}^N$, and on real tensors $S_{N;\infty}(\otimes_a \varphi_a) = \sum_\pi \prod_B S_{|B|;\infty}^T(\otimes_{a \in B} \varphi_a)$. A partition in $\Pi_{\geq 2}[N]$ has at most $\lfloor N/2 \rfloor$ blocks, so (4.16) holds with $a = 5 \lfloor N/2 \rfloor$ and $G = \sum_\pi \prod_B G_{|B|}$, and Proposition 4.16 applies. Reality: the values on real tensors are real; in the expansion of Lemma 4.7 the coefficients of $\bar{\Phi}$ are the conjugates of those of Φ with conjugate tensors, so $S(\bar{\Phi}) = \overline{S(\Phi)}$ for Φ supported in a product of small separated cubes, hence (by a real partition of unity) on $C_c^\infty((\mathbb{R}^4)_{\neq}^n)$, and this extends to ${}^0\mathcal{S}$ by density and continuity.

(E1), (E3) For $\gamma \in \mathbb{R}^4 \rtimes W_4$ and a permutation σ of the arguments, the maps $\varphi \mapsto \varphi \circ \gamma^{-1}$ (diagonally) and $\varphi \mapsto \varphi \circ \sigma$ are continuous on ${}^0\mathcal{S}$ and preserve $C_c^\infty((\mathbb{R}^4)_{\neq}^n)$; on separated tensors invariance is Theorem 4.10(c) and symmetry is Theorem 4.10(a). By the uniqueness in Step 5 and Proposition 4.16, both pass to the distributions and to their extensions.

(E2) Case (i) at $\{x^0 = 0\}$ is Step 7 on $\mathbb{R}^4$ (see the paragraph following the proof of Theorems 4.8 and 4.10). Case (ii) at $\{x^0 = 0\}$: let $F_N \in C_c^\infty(\Omega_{0,0}^{(N)})$. Each point of $\text{supp } F_N$ has a neighbourhood $\prod_a (y_a + (-\frac{\sigma}{2}, \frac{\sigma}{2})^4)$, with the cubes $y_a + [-\sigma, \sigma]^4$ pairwise disjoint and contained in $\{x^0 > 0\}$, $\sigma \leq \frac{1}{32}$; a finite partition of unity splits F_N into pieces supported in such products, and by Lemma 4.7 each piece is the limit in C_c^∞ of the Fourier partial sums, which are finite sums of tensors whose real and imaginary parts give elements $F_N^{(\ell)}$ of $\mathcal{T}_N(\{x^0 > 0\})$. The map $(\varphi, \psi) \mapsto \Theta \bar{\varphi} \otimes \psi$ is continuous from $C_c^\infty(W) \times C_c^\infty(W')$ to $C_c^\infty(\Theta W \times W')$ for open $W \subset \Omega_{0,0}^{(N)}$, $W' \subset \Omega_{0,0}^{(N')}$, and $\Theta W \times W' \subset (\mathbb{R}^4)_{\neq}^{N+N'}$ because reflected points have negative times; $S_{N+N';\infty}$ is a distribution there, equal on separated smooth tensors to the functional of Step 4. Hence the left side of (4.15) is the limit of the non-negative numbers $(F^{(\ell)}, F^{(\ell)})_{\text{OS}}$. Case (iii) at $\{x^0 = 0\}$: $\Omega_{0,0}^{(N)}$ is of the form of Lemma 4.13, so by Lemma 4.14 there are $F_N^{(\ell)} \in C_c^\infty(\Omega_{0,0}^{(N)})$ converging to F_N in $\mathcal{S}$. For $f \in \mathcal{S}_{\Omega_{0,0}^{(N)}}$ and $g \in \mathcal{S}_{\Omega_{0,0}^{(N'')}}$ every derivative of $\Theta \bar{f} \otimes g$ vanishes at (x, y) unless $\theta x \in \Omega_{0,0}^{(N)}$ and $y \in \Omega_{0,0}^{(N')}$, in which case all $x_a^0 < 0 < y_b^0$ and $(x, y) \in (\mathbb{R}^4)_{\neq}^{N+N'}$; so $\Theta \bar{f} \otimes g \in {}^0\mathcal{S}(\mathbb{R}^{4(N+N')})$. Since $\|u \otimes v\|_{k,p} \leq \|u\|_{k,p} \|v\|_{k,p}$ and Θ and conjugation are isometries of every $\|\cdot\|_{k,p}$, $\Theta \bar{F}_N^{(\ell)} \otimes F_{N'}^{(\ell)} \rightarrow \Theta \bar{F}_N \otimes F_{N'}$ in ${}^0\mathcal{S}$, and (E0) together with case (ii) gives (4.15). If $f \in \mathcal{S}$ is supported in the closed time-ordered region $\{0 \leq x_1^0 \leq \dots \leq x_N^0\}$,

then all derivatives of f vanish off its interior, which is contained in $\Omega_{0,0}^{(N)}$, because every boundary point of that region is a limit of points outside it; so case (iii) covers these functions. Finally, for a general hyperplane $\{x^\mu = c\}$ let $\gamma := \tau_{ce_\mu} \circ w_{0\mu}$ as in Step 7; then $\gamma\theta\gamma^{-1} = \theta_{\mu,c}$, γ maps the three test-function classes at $\{x^0 = 0\}$ onto those at $\{x^\mu = c\}$, and (E1) transfers (4.15). $\square$

4.4 Reconstruction

We follow the construction of Osterwalder and Schrader [21]: a Hilbert space from the positivity (E2), and a contraction semigroup of time translations from positivity and invariance by iterating the Cauchy–Schwarz inequality. The one input that differs is the growth bound needed for the iteration: instead of the regularity axiom we use the per-ball bound (4.10), which gives $\sup_{t \geq 0} (F, \mathbb{T}_t F)_{\text{OS}} < \infty$ directly (step (b) below). Let $\mathcal{P}_+ := \mathcal{P}(\{x^0 > 0\})$, with the form (4.9) built from the $S_{N;\infty}$ of a limit as in Theorem 4.10. For $t \geq 0$ and $\vec{a} \in \mathbb{R}^3$ let $(\mathbb{T}_t F)_N(x) := F_N(x_1 - te_0, \dots, x_N - te_0)$ and $(S_{\vec{a}} F)_N(x) := F_N(x_1 - (0, \vec{a}), \dots, x_N - (0, \vec{a}))$, and for $w \in W_3$ let $(w \cdot F)_N(x) := F_N(w^{-1}x_1, \dots, w^{-1}x_N)$; all three map $\mathcal{P}_+$ into itself.

Theorem 4.17 (reconstruction). *For every limit as in Theorem 4.10:*

- (R1) $(\cdot, \cdot)_{\text{OS}}$ is a positive semidefinite Hermitian form on $\mathcal{P}_+$; with $\mathcal{N} := \{F : (F, F)_{\text{OS}} = 0\}$, the completion $\mathcal{H}$ of $\mathcal{P}_+/\mathcal{N}$ is a separable Hilbert space, and $\Omega := [(1, 0, 0, \dots)]$ is a unit vector.
- (R2) $\mathbb{T}_t$ induces a bounded operator T_t on $\mathcal{H}$; $(T_t)_{t \geq 0}$ is a strongly continuous semigroup of positive self-adjoint contractions with $T_t \Omega = \Omega$ and $\ker T_t = \{0\}$, and $T_t = e^{-tH}$ for a unique self-adjoint operator $H \geq 0$ on $\mathcal{H}$, with $H\Omega = 0$.
- (R3) $S_{\vec{a}}$ induces a strongly continuous unitary representation $\vec{a} \mapsto U(\vec{a})$ of $\mathbb{R}^3$ on $\mathcal{H}$ with $U(\vec{a})\Omega = \Omega$ and $U(\vec{a})T_t = T_t U(\vec{a})$; its self-adjoint generators $\vec{P} = (P_1, P_2, P_3)$ commute strongly with each other and with H , and the joint spectrum of $(H, \vec{P})$ is contained in $[0, \infty) \times \mathbb{R}^3$.
- (R4) W_3 acts on $\mathcal{H}$ by unitaries $U(w)$ fixing Ω and commuting with T_t , with $U(w)U(\vec{a})U(w)^* = U(w\vec{a})$; the joint spectrum of $(H, \vec{P})$ is invariant under W_3 .

Proof. Write every $F \in \mathcal{P}_+$ as a finite combination of tensors of real C_c^1 functions, and put $\|F\| := (F, F)_{\text{OS}}^{1/2}$.

(a) *The space.* By (E2)(i) and (4.13) the form is positive semidefinite and Hermitian; by the Cauchy–Schwarz inequality $\mathcal{N}$ is a subspace and the form descends to an inner product on $\mathcal{P}_+/\mathcal{N}$; $\|(1, 0, \dots)\|^2 = S_{0;\infty} = 1$. Separability: let $\mathcal{P}_+^{\mathbb{Q}}$ be the countable set of finite combinations with coefficients in $\mathbb{Q} + i\mathbb{Q}$ of tensors $g_1 \otimes \dots \otimes g_N$, $g_a \in D_{Q_a}$, with $Q_1, \dots, Q_N \in \mathcal{Q}$ pairwise disjoint and contained in $\{x^0 > 0\}$ (Step 2). Given F and $\eta > 0$, replace each function f occurring in F by $g \in D_{Q(f)}$ with $\|f - g\|_{C^1} \leq \eta$, where $\text{supp } f \subset Q(f) \in \mathcal{Q}$ and $Q(f)$ is so close to $\text{supp } f$ that within each tensor the sets $Q(f)$ stay pairwise at distance at least half the original distance and inside $\{x^0 > 0\}$, and each coefficient by a rational one within η ; call the result $G \in \mathcal{P}_+^{\mathbb{Q}}$. Telescoping each tensor as in Step 1, $(F - G, F - G)_{\text{OS}}$ is a finite sum of values of $S_{N+N';\infty}$ at separated tuples, each containing a factor of $\sharp$ -norm at most $c_{\sharp}\eta$ or a coefficient of modulus at most η , with radii and separations controlled by F ; by (4.10) it is at most $C_F\eta$ for $\eta \leq 1$. Hence $\{[G] : G \in \mathcal{P}_+^{\mathbb{Q}}\}$ is dense in $\mathcal{H}$.

(b) *The growth bound.* Let R_F be such that every function occurring in F is supported in a ball of radius R_F , let $d_F > 0$ be the least distance between the supports of two factors of one tensor term, and $\delta_F > 0$ the least distance of an occurring support from $\{x^0 = 0\}$. For $t \geq 0$, $(F, \mathbb{T}_t F)_{\text{OS}}$ is a finite sum, with a number of terms independent of t , of products of values of $S_{n;\infty}^T$ at tuples of

functions $f \circ \theta$ and $\tau_{te_0} f'$, with f, f' occurring in F . Any two functions in such a tuple are supported in balls of radius R_F at distance at least $d_0 := \min\{d_F, 2\delta_F\}$: two reflected, or two translated, functions from one tensor term are at distance $\geq d_F$, and a reflected one (supported in $\{x^0 \leq -\delta_F\}$) and a translated one (supported in $\{x^0 \geq \delta_F + t\}$) are at distance $\geq 2\delta_F + t$. By (4.10), applied with the t -independent data (d_0, R_F) ,

$$M_F := \sup_{t \geq 0} (F, \mathbb{T}_t F)_{\text{OS}} < \infty.$$

This is where the per-ball form of clause (iii)(b) matters: the radius of a single ball containing all the supports grows linearly in t , and the constant of (iii)(b) grows faster than any power of the radius, so a bound in terms of such a radius would grow too fast in t for the iteration in (c) to close.

(c) *Contractions.* Since $\theta(y + te_0) = \theta y - te_0$, $\Theta \overline{\mathbb{T}_t F} = \mathbb{T}_{-t} \Theta \bar{F}$; invariance of the $S_{N;\infty}$ under the translation of all arguments by te_0 gives $(\mathbb{T}_t F, G)_{\text{OS}} = (F, \mathbb{T}_t G)_{\text{OS}}$. Then $\|\mathbb{T}_t F\|^2 = (F, \mathbb{T}_{2t} F)_{\text{OS}} \leq \|F\| \|\mathbb{T}_{2t} F\|$, so $\mathbb{T}_t \mathcal{N} \subset \mathcal{N}$. If $\|F\| > 0$, the numbers $b_k := \|\mathbb{T}_{2^k t} F\|/\|F\|$ satisfy $b_k^2 \leq b_{k+1}$ and $b_k^2 \leq M_F/\|F\|^2$ by (b), so

$$b_0 \leq b_k^{2^{-k}} \leq (M_F/\|F\|^2)^{2^{-k-1}} \longrightarrow 1 \quad (k \rightarrow \infty).$$

Thus $\|\mathbb{T}_t F\| \leq \|F\|$, and $\mathbb{T}_t$ induces a contraction T_t of $\mathcal{H}$, symmetric and bounded, hence self-adjoint, with $T_s T_t = T_{s+t}$, $T_0 = 1$, $T_t \Omega = \Omega$, and $(v, T_t v) = \|T_{t/2} v\|^2 \geq 0$.

(d) *Strong continuity.* Let $F, G \in \mathcal{P}_+$ and $0 \leq s, t$ with $|t - s| \leq \min\{1, d_G/4\}$. Telescoping in the translated factors as in Step 1, $(F, \mathbb{T}_t G)_{\text{OS}} - (F, \mathbb{T}_s G)_{\text{OS}}$ is a finite sum of values of $S_{N+N';\infty}$ at separated tuples (radii $\leq R_F + R_G + 1$, separations $\geq \min\{d_F, 3d_G/4, \delta_F + \delta_G\}$), each containing one entry $\tau_{te_0} g - \tau_{se_0} g$, whose $\sharp$ -norm is at most $\varpi_g(|t - s|)$ by Lemma 4.5. By (4.10), $t \mapsto (F, \mathbb{T}_t G)_{\text{OS}}$ is continuous. Hence $\|T_t[F] - [F]\|^2 \leq 2\|F\|^2 - 2\text{Re}(F, \mathbb{T}_t F)_{\text{OS}} \rightarrow 0$ as $t \downarrow 0$; since $\|T_t\| \leq 1$ and the classes $[F]$ are dense, $T_t \rightarrow 1$ strongly as $t \downarrow 0$, and the semigroup property gives strong continuity on $[0, \infty)$.

(e) *The generator.* If $T_t v = 0$ then $\|T_{t/2} v\|^2 = (v, T_t v) = 0$; iterating, $T_{2^{-k}t} v = 0$ for all k , and $v = \lim_k T_{2^{-k}t} v = 0$. So T_1 is a positive self-adjoint contraction with trivial kernel, and $H := -\log T_1$, defined by the spectral theorem, is self-adjoint and ≥ 0 . Since $T_{1/2}$ is a positive square root of T_1 , uniqueness of positive square roots gives $T_{1/2} = T_1^{1/2}$, and inductively $T_q = T_1^q = e^{-qH}$ for dyadic $q \geq 0$; both sides are strongly continuous in t , so $T_t = e^{-tH}$ for all $t \geq 0$. Uniqueness of H follows from $H = -\log T_1$, and $T_1 \Omega = \Omega$ gives $H \Omega = 0$.

(f) *Space translations and momenta.* $S_{\vec{a}}$ maps $\mathcal{P}_+$ onto itself (with inverse $S_{-\vec{a}}$), commutes with Θ (since θ fixes $\vec{x}$) and with $\mathbb{T}_t$, and preserves the form by the translation invariance of the $S_{N;\infty}$; so it induces a unitary $U(\vec{a})$ on $\mathcal{H}$, with $U(\vec{a})U(\vec{b}) = U(\vec{a} + \vec{b})$, $U(\vec{a})\Omega = \Omega$ and $U(\vec{a})T_t = T_t U(\vec{a})$. Strong continuity follows as in (d), with spatial translations in place of time translations. By the theorem of Stone, Naimark, Ambrose and Godement, $U(\vec{a}) = \int e^{i\vec{a} \cdot \vec{k}} dE(\vec{k})$ for a projection-valued measure E on $\mathbb{R}^3$, and $P_l := \int k_l dE$ are self-adjoint and strongly commuting. Each $U(\vec{a})$ commutes with T_1 , hence with the spectral projections of T_1 , which are those of H ; therefore E commutes with the spectral measure of H , the operators H, P_1, P_2, P_3 commute strongly, and their joint spectral measure is carried by $\sigma(H) \times \mathbb{R}^3 \subset [0, \infty) \times \mathbb{R}^3$.

(g) *The cubic group.* For $w \in W_3$, $F \mapsto w \cdot F$ maps $\mathcal{P}_+$ onto itself (as w fixes x^0), commutes with Θ and $\mathbb{T}_t$, satisfies $w S_{\vec{a}} w^{-1} = S_{w\vec{a}}$, and preserves the form by (E1). This gives unitaries $U(w)$ with the stated properties, and $U(w)E(\cdot)U(w)^* = E(w \cdot)$. $\square$

Remark 4.18 (non-degeneracy, from clause (iv)). Clause (iv), in the form valid in every volume (Theorem 6.24), gives, for g below the coupling ceiling $\gamma_W^{\mathbb{T}}$ of that clause for the given ϑ , for the

separations $\mathbf{d}$ it allows, and for test pairs (f_1, f_2) in the class $\mathfrak{T}(\mathbf{d}; \vartheta)$, a lower bound $S_{2;K,m}^T(f_1, f_2) \geq c_0 \|f_1\|_\infty \|f_2\|_\infty$ with $c_0 = c_0(\mathbf{d}, \vartheta, g) > 0$ independent of K, g_0, m and of the positions. Let $f(x) := \psi(|x - 2\mathbf{d}e_0|/\mathbf{d})$, where $\psi(r) = (1 - r^2)^2$ for $r \leq 1$ and $\psi(r) = 0$ for $r > 1$. Then $f \in C_c^1(\mathbb{R}^4)$ is non-negative, $\|f\|_\infty = 1$, and f is supported in the closed ball of radius $\mathbf{d}$ about $2\mathbf{d}e_0$; the supports of $f \circ \theta$ and f are at distance $2\mathbf{d}$, and $(f \circ \theta, f) \in \mathfrak{T}(\mathbf{d}; \vartheta)$ for a $\vartheta \in (0, 1]$ depending only on ψ . With $F := (0, f, 0, \dots) \in \mathcal{P}_+$, Theorem 4.10 gives $\|[F]\|^2 = S_{2;\infty}(f \circ \theta, f) = S_{2;\infty}^T(f \circ \theta, f) \geq c_0 > 0$, while $([F], \Omega) = S_{1;\infty}(f) = 0$. Hence, under the hypotheses of clause (iv), $\dim \mathcal{H} \geq 2$.

4.5 What is not obtained

Remark 4.19. The following are not asserted.

- (1) *The growth condition (E0'), field operators, Wightman functions.* The bounds of Theorems 4.10 and 4.12 hold for each n separately, with constants inherited from clause (iii)(b). In the form proved in Section 3 these constants grow faster than any power of $n!$: the bound of Theorem 3.10 rests on a chessboard estimate whose cells were taken large enough to contain n separated balls, and it pays a stability factor exponential in the volume of a cell. A variant of that estimate, with a grid independent of n and the test functions cut into pieces by multilinearity, gives constants of factorial type, at most $(n!)^c C^n$ for constants c, C ; it is proved in the companion account and is not part of Theorem 0. With it, and with the order of the seminorms in (4.14), which is linear in n , the growth condition (E0') of [22] reduces to a bookkeeping that we do not carry out here. That condition is what allows one to define field operators on the reconstructed space and to continue the Schwinger functions to Wightman distributions; the remaining obstruction to field operators in the sense of Wightman is invariance under all rotations, which Theorem 0 gives only in clause (vi), under its hypothesis. We assert neither field operators nor Wightman distributions.
- (2) *Coinciding arguments and $n = 1$.* Our Schwinger functions are functionals on ${}^0\mathcal{S}$. Extensions to $\mathcal{S}(\mathbb{R}^{4n})$ exist by the Hahn–Banach theorem but are determined only up to distributions supported on the diagonals; we assert no canonical one. Already for free fields the two-point function of the action density is a multiple of $|x - y|^{-8}$, which is not locally integrable in four dimensions. For $n = 1$ we use only the centring $S_1 = 0$; no cutoff-uniform statement about $\langle \mathcal{O}(f) \rangle_{K,m}$ or about coinciding or overlapping supports is made.
- (3) *The cluster property (E4), uniqueness of the vacuum, a mass gap.* The constants in our bounds do not decay as the separation of the supports grows, so no decay of the connected functions is available. Accordingly we do not assert that Ω is the only vector invariant under T_t , ergodicity of the time translations, or a gap in the spectrum of H . The hypothesis of clause (vi) is a decay statement of a different kind: it concerns, uniformly in the cutoff, the correlations with one insertion of the stress tensor at large distance, and we derive from it neither (E4) nor the uniqueness of Ω .
- (4) *Rotations and the relativistic spectral condition.* Only invariance under $\mathbb{R}^4 \rtimes W_4$ is proved here; invariance under $O(4)$ is clause (vi), under its infrared hypothesis (Section 8). The spectral condition $H \geq |\vec{P}|$ and Lorentz boosts would require rotations mixing time and space; only $H \geq 0$ and the joint spectrum in $[0, \infty) \times \mathbb{R}^3$ are asserted.
- (5) *Uniqueness.* The limits on $\mathbb{R}^4$ are obtained along subsequences of cutoffs and volumes; independence of the subsequence, convergence without subsequences in the volume, independence of the regularisation, and convergence for merely Lipschitz test functions on $\mathbb{R}^4$ are not asserted. On each torus the limits are described further by clause (v) (Section 7).

5 Clause (iv) in finite volume: the curvature two-point witness

Throughout this section $N = 2$, so that $b_0 = (N^2 - 1)/2 = \frac{3}{2}$, and (H1)–(H7) and the notation of Section 1 are in force. We prove Theorem 5.8: on every torus of the family, the connected two-point function of the smeared action densities of two non-negative bumps of size $\mathbf{d}$, placed at distance between $\mathbf{d}$ and $6\mathbf{d}$, equals b_0 times an explicit positive and scale-free kernel of the free lattice Maxwell field, times the fourth power of the running coupling at the depth of the scale $\mathbf{d}$, times the square of the source factor of clause (iii)(c), up to a relative error of at most $\frac{1}{4}$. This holds uniformly in the cutoff, the bare coupling and the position. On every torus below a horizon, $m \leq m_V(g)$ with $m_V(g) \rightarrow \infty$ as $g \downarrow 0$, it holds uniformly in the volume as well; on larger tori the volume enters through one threshold $s_{\text{vol}}(g, m)$ on the depth and through nothing else. Section 6 passes from one small torus to every volume by a different argument; the present section supplies its input.

The proof (Steps 1–5) writes the two-point function exactly, by the law of total covariance, on Balaban’s history space at the level $K - s$, $s = s(\mathbf{d})$ the depth of the scale $\mathbf{d}$, and extracts its main term, two Gaussian covariance lines between action densities of the background field; the remainders are controlled by the errors of the extraction, by the room that the separation of the supports provides, and, for live large-field regions near the supports, by masses of events concerning one cell; the last is where the volume enters (Step 4). Section 5.8 lists what is proved here, what is taken from Balaban’s papers or proved in the companion account, and the readings that are used; no statement of this section is deferred. The parts of the companion account used here are: [T-W], on the decomposition of the renormalised insertions at the witness level and the bounds on the contributions of live large-field components; [T-GX], on the Gaussian extraction with one and two inserted observables; [T-LF], on live large-field regions on a torus; and a companion note on the free kernel, which proves its two-sided bounds, its comparison between tori and its continuum limit. The correlation theorem with sources is that of Section 3, whose status is stated there.

5.1 Notation, the test class, the depth of a scale, the kernel

Let $g \leq \gamma_J$, $g_0 \in (0, g_-(g)]$, $K = K(g_0, g)$, and let $m \geq m_{\mathbb{T}}(g_-(g))$ as in (H5). The torus $\mathbb{T}_m$ has side $2L^m$ and bare spacing $\varepsilon = L^{-K}$; $x_k = g_k^{-2}$ are the running inverse couplings. By clause (0), for $0 \leq s \leq K$,

$$\underline{x}_s := g^{-2} + \lambda_{\sharp}s - c_5 \leq x_{K-s} \leq \bar{x}_s := g^{-2} + B_{\sharp}(s+1), \quad (5.1)$$

and $\lambda_{\sharp} \leq B_{\sharp}$. Balaban’s threshold function is $p_0(g') := A_0(\log g'^{-2})^{\mathfrak{p}_0}$; the small-field threshold at level k is $\varepsilon_k = g_k p_0(g_k)$, and R_k , the least power of L not smaller than $(\log x_k)^{r_B}$, is both his large-cube scale at level k [9, §1] and the number of steps after which a small large-field region in which no new large fields were created is removed [10, §1, (i)–(ii)], [11, (1.80)]. Here A_0 , the cube size M , $r_B \geq 2$ and $\mathfrak{p}_0$ are among Balaban’s parameters fixed in (H3); his conditions on the exponent are $\mathfrak{p}_0 \geq 5r_B$ [9, p. 246] and that p_0 be large [11, (1.80)–(1.88)], and we fix the floor $\mathfrak{p}_0 \geq 6r_B + 1$, which implies the first and is used in the prepayment of age in Step 4 (a lower bound on one exponent, of the kind (H3) allows). $\check{R}$ denotes a majorant of R that is non-decreasing in its argument, with $\check{R}(x) \leq C_R(\log x)^{r_B} + 1$. For a coupling g' , $b_*(g')$ is the integer of Theorem 3.1 for the target g' : the least depth below the cutoff at which its run with sources may be stopped; it is non-increasing in g' and of order $\log_L \log g'^{-2}$. For a depth s , $k_0 := K - s$ is the *witness level*, and the cubes of side L^{-s} of its lattice are the *witness cells*; measured in witness cells the bare spacing is $\eta := L^{-k_0}$. For a function f on the bare lattice, $\mathfrak{n}(f) := \eta^4 \sum_x f(x)$ is its mass in witness-cell units, $\text{Lip}_{k_0} f$ its Lipschitz constant in witness-cell units, and $\|f\|_{\sharp} = \max\{\|f\|_{\infty}, 40M \text{Lip}_{k_0} f\}$ is the norm of clause (iii) read at the witness level. Constants written C or c with an index depend

only on L , on Balaban's parameters, on N and on ϑ unless the contrary is said, never on K , g_0 , m , s or positions. We write $\mathbf{1}$ for the unit of $\text{SU}(2)$ and for indicator functions.

Definition 5.1 (test class). For $\mathbf{d} > 0$ and $\vartheta \in (0, 1]$, $\mathfrak{T}(\mathbf{d}; \vartheta)$ is the set of pairs (f_1, f_2) of functions on $\mathbb{T}_m$ such that, for $i = 1, 2$, $f_i \geq 0$ is Lipschitz, $f_i \not\equiv 0$, $\text{supp } f_i \subset B(x_i, \mathbf{d})$ for some $x_i \in \mathbb{T}_m$, $\int f_i \geq \vartheta \|f_i\|_\infty \mathbf{d}^4$ and $\text{Lip } f_i \leq \|f_i\|_\infty / (\vartheta \mathbf{d})$; and $\mathbf{d} \leq \text{dist}(\text{supp } f_1, \text{supp } f_2) \leq 6\mathbf{d}$.

The sup-normalised dilates of two fixed bumps lie in $\mathfrak{T}(\mathbf{d}; \vartheta)$ with one ϑ for all $\mathbf{d}$. The window $[\mathbf{d}, 6\mathbf{d}]$ restricts the test functions of the conclusion; it is not a condition on any parameter. We use the mass condition in its lattice form $\mathbf{n}(f_i) \geq \vartheta \rho^4 \|f_i\|_\infty$ (ρ as below), which differs from the continuum form by the Riemann-sum factor $1 + O(\eta/(\vartheta \rho))$ and follows from it once ϑ is replaced by a fixed fraction of itself; and $\mathbf{n}(f_i) \leq 10\rho^4 \|f_i\|_\infty$.

Definition 5.2 (room; depth of a scale). Let $C_{\text{room}} \geq 1$ be the constant fixed in Step 5, and put $C_W(s) := C_{\text{room}} p_0 (\bar{x}_s^{-1/2})^2 = C_{\text{room}} A_0^2 (\log \bar{x}_s)^{2\mathbf{p}_0}$. The *depth* of a scale $\mathbf{d} > 0$ is $s(\mathbf{d}) := \min\{s \geq 0 : C_W(s)L^{-s} \leq \mathbf{d}\}$. For $s = s(\mathbf{d}) \geq 1$ let $\rho := L^s \mathbf{d}$ be the length $\mathbf{d}$ in witness cells, so that $C_W(s) \leq \rho < LC_W(s)$. The *room count* is

$$r(s) := \lceil \log_L(20LC_W(s)) \rceil + \lceil (\tfrac{1}{2}\mathbf{p}_0 + 1) \log_L \log \bar{x}_s \rceil + r_0,$$

with r_0 fixed in Step 5; we also put $r_1 := \lceil \log_L(20L\rho) \rceil$.

The depth $s(\mathbf{d})$ does not depend on K and tends to infinity as $\mathbf{d} \downarrow 0$, because $C_W(s)$ grows only polylogarithmically in s . The supports of a pair in $\mathfrak{T}(\mathbf{d}; \vartheta)$ are at least $\rho \geq C_W(s)$ witness cells apart: this is the room used in Steps 2–4. The coupling g_{K-s} is the running coupling at the length $L^{-s} \in [\mathbf{d}/(LC_W(s)), \mathbf{d}/C_W(s)]$.

Definition 5.3 (kernel). Let $\mathcal{A}$ be the free lattice Maxwell field at unit coupling on the bare lattice of $\mathbb{T}_m$, with values in $\mathfrak{su}(2)$, modulo gauge transformations and harmonic forms: colour by colour, in an orthonormal basis for $\langle X, Y \rangle = -\text{tr } XY$, its curvature $d\mathcal{A}$ is the standard Gaussian on the exact 2-forms, with covariance the orthogonal projection P_{ex} onto them. Let Q_{k_0} be the linearisation at $\mathbf{1}$ of Balaban's k_0 -fold averaging map [1, (1.17)–(1.18)], $B := Q_{k_0}\mathcal{A}$, $A_B := \mathbb{E}[\mathcal{A} | B]$ (the linear minimiser of [1, (1.19)]), and $F_B := dA_B$, the curvature of the background field at the witness level. For bare plaquettes p, q let $C^{\text{bg}}(p, q) := \mathbb{E}[F_B^a(p)F_B^a(q)]$, for any colour a , and

$$\mathcal{K}_s(f_1, f_2) := \sum_{p, q} f_1(x(p)) f_2(x(q)) C^{\text{bg}}(p, q)^2, \quad (5.2)$$

the sum running over all bare plaquettes of all six orientations, $x(p)$ the barycentre of p .

The kernel depends on K , s and the torus; its bounds do not. Since $\mathcal{A} - A_B$ is independent of B , $P_{\text{ex}} = C^{\text{bg}} + C^{\text{fl}}$, with C^{fl} the curvature covariance of the fluctuations between the bare and the witness level; and by the law of total covariance along $\mathcal{G}_i := \sigma(Q_{k_0+i}\mathcal{A})$, C^{bg} is the sum of the contributions C_i^{step} of the coarser levels $k_0 + i$, $i \geq 0$, which the Gaussian extraction of Step 1 produces one level at a time.

Proposition 5.4 (kernel bounds). *There are $C_u = C_u(L)$, an integer $k_W = k_W(L) \geq 1$ and $C^{\text{ker}} = C^{\text{ker}}(L)$ such that, if $K - s \geq k_W$ and distances are measured in witness cells:*

$$(a) \quad |C^{\text{bg}}(p, q)| \leq C_u \eta^4 (1 + |x(p) - x(q)|)^{-4} \text{ for all bare plaquettes } p, q;$$

(b) for every $\mathbf{d}$ with $s = s(\mathbf{d})$ and $\rho \geq C^{\ker}$ and every $(f_1, f_2) \in \mathfrak{T}(\mathbf{d}; \vartheta)$,

$$c_1 \rho^{-8} \mathbf{n}(f_1) \mathbf{n}(f_2) \leq \mathcal{K}_s(f_1, f_2) \leq C_1 \rho^{-8} \mathbf{n}(f_1) \mathbf{n}(f_2), \quad c_1 := 2 \cdot 10^{-8} \pi^{-4}, \quad C_1 := 36 C_u^2,$$

hence $c_K \|f_1\|_\infty \|f_2\|_\infty \leq \mathcal{K}_s(f_1, f_2) \leq C_K \|f_1\|_\infty \|f_2\|_\infty$ with $c_K := c_1 \vartheta^2$, $C_K := 100 C_1$.

It concerns the free field only. The corresponding statements for the free, unconditioned kernel—the kernel (5.2) with P_{ex} in place of C^{bg} —are proved in the companion note on the free kernel: its two-sided bounds, by a comparison of the kernel, summed over orientations, with η^8 times the continuum germ of Remark 5.7 through the asymptotics $G(n) = (4\pi^2 |n|^2)^{-1} + O(|n|^{-4})$ of the Green's function of $\mathbb{Z}^4$, its comparison between tori, and its continuum limit. The passage to the block-conditioned covariance $C^{\text{bg}} = P_{\text{ex}} - C^{\text{fl}}$, that is, the bound (a) and the two-sided comparison (b) for $\mathcal{K}_s$, is proved there as an implication from Balaban's propagator decay [1, Proposition 1.2], [2, Propositions 2.5–2.6], used by statement, through the decay of C^{fl} beyond the witness scale; (a) uses also the regularity of the linear minimiser [6, Theorem 1, (9)–(10)]. The condition $K - s \geq k_W$ makes the bare lattice fine at the witness scale.

Lemma 5.5 (colour factor). *If $(\xi_a)_{a \leq n}$, $(\eta_a)_{a \leq n}$, $n = N^2 - 1$, are centred and jointly Gaussian with $\text{Cov}(\xi_a, \eta_b) = \delta_{ab} c$, then $\text{Cov}(\frac{1}{2} \sum_a \xi_a^2, \frac{1}{2} \sum_a \eta_a^2) = b_0 c^2$.*

Proof. By Isserlis' formula $\text{Cov}(\xi_a^2, \eta_b^2) = 2 \text{Cov}(\xi_a, \eta_b)^2$, so the covariance is $\frac{1}{4} \sum_{a,b} 2 \delta_{ab} c^2 = \frac{n}{2} c^2$. $\square$

Lemma 5.6 (no term of order one; no exchange of one line). *Let $N \geq 2$. (a) A C^1 function of $\mathfrak{su}(N)$ -valued 1-forms on a finite lattice that is invariant under constant gauge transformations has vanishing gradient at the zero form. (b) If Ψ is real, ξ is $\mathfrak{su}(N)$ -valued, $\mathbb{E}|\Psi\xi| < \infty$, and the law of (Ψ, ξ) is invariant under $(\Psi, \xi) \mapsto (\Psi, \text{Ad}_u \xi)$ for all $u \in \text{SU}(N)$, then $\mathbb{E}[\Psi\xi] = 0$.*

Proof. The adjoint representation has no non-zero fixed vector, since the centre of $\mathfrak{su}(N)$ is trivial. (a) The gradient at 0 is $A \mapsto \sum_b \langle c_b, A_b \rangle$ with each c_b fixed by Ad , hence 0. (b) $\text{Ad}_u \mathbb{E}[\Psi\xi] = \mathbb{E}[\Psi \text{Ad}_u \xi] = \mathbb{E}[\Psi\xi]$ for all u . $\square$

This is why the leading power is g_{K-s}^4 . Since $1 - \text{Re tr } e^Y = \frac{1}{2} |Y|^2 - \frac{1}{24} \text{tr } Y^4 + O(|Y|^6)$ for $Y \in \mathfrak{su}(N)$, $|Y|^2 = -\text{tr } Y^2$, the smeared action density of the background at the witness level is, at leading order, $\frac{1}{2} g_{k_0}^2 \sum_p f(x(p)) \sum_a F_B^a(p)^2$. Gauge invariance removes the term of order one in the field and the exchange of one covariance line between two invariant insertions (Lemma 5.6); it does not remove the exchange of two lines, whose value is $b_0 g_{k_0}^4 \sum_{p,q} f_1(x(p)) f_2(x(q)) C^{\text{bg}}(p, q)^2$ (Lemma 5.5). Proposition 5.15 below is the non-perturbative form of this computation.

Remark 5.7 (the continuum germ). For the free Maxwell field on $\mathbb{R}^4$ with covariance $\delta_{\mu\nu} (4\pi^2 |x|^2)^{-1}$,

$$\sum_{\mu < \nu, \rho < \sigma} \langle F_{\mu\nu}(x) F_{\rho\sigma}(0) \rangle^2 = \frac{6}{\pi^4 |x|^8}. \quad (5.3)$$

Indeed $\langle F_{\mu\nu}(x) F_{\rho\sigma}(0) \rangle = -(\delta_{\nu\sigma} H_{\mu\rho} - \delta_{\nu\rho} H_{\mu\sigma} - \delta_{\mu\sigma} H_{\nu\rho} + \delta_{\mu\rho} H_{\nu\sigma})$ with $H_{\mu\rho} := \partial_\mu \partial_\rho (4\pi^2 |x|^2)^{-1}$; by rotation invariance of the sum we may take $x = |x| e_3$, with e_3 the unit vector of the coordinate x^3 (Section 4), where $H = (2\pi^2 |x|^4)^{-1} \text{diag}(-1, -1, -1, 3)$ in the coordinates (x^0, x^1, x^2, x^3) , so only the six diagonal terms survive, with values $\pm \pi^{-2} |x|^{-4}$. The continuum counterpart of the main term of Theorem 5.8 is thus $3(N^2 - 1) \pi^{-4} \iint f_1(x) f_2(y) |x - y|^{-8} dx dy$, invariant under the sup-normalised dilations $f_i \mapsto f_i(\cdot/\lambda)$; so the main term carries no power of $\mathbf{d}$, and depends on $\mathbf{d}$ through $g_{K-s(\mathbf{d})}^4$. The convergence of b_0 times the free, unconditioned kernel to this value, as the bare lattice becomes fine on the scale of the supports and the torus large against it, and that of $b_0 \mathcal{K}_{s(\mathbf{d})}$ along the sup-normalised dilates of a fixed pair as $\mathbf{d} \downarrow 0$ and $K - s(\mathbf{d}) \rightarrow \infty$, are proved in the companion note on the free kernel, the second with Balaban's propagator decay used by statement; neither is used in this section, where only Proposition 5.4 enters.

5.2 The statement

Theorem 5.8 (the curvature two-point witness on a torus). *Let $N = 2$ and assume (H1)–(H7). For every $\vartheta \in (0, 1]$ there exist*

- a coupling $\gamma_*^\dagger \in (0, \gamma_J]$ depending only on $(L, c_{\text{age}}, \mathbf{p}, \gamma; \varrho)$, and a coupling

$$\gamma_W^\mathbb{T} = \gamma_W^\mathbb{T}(L, \mathbf{p}, \gamma, \vartheta; \varrho) \in (0, \gamma_*^\dagger);$$

- the non-decreasing function $C_W(s) = C_{\text{room}} A_0^2 (\log \bar{x}_s)^{2p_0} \geq 1$ of Definition 5.2, with C_{room} depending on ϑ ;
- a depth floor $s_W(g) \geq b_*(g)$, depending on g (and on L and Balaban's parameters) only, non-decreasing in g^{-2} , with $s_W(g) = O(\log_L \log g^{-2})$;
- an integer $k_W = k_W(L) \geq 1$, and constants $c_K, C_K > 0$ depending on ϑ ;
- an integer $m_V(g)$, depending only on g and on $(L, c_{\text{age}}, \mathbf{p}, \gamma; \varrho)$, with $m_V(g) \rightarrow \infty$ as $g \downarrow 0$;
- a function $s_{\text{vol}}(g, m) \geq 0$, finite for every (g, m) and non-decreasing in m , with $s_{\text{vol}}(g, m) = 0$ for $m \leq m_V(g)$, depending only on (g, m) and on the constants fixed before it;

none of which depends on $K, g_0, \mathbf{d}$ or positions, with the following property. Let $g \leq \gamma_W^\mathbb{T}$, $g_0 \in (0, g_-(g)]$, $K = K(g_0, g)$, let m be any torus exponent allowed by (H5), and let $\mathbf{d} > 0$ satisfy

$$\max\{s_W(g), s_{\text{vol}}(g, m)\} \leq s(\mathbf{d}) \quad \text{and} \quad K - s(\mathbf{d}) \geq k_W. \quad (5.4)$$

Put $s := s(\mathbf{d})$. Then for every $(f_1, f_2) \in \mathfrak{T}(\mathbf{d}; \vartheta)$

$$S_{2;K,m}^T(f_1, f_2) = (\zeta'_{K-s})^2 b_0 g_{K-s}^4 \mathcal{K}_s(f_1, f_2) (1 + R_K(f_1, f_2)), \quad |R_K(f_1, f_2)| \leq \frac{1}{4}, \quad (5.5)$$

where ζ'_{K-s} is the source factor of clause (iii)(c), $\mathcal{K}_s$ is the kernel (5.2), and

$$\begin{aligned} \zeta'_{K-s} &\in [\tfrac{2}{3}, \tfrac{4}{3}], \quad |\zeta'_{K-s} - 1| \leq C_\zeta g_{K-s} (\log g_{K-s}^{-2})^{p_0}, \\ c_K \|f_1\|_\infty \|f_2\|_\infty &\leq \mathcal{K}_s(f_1, f_2) \leq C_K \|f_1\|_\infty \|f_2\|_\infty, \end{aligned} \quad (5.6)$$

with C_ζ independent of K, g_0, m and s . In particular $S_{2;K,m}^T(f_1, f_2) > 0$,

$$\tfrac{2}{9} b_0 g_{K-s}^4 \mathcal{K}_s(f_1, f_2) \leq S_{2;K,m}^T(f_1, f_2) \leq \tfrac{8}{3} b_0 g_{K-s}^4 \mathcal{K}_s(f_1, f_2), \quad (5.7)$$

and, by (5.1),

$$\frac{\frac{2}{9} b_0 c_K \|f_1\|_\infty \|f_2\|_\infty}{(g^{-2} + B_\#(s+1))^2} \leq S_{2;K,m}^T(f_1, f_2) \leq \frac{\frac{8}{3} b_0 C_K \|f_1\|_\infty \|f_2\|_\infty}{(g^{-2} + \lambda_\# s - c_5)^2}. \quad (5.8)$$

These bounds are uniform in K, g_0 and the position of the pair. For $m_\mathbb{T}(g_-(g)) \leq m \leq m_V(g)$ the conditions (5.4) do not involve m , so the statement is uniform in m as well; for $m > m_V(g)$ the volume enters through $s_{\text{vol}}(g, m)$ and through nothing else.

Remark 5.9. (1) *Form of the conditions.* All conditions are one-sided: floors on the depth, a floor on the number $K - s$ of levels below the witness level, and a ceiling on the coupling chosen after ϑ . Nothing bounds L , K , m , a number of steps, an age or a depth from above, and no coupling is bounded from below; the only two-sided condition is the window of the test class. The three members of the floor $s_W(g)$ are explained in Step 5; we do not claim that s_W can be removed. The constants $\frac{2}{9}$ and $\frac{8}{3}$ in (5.7) follow from $|R_K| \leq \frac{1}{2}$; from $|R_K| \leq \frac{1}{4}$ one obtains $\frac{1}{3}$ and $\frac{20}{9}$.

(2) *The horizon.* $m_V(g)$ is not a hypothesis: the theorem covers every torus allowed by (H5), and $m_V(g)$, defined in (5.20), names the range on which s_{vol} vanishes. The condition $m \leq m_V(g)$ is sufficient, and free of K , for Balaban's run, continued on the same bare lattice beyond the level K , to use up the torus before its running coupling reaches the fixed value $\gamma_*^\dagger$; physically, these are the tori smaller, up to a fixed number of factors L , than the length at which the running coupling first reaches $\gamma_*^\dagger$. The horizon grows like $g^{-2}/B_\#$, while the floor of (H5) is $m_{\mathbb{T}}(g_-(g)) = O(\log_L \log g^{-2})$; so for every $m' \geq 0$ there is $\gamma_V(m') > 0$ such that $s_{\text{vol}}(g, m) = 0$ whenever $m \leq m_{\mathbb{T}}(g_-(g)) + m'$ and $g \leq \gamma_V(m')$. Step 4 explains why the volume enters beyond the horizon.

(3) *Not asserted.* Nothing about one source, overlapping or signed sources, separations beyond $6d$, shapes with $\vartheta \rightarrow 0$, or depths outside (5.4); no faster convergence $\zeta'_{K-s} \rightarrow 1$; no determination of b_0 beyond the fixed factor in (5.7); nothing about non-Gaussianity, $O(4)$, a mass gap, block observables, Wilson loops, or $N \geq 3$; not that s_{vol} is needed, nor that the admissible separations are bounded below uniformly in m (uniformity in the volume is obtained in Section 6 by another route). Since (5.8) does not involve K , it passes, for fixed d and fixed test functions, to every limit $K \rightarrow \infty$ on a fixed torus.

5.3 Step 1: exact insertion, total covariance, Gaussian extraction

In the proof, g, g_0, K, m, d and $(f_1, f_2) \in \mathfrak{T}(d; \vartheta)$ are as in Theorem 5.8, $s := s(d)$, $k_0 := K - s$, $\rho = L^s d$.

Exact insertion. For $z = (z_1, z_2) \in \mathbb{C}^2$ let $w(p) := z_1 f_1(x(p)) + z_2 f_2(x(p))$ and $\rho_0^w(U) := \exp[-\sum_p (x_0 - w(p))(1 - \text{Re tr } U(\partial p)) - E]$, $x_0 = g_0^{-2}$, E a constant: a curvature source is a plaquette-dependent shift of the bare inverse coupling, and $\int \rho_0^w dU = e^{-E} Z(z)$. Since $s \geq b_*(g)$, Theorem 3.1 applies with this s in its part (c): its run with sources stops at the witness level, $K' = k_0 = K - s$, and its norm $\|\cdot\|_\#$, its classes of sources and its polydiscs are those of the lattice of level k_0 , with the source radius ϱ . It gives the density of the block variable V of level k_0 after k_0 of Balaban's steps as

$$\rho_{k_0}^w = \sum_h \tau_h^w, \quad \sum_h \int \tau_h^w dV = \int \rho_0^w dU \quad (z \in \mathbb{D} := \{|z_1| \|f_1\|_\# + |z_2| \|f_2\|_\# < \varrho\}). \quad (5.9)$$

The index h runs over Balaban's large-field histories at level k_0 in the form produced by his large-field operation **R**: the components removed by **R** are summed out [11, (1.71)–(1.72)], their contribution being carried by whole-lattice terms of the effective action [9, (2.30)–(2.31)] and by the coupling flow, so that h lists the large-field components that are live at level k_0 , of any age. Each τ_h^w is analytic in $z \in \mathbb{D}$, its characteristic functions do not depend on z , and $\tau_h^0 \geq 0$, all expressions in Balaban's integrals being real and the exponential densities positive [9, (2.18)–(2.22)], [11, (1.73)–(1.75)]. The identity in (5.9) is the exact insertion of clause (i-d). For $(f_1, f_2) \in \mathfrak{T}$, $\text{Lip}_{k_0} f_i \leq \|f_i\|_\infty / (\vartheta \rho)$, so $\|f_i\|_\# = \|f_i\|_\infty$ once $\rho \geq 40M/\vartheta$.

Let $\mathcal{Z} := \sum_h \int \tau_h^0 dV$ and let $\tilde{\nu}$ be the probability on pairs (h, V) with density $\tau_h^0(V)/\mathcal{Z}$; its V -marginal ν^B is the law of Balaban's block variable of level k_0 under his effective density. We

never compare ν^B with the law of the block average of the bare field: only total integrals occur, and they are preserved exactly. On $\{\tau_h^0 > 0\}$ put $\Phi_i^h := \partial_{z_i} \log \tau_h^w|_{z=0}$ and $\Psi^h := \partial_{z_1} \partial_{z_2} \log \tau_h^w|_{z=0}$.

Lemma 5.10 (total covariance). $S_{2;K,m}^T(f_1, f_2) = \mathbb{E}_{\bar{\nu}}[\Psi^h] + \text{Cov}_{\bar{\nu}}(\Phi_1^h, \Phi_2^h)$.

Proof. By (5.9), $\log Z(z) = E + \log \sum_h \int \tau_h^w dV$ on $\mathbb{D}$; the sum is finite, the integrals are over a compact space and the integrands are analytic, so we may differentiate under them. Since the characteristic functions of h do not depend on z , on $\{\tau_h^0 > 0\}$ we have $\partial_{z_i} \tau_h^w = \tau_h^w \partial_{z_i} \log \tau_h^w$ and $\partial_{z_1} \partial_{z_2} \tau_h^w = \tau_h^w (\partial_{z_1} \partial_{z_2} \log \tau_h^w + \partial_{z_1} \log \tau_h^w \partial_{z_2} \log \tau_h^w)$. Hence at $z = 0$, $\partial_{z_1} \partial_{z_2} \log Z = \mathbb{E}_{\bar{\nu}}[\Psi^h + \Phi_1^h \Phi_2^h] - \mathbb{E}_{\bar{\nu}}[\Phi_1^h] \mathbb{E}_{\bar{\nu}}[\Phi_2^h]$. $\square$

Lemma 5.11 (translations). $S_{2;K,m}^T(f_1(\cdot - v), f_2(\cdot - v)) = S_{2;K,m}^T(f_1, f_2)$ for every translation v of the bare lattice.

Proof. The Wilson measure on $\mathbb{T}_m$ is invariant under these translations, and $\mathcal{O}(f(\cdot - v)) = \mathcal{O}(f) \circ T_v$ for the induced map T_v on configurations. $\square$

We write $\langle\langle \cdot \rangle\rangle$ for the average over the translations of the pair by $S_0 \mathbb{Z}^4$, $S_0 := L^2 M \check{R}(x_{k_0})$ witness cells; by Lemma 5.11 the two-point function equals its average, and all bounds below are uniform in the position.

Lemma 5.12 (source factor). Under clause (iii)(c), ζ'_k is real and $\zeta'_k \in [\frac{2}{3}, \frac{4}{3}]$ for every $k \leq K$.

Proof. $\phi(\zeta) := \zeta_k(\zeta) - \zeta$ is analytic on $|\zeta| < 2\rho$, real on reals, $\phi(0) = 0$ and $|\phi(\zeta)| \leq |\zeta|/3$; so $\phi(\zeta)/\zeta$ is analytic of modulus at most $\frac{1}{3}$, and $\zeta'_k - 1 = \phi'(0)$ is real with $|\phi'(0)| \leq \frac{1}{3}$. $\square$

Write $\mathfrak{d}_W(x) := C_\zeta x^{-1/2}(\log x)^{p_0}$, with C_ζ the constant of Corollary 3.3 (p_0 is the exponent written p_0 in Section 3). The modulus $|\zeta'_{k_0} - 1| \leq \mathfrak{d}_W(x_{k_0})$ in (5.6) is the last bound of that corollary at $k = k_0$; it is proved in the companion account, and its structure is as follows: $\zeta'_{k_0} = 1 + \sum_{j \leq k_0} b'_j(0)$, b_j the coupling renormalisations of the correlation theorem, and its flow bounds for $b'_j(0)$ are summed over $j \leq k_0$ only, where $x_j - x_{k_0} \geq \lambda_\#(k_0 - j) - c_5$ by clause (0). A ceiling on g gives $\mathfrak{d}_W \leq \frac{1}{3}$.

Live components and the slopes.

Definition 5.13 (live components). A *live component* of a history h at level k_0 is a connected component of its large-field region at level k_0 not removed by **R** [10, §1]. For a live component Z , $j(Z)$ is the smallest level at which a point of Z lay in a large-field region (the oldest layer of the region it has occupied), and $\text{age}(Z) := k_0 - j(Z)$. Let $S_i := \text{supp } f_i$, S_i^+ its $C_h M$ -neighbourhood (C_h a constant of Balaban's partitions of unity), N_i its $\rho/4$ -neighbourhood. Z is *near* S_i if the domains on which its operator and its attached terms depend come within $\rho/8$ witness cells of S_i^+ ; under the floors $C_{\text{room}} \geq 16C_h M$, $C_{\text{room}} \geq 3200M$ such a Z meets N_i . B_i is the set of histories with a live component near S_i . With $\mathfrak{l}(x) := L\check{R}(x) + c_L$, c_L a constant, $\mathfrak{l} := \mathfrak{l}(x_{k_0})$, let $A_i(h) := \max\{\mathfrak{l}, \text{age}(Z) : Z \text{ live, near } S_i\}$.

A large-field component contained in a cube of size $100MR$ in which no new large fields were created during R steps is removed by **R** [10, §1, (i)–(ii)], well within $\mathfrak{l}$ levels; but components born large, or in which new large fields keep appearing, stay live, and a merged component inherits the oldest layer of its parts. Their age enters the bounds below explicitly.

Proposition 5.14 (decomposition of the slopes). *There are constants $C_G, C_{\text{live}}, C_{\text{src}}, C_{\text{far}}, \kappa' > 0$ and a polynomial $\mathfrak{p}_{\text{src}}$ such that, with $\varepsilon := g_{k_0} p_0(g_{k_0})$, $\bar{\Theta} := L^{4(\mathfrak{l}(\bar{x}_s)+1)}$ and*

$$\mathfrak{s}_i^\# := \frac{10C_{\text{live}}\bar{\Theta}\mathfrak{p}_{\text{src}}(\log \bar{x}_s)\rho^8\|f_i\|_\infty}{\vartheta}, \quad \mathfrak{s}_i^{(a)} := \mathfrak{s}_i^\# L^{4(a-\mathfrak{l})} \quad (a \geq \mathfrak{l}), \quad \mathfrak{f}_i := C_{\text{src}}C_{\text{far}}\|f_i\|_\infty\rho^8 e^{-\kappa'\rho/16}, \quad (5.10)$$

the following holds $\tilde{\nu}$ -almost surely, for $i = 1, 2$.

- (a) $\Phi_i^h = c_i^0 + G_i(V) + \mathfrak{L}_i^h(V)$, c_i^0 a number depending neither on h nor on V , $G_i := \zeta'_{k_0} \mathcal{O}^{(k_0)}(\tilde{f}_i) + \text{Irr}_i$ a function of V only. Here $\mathcal{O}^{(k_0)}(f)(V) := \sum_p f(x(p))[1 - \text{Re tr } U_{k_0}(V)(\partial p)]$, $U_{k_0}(V)$ Balaban's background field [6, Theorem 1]; $\tilde{f}_i := f_i + \delta_i/\zeta'_{k_0}$ with δ_i Lipschitz, $\text{supp } \delta_i \subset S_i^+$, $|\delta_i| \leq \mathfrak{d}_W(x_{k_0})C_h M\|f_i\|_\infty/(\vartheta\rho)$; Irr_i is the z_i -derivative at 0 of the irrelevant terms of Balaban's effective action at level k_0 [7, (0.28)–(0.29)], minus its value at $V = \mathbf{1}$; and $\text{sup}_V |G_i| \leq C_G \mathfrak{n}(f_i)\varepsilon^2$.
- (b) $\mathfrak{L}_i^h = \mathfrak{L}_i^{\text{near},h} + \mathfrak{F}_i^h$ with $\text{sup}_V |\mathfrak{F}_i^h| \leq \mathfrak{f}_i$ for every h , $\mathfrak{L}_i^{\text{near},h} = 0$ for $h \notin B_i$, and $\text{sup}_V |\mathfrak{L}_i^{\text{near},h}| \leq P_h \mathfrak{s}_i^{(A_i(h))}$ for $h \in B_i$, where $P_h = 1$ if $A_i(h) = \mathfrak{l}$ and, for older components, $P_h = P(A_i(h), \varepsilon^{-1})$ with a fixed polynomial $P \geq 1$ whose coefficients depend only on L and Balaban's parameters.
- (c) $\Psi^h = \Psi^{\text{span},h} + \Psi^{\text{live},h}$ with $|\Psi^{\text{span},h}| \leq \mathfrak{t}^{\text{loc}}$ (Lemma 5.19) for every h , and $|\Psi^{\text{live},h}| \leq 2(\mathfrak{s}_1^{(A_1(h))} \mathbf{1}_{B_1}(h) + \mathfrak{f}_1)(\mathfrak{s}_2^{(A_2(h))} \mathbf{1}_{B_2}(h) + \mathfrak{f}_2)$.

This is proved in the companion account [T-W]; we give the statement and the structure of the proof. It rests in addition on an identification of the format of the operators of live components in the representation of Section 3 (all dependence on the source sits in the exponent of a positive integral; inside a live component the configuration is regular at the scale of its oldest layer, as in [9, (2.17)–(2.22)]), read off Balaban's displays (item (e)(ii) of §5.8). The factor P comes from the number of levels inside an old component and from the ratio of the small-field thresholds at its birth level and at the witness level. It is absorbed in Step 4 as follows: $P(a, \varepsilon^{-1})^2 \leq C_P L^{a+1} \varepsilon^{-2d_P}$ for constants C_P, d_P ; the power L^{a+1} is the surplus of (5.22) over (5.17), and $\varepsilon^{-2d_P} \leq C'_P e^{\frac{1}{2}c_{\text{lv}}p_0(g_{k_0})}$. So Theorem 5.20 holds with P inserted, with A_{lv} multiplied by a constant and with c_{lv} replaced by $\frac{1}{2}c_{\text{lv}}$ in (5.19); we write the proof for $P \equiv 1$, the general case differing only in these constants.

Structure. (i) The source enters the effective action at level k_0 as a position-dependent shift of the inverse coupling: $w(p)$ plus the coupling renormalisations of the levels $j \leq k_0$, evaluated at the local values of w and smeared with Balaban's partitions of unity. Its z_i -derivative is $\zeta'_{k_0} f_i + \delta_i$ plus terms in live regions (part of $\mathfrak{L}_i^h$); the smearing defect δ_i is bounded by $\text{Lip } f_i$ and the partial sums of the $b'_j(0)$. (ii) The whole-lattice irrelevant terms give Irr_i , independent of h ; their level sums are uniform in K [7, (0.30)]. (iii) The operator of the live components is a positive integral whose exponent carries all the dependence on the source; differentiating its logarithm gives an expectation (one source) and a covariance (two sources) under the probability defined by that joint integral, bounded by suprema through positivity alone, with no lower bound on the integral and no factorisation over components. (iv) Inside a live component Z the configuration is regular only at the scale of its oldest layer, where each cell carries action at most $C\varepsilon^2$ [6, Theorem 1]; a witness cell contains $L^{4(\text{age}(Z)+1)}$ such cells, whence $\mathfrak{s}_i^{(a)}$, and $\bar{\Theta}$ for ages at most $\mathfrak{l}$. Components not near S_i reach S_i only through terms with long localisation domains; counted by their anchors in S_i^+ they give at most $\mathfrak{f}_i$, whatever their number. (v) Ψ^{span} collects the terms that depend on both sources (Lemma 5.19).

The Gaussian extraction. The segment $[k_0, k_0 + r]$ of the history decomposition carries an observable F of the block variable of level k_0 to $F^{(h)}(W)$, its conditional expectation given the segment of h and the block variable W of level $k_0 + r$. The observables are those of the class of [T-GX]: real, local, gauge-invariant, analytic and bounded on a fixed complex neighbourhood of the small-field configurations; it contains $\mathcal{O}^{(k_0)}(f)$. A history is *clean* near a support if no live large-field region meets the boxes of radius $D_\square$ (in cells of the respective level) around the images of the support at the levels $k_0, \dots, k_0 + r$, and the terms attached to older large-field regions under these boxes are small in sum. We use the box radius

$$D_\square(s) := \max\{\lceil c_{\text{gx}}^{-1}(6\mathfrak{p}_0 \log \log \bar{x}_s + 8 \log(LC_W(s)) + \log C_D) \rceil, D_0(s)\},$$

where c_{gx} is the constant of Proposition 5.15 (which holds for every box radius $D \geq D_0(s)$, with constants independent of D), $C_D \geq 1$ is fixed in Step 5, and $D_0(s)$, affine in $r(s)$, is the lower bound on the box radius required in [T-GX]; $D_\square$ is free of K and polylogarithmic in $\bar{x}_s$. At each level of the segment the carried observables are built with Balaban's domain-local minimisers [9, (2.13)] on the half-box—the union of the cubes of side M that meet the neighbourhood, of half the box radius, of the image of the support—whose regularity constants depend on the dimension and on L only, whatever the domain [6, Theorem 1]; the localisation is applied across the margin, of at least a quarter of the box radius, between the half-box and the boundary of the box.

Proposition 5.15 (Gaussian extraction). *There are $C_{\text{gx}}, c_{\text{gx}} > 0$, $\sigma_1 \in (0, 1]$ and a depth $s_* \geq 0$ (depending only on L and on the parameters fixed before it in (H3)) such that the following holds for every K and every s with $C_W(s) \leq \rho < LC_W(s)$, $r := r(s)$, $s - r \geq s_*$ and $2B_\sharp(r+1) \leq x_{k_0}$, provided the one-cell bound (5.22) holds at the levels $j \leq k_0 + r$. With $D_\square = D_\square(s)$ and $\Pi_r := r D_\square^4 L^4 p_0(g_{k_0})^2$,*

$$\langle\langle \text{Cov}_{\nu^B}(\mathcal{O}^{(k_0)}(\tilde{f}_1^v), \mathcal{O}^{(k_0)}(\tilde{f}_2^v)) \rangle\rangle = b_0 g_{k_0}^4 \mathcal{K}_s(\tilde{f}_1, \tilde{f}_2) + \mathcal{E}_a, \quad f^v := f(\cdot - v), \quad (5.11)$$

$$|\mathcal{E}_a| \leq C_{\text{gx}} g_{k_0}^4 \mathfrak{n}(\tilde{f}_1) \mathfrak{n}(\tilde{f}_2) \left\{ g_{k_0}^{\sigma_1} \Pi_r + g_{k_0}^2 (1+r) + D_\square^4 (1+r) \rho^4 p_0(g_{k_0})^4 e^{-c_{\text{gx}} p_0(g_{k_0})} \right. \\ \left. + p_0(g_{k_0})^4 e^{-c_{\text{gx}} D_\square} + \rho^{-8} [L^{-4(r-r_1)} + L^{-8(r-r_1)} p_0(g_{k_0+r})^4] \right\}. \quad (5.12)$$

Here $\tilde{f}_i$ are the profiles of Proposition 5.14; under the floor $C_{\text{room}} \geq 160 C_h M / \vartheta^2$, $\frac{1}{2} \mathfrak{n}(f_i) \leq \mathfrak{n}(\tilde{f}_i) \leq 2 \mathfrak{n}(f_i)$ and the supports of $\tilde{f}_1, \tilde{f}_2$ are at least $\rho/2$ apart. Proposition 5.15 is proved in the companion account [T-GX]; we give the statement and the structure of the proof. Its inputs are Balaban's one-step formula, inductive representation, minimiser regularity, propagator bounds and bounds on the terms left by removed regions, the flow of clause (0), and the reading listed in §5.8, item (e)(i).

Structure. (1) *Segment.* Restricted to $[k_0, k_0 + r]$, the history decomposition is a composition of multiplications by members of partitions of unity, measure-preserving changes of variables, integrations, and multiplications by positive constants [9, (2.18)–(2.22)]; so $F \mapsto F^{(h)}$ is positive and normalised, and the covariance under ν^B splits, by total covariance along the segment, into a part within the segment and the covariance at depth $s - r$ of the carried one-point functions. (2) *Clean histories.* On them the steps near the supports are Balaban's small-field steps at the couplings g_{k_0+i} [9, (2.24)]. The other histories have weight at most $C D_\square^4 (1+r) \rho^4 e^{-c p_0(g_{k_0})}$ by (5.22), and on them the integrands are bounded: $0 \leq \mathcal{O}^{(k_0)}(f_i)(V) \leq C \mathfrak{n}(f_i) / \vartheta$ wherever $U_{k_0}(V)$ is defined near S_i [6, Theorem 1]. (3) *One step with an inserted observable.* Balaban's computation of a small-field step [7, (2.1)–(2.13)] stays an identity with an observable inserted, each operation being a change of variables or a rewriting free of the next block variable; so the conditional expectation is a ratio

of integrals against the Gaussian with covariance $C^{(k)} = (\mathbf{C}^* \Delta^{(k)} \mathbf{C})^{-1}$, restricted to his window and perturbed by a function bounded by a second-order polynomial in the fluctuation field with coefficients $O(g_k)$ [7, §§2–3]. (4) *Box localisation*. On a clean history the step law near the support is uniformly log-concave on a product of convex windows, one per bond ([7, (2.9)], [9, (3.16)], and the lower bound on $\mathbf{C}^* \Delta^{(k)} \mathbf{C}$ in a background field [5, (3.157)–(3.158)], stated there with its proof indicated); with Combes–Thomas weights this gives the box-marginal Gaussian, the box window and the box part of the perturbation, up to $e^{-cD_{\square}L}$, uniformly in the exterior. (5) *Gaussian evaluation*. A finite-dimensional Laplace estimate compares the covariances with Gaussian ones, with an error linear in the size of the perturbation on the window and in the square root of the Gaussian mass outside it, times fourth moments. The Gaussian covariance of two invariant observables has no single-line term at flat background (Lemma 5.6), and its two-line term is $2n \operatorname{tr}(H^{F_1} c H^{F_2} c)$, H^F the invariant Hessian and c the step covariance; for smeared action densities ($H = \frac{1}{2}$ per plaquette) this is $b_0 \sum f_1 f_2 c^2$ (Lemma 5.5). (6) *The r steps*. At flat background the step $k_0 + i$ produces C_i^{step} , and the terms between levels combine into $b_0 \sum f_1 f_2 (\sum_{i < r} C_i^{\text{step}})^2$. The levels $i \geq r$ contribute at most $\rho^{-8} L^{-4(r-r_1)}$ relative to the main term, the kernel between supports ρ cells apart being carried by the levels up to r_1 . On a clean history the block variable of level $k_0 + r$ lies in Balaban’s small-field window near the supports [9, §1], and the response of the background at the witness scale is damped by L^{-2r} [6, Theorem 1], which bounds the covariance of the carried one-point functions by $\rho^{-8} L^{-8(r-r_1)} p_0(g_{k_0+r})^4$ relative to the main term. The variation of the coupling over the segment gives $g_{k_0}^2(1+r)$.

5.4 Step 2: the main term

By Lemma 5.10 and Proposition 5.14, since c_1^0, c_2^0 are constants and G_1, G_2 depend on V only, $\operatorname{Cov}_{\bar{\nu}}(\Phi_1^h, \Phi_2^h) = \operatorname{Cov}_{\nu^B}(G_1, G_2) + R^{\mathcal{L}}$ with $R^{\mathcal{L}} := \operatorname{Cov}_{\bar{\nu}}(\mathfrak{L}_1, G_2) + \operatorname{Cov}_{\bar{\nu}}(G_1, \mathfrak{L}_2) + \operatorname{Cov}_{\bar{\nu}}(\mathfrak{L}_1, \mathfrak{L}_2)$. Expanding $G_i = \zeta'_{k_0} \mathcal{O}^{(k_0)}(f_i) + \operatorname{Irr}_i$ and averaging over translations, we obtain the exact decomposition

$$S_{2;K,m}^T(f_1, f_2) = \mathfrak{M} + R^{\text{prof}} + R^{\text{gx}} + R^{\text{irr}} + R^{\text{loc}} + R^{\text{lf}}, \quad (5.13)$$

where $\mathfrak{M} := (\zeta'_{k_0})^2 b_0 g_{k_0}^4 \mathcal{K}_s(f_1, f_2)$ is the main term of (5.5), $R^{\text{prof}} := (\zeta'_{k_0})^2 b_0 g_{k_0}^4 [\mathcal{K}_s(\tilde{f}_1, \tilde{f}_2) - \mathcal{K}_s(f_1, f_2)]$, $R^{\text{gx}} := (\zeta'_{k_0})^2 \mathcal{E}_a$, R^{irr} is the average of the three covariances under ν^B that contain Irr_1 or Irr_2 , $R^{\text{loc}} := \langle \mathbb{E}_{\bar{\nu}} \Psi^{\text{span}} \rangle$ and $R^{\text{lf}} := \langle \mathcal{R}^{\mathcal{L}} + \mathbb{E}_{\bar{\nu}} \Psi^{\text{live}} \rangle$. By Lemma 5.12 and Proposition 5.4,

$$\mathfrak{M} \geq c_M \mathfrak{S} \geq c_M \vartheta^2 g_{k_0}^4 \|f_1\|_{\infty} \|f_2\|_{\infty}, \quad \mathfrak{S} := g_{k_0}^4 \rho^{-8} \mathfrak{n}(f_1) \mathfrak{n}(f_2), \quad c_M := \frac{4}{9} b_0 c_1. \quad (5.14)$$

We show that each of the five remainders is at most $\mathfrak{M}/32$.

Lemma 5.16 (profile). *There is C_{prof} such that $|R^{\text{prof}}| \leq \mathfrak{M}/32$ once*

$$\rho \geq C^{\text{prof}} := 32 C_{\text{prof}} (C_1/c_1) C_h M / \vartheta^2.$$

Proof. With $\hat{\delta}_i := \delta_i / \zeta'_{k_0}$, $|\hat{\delta}_i| \leq \frac{3}{2} \mathfrak{d}_W C_h M \|f_i\|_{\infty} / (\vartheta \rho)$, $\mathfrak{d}_W \leq \frac{1}{3}$, $\operatorname{supp} \hat{\delta}_i \subset S_i^+$, and bilinearity gives $\mathcal{K}_s(\tilde{f}_1, \tilde{f}_2) - \mathcal{K}_s(f_1, f_2) = \mathcal{K}_s(\hat{\delta}_1, f_2) + \mathcal{K}_s(f_1, \hat{\delta}_2) + \mathcal{K}_s(\hat{\delta}_1, \hat{\delta}_2)$. The supports involved are at least $\rho/2$ apart, so Proposition 5.4(a), summed over orientations, gives $|\mathcal{K}_s(\hat{\delta}_1, f_2)| \leq 2^8 C_1 \rho^{-8} \mathfrak{n}(|\hat{\delta}_1|) \mathfrak{n}(f_2)$, and similarly for the others. As S_i^+ has at most $80\rho^4$ witness cells, $\mathfrak{n}(|\hat{\delta}_i|) \leq 160\rho^4 \mathfrak{d}_W C_h M \|f_i\|_{\infty} / (\vartheta \rho)$, while $\mathcal{K}_s(f_1, f_2) \geq c_1 \rho^{-8} \mathfrak{n}(f_1) \mathfrak{n}(f_2)$ and $\mathfrak{n}(f_i) \geq \vartheta \rho^4 \|f_i\|_{\infty}$. Hence $|R^{\text{prof}}| / \mathfrak{M} \leq C_{\text{prof}} (C_1/c_1) C_h M / (\vartheta^2 \rho) \leq \frac{1}{32}$. $\square$

The errors of the extraction. Since $\mathfrak{n}(\tilde{f}_i) \leq 2\mathfrak{n}(f_i)$, $(\zeta'_{k_0})^2 \leq \frac{16}{9}$ and $\mathfrak{M} \geq c_M \mathfrak{S}$, (5.12) gives, with $p_0 = p_0(g_{k_0})$ and $r = r(s)$,

$$\begin{aligned} \frac{|R^{\text{gx}}|}{\mathfrak{M}} &\leq \frac{64C_{\text{gx}}}{9c_M} \left\{ \rho^8 [g_{k_0}^{\sigma_1} \Pi_r + g_{k_0}^2 (1+r) + D_{\square}^4 (1+r) \rho^4 p_0^4 e^{-c_{\text{gx}} p_0} + p_0^4 e^{-c_{\text{gx}} D_{\square}}] \right. \\ &\quad \left. + L^{-4(r-r_1)} + L^{-8(r-r_1)} p_0 (g_{k_0+r})^4 \right\}. \end{aligned} \quad (5.15)$$

Since $\rho < LC_W(s)$, $r - r_1 \geq \lceil (\frac{1}{2}\mathfrak{p}_0 + 1) \log_L \log \bar{x}_s \rceil + r_0 - 1$; with $x_{k_0+r} \leq \bar{x}_s$ this gives $L^{-4(r-r_1)} \leq L^{4-4r_0}$ and $L^{-8(r-r_1)} p_0 (g_{k_0+r})^4 \leq L^{8-8r_0} A_0^4$. The choice of $D_{\square}$ gives $\rho^8 p_0^4 e^{-c_{\text{gx}} D_{\square}} \leq A_0^4 C_D^{-1}$. Each of the three remaining terms is a polynomial in $\log \bar{x}_s$ times $\underline{x}_s^{-\sigma_1/2}$, $\underline{x}_s^{-1}$ or $e^{-c_{\text{gx}} A_0 (\log \underline{x}_s)^{\mathfrak{p}_0}}$; as $\bar{x}_s \leq \bar{x}(\underline{x}_s)$ with $\bar{x}(y) := (B_{\#}/\lambda_{\#})(y + c_5) + B_{\#}$, each is at most $\sup_{y \geq g^{-2-c_5}} \Phi(y)$ for some Φ decreasing to 0. Hence, for r_0 and C_D large and g below a ceiling chosen last, each of the six terms is at most $1/(6 \cdot 32)$, and $|R^{\text{gx}}| \leq \mathfrak{M}/32$.

5.5 Step 3: the room

Proposition 5.17 (room estimate). *There is C_b such that, under the conditions of Proposition 5.15, for $G_1 = \text{Irr}_1$ and $G_2 \in \{\zeta'_{k_0} \mathcal{O}^{(k_0)}(\tilde{f}_2), \text{Irr}_2\}$, and symmetrically,*

$$|\langle \text{Cov}_{\nu^B}(G_1^v, G_2^v) \rangle| \leq C_b \rho^{-2} p_0 (g_{k_0})^4 b_0 g_{k_0}^4 \mathcal{K}_s(f_1, f_2) + \mathcal{E}'_a,$$

where G^v is G computed for the translated pair and $\mathcal{E}'_a$ obeys the bound (5.12).

This is proved in the companion account [T-GX] as an implication from the correlation theorem of Section 3 and its inputs, and so has the same status as that theorem; we give the statement and the mechanism. The non-marginal terms of Balaban's effective action are separated from the marginal ones by Ward identities: their part of scaling dimension four is the coupling renormalisation b_j , absorbed into the running coupling and into ζ'_{k_0} , and the rest obeys the irrelevance bound $O(1)(L^j \eta)^{4+\alpha} e^{-\kappa d_j(X)}$, $\alpha > 0$ [7, (0.28)–(0.30)]. In the Gaussian approximation the gain is proved in [T-W]: a quadratic form in the curvature localised within ℓ witness cells of a point, with vanishing moment against F^2 , has covariance with $\sum_q f_2(x(q)) F(q)^2$ at distance $D \geq 4\ell$ at most $C \mathfrak{n}(f_1) \|f_2\|_1 \ell D^{-9}$, one factor $\ell/D \leq C/C_W(s)$ below the main term. This cancellation is carried from the Gaussian approximation to ν^B by the cutoff-uniform smallness of the second-order response of the extracted irrelevant source families against the main Gaussian term, in the averaged form that Theorem 5.8 uses. It is proved in [T-GX] by a direct argument: the source-dependent part of every term of the effective action is quadratically small in the field radius times a negative power of $\log g^{-2}$; on every real regular background near the supports the response of the irrelevant families is then small outright; and a covariance is at most twice the product of the suprema, under any measure, for every translate, every volume and every cutoff. No regularity of the kernel, no enlargement of the radius of the sources, no restriction to clean histories, no rarity of large fields and no localisation to boxes enter. The one price is the lower bound $q \geq 21\mathfrak{p}_0 + 1$ of (1.9) on the integer power q of the logarithm in Balaban's analyticity radii [9, (2.28)], which his text leaves free ("integers greater than 1"); it raises one existing lower bound on the block size and lowers the coupling ceiling, and it bounds nothing from above.

Lemma 5.18 (irrelevant part). *If $C_{\text{room}}^2 \geq 448 C_b$, then $|R^{\text{irr}}| \leq \mathfrak{M}/32$.*

Proof. As $b_0 g_{k_0}^4 \mathcal{K}_s(f_1, f_2) = \mathfrak{M}/(\zeta'_{k_0})^2 \leq \frac{9}{4} \mathfrak{M}$, the three covariances contribute at most

$$\begin{aligned} &3 \cdot \frac{9}{4} C_b \rho^{-2} p_0 (g_{k_0})^4 \mathfrak{M} \\ &\leq 7 C_b \rho^{-2} p_0 (g_{k_0})^4 \mathfrak{M}. \end{aligned}$$

Since $\rho \geq C_W(s) = C_{\text{room}} p_0(\bar{x}_s^{-1/2})^2$ and $p_0(g_{k_0}) \leq p_0(\bar{x}_s^{-1/2})$ ($x_{k_0} \leq \bar{x}_s$, and $p_0(g')$ increases with g'^{-2}), this is at most $7C_b C_{\text{room}}^{-2} \mathfrak{M} \leq \mathfrak{M}/64$. The three terms $\mathcal{E}_a'$ are at most $\mathfrak{M}/64$ by Step 2, after adjusting r_0 , C_D and the ceiling there by fixed factors. $\square$

Lemma 5.19 (bi-sourced local terms). *There are C_{loc} and $\kappa'' \geq \kappa/2$, κ Balaban's localisation exponent, such that for $\rho \geq 24M$ and every h*

$$|\Psi^{\text{span},h}| \leq \mathfrak{t}^{\text{loc}} := C_{\text{loc}} \varrho^{-2} \|f_1\|_{\infty} \|f_2\|_{\infty} (\rho/M)^4 L^4 e^{-\kappa'' \rho/(32M)}.$$

Consequently $|R^{\text{loc}}| \leq \mathfrak{M}/32$ once

$$\begin{aligned} \rho \geq (32M/\kappa'') \Big[& 2 \log \bar{x}_s + 4 \log(\rho/M) \\ & + 4 \log L + \log(32C_{\text{loc}}/(\vartheta^2 \varrho^2 c_M)) \Big], \end{aligned}$$

which holds for all s once $C_{\text{room}} \geq C^{\text{loc}}$.

The bound on Ψ^{span} is proved in the companion account [T-W] from the bounds of the correlation theorem of Section 3; we give the statement and the structure of the proof. (i) The second derivative of the shifted inverse coupling is a combination of products $f_1(y)f_2(y)$ at single points, which vanish since the supports are disjoint. (ii) A local term T of level $j \leq k_0$ depends on both sources only if its localisation domain X meets both supports; then its tree length is $d_j(X) \geq [\rho L^{\nu}/(4M) - 3]_+$, $\nu := k_0 - j$; Cauchy's inequality on $\mathbb{D}$ gives $|\partial_{z_1} \partial_{z_2} T|_{z=0} \leq 4 \sup |T| \|f_1\|_{\infty} \|f_2\|_{\infty} / \varrho^2$; and the correlation theorem gives $\sup |T| \leq C e^{-\kappa'' d_j(X)}$ for regular terms, terms near large-field regions and terms left by removed regions alike (compare [9, (2.27)–(2.31)]). (iii) The domains are summed with the tree bound [8, (1.26)] from at most $(4L\rho L^{\nu}/M + 3)^4$ starting cells; the sum over ν is dominated by $\nu = 0$. The consequence follows from (5.14) and $x_{k_0} \leq \bar{x}_s$, and the last assertion because $C_W(s) = C_{\text{room}} A_0^2 (\log \bar{x}_s)^{2p_0}$ grows faster than the right-hand side. This estimate is what forces the separation in witness cells to grow with $\log \bar{x}_s$: the main term is of size $x_{k_0}^{-2}$, while the terms that see both sources are only exponentially small in the separation.

5.6 Step 4: live large-field regions, and the horizon

By Proposition 5.14, live components near the supports enter only through $\mathfrak{L}_i^{\text{near}}$, $\mathfrak{f}_i$ and Ψ^{live} , and their age only through the factor $L^{4(a-l)}$ in $\mathfrak{s}_i^{(a)}$. We bound R^{lf} by masses of events concerning one cell (Theorem 5.20, proved here), and then bound these masses on a torus (Proposition 5.21); this is where the volume enters.

For a level $j \leq K$, a cell $\square$ of level j and $a \geq 0$, let $\text{LV}_j(\square)$ be the event that $\square$ lies in a live component at level j , and $\text{LV}_j^{\bar{a}}(\square)$ the event that it lies in a live component Z at level j with $j - j(Z) = a$; $\tilde{\nu}$ is the history probability of the run stopped at level j (by the conservation of total integrals, the mass of an event concerning levels at most j does not depend on where the run is stopped [T-LF]). Let $q_j^{\bar{a}} := \max_{\square} \tilde{\nu}(\text{LV}_j^{\bar{a}}(\square))$, let $n_i \leq (8\rho + 2)^4$ be the number of cells of level k_0 meeting N_i , and consider

$$(\text{LP}_1) \quad \tilde{\nu}(\text{LV}_{k_0}(\square)) \leq q_{k_0}^{\text{lv}} := C_{\text{lv}} e^{-c_{\text{lv}} p_0(g_{k_0})} \quad \text{for every cell } \square \text{ of level } k_0, \quad (5.16)$$

$$(\text{LP}_1^{\text{age}}) \quad \sum_{a \geq l} L^{8(a+1)} q_{k_0}^{\bar{a}} \leq A_{\text{lv}} L^{8(l+1)} q_{k_0}^{\text{lv}}; \quad \text{put } \mathfrak{q}^{\text{add}} := 2(n_1 + n_2)(1 + A_{\text{lv}}) q_{k_0}^{\text{lv}}. \quad (5.17)$$

Theorem 5.20 (additive bound on the live remainder). *Assume Proposition 5.14, (5.16) and (5.17). Then $\mathbb{E}_{\tilde{\nu}}|\mathfrak{L}_i^{\text{near}}| \leq \mathfrak{s}_i^{\sharp} \mathfrak{q}^{\text{add}}$, $\mathbb{E}_{\tilde{\nu}}|\mathfrak{L}_i| \leq \mathfrak{s}_i^{\sharp} \mathfrak{q}^{\text{add}} + \mathfrak{f}_i$, $\mathbb{E}_{\tilde{\nu}}[\mathfrak{s}_1^{(A_1)} \mathfrak{s}_2^{(A_2)} \mathbf{1}_{B_1 \cap B_2}] \leq \mathfrak{s}_1^{\sharp} \mathfrak{s}_2^{\sharp} \mathfrak{q}^{\text{add}}$, and*

$$|R^{\text{lf}}| \leq 2C_G \varepsilon^2 [\mathfrak{n}(f_2)(\mathfrak{s}_1^\# \mathfrak{q}^{\text{add}} + \mathfrak{f}_1) + \mathfrak{n}(f_1)(\mathfrak{s}_2^\# \mathfrak{q}^{\text{add}} + \mathfrak{f}_2)] \\ + \mathfrak{s}_1^\# \mathfrak{s}_2^\# (3\mathfrak{q}^{\text{add}} + (\mathfrak{q}^{\text{add}})^2) + 4(\mathfrak{s}_1^\# \mathfrak{f}_2 + \mathfrak{f}_1 \mathfrak{s}_2^\#) \mathfrak{q}^{\text{add}} + 4\mathfrak{f}_1 \mathfrak{f}_2, \quad (5.18)$$

Moreover $|R^{\text{lf}}| \leq \mathfrak{M}/32$ under the ceiling (5.19) on g and a floor $C_{\text{room}} \geq C^{\text{src}}$.

Proof. The only integration inequality used is $\mathbb{E}_{\tilde{\nu}}|X| \leq \sum_h \tilde{\nu}(h) \sup_V |X^h|$, valid since $\tau_h^0 \geq 0$; no ratio of a configuration with a live component to one without it is ever formed. *Union bounds.* A live component near S_i contains a cell of level k_0 meeting N_i ; hence $\tilde{\nu}(B_i) \leq n_i q_{k_0}^{\text{lv}}$, $\tilde{\nu}(A_i = a) \leq n_i q_{k_0}^{\text{lv}}$ and $\tilde{\nu}(A = a) \leq (n_1 + n_2) q_{k_0}^{\text{lv}}$ for $a > \mathfrak{l}$, $A := \max\{A_1, A_2\}$, and $\tilde{\nu}(B_1 \cap B_2) \leq \min\{n_1, n_2\} q_{k_0}^{\text{lv}}$ (an intersection bounded by one of its members: no joint presence of two components is used). For $a > \mathfrak{l}$, $L^{4(a-\mathfrak{l})} \leq L^{8(a+1)} L^{-8\mathfrak{l}-12}$ and $L^{8(a-\mathfrak{l})} = L^{8(a+1)} L^{-8\mathfrak{l}-8}$, so (5.17) gives $\sum_{a>\mathfrak{l}} L^{4(a-\mathfrak{l})} q_{k_0}^{\text{lv}} \leq A_{\text{lv}} q_{k_0}^{\text{lv}}$ and $\sum_{a>\mathfrak{l}} L^{8(a-\mathfrak{l})} q_{k_0}^{\text{lv}} \leq A_{\text{lv}} q_{k_0}^{\text{lv}}$. *First moments.* By Proposition 5.14(b), $|\mathfrak{L}_i^{\text{near},h}| \leq \mathfrak{s}_i^\# [\mathbf{1}_{B_i \cap \{A_i=\mathfrak{l}\}} + \sum_{a>\mathfrak{l}} L^{4(a-\mathfrak{l})} \mathbf{1}_{\{A_i=a\}}]$; integrating gives $\mathbb{E}|\mathfrak{L}_i^{\text{near}}| \leq \mathfrak{s}_i^\# n_i (1 + A_{\text{lv}}) q_{k_0}^{\text{lv}} \leq \mathfrak{s}_i^\# \mathfrak{q}^{\text{add}}$. *Second moments.* $\mathfrak{s}_1^{(A_1)} \mathfrak{s}_2^{(A_2)} \leq \mathfrak{s}_1^\# \mathfrak{s}_2^\# L^{8(A-\mathfrak{l})}$, so the expectation over $B_1 \cap B_2$ is at most $\mathfrak{s}_1^\# \mathfrak{s}_2^\# [\min\{n_1, n_2\} + (n_1 + n_2) A_{\text{lv}}] q_{k_0}^{\text{lv}} \leq \mathfrak{s}_1^\# \mathfrak{s}_2^\# \mathfrak{q}^{\text{add}}$, and the same bounds $\mathbb{E}|\mathfrak{L}_1^{\text{near}} \mathfrak{L}_2^{\text{near}}|$. *The members.* With $|\text{Cov}(X, Y)| \leq 2 \sup |Y| \mathbb{E}|X|$ and $|\text{Cov}(X, Y)| \leq \mathbb{E}|XY| + \mathbb{E}|X| \mathbb{E}|Y|$: $|\text{Cov}(\mathfrak{L}_1, G_2)| \leq 2C_G \mathfrak{n}(f_2) \varepsilon^2 (\mathfrak{s}_1^\# \mathfrak{q}^{\text{add}} + \mathfrak{f}_1)$ and symmetrically; $|\text{Cov}(\mathfrak{L}_1, \mathfrak{L}_2)| \leq \mathfrak{s}_1^\# \mathfrak{s}_2^\# (\mathfrak{q}^{\text{add}} + (\mathfrak{q}^{\text{add}})^2) + 2(\mathfrak{s}_1^\# \mathfrak{f}_2 + \mathfrak{f}_1 \mathfrak{s}_2^\#) \mathfrak{q}^{\text{add}} + 2\mathfrak{f}_1 \mathfrak{f}_2$; and, by Proposition 5.14(c), $\mathbb{E}|\Psi^{\text{live}}| \leq 2[\mathfrak{s}_1^\# \mathfrak{s}_2^\# \mathfrak{q}^{\text{add}} + (\mathfrak{s}_1^\# \mathfrak{f}_2 + \mathfrak{f}_1 \mathfrak{s}_2^\#) \mathfrak{q}^{\text{add}} + \mathfrak{f}_1 \mathfrak{f}_2]$. Summing the two cross-source covariance bounds without taking maxima gives (5.18). All its terms are separately homogeneous of degree one in f_1 and f_2 ; since $\mathfrak{M}$ has the same homogeneity, the following relative estimates do not require a bound on their amplitude ratio. *Relative size.* By (5.10), $\mathfrak{n}(f_i) \leq 10\rho^4 \|f_i\|_\infty$, $\varepsilon^2 = g_{k_0}^2 p_0(g_{k_0})^2$, $\mathfrak{q}^{\text{add}} \leq 4 \cdot 10^4 (1 + A_{\text{lv}}) C_{\text{lv}} \rho^4 e^{-c_{\text{lv}} p_0(g_{k_0})}$ and (5.14), each term of (5.18) without $\mathfrak{f}$ is at most $C_{\text{tot}} \bar{\Theta}^2 \mathfrak{p}_{\text{tot}} (\log \bar{x}_s) \bar{x}_s^2 e^{-c_{\text{lv}} p_0(g_{k_0})} \mathfrak{M}/\vartheta^4$, $\mathfrak{p}_{\text{tot}}$ a polynomial (e.g. $\mathfrak{s}_1^\# \mathfrak{s}_2^\# \mathfrak{q}^{\text{add}} \leq C \bar{\Theta}^2 \mathfrak{p}_{\text{src}}^2 \rho^{20} \bar{x}_s^2 e^{-c_{\text{lv}} p_0(g_{k_0})} \mathfrak{M}/\vartheta^4$), and each term with $\mathfrak{f}$ at most $C'_{\text{tot}} \bar{\Theta} \mathfrak{p}'_{\text{tot}} (\log \bar{x}_s) \rho^{24} \bar{x}_s^2 e^{-\kappa' \rho/16} \mathfrak{M}/\vartheta^3$, $\mathfrak{p}'_{\text{tot}}$ a polynomial. With $y := \bar{x}_s$, $\bar{x}_s \leq \bar{x}(y)$, $p_0(g_{k_0}) \geq A_0 (\log y)^{p_0}$ and $\bar{\Theta}^2 \leq L^{8(\mathfrak{l}(\bar{x}(y))+1)}$ by monotonicity of $\bar{R}$; so the first kind contributes at most $\mathfrak{M}/64$ under the ceiling

$$\sup_{y \geq g^{-2-c_5}} L^{8(\mathfrak{l}(\bar{x}(y))+1)} \mathfrak{p}_{\text{tot}} (\log \bar{x}(y)) \bar{x}(y)^2 e^{-\frac{1}{2} c_{\text{lv}} A_0 (\log y)^{p_0}} \leq \frac{\vartheta^4 c_M}{64 C_{\text{tot}}}, \quad (5.19)$$

a condition $g \leq \gamma_{\text{src}}$, since the logarithm of the function of y is $8 \log L \cdot \mathfrak{l}(\bar{x}(y)) + O(\log y) - \frac{1}{2} c_{\text{lv}} A_0 (\log y)^{p_0}$, $\mathfrak{l}(x) \leq L(C_R (\log x)^{r_B} + 1) + c_L$ and $\mathfrak{p}_0 > r_B$. The second kind contributes at most $\mathfrak{M}/64$ for all s once $C_{\text{room}} \geq C^{\text{src}}$, because $\rho \geq C_{\text{room}} A_0^2 (\log \bar{x}_s)^{2p_0}$ while $\log(\bar{\Theta} \mathfrak{p}'_{\text{tot}} \rho^{24} \bar{x}_s^2) = O(L(\log \bar{x}_s)^{r_B}) + O(\log C_{\text{room}} + \log \bar{x}_s)$ and $2\mathfrak{p}_0 > r_B$. $\square$

Live regions on a torus; the horizon. Let $\gamma_*^\dagger$ be the coupling of Proposition 5.21, $X_* := (\gamma_*^\dagger)^{-2}$, and let $K_* := K(g_0, \gamma_*^\dagger)$ be the first-passage level of clause (0) for the target $\gamma_*^\dagger$; it exists, since $g_0 \leq g_-(g) \leq g_-(\gamma_*^\dagger)$, and the flow bounds of clause (0) hold up to it for that target. Put

$$m_V(g) := \left\lfloor \frac{g^{-2} - X_*}{B_\#} \right\rfloor - 3, \quad p_C(g) := \max\{0, m_V(g) + 1 - c_{\text{gap}} - b_*(g)\}, \quad (5.20)$$

where $c_{\text{gap}} := \lfloor c_5/\lambda_\# \rfloor + 1$. Among any c_{gap} consecutive levels $i+1, \dots, i+c_{\text{gap}}$ there is one at which x_k is smaller than at every earlier level, that is, a first-passage level of some target: if k^* is the first index at which $\min_{j \leq i+c_{\text{gap}}} x_j$ is attained, the lower flow bound of clause (0) gives $x_j - x_{i+c_{\text{gap}}} \geq \lambda_\# c_{\text{gap}} - c_5 > 0$ for $j \leq i$, so $k^* > i$. Let $c_{\text{age}} \geq 9 \log L + \log 2$ be the age rate, fixed after L , and $q := e^{-c_{\text{age}}}$, so $L^9 q \leq \frac{1}{2}$. For a target coupling g' , $B(g') := E_+ + C_{\text{low}}$ is the sum

of the exponents of the two global stability bounds of clause (i-b) at the last lattice of the run, $e^{-C_{\text{low}}N^4} \leq \mathcal{Z} \leq e^{E_+N^4}$, where $\mathcal{Z}$ is the integral of the effective density at the last lattice, with the normalisation of clause (i-b), and N the number of sites per side there; $B(g') \leq C_B(1 + \log g'^{-2})$. Let $s_{\text{vol}}(g, m) := 0$ if $m \leq m_V(g)$; otherwise let $s_{\text{vol}}(g, m)$ be the least $s_0 \geq 0$ such that

$$c_{\text{age}}(s - r(s) + p_C(g)) \geq C_B(1 + \log g^{-2}) (2L^{m-p_C(g)})^4 \quad \text{for every } s \geq s_0. \quad (5.21)$$

It is finite ($s - r(s) \rightarrow \infty$), non-decreasing in m , a floor on the depth, and free of K , g_0 , $\mathbf{d}$ and positions.

Proposition 5.21 (live regions on a torus). *There are $\gamma_*^\dagger \in (0, \gamma_J]$, $C_{\text{lv}} \geq 1$ and $c_{\text{lv}} > 0$, depending only on $(L, c_{\text{age}}, \mathbf{p}, \gamma; \varrho)$, such that for every $g < \gamma_*^\dagger$, $g_0 \in (0, g_-(g)]$, every m allowed by (H5), every depth $s \leq K$ with $s \geq \max\{s_{\text{vol}}(g, m), r(s)\}$, and every level $j \leq K - s + r(s)$,*

$$\sum_{a \geq 0} L^{9(a+1)} \max_{\square} \tilde{\nu}(\text{LV}_j^{\bar{a}}(\square)) \leq C_{\text{lv}} e^{-c_{\text{lv}} p_0(g_j)}. \quad (5.22)$$

Hence (5.16) holds, and (5.17) holds with $A_{\text{lv}} := L^{-8(l+1)} \leq 1$. The bound is uniform in K , g_0 and positions, and in m on $\{m \leq m_V(g)\}$, where $s_{\text{vol}} = 0$.

This is proved in the companion account [T-LF]; we give the statement and the structure of the proof. The consequences hold because $\text{LV}_{k_0}(\square) = \bigsqcup_a \text{LV}_{k_0}^{\bar{a}}(\square)$ and $\sum_{a \geq l} L^{8(a+1)} q_{k_0}^{\bar{a}} \leq \sum_a L^{9(a+1)} q_{k_0}^{\bar{a}}$.

Structure. (1) *Prolongation.* No renormalisation step reads the target coupling, the cutoff or the size of the torus except through the level at which the run stops: every object of the k -th step is determined by the bare datum and k [9, §§2–3], [10, §1]. So we may continue the run on the same bare lattice up to the largest level $k^\dagger \leq \hat{\kappa} := \min\{K + m, K_* - 1\}$ that is a first-passage level of some target (among any c_{gap} consecutive levels there is one). This is a run of the family with a target $g^\dagger \in [g, \gamma_*^\dagger]$; its last lattice is at level $K^{\dagger'} := k^\dagger - b_*(g^\dagger)$ and has $N^\dagger$ sites per side. That $K^{\dagger'} \geq K$ is proved in [T-LF] from the flow bounds of clause (0) and (H5): the level at which the continued run has used up its own torus is itself a first-passage level and does not exceed $\hat{\kappa}$, and the stopping depth does not increase along the continued run. Events of levels at most K keep their mass. (2) *Fate.* A component live at level j through $\square$ is either removed by $\mathbf{R}$ later in the continued run, or still live at $K^{\dagger'}$. (3) *Removed later.* Mass at most $C_H q^a e^{-\frac{3}{2}p_0(g_j)}$, at every level, with no condition on depth or volume: removal by $\mathbf{R}$ goes through a quotient that Balaban forms and bounds himself [10, (0.3)–(0.5)], [11, (1.89)]; positivity turns it into a bound on the mass, and (4) supplies q^a . (4) *Prepayment of age.* Balaban's inductive bound on the factors carried by large-field regions [11, (1.79)–(1.88)] remains valid when its entropy constant is increased by c_{age} (this is where the floor $\mathbf{p}_0 \geq 6r_B + 1$ of §5.1 is used); the surplus is a factor q^{a+1} on the un-normalised weight of every component whose first large-field region is a levels old. This uses the statement on second-class large-field regions proved in the companion account (§5.8, item (d)). (5) *The one conversion.* Multiply the operators of the components live at the last lattice with first index at most $j - a$ by a number $\mu \geq 1$. For $|\mu| \leq \bar{\mu}$, where $\bar{\mu}$ is $q^{-(K^{\dagger'}-j+a)}$ times the slack between the reserve of a creation and what Balaban's upper bound at the last lattice requires of a live factor, that bound persists by (4), so the partition function with μ inserted is at most $e^{E_+N^{\dagger 4}}$. At zero source each term of the event gains at least the factor $\bar{\mu}$ at $\mu = \bar{\mu}$ and the others are unchanged; by positivity the un-normalised weight of the event is at most $2\bar{\mu}^{-1} e^{E_+N^{\dagger 4}}$. Dividing by $\mathcal{Z} \geq e^{-C_{\text{low}}N^{\dagger 4}}$ gives, for the event $T_j^{\bar{a}}(\square)$ that the component is still live at $K^{\dagger'}$,

$$\tilde{\nu}(T_j^{\bar{a}}(\square)) \leq C_q q^{a+K^{\dagger'}-j} e^{-2\theta p_0(g_j)} e^{B(g^\dagger)N^{\dagger 4}}, \quad (5.23)$$

with C_q and $\theta \in (0, 1)$ from Balaban's bounds. (The lower bound on $\mathcal{Z}$ follows from clause (i-b) by restricting to block configurations within g_k of $\mathbf{1}$, where the small-field characteristic functions equal 1 and the background action is $O(g_k^2)$ per site [6, Theorem 1]; so $C_{\text{low}} = O(1 + \log g_k^{-2})$.) (6) *Volume scale or coupling ceiling.* *Case V*, $K + m \leq K_* - 1$: the continued run uses up the torus first, $N^\dagger \leq 2L^{c_{\text{gap}}} L^{b_*(g^\dagger)}$ is polylogarithmic in $g^{\dagger-2}$ of degree r_B , and $B(g^\dagger)N^{\dagger 4}$ has degree $4r_B + 1 < p_0$ in $\log g^{\dagger-2}$; a ceiling on $\gamma_*^\dagger$ (namely $C_B(1 + \log X')(2L^{c_{\text{gap}}} L^{b_*(X'^{-1/2})})^4 \leq \theta(p_0(X'^{-1/2}) - 1)$ for all $X' \geq X_*$) gives $B(g^\dagger)N^{\dagger 4} \leq \theta p_0(g_j)$: half the reserve of one creation pays the conversion, with no condition on the depth. The condition $m \leq m_V(g)$ implies *Case V*: clause (0) for the target $\gamma_*^\dagger$ gives $B_\sharp(K_* - K) \geq x_K - x_{K_*} > g^{-2} - X_* - B_\sharp$, so $K_* - K \geq m_V(g) + 3$. *Case C*, otherwise: the coupling ceiling is reached first, $k^\dagger \geq K_* - 1 - c_{\text{gap}}$, hence $K^{\dagger'} - K \geq p_C(g)$ and $N^\dagger \leq 2L^{m-p_C(g)}$, which grows with the volume. For $j \leq k_0 + r(s)$ the survival factor $q^{K^{\dagger'}-j} \leq e^{-c_{\text{age}}(s-r(s)+p_C(g))}$ pays $e^{B(g^\dagger)N^{\dagger 4}}$, since $B(g^\dagger) \leq C_B(1 + \log g^{-2})$, as soon as (5.21) holds, that is for $s \geq s_{\text{vol}}(g, m)$. In both cases $\tilde{\nu}(T_j^a(\square)) \leq C_\dagger(2L^9)^{-a} e^{-c_\dagger p_0(g_j)}$. (7) *Summation.* As $L^9 q \leq \frac{1}{2}$, the series over a converges, with $C_{\text{lv}} := 2L^9(C_H + C_\dagger)$ and $c_{\text{lv}} := \min\{\frac{3}{2}, c_\dagger\}$.

Why the volume enters. Balaban's bounds control a live large-field region in his un-normalised weights: its factor is small relative to the majorant of the corresponding small-field term, not relative to the partition function [11, (1.79)]. The only conversion of these weights into probabilities that needs no further estimate is the quotient of the two global stability bounds at the last lattice, at the price e^{BN^4} . Continuing the run makes the last lattice as coarse as the torus and the coupling ceiling allow. If the torus is used up first, as it is for $m \leq m_V(g)$, the last lattice has polylogarithmically many sites and one creation reserve pays the price. If the ceiling is reached first, N grows like $L^{m-m_V(g)}$, and only the survival factor of a component still live at the end of the continued run can pay; this costs the depth threshold $s_{\text{vol}}(g, m)$. We do not decide here whether s_{vol} can be removed by an estimate on large-field regions uniform in the volume; no such estimate is proved or used in this paper, and Section 6 makes the conclusion uniform in the volume by a different route.

5.7 Step 5: the order of the constants, and the budget

The constants are fixed in this order, each from those before it (the exponents of (H2), among them $q \geq 21p_0 + 1$, being fixed before L): L ; c_{age} ; Balaban's parameters; γ , γ_H ; ϱ ; γ_J ; $\gamma_*^\dagger$ (Proposition 5.21, with the ceiling of Case V and those of steps (3)–(4) there); ϑ ; C_{room} , above the floors $40M/\vartheta$, $16C_h M$, $3200M$, $160C_h M/\vartheta^2$, $24M$, C^{ker} , C^{prof} , C^{loc} , C^{src} , $(448C_b)^{1/2}$ and those of Propositions 5.14, 5.15 and 5.17; r_0 , C_D ; k_W ; then γ_W , the largest $g \leq \gamma_J$ satisfying the finitely many ceilings of Steps 1–4 (that of Step 2, (5.19), the one giving $\mathfrak{d}_W \leq \frac{1}{3}$, and those of Propositions 5.14, 5.15 and 5.17); and $\gamma_W^\mathbb{T} := \min\{\gamma_W, \gamma_*^\dagger/2\}$. Each ceiling reads $\sup_{y \geq g^{-2}-c} \Phi(y) \leq \text{const}$ with Φ decreasing to 0, and holds for all depths at once, because each relative bound in Steps 2–4 is a function decreasing in $x_{k_0} \geq \underline{x}_s$ times a polynomial in $\log \bar{x}_s$, and $\bar{x}_s \leq \bar{x}(\underline{x}_s)$. For given g , $s_W(g)$ is the least s_0 such that for every $s \geq s_0$: $s - r(s) \geq s_*$ (the extraction uses the $r(s)$ levels above the witness level and starts at the depth s_*), $s \geq b_*(g)$ (Theorem 3.1 can be stopped at the level $K - s$), and $2B_\sharp(r(s) + 1) \leq \underline{x}_s$ (the running coupling varies little over the extracted levels); as $r(s) = O(\log_L \log \bar{x}_s)$ and $b_*(g) = O(\log_L \log g^{-2})$, $s_W(g) = O(\log_L \log g^{-2})$; replacing s_W by $\max_{g' \geq g} s_W(g')$, which is finite and of the same order, makes it non-decreasing in g^{-2} , and we do so. For given (g, m) , $m_V(g)$ and $s_{\text{vol}}(g, m)$ are given by (5.20)–(5.21). Nothing chosen later enters the choice of an earlier quantity, and nothing bounds L , K , m or s from above.

Proof of Theorem 5.8. Let $g \leq \gamma_W^\mathbb{T}$, g_0 , K , m , $\mathfrak{d}$ be as in the theorem, $s = s(\mathfrak{d})$, $k_0 = K - s$. As $s \geq s_W(g)$: $s - r(s) \geq s_*$, $s \geq b_*(g) \geq 1$ (so $C_W(s) \leq \rho < LC_W(s)$), and $2B_\sharp(r(s) + 1) \leq \underline{x}_s \leq x_{k_0}$.

As $K - s \geq k_W$, Proposition 5.4 applies. As $s \geq s_{\text{vol}}(g, m)$ and $g < \gamma_*^\dagger$, Proposition 5.21 gives (5.22) at all levels $j \leq k_0 + r(s)$, hence (5.16) and (5.17) with $A_{\text{lv}} \leq 1$. So all results of Steps 1–4 apply, and (5.13), Lemma 5.16, Step 2, Lemmas 5.18, 5.19 and Theorem 5.20 give

$$|S_{2;K,m}^T(f_1, f_2) - \mathfrak{M}| \leq |R^{\text{prof}}| + |R^{\text{gx}}| + |R^{\text{irr}}| + |R^{\text{loc}}| + |R^{\text{lf}}| \leq \frac{5}{32} \mathfrak{M} < \frac{1}{4} \mathfrak{M}.$$

As $\mathfrak{M} > 0$, $R_K := S_{2;K,m}^T(f_1, f_2)/\mathfrak{M} - 1$ satisfies (5.5) with $|R_K| \leq \frac{1}{4}$. The window (5.6) is Lemma 5.12, the modulus of Corollary 3.3 (Step 1) and Proposition 5.4. Since $(\zeta'_{K-s})^2 \in [\frac{4}{9}, \frac{16}{9}]$, $S_{2;K,m}^T(f_1, f_2)$ lies between $\frac{1}{3}$ and $\frac{20}{9}$ times $b_0 g_{K-s}^4 \mathcal{K}_s(f_1, f_2) > 0$, which gives positivity and (5.7); (5.8) follows with $g_{K-s}^4 = x_{K-s}^{-2}$ and (5.1). No constant depends on K , g_0 or positions, and the conditions depend on m only through $s_{\text{vol}}(g, m)$, which vanishes for $m \leq m_V(g)$. $\square$

The deduction never compares Balaban’s block law ν^B with the law of the block average of the bare field, and it uses no statement on the rarity of large-field regions in the normalised measure that is uniform in the volume, no clustering, no gap and no uniqueness of limits in the volume.

5.8 What the proof rests on

- (a) *Proved in this section:* Lemmas 5.5, 5.6, 5.10, 5.11, 5.12, 5.16, 5.18; Remark 5.7; the reduction of R^{gx} in Step 2; Theorem 5.20; the fact that $m \leq m_V(g)$ implies Case V; the order of the constants and the assembly in Step 5.
- (b) *From Balaban’s papers, used by statement:* the inductive representation of the effective densities and its bounds [9, (2.17)–(2.31)], valid at every level by [11, Theorem 1], with the integrated stability bounds in the form of clause (i-b) (Section 2); the small-field step [7, (2.1)–(2.15)] and the irrelevance bound [7, (0.28)–(0.30)]; the regularity of minimisers [6, Theorem 1]; the block propagators [1, (1.17)–(1.19)], [2], their exponential decay [1, Proposition 1.2], [2, Propositions 2.5–2.6], and their positivity in a background field [5, (3.157)–(3.158)], stated there with its proof indicated; the large-field operation, its summation, positivity and inductive bounds [10, §1], [11, (1.71)–(1.89)]; the tree bound [8, (1.26)]. Those of his statements that the construction uses as he asserts them, without full proof in print, are named in Section 9.
- (c) *From earlier sections, as qualified there:* clauses (0) and (i), in particular the global stability bounds (i-b) and the exact insertion (i-d) (Section 2); the correlation theorem with sources and the running shift (iii)(c) (Section 3).
- (d) *Proved in the companion account; we give the statement and the structure of the proof:* Proposition 5.14 and the bound on Ψ^{span} in Lemma 5.19 [T-W]; the modulus in (5.6) (Corollary 3.3); Proposition 5.15 [T-GX]; Proposition 5.21, with the prolongation, the prepayment of age, the conversion at the last lattice and the bound on components removed later [T-LF]; Proposition 5.17, the room estimate, proved in [T-GX] as an implication from the correlation theorem, under the lower bound $q \geq 21\mathfrak{p}_0 + 1$; the statements of Proposition 5.4 on the free, unconditioned kernel, in the companion note on the free kernel, the passage to the block-conditioned kernel $\mathcal{K}_s$ being from Balaban’s propagator decay, used by statement (item (b)); and the statement on second-class large-field regions used in the prepayment of age: each second-class outcome of Balaban’s large-field operation is marked by a plaquette of the block field of its level deviating from $\mathbf{1}$ by at least a fixed fraction of the small-field threshold, and the small factor so produced pays for the combinatorial entropy of the outcome; it is also used for clause (v) (Section 7).

- (e) *Readings, declared here; they involve no new estimate:* (i) on clean histories the step near the supports is the small-field step of [7, §2] inside [9, (2.18)–(2.24)], the small-field exponent is analytic, domain by domain, on a fixed complex neighbourhood of the window, and a segment of the history decomposition is a composition of the four kinds of operations of Step 1 (used in [T-GX]); (ii) the format of the operators of live components stated after Proposition 5.14 (used in [T-W]); (iii) the termwise form of Balaban’s bound on the reserve carried by large-field regions, used in the proof of the bound on components removed later given in [T-LF]; (iv) in the room estimate, [6, Theorem 1] read on its complex neighbourhood with absolute amplitudes, and the irrelevance lemma of the correlation theorem read as depending on the background field only through the size and the Hölder structure of its block data (used in [T-GX]).

No statement in this list is deferred, and none concerns the rarity of large-field regions uniformly in the volume, clustering, a mass gap or uniqueness in the volume. Theorem 5.8 is thus a complete deduction from the statements on which clauses (0), (i) and (iii) rest, together with (b), (d) and the readings (e).

6 Clause (iv) in every volume: volume transfer

In this section we prove clause (iv) of Theorem 0 on every torus of the family $\{\mathbb{T}_m : m \geq m_{\mathbb{T}}(g_{-}(g))\}$, uniformly in m , and in every infinite-volume limit of clause (ii). The finite-volume statement—the curvature two-point witness on one torus—is the theorem of Section 5. We use it on one torus only, the smallest torus of the family, $\mathbb{T}_{m'}$ with $m' = m_{\mathbb{T}}(g_{-}(g))$, and we compare the connected two-point functions of any two tori of the family at the same cutoff and the same bare coupling. The comparison (Corollary 6.19) reads

$$|S_{2;K,m}^T(f_1, f_2) - S_{2;K,m'}^T(f_1, f_2)| \leq C_T(g) \mathbf{d}^2 \|f_1\|_{\infty} \|f_2\|_{\infty} \quad (6.1)$$

for all $m \geq m' \geq m_{\mathbb{T}}(g_{-}(g))$ and all real Lipschitz f_1, f_2 of scale $\mathbf{d}$ in one ball of radius $5\mathbf{d}$, with $C_T(g)$ independent of the cutoff, the bare coupling, the volumes and the position. The main term of the witness is of order $g_{K-s(\mathbf{d})}^4$, that is, of order $(g^{-2} + \log(1/\mathbf{d}))^{-2}$; it dominates $\mathbf{d}^2$ below a separation threshold depending on g only, which is a floor on the depth of the kind clause (iv) carries already.

The comparison has two parts: the volume-transfer theorem (Theorem 6.13), a bound by a constant times $\|F_1\|_{\#} \|F_2\|_{\#}$ for all real Lipschitz F_1, F_2 in one ball of fixed radius, independent of the cutoff, the volumes, the position, and the distance and overlap of the supports; and the averaging lemma (Lemma 6.18), which turns such a bound for a translation-invariant bilinear form into one with a factor $(32r_0/R)^4$ on a ball of radius r_0 . Of the large torus the proof uses reflection positivity of the bare Wilson measure (the chessboard estimate, and a trace inequality for mirror-periodic partition functions, Lemma 6.4), the lower stability bound at zero source, and the extensivity of Balaban’s vacuum constant (Lemma 6.8); of the small torus, the upper stability bound with sources. One holomorphic counterterm serves both volumes, and Schwarz’s lemma on a complex polydisc turns these one-sided bounds into a two-sided bound on the difference (Remark 6.15).

All proofs are given in full; two optional variants, which the argument does not use, are stated without proof here: the first is proved in the companion account (Remark 6.10), and of the second the free-field part is proved in the companion note on the free kernel (Remark 6.26). Used without proof are: the statements of Sections 2, 3 and 5 recalled in Propositions 6.1 and 6.2 and Theorem 6.21; the torus-independence lemma of the companion account (Lemma 6.7); and Balaban’s lattice counts and bounds at the unit configuration, cited by equation.

6.1 Setting, notation, and the statements used

Standing assumptions. Throughout this section the hypotheses (H1)–(H7) of Section 1 hold with $N = 2$ (gauge group $\text{SU}(2)$); $\varrho > 0$ is the radius of (H7), and $g \leq \gamma_J$, the coupling ceiling of clause (iii). We fix $g_0 \in (0, g_-(g)]$ and let $K = K(g_0, g)$ be the first-passage index of clause (0); $x_0 := g_0^{-2}$, and $\varepsilon = L^{-K}$ is the bare spacing. By clause (0) (Section 2), K and the running couplings $g_0, \dots, g_K$ are the same for every torus of the family. The trace inequality of §6.2 holds more generally, for every compact gauge group and every $x_0 > 0$, and the averaging lemma of §6.5 for every bilinear form with the stated properties.

Tori, plaquettes, generating functions. For $m \geq m_{\mathbb{T}}(g_-(g))$, $\mathbb{T}_m = \mathbb{R}^4/2L^m\mathbb{Z}^4$ carries the bare lattice of spacing ε . We use Balaban’s centred lattice: the bare sites are the points of $\varepsilon(\mathbb{Z} + \frac{1}{2})^4$ [7, (0.1)]. Hence a hyperplane $\{x_\mu = c\}$ with $c \in \varepsilon\mathbb{Z}$ contains no site; it meets only bonds parallel to e_μ , which it crosses. We write $|\mathbb{T}_m| := (2L^m)^4$ for the volume of $\mathbb{T}_m$. Of the floor (H5), $m \geq m_{\mathbb{T}}(g_-(g))$, the arguments written out in this section use only the inequality $L^m \geq L^{m_{\mathbb{T}}(g_-(g))} \geq L^{11}M$ (Section 1), where M is Balaban’s block size; the statements recalled from Sections 2, 3 and 5, and Lemma 6.7, use (H5) as stated there. For a bare plaquette p let x_p be its barycentre and $a_p(U) := 1 - \text{Re tr } U(\partial p) \in [0, 2]$, so that $A(U) = \sum_p a_p(U)$. For a complex function v on $\mathbb{T}_m$ put

$$\mathcal{O}(v) := \sum_p v(x_p) a_p(U), \quad Z_m(v) := \int e^{-x_0 A(U) + \mathcal{O}(v)} dU, \quad (6.2)$$

the sum over all bare plaquettes and dU the normalised Haar measure; no additive constant is included in Z_m . We write $\langle \cdot \rangle_m$ for the expectation of the bare Wilson measure on $\mathbb{T}_m$, so that $\langle e^{\mathcal{O}(v)} \rangle_m = Z_m(v)/Z_m(0)$, and, for real Lipschitz f_1, f_2 ,

$$S_{2;K,m}^T(f_1, f_2) := \partial_{z_1} \partial_{z_2} \log Z_m(z_1 f_1 + z_2 f_2) \Big|_{z=0} = \langle \mathcal{O}(f_1) \mathcal{O}(f_2) \rangle_m - \langle \mathcal{O}(f_1) \rangle_m \langle \mathcal{O}(f_2) \rangle_m.$$

It is defined at every finite K for every pair, and it is bilinear in (f_1, f_2) ; clause (iv) restricts what is asserted, not what is defined. For a lattice isometry ω and a configuration U put $(U \circ \omega)(b) := U(\omega b)$ if ω preserves the orientation of the bond b , and $(U \circ \omega)(b) := U(\omega b)^{-1}$ otherwise. Then $a_p(U \circ \omega) = a_{\omega p}(U)$, because Re tr of a holonomy is invariant under cyclic relabelling and inversion; and the bare Wilson measure is invariant under $U \mapsto U \circ \omega$, because Haar measure is invariant under relabelling and inversion and ω permutes the plaquettes.

Transport. A compactly supported function F on $\mathbb{R}^4$ is transported to $\mathbb{T}_m$ by periodisation, $F^{(m)}(x) := \sum_{n \in \mathbb{Z}^4} F(x + 2L^m n)$. This map is linear and commutes with translations. Suppose that $\text{supp } F$ lies in a closed ball B of radius r with $2r < L^m$. Then at most one term of the sum is non-zero at each point, $\sup |F^{(m)}| = \sup |F|$, and $\text{Lip } F^{(m)} \leq \text{Lip } F$ for the distance of the torus. For the last claim let $x, y \in \mathbb{T}_m$. If $F^{(m)}(x) = F(a) \neq 0$ with $a \in B$ and $F^{(m)}(y) = 0$, then $F(b') = 0$ for every lift b' of y (at most one term of the periodisation sum is non-zero at y , and the sum vanishes), so $|F(a)| = |F(a) - F(b')| \leq \text{Lip } F \cdot |a - b'|$ for every lift b' of y , and the minimum of $|a - b'|$ over these lifts is the distance of x and y on the torus. If both values are non-zero, take lifts $a, b \in B$; then $|a - b| \leq 2r$, while every other difference $a - b + 2L^m n$, $n \neq 0$, has length at least $2L^m - 2r > 2r$; so $|a - b|$ is the distance of x and y on the torus, and $|F^{(m)}(x) - F^{(m)}(y)| = |F(a) - F(b)| \leq \text{Lip } F \cdot |a - b|$. If both values vanish there is nothing to prove. For functions on $\mathbb{R}^4$ supported in such a ball we write $S_{2;K,m}^T(F_1, F_2) := S_{2;K,m}^T(F_1^{(m)}, F_2^{(m)})$.

Sources. For a complex Lipschitz function v , on $\mathbb{R}^4$ or on a torus, $\|v\|_{\#} := \max\{\|v\|_{\infty}, 40M \text{Lip } v\}$, and $\mathcal{W}$ is the class of complex Lipschitz functions v on a torus with $\|v\|_{\#} < \varrho$. For real Lipschitz functions $F_1, \dots, F_n$ on $\mathbb{R}^4$ we write $\rho(z) := \varrho^{-1} \sum_i |z_i| \|F_i\|_{\#}$ and, for $\mathfrak{r} \in \{1, 2\}$, $\mathbb{D}_{\mathfrak{r}} := \{z \in \mathbb{C}^n : \rho(z) < 1/\mathfrak{r}\}$; for a finite family $h = (h_l)$ of real Lipschitz functions on a torus, $\mathbb{D}(h) := \{\zeta : \sum_l |\zeta_l| \|h_l\|_{\#} < \varrho\}$. By the preceding paragraph, the norms on a torus of transported functions are at most their norms on $\mathbb{R}^4$.

Cells and reflections. Let $\ell \geq 1$ be a power of L . The cells of $\mathbb{R}^4$ are the cubes $c_k := \ell k + [0, \ell]^4$, $k \in \mathbb{Z}^4$. Their faces lie on the hyperplanes $\{x_{\mu} \in \ell\mathbb{Z}\}$, which contain no bare site, since $\ell \in \varepsilon\mathbb{Z}$. For a direction $\mu \in \{0, 1, 2, 3\}$ and $a \in \mathbb{Z}$ let $\theta_{\mu,a}$ be the reflection $x_{\mu} \mapsto 2a\ell - x_{\mu}$ (the other coordinates fixed), and let $\mathfrak{G}_{\ell}$ be the group generated by all $\theta_{\mu,a}$. Each $\theta_{\mu,a}$ maps the bare lattice onto itself ($\varepsilon(i + \frac{1}{2}) \mapsto \varepsilon(2a\ell/\varepsilon - i - 1 + \frac{1}{2})$), hence bonds to bonds, plaquettes to plaquettes and cells to cells. On cell indices $\theta_{\mu,a}$ acts by $k_{\mu} \mapsto 2a - 1 - k_{\mu}$; $\theta_{\mu,a+t}\theta_{\mu,a}$ is the translation by $2t\ell e_{\mu}$; reflections in different directions commute. In each direction the generated group is $\{k_{\mu} \mapsto k_{\mu} + 2t\} \cup \{k_{\mu} \mapsto 2a - 1 - k_{\mu}\}$, which acts simply transitively on $\mathbb{Z}$; hence $\mathfrak{G}_{\ell}$ acts simply transitively on the cells. A reflection in direction μ changes the parity of k_{μ} , and no other generator changes k_{μ} ; therefore, in any word representing $\omega \in \mathfrak{G}_{\ell}$ with $\omega c_k = c_{k'}$, the number of reflections in direction μ is congruent to $k'_{\mu} - k_{\mu}$ modulo 2, and

$$\text{sg}(\omega) := (-1)^{\text{number of reflections in a word for } \omega} = (-1)^{\sum_{\mu} (k'_{\mu} - k_{\mu})}$$

is a well-defined homomorphism $\mathfrak{G}_{\ell} \rightarrow \{\pm 1\}$. If $m'' \geq \log_L \ell$, the side $2L^{m''}$ of $\mathbb{T}_{m''}$ is an even multiple of ℓ ; the translations by $2L^{m''}\mathbb{Z}^4$ belong to $\mathfrak{G}_{\ell}$ and have $\text{sg} = +1$. So $\mathfrak{G}_{\ell}(\mathbb{T}_{m''}) := \mathfrak{G}_{\ell}/2L^{m''}\mathbb{Z}^4$ acts on $\mathbb{T}_{m''}$, simply transitively on its set $\mathcal{C}_{m''}$ of cells, sg descends to it, and

$$|\mathcal{C}_{m''}| = (2L^{m''}/\ell)^4, \quad |\mathbb{T}_{m''}|/|\mathcal{C}_{m''}| = \ell^4.$$

For a number or a function ξ and $s \in \{\pm 1\}$ we put $\xi^{(s)} := \xi$ if $s = 1$ and $\xi^{(s)} := \bar{\xi}$ if $s = -1$.

The decomposition with sources. Let $m'' \geq m_{\mathbb{T}}(g_{-}(g))$. Balaban's bare density on $\mathbb{T}_{m''}$ is $\exp[-x_0 A(U) - E_{m''}]$, where the vacuum constant $E_{m''}$ is the number defined in §6.3; it does not depend on any source. We write $\mathcal{D}_j = \mathcal{D}_j(\mathbb{T}_{m''})$ for Balaban's localisation domains at level j , the connected unions of cubes of side M in level- j units [7]; $d_j(X)$ for the length, in units of M , of a shortest tree graph contained in X that meets all its cubes; X^+ for X enlarged by one layer of such cubes; and $\text{plaq}(X^+)$ for the set of bare plaquettes that meet X^+ .

Proposition 6.1 (Statements used from Sections 2 and 3). *Let $m'' \geq m_{\mathbb{T}}(g_{-}(g))$, let $b_* = b_*(g)$ be the integer of Theorem 3.1, assume $K \geq b_*$ (the case $K < b_*$ is treated after Theorem 6.13), and let $K' := K - b_*$ be the level at which the run with sources of that theorem stops when $s = b_*$ in its part (c). For $j \leq K'$ and $X \in \mathcal{D}_j$ there are complex numbers $e_j(X; v)$, depending on the weight v only, with $e_j(X; 0) = 0$. Put*

$$\mathcal{E}_{m''}(v) := \sum_{j \leq K'} \sum_{X \in \mathcal{D}_j} e_j(X; v), \quad \tilde{Z}_{m''}(v) := e^{-E_{m''} - \mathcal{E}_{m''}(v)} Z_{m''}(v).$$

For every finite family $h = (h_l)$ of real Lipschitz functions on $\mathbb{T}_{m''}$, every $\zeta \in \mathbb{D}(h)$, and the weight $v = \sum_l \zeta_l h_l$:

(L) $e_j(X; v)$ depends only on the values of v at the barycentres of the plaquettes of $\text{plaq}(X^+)$;

(A) $\zeta \mapsto e_j(X; \sum_l \zeta_l h_l)$ is holomorphic on $\mathbb{D}(h)$;

(B) $|e_j(X; v)| \leq C_e e^{-(1-\delta)\kappa d_j(X)}$;

(S) $|\tilde{Z}_{m''}(v)| \leq e^{E_+^\sharp |\mathbb{T}_{m''}|}$.

Moreover

(F) $\tilde{Z}_{m''}(0) \geq e^{-C_{\text{low}} |\mathbb{T}_{m''}|}$.

The numbers C_e , $\delta \in (0, 1)$, κ , $E_+^\sharp$ and C_{low} depend only on L , Balaban's parameters, g and ϱ ; they depend neither on K , nor on g_0 , nor on m'' .

Statements (L), (A), (B) are parts (d1)–(d3) of Theorem 3.1, with C_e its constant; we use them as qualified there. (S) and (F) follow from part (d4) of that theorem and from the two stability bounds of clause (i-b) (Proposition 2.4) at level K . By clause (i-c), $\tilde{Z}_{m''}(0) = e^{-E_{m''}} Z_{m''}(0)$ is the integral of the effective density at level K , so (i-b) gives $\tilde{Z}_{m''}(0) \leq e^{E_+ |\mathbb{T}_{m''}|}$; and restricting that integral to the history without large fields and to a neighbourhood of the trivial configuration, and using $x_K \leq g^{-2} + B_\sharp$, gives (F), with C_{low} depending only on L , Balaban's parameters and g ; this holds on every torus of the family. With (d4) at $s = b_*$, $|\tilde{Z}_{m''}(v)| = e^{-E_{m''}} |e^{-\mathcal{E}_{m''}(v)} Z_{m''}(v)| \leq e^{c_{\text{vol}} |\Lambda_{K'}|} \tilde{Z}_{m''}(0) \leq e^{(E_+ + c_{\text{vol}} L^{4b_*}) |\mathbb{T}_{m''}|}$, since $|\Lambda_{K'}| = L^{4b_*} |\mathbb{T}_{m''}|$; this is (S) with $E_+^\sharp := E_+ + c_{\text{vol}} L^{4b_*}$. The proof of (d4) bounds each term of Balaban's expansion by its modulus before integration, as his proof of [9, Corollary 3] does; no pointwise upper bound on the density is used. At $v = 0$, (S) and (F) give $e^{-C_{\text{low}} |\mathbb{T}_{m''}|} \leq \tilde{Z}_{m''}(0) \leq e^{E_+^\sharp |\mathbb{T}_{m''}|}$, hence

$$\mathbf{B} := E_+^\sharp + C_{\text{low}} \geq 0.$$

In Step 3 of the proof of Theorem 6.13, (B) and (S) are used at a weight that is the weight of a family of four real functions at a parameter in its polydisc whenever $z \in \mathbb{D}_2$; we therefore put $\mathfrak{r} := 2$. If (B) and (S) hold for every $v \in \mathcal{W}$, one may take $\mathfrak{r} := 1$. The constants below are written with $\mathfrak{r}$.

Proposition 6.2 (Chessboard estimate and face bound; Section 3). *Let ℓ be a power of L and $m'' \geq \log_L \ell$.*

- (i) *For a cell c of $\mathbb{T}_{m''}$ let $\mathfrak{A}_c$ be the set of bounded measurable functions of the bond variables of the bonds with both end points in the open cell c° . For $F \in \mathfrak{A}_c$ and $c' \in \mathcal{C}_{m''}$ put $F^{[c']}(U) := (F(U \circ \omega))^{\langle \text{sg}(\omega) \rangle}$, where ω is the element of $\mathfrak{G}_\ell(\mathbb{T}_{m''})$ with $\omega c = c'$. Then $\langle \prod_{c' \in \mathcal{C}_{m''}} F^{[c']} \rangle_{m''}$ is real and non-negative, and*

$$|\langle F \rangle_{m''}| \leq \left\langle \prod_{c' \in \mathcal{C}_{m''}} F^{[c']} \right\rangle_{m''}^{1/|\mathcal{C}_{m''}|}.$$

- (ii) *For $X \in \mathcal{D}_j$ write $X \in \mathcal{X}_{c'}$ if $X^+ \subset (c')^\circ$, and $X \in \mathcal{X}_\partial$ if X belongs to no $\mathcal{X}_{c'}$. Let $R > 0$, and let $v = \sum_{i \in I} v_i$ be a finite sum of weights, each supported (at barycentres) in a closed ball of radius R at distance at least $R + 2$ from all faces of the cells, such that (B) holds at v . Then*

$$\sum_{j \leq K'} \sum_{X \in \mathcal{X}_\partial} |e_j(X; v)| \leq |I| C_{\text{face}}(R),$$

where $C_{\text{face}}(R)$ depends only on L , Balaban's parameters, g , ϱ and R , and depends neither on K , g_0 , m'' and ℓ nor on the positions of the balls.

Part (i) is the chessboard estimate for complex cell observables, a consequence of reflection positivity of the Wilson action at the face hyperplanes in all four directions [23]; part (ii) bounds the local energies of the domains that meet a face hyperplane, using (L), (B) and the length of such domains. Both are proved in Section 3.

6.2 Transfer operators and the trace inequality

In this subsection G is any compact group with a finite-dimensional unitary representation, tr is the normalised trace in that representation, $x_0 > 0$, and ℓ is a power of L . For $\mathbf{n} = (\mathbf{n}_0, \dots, \mathbf{n}_3)$ with all $\mathbf{n}_\mu$ even positive integers let $\mathbb{T}[\mathbf{n}] := \prod_\mu \mathbb{R}/\mathbf{n}_\mu \ell \mathbb{Z}$, with bare sites $\varepsilon(\mathbb{Z} + \frac{1}{2})^4$, and $Z_{\mathbf{n}}(W) := \int e^{-x_0 A(U) + \mathcal{O}(W)} dU$ on $\mathbb{T}[\mathbf{n}]$. For $m'' \geq \log_L \ell$, $\mathbb{T}_{m''} = \mathbb{T}[(2L^{m''}/\ell)(1, 1, 1, 1)]$.

Definition 6.3 (Mirror-periodic weights). A bounded function $W : \mathbb{R}^4 \rightarrow \mathbb{C}$ is *mirror-periodic* if

- (M1) $W \circ \theta_{\mu,a} = \overline{W}$ for every direction μ and every $a \in \mathbb{Z}$;
- (M2) $W(x_p) = 0$ for every plaquette p that contains a bond crossing one of the hyperplanes $\{x_\mu \in \ell \mathbb{Z}\}$.

Since $\theta_{\mu,a+1}\theta_{\mu,a}$ is the translation by $2\ell e_\mu$, (M1) applied twice gives $W(x + 2\ell e_\mu) = W(x)$. Hence a mirror-periodic W descends to every $\mathbb{T}[\mathbf{n}]$ with $\mathbf{n} \in (2\mathbb{N})^4$.

Transfer operators. We use the transfer-operator form of reflection positivity of the Wilson action, due to Osterwalder and Seiler [23]; we prove what we need. Single out the direction $\mu = 0$ and call x^0 time. The layers are the sets of bare sites with $x^0 = \varepsilon(i + \frac{1}{2})$, $i \in \mathbb{Z}/t\mathbb{Z}$, $t := \mathbf{n}_0 \ell / \varepsilon$. Let Σ be the spatial cross-section, a three-dimensional periodic lattice with bond set $\mathbf{E}$, and $\mathcal{H} := L^2(G^{\mathbf{E}})$ with Haar measure; a vector of $\mathcal{H}$ is a function of the spatial bond variables of one layer. Every plaquette is either spatial, with its four bonds in one layer, or time-like, between two consecutive layers $i, i+1$, with two spatial bonds over one bond $b \in \mathbf{E}$ and the two time-like bonds over the end points $b_\pm$ of b (b oriented from b_- to b_+). Let $\mathbf{M}_i$ be the operator of multiplication by $\exp[-\sum_p (x_0 - W(x_p))a_p(U)]$, the sum over the spatial plaquettes of layer i , and let $\mathbf{K}_i$ be the integral operator with kernel

$$\mathbf{K}_i(U, U') := \int \prod_{x \in \Sigma} dV_x \prod_{b \in \mathbf{E}} \exp \left\{ - (x_0 - W(x_{p(b,i)})) \left[1 - \text{Re tr} (U_b V_{b_+} U_b'^{-1} V_{b_-}^{-1}) \right] \right\}, \quad (6.3)$$

where U, U' are the spatial configurations of layers i and $i+1$, V_x is the time-like bond from x in layer i to x in layer $i+1$, and $p(b, i)$ is the time-like plaquette over b between the two layers. All kernels are continuous functions on compact spaces. By Fubini's theorem,

$$Z_{\mathbf{n}}(W) = \text{Tr} (\mathbf{M}_1 \mathbf{K}_1 \mathbf{M}_2 \mathbf{K}_2 \cdots \mathbf{M}_t \mathbf{K}_t), \quad (6.4)$$

the ordered product once around the time circle: each $\mathbf{M}_i \mathbf{K}_i$ is an integral operator with continuous kernel, hence Hilbert–Schmidt; a cyclic product of $t \geq 2$ such operators is trace class, and its trace is the integral of the cyclic product of the kernels. No gauge is fixed: the time-like bonds are integrated inside the kernels.

Lemma 6.4 (Trace inequality). *Let W be mirror-periodic (complex, and of any size), $\mathbf{n} \in (2\mathbb{N})^4$, μ a direction and $k \geq 1$ an integer, and let $\mathbf{n}^{(\mu,k)}$ be $\mathbf{n}$ with $\mathbf{n}_\mu$ replaced by $k\mathbf{n}_\mu$. Then*

$$0 \leq Z_{\mathbf{n}^{(\mu,k)}}(W) \leq Z_{\mathbf{n}}(W)^k, \quad \text{hence} \quad 0 \leq Z_{k\mathbf{n}}(W) \leq Z_{\mathbf{n}}(W)^{k^4}. \quad (6.5)$$

Proof. Take $\mu = 0$.

(1) *Adjoints.* $\mathbf{M}_i[W]^* = \mathbf{M}_i[\overline{W}]$, since a_p is real. The kernel (6.3) is symmetric in (U, U') : substituting $V_x \mapsto V_x^{-1}$ (Haar measure is invariant under inversion) and using $\text{Re tr } Y^{-1} = \text{Re tr } Y$ for unitary Y together with the cyclicity of the trace,

$$\text{Re tr} (U_b' V_{b_+}^{-1} U_b^{-1} V_{b_-}) = \text{Re tr} (U_b V_{b_+} U_b'^{-1} V_{b_-}^{-1}).$$

As the bracket in (6.3) is real, $\overline{K_i[W](U', U)} = K_i[\overline{W}](U', U) = K_i[\overline{W}](U, U')$, that is, $K_i[W]^* = K_i[\overline{W}]$.

(2) *The face operator.* A face hyperplane $\{x^0 = a\ell\}$ contains no site and lies between two consecutive layers. Every time-like plaquette between these layers contains a bond crossing it, so by (M2) the operator between them is the source-free operator K^0 , with kernel (6.3) at $W = 0$. Let $\varphi(Y) := e^{-x_0(1 - \text{Re tr } Y)}$ on G and $\Phi(U) := \prod_{b \in E} \varphi(U_b)$ on G^E ; let C be the convolution $(C\psi)(U) := \int \Phi(UU''^{-1})\psi(U'') dU''$ (products taken bond by bond), and let $P := \int \prod_x dV_x \mathcal{R}_V$, where $(\mathcal{R}_V\psi)(U) := \psi((V_{b_-}^{-1}U_b V_{b_+})_b)$. Substituting $U'_b = V_{b_-}^{-1}U''_b V_{b_+}$ in $\int K^0(U, U')\psi(U') dU'$ gives $K^0 = CP$. The function φ is a positive-definite class function on G : Re tr is the half-sum of the normalised characters of the representation and of its conjugate, characters are positive definite, and sums with non-negative coefficients, products and uniform limits of positive-definite functions are positive definite, so that $e^{x_0 \text{Re tr}} = \sum_{n \geq 0} (x_0 \text{Re tr})^n / n!$ is positive definite. Hence Φ is a positive-definite function on the compact group G^E , real and invariant under inversion, and C is self-adjoint and non-negative. Since Φ is a class function, C commutes with every $\mathcal{R}_V$, hence with the orthogonal projection P onto gauge-invariant functions. Therefore

$$K^0 = CP = PCP \geq 0.$$

Its kernel is continuous on a compact space, so K^0 is trace class (Mercer's theorem) and $S := (K^0)^{1/2}$ is Hilbert–Schmidt.

(3) *Mirror structure.* Number the face hyperplanes $\{x^0 = a\ell\}$ by $a \in \mathbb{Z}/n_0\mathbb{Z}$. Let $Y_a := \text{MKM} \cdots \text{KM}$ be the ordered product over the ℓ/ε layers strictly between the hyperplanes $a\ell$ and $(a+1)\ell$, a bounded operator, and $A_a := SY_aS$, a trace-class operator. By (2) and the cyclicity of the trace, (6.4) becomes $Z_n(W) = \text{Tr}(A_1 A_2 \cdots A_{n_0})$. The reflection $\theta_{0,a+1}$ maps the layers and the time-like plaquettes of the slab between $a\ell$ and $(a+1)\ell$ onto those of the next slab in the reverse order, maps spatial bonds to spatial bonds preserving their orientation, and, by (M1), replaces the weights by their complex conjugates. Together with (1) and the symmetry of the kernels this gives $Y_{a+1} = Y_a^*$, hence $A_{a+1} = A_a^*$. Since n_0 is even—the only place where evenness is used—

$$A_1 A_2 \cdots A_{n_0} = (A_1 A_1^*)^{n_0/2} =: \mathcal{T} \geq 0,$$

a non-negative trace-class operator. The same operator A_1 (same cross-section, same 2ℓ -periodic W) serves the time extent $kn_0\ell$. Therefore $Z_{n(0,k)}(W) = \text{Tr } \mathcal{T}^k$ and $Z_n(W) = \text{Tr } \mathcal{T} \geq 0$.

(4) *Powers.* If $\lambda_i \geq 0$ are the eigenvalues of $\mathcal{T}$, then $\sum_i \lambda_i^k \leq (\sup_i \lambda_i)^{k-1} \sum_i \lambda_i \leq (\sum_i \lambda_i)^k$. Hence $0 \leq Z_{n(0,k)}(W) \leq Z_n(W)^k$.

(5) *All directions.* The hypotheses are symmetric in the four directions, and the intermediate tori may be rectangular. Passing from n to kn one direction at a time, every intermediate partition function is non-negative by (3), so taking powers preserves the inequalities:

$$Z_{kn}(W) \leq Z_{(n_0, kn_1, kn_2, kn_3)}(W)^k \leq Z_{(n_0, n_1, kn_2, kn_3)}(W)^{k^2} \leq \cdots \leq Z_n(W)^{k^4}. \quad \square$$

Remark 6.5. (i) Lemma 6.4 is a property of the bare Wilson measure at every cutoff, for every compact gauge group; nothing of Balaban's construction is used. W may be complex and large; positivity is used only for the source-free face operator K^0 and for $A_1 A_1^*$. (ii) At $W = 0$ the lemma says that the raw pressure $\pi_m := |\mathbb{T}_m|^{-1} \log Z_m(0)$ does not increase along the family: $\pi_m \leq \pi_{m'}$ for $m \geq m' \geq \log_L \ell$. This is not the direction that Theorem 6.13 needs at zero source; that one is Lemma 6.8. (iii) Evenness is asked of the cell counts $2L^{m''}/\ell$, never of k : in the application $k = L^{m-m'}$ is odd, every pair $m \geq m' \geq \log_L \ell$ is reached in one step, and the intermediate rectangular tori carry the bare measure only. (iv) If $Z_n(W) = 0$, all inequalities read $0 \leq 0$.

Remark 6.6 (Monotonicity, convexity, and the tori beyond the next one). Write $|\mathbf{n}| := \prod_{\mu} n_{\mu}$. The proof of Lemma 6.4 gives more. With $\mathcal{S} := \mathbf{A}_1 \mathbf{A}_1^* \geq 0$ we have $Z_{\mathbf{n}}(W) = \text{Tr } \mathcal{S}^{n_0/2}$, and $p \mapsto (\text{Tr } \mathcal{S}^p)^{1/p}$, a Schatten norm, is non-increasing on $[1, \infty)$. Direction by direction this gives, for $\mathbf{n} \leq \mathbf{n}'$ componentwise in $(2\mathbb{N})^4$ and without any divisibility,

$$Z_{\mathbf{n}'}(W)^{1/|\mathbf{n}'|} \leq Z_{\mathbf{n}}(W)^{1/|\mathbf{n}|}.$$

Moreover $p \mapsto \log \text{Tr } \mathcal{S}^p$ is convex, by Hölder's inequality on the eigenvalues. At $W = 0$ put $\varpi(\mathbf{n}) := |\mathbf{n}|^{-1} \log Z_{\mathbf{n}}(0)$, and let $a < b < r$ be even values of one entry, the other entries fixed. Convexity of $x \mapsto x\varpi$ at b , between a and r , reads $r(b-a)\varpi_r \geq (r-a)b\varpi_b - (r-b)a\varpi_a$; with $\varpi_a \geq \varpi_b$ this gives

$$\varpi_a - \varpi_r \leq \frac{b(r-a)}{r(b-a)} (\varpi_a - \varpi_b) \leq c(\varpi_a - \varpi_b), \quad c := \frac{b}{b-a}.$$

Let $f_i := \varpi(a, a, a, a) - \varpi(r, \dots, r, b, \dots, b)$, with i entries r and $4-i$ entries b . Splitting f_{i+1} at the point with entries $(r, \dots, r, a, b, \dots, b)$ and using monotonicity in each entry and the last display in the $(i+1)$ -st entry, $f_{i+1} \leq f_i + cf_i$; hence $\varpi(a^4) - \varpi(r^4) = f_4 \leq (1+c)^4 f_0$. With $a = 2L^{m'}/\ell$, $b = La$, $r = L^{m-m'}a$ and $\varpi(a^4) = \ell^4 \pi_{m'}$, this reads

$$0 \leq \pi_{m'} - \pi_m \leq c_L (\pi_{m'} - \pi_{m'+1}), \quad c_L := \left(\frac{2L-1}{L-1} \right)^4, \quad m > m' \geq \log_L \ell. \quad (6.6)$$

Thus reflection positivity alone turns a bound on the drop of the raw pressure between one pair of adjacent tori into a bound on the drop to every larger torus. It does not bound the first drop; that is where Balaban's construction is used (§6.3). Remark 6.17 draws the consequence for Theorem 6.13.

6.3 The vacuum constant

Definition. Let $\mathbb{T}$ be a torus of the family and $K = K(g_0, g)$. Each step $j < K$ of Balaban's construction on $\mathbb{T}$ produces field-independent numbers of three kinds:

- (α) the additive normalisation numbers $\mathcal{N}^{(j)}(\mathbb{T})$ of step j , built from $\log g_j$, from the logarithm of the gauge-fixing normalisation integral z_j , from the normalisation of the Gaussian integral and from constants of G , each multiplied by a lattice count of $\mathbb{T}$ [9, (1.15), (3.15), (3.33)];
- (β) the values $\mathbf{E}_{\mathbb{T}}^{(j+1)}(X; \mathbf{1}, z)$ at the unit configuration of the regular local terms of step $j+1$, that is, of the terms, localised in $X \in \mathcal{D}_{j+1}(\mathbb{T})$ with base point $z \in X$, into which the logarithm of the fluctuation integral of step $j+1$ on the whole torus is expanded [9, (2.26)–(2.27), (3.28)–(3.38)];
- (γ) the values $\mathbf{R}_{\mathbb{T}}^{(j+1)}(X; \mathbf{1})$ at the unit configuration of the remainders of step $j+1$ at the empty large-field label, $X \in \mathcal{D}_{j+1}(\mathbb{T})$ [9, (2.30)–(2.33)].

The vacuum constant is

$$E_{\mathbb{T}} := \sum_{j=0}^{K-1} \left[\mathcal{N}^{(j)}(\mathbb{T}) + \sum_{X \in \mathcal{D}_{j+1}} \sum_{z \in X} \mathbf{E}_{\mathbb{T}}^{(j+1)}(X; \mathbf{1}, z) + \sum_{X \in \mathcal{D}_{j+1}} \mathbf{R}_{\mathbb{T}}^{(j+1)}(X; \mathbf{1}) \right], \quad (6.7)$$

and Balaban's bare density is $\exp[-x_0 A(U) - E_{\mathbb{T}}]$. This makes explicit Balaban's description of the initial constant, which “is defined in fact as a sum of all such expressions for all the lattices $T^{(k)}$ as the small field regions” [9, p. 262]. The constant enters the effective densities only through the factor $e^{-E_{\mathbb{T}}}$; the construction may be run with $E_{\mathbb{T}} = 0$ and the constant fixed afterwards. We write $E_m := E_{\mathbb{T}_m}$ and $e_m := E_m/|\mathbb{T}_m|$.

The Gaussian normaliser. In Balaban's notation, let A be the fluctuation field of step k , $C^* \Delta^{(k)} C$ its quadratic form at the background U_{k+1} , and $z^{(k)}$ the factor produced by the elimination of the gauge-fixing δ -functions. The logarithm of the Gaussian integral of step k is split as [9, (3.33)]

$$\log \left[z^{(k)} \int dA e^{-\frac{1}{2} \langle A, C^* \Delta^{(k)} C A \rangle} \right] = \sum_c \log z^{(k)}(c) - \frac{1}{2} \log \det(C^* \Delta^{(k)} C) + \log \int dA e^{-\frac{1}{2} |A|^2},$$

the first sum running over the bonds c of the coarse lattice, $z^{(k)}$ being the product of one factor $z^{(k)}(c)$ per coarse bond; the last term is a lattice count times a constant and is put into the normalisation numbers. The determinant is written as [9, (3.36)]

$$-\frac{1}{2} \log \det(C^* \Delta^{(k)} C) = -\frac{1}{2} (\log \lambda_0) n_f - \frac{1}{2} \int_0^\infty \frac{d\lambda}{\lambda_0 + \lambda} \operatorname{Tr} (C^* \Delta^{(k)} C - \lambda_0 I) (C^* \Delta^{(k)} C + \lambda I)^{-1},$$

where $\lambda_0 > 0$ is a fixed number and n_f is the number of fluctuation bonds; the trace is distributed over the fluctuation bonds as a pointed density [9, (3.37)] and localised, together with the other contributions of the step, into the terms of kind (β) [9, (3.38) and p. 278]. Thus, apart from lattice counts multiplied by numbers that do not depend on the torus, no torus-wide number occurs among the constants (6.7).

Torus-independence of local terms. Fix $1 \leq j \leq K$ and work in level- j units, in which $\mathbb{T}_m$ has side $s = 2L^{m+K-j}$; let $p_s : \mathbb{R}^4 \rightarrow \mathbb{T}_m$ be the covering map. A set $X \subset \mathbb{T}_m$ is *non-wrapping* if it lies in a closed cube of side at most $s/4$; otherwise it *wraps*. A *lift* of a non-wrapping X is a set $\hat{X} \subset \mathbb{R}^4$, contained in a cube of side less than $s/2$, that p_s maps bijectively onto X ; two lifts differ by a translation in $s\mathbb{Z}^4$. We write $\mathcal{D}_j^\infty$ for Balaban's localisation domains of $\mathbb{R}^4$.

Lemma 6.7 (Torus-independence of local terms). *Let $1 \leq j \leq K$. There are functions $\mathbf{E}^{(j)}(\hat{X}; \cdot, \hat{z})$ and $\mathbf{r}^{(j)}(\hat{X}; \cdot)$, for $\hat{X} \in \mathcal{D}_j^\infty$ and $\hat{z} \in \hat{X}$, determined by $g_0, \dots, g_{j-1}$ and Balaban's parameters alone, such that for every $m \geq m_{\mathbb{T}}(g_-(g))$ and every non-wrapping $X \in \mathcal{D}_j(\mathbb{T}_m)$ with lift $\hat{X}$:*

- (i) *the regular local term $\mathbf{E}_{\mathbb{T}_m}^{(j)}(X; \cdot, z)$ of the fluctuation integral of step j on the whole torus, read on $\hat{X}$, equals $\mathbf{E}^{(j)}(\hat{X}; \cdot, \hat{z})$, where $\hat{z} \in \hat{X}$ is the lift of z ;*
- (ii) *the remainder $\mathbf{R}_{\mathbb{T}_m}^{(j)}(X; \cdot)$ at the empty large-field label, read on $\hat{X}$, equals $\mathbf{r}^{(j)}(\hat{X}; \cdot)$.*

In particular, the values of these terms at the unit configuration are the same numbers on every torus of the family.

Lemma 6.7 is proved in the companion account as part of clause (0); we give the statement and the structure of the proof. It is the lemma that makes the coefficient of each coupling increment, hence each running coupling g_j and the index $K(g_0, g)$, the same on every torus; for every $m \geq m_{\mathbb{T}}(g_-(g))$ and $K = K(g_0, g)$ it applies on $\mathbb{T}_m$ at every level $j \leq K$, because under (H5) the torus holds Balaban's coarsest cubes at every scale (Section 2). Each local term is a convergent cluster series whose index set and coefficients are determined by incidences among standard cubes of X . Its ingredients—random-walk expansions of the propagators and of the minimiser operators, built from inverses with Dirichlet conditions on cubes, local solutions of the nonlinear equations, characteristic functions on bonds, local densities, and the interpolation parameters attached to sub-domains—are each the solution of a problem posed inside X together with a collar of width at most $s/16$, with data restricted there. A non-wrapping X with such a collar lies in a cube of side less than $s/2$, on which p_s is injective and isometric and maps the standard lattices and partitions onto those

of the torus; hence every ingredient, and the term itself, is carried to the corresponding object on $\mathbb{R}^4$. Among the inputs of the proof is the statement that the localised terms of the fluctuation expansion in the small-field region coincide with those of the expansion on the whole lattice, which Balaban asserts without proof [9, p. 278] and which is proved in the companion account (group (B) of Subsection 2.4); for part (ii) the proof also uses his statement on the locality of the terms produced by the large-field renormalisation operation [11, pp. 379, 390], used as printed (Section 9). The lemma is proved together with clause (0), from the same inputs (Section 2).

Geometry of domains. We use three facts, in level- j units on a torus of side s :

- (g1) there are $\kappa(d)$ and K_0 , depending on the dimension only, such that $\sum_{X \in \mathcal{D}_j, X \ni z} e^{-\kappa(d)d_j(X)} \leq K_0$ for every point z , on every torus and on $\mathbb{R}^4$ [8, (1.26)]; Balaban's parameters satisfy $\kappa \geq 2\kappa(d)$;
- (g2) every $X \in \mathcal{D}_j^\infty$ lies in a closed cube of side $M(d_j(X) + 2)$, and every $X \in \mathcal{D}_j$ on a torus lies in the image under p_s of such a cube; a lift of a non-wrapping X has the same tree length as X ;
- (g3) if X wraps, then $d_j(X) > s/(4M) - 2$.

For (g2), let $\Upsilon \subset X$ be a shortest tree graph meeting all the cubes of X ; its length is $Md_j(X)$. Being contractible, it lifts to a connected set of the same length in $\mathbb{R}^4$, which lies in a closed cube of side $Md_j(X)$; every cube of side M of X meets Υ , so X lies in the image of the concentric closed cube of side $Md_j(X) + 2M$. A lift of a non-wrapping X lies in a cube of side less than $s/2$, on which p_s is isometric and maps the cubes of side M onto those of the torus; so tree lengths are preserved. For (g3): if $M(d_j(X) + 2) \leq s/4$, the image of that cube is a closed cube of side at most $s/4$ containing X , and X does not wrap.

Bounds at the unit configuration. By clause (i) (Section 2) and Balaban's inductive bounds [7, (1.18)], [9, (2.31)], for $1 \leq j \leq K$ and on every torus of the family,

$$|\mathbf{E}^{(j)}(X; \mathbf{1}, z)| \leq E_0 e^{-\kappa d_j(X)}, \quad |\mathbf{R}^{(j)}(X; \mathbf{1})| \leq g_j^{\kappa_0} e^{-\kappa d_j(X)} \leq e^{-\kappa d_j(X)}, \quad (6.8)$$

with E_0 the constant of [7, (1.18)], $\kappa_0 > 0$, and $g_j \leq \gamma < 1$.

Lemma 6.8 (Extensivity of the vacuum constant, one-sided). *For all $m \geq m' \geq m_{\mathbb{T}}(g_-(g))$,*

$$e_{m'} - e_m = \frac{E_{m'}}{(2L^{m'})^4} - \frac{E_m}{(2L^m)^4} \leq \bar{\delta}, \quad \bar{\delta} := 2(E_0 + 1)K_0 \sum_{u \geq 0} L^{4u} \exp \left[-\frac{\kappa}{2} \left(\frac{1}{2} L^{11} L^u - 2 \right) \right] < \infty. \quad (6.9)$$

The number $\bar{\delta}$ depends on L and Balaban's parameters only; it depends neither on K , g_0 , g , nor on m , m' .

In words: the density of the zero-source vacuum energy of the large torus is not smaller than that of the small torus by more than a constant free of the cutoff. Only this inequality is used below, and only the facts that $\bar{\delta}$ is finite and independent of K ; its size does not matter.

Proof. We prove the two-sided bound $|E_m - q^4 E_{m'}| \leq \bar{\delta} |\mathbb{T}_m|$, $q := L^{m-m'}$; dividing by $|\mathbb{T}_m| = q^4 |\mathbb{T}_{m'}|$ gives (6.9). Shifting the index in (6.7),

$$E_{\mathbb{T}} = \sum_{j=1}^K \left[\mathcal{N}^{(j-1)}(\mathbb{T}) + \sum_{X \in \mathcal{D}_j} \sum_{z \in X} \mathbf{E}_{\mathbb{T}}^{(j)}(X; \mathbf{1}, z) + \sum_{X \in \mathcal{D}_j} \mathbf{R}_{\mathbb{T}}^{(j)}(X; \mathbf{1}) \right].$$

(α) *Counts.* On a torus every lattice count entering [9, (1.15), (3.15), (3.33)]—of sites, bonds and blocks of a given level—is the number of sites of some level times an integer depending only on L ; a torus has no boundary, and every block partition divides it. The number of sites of any fixed level of $\mathbb{T}_m$ is q^4 times that of $\mathbb{T}_{m'}$. The coefficients are built from g_j, z_j and constants of G , and are the same on both tori by clause (0). Hence $\mathcal{N}^{(j-1)}(\mathbb{T}_m) = q^4 \mathcal{N}^{(j-1)}(\mathbb{T}_{m'})$ exactly.

(β) *Local terms.* Work at level j in level- j units: the sides are $s = 2L^{m+K-j} = qs'$ and $s' = 2L^{m'+K-j}$, and the covering maps satisfy $p_{s'} = \mathbf{q} \circ p_s$, with $\mathbf{q} : \mathbb{T}_m \rightarrow \mathbb{T}_{m'}$ the q^4 -fold covering. Call $\hat{X} \in \mathcal{D}_j^\infty$ *short* if it lies in a closed cube of side at most $s'/4$. If $\hat{X}$ is short, then $p_s \hat{X}$ and $p_{s'} \hat{X}$ are non-wrapping (as $s'/4 \leq s/4$), and Lemma 6.7(i) gives

$$\mathbf{E}_{\mathbb{T}_m}^{(j)}(p_s \hat{X}; \mathbf{1}, p_s \hat{z}) = \mathbf{E}^{(j)}(\hat{X}; \mathbf{1}, \hat{z}) = \mathbf{E}_{\mathbb{T}_{m'}}^{(j)}(p_{s'} \hat{X}; \mathbf{1}, p_{s'} \hat{z}).$$

The pairs (X', z') on $\mathbb{T}_{m'}$ with X' non-wrapping are the images under $p_{s'}$ of the short pairs $(\hat{X}, \hat{z})$, each image coming from exactly one $s'\mathbb{Z}^4$ -orbit of short pairs. An $s'\mathbb{Z}^4$ -orbit consists of exactly q^4 orbits of $s\mathbb{Z}^4$, and each of these is the set of lifts of exactly one pair $(p_s \hat{X}, p_s \hat{z})$ on $\mathbb{T}_m$. Therefore the sum over the pairs on $\mathbb{T}_m$ whose domain has a short lift is exactly q^4 times the sum over the pairs on $\mathbb{T}_{m'}$ whose domain does not wrap; no covariance of the terms is used. What remains is: on $\mathbb{T}_{m'}$, the wrapping X' , which have $d_j(X') > s'/(4M) - 2 =: t_j$ by (g3); on $\mathbb{T}_m$, the X that wrap, so that $d_j(X) > s/(4M) - 2 \geq t_j$, and the non-wrapping X without a short lift: by (g2) a lift of such an X lies in a closed cube of side $M(d_j(X) + 2)$, so $M(d_j(X) + 2) > s'/4$, that is, again $d_j(X) > t_j$. By (6.8), (g1), and $\kappa \geq 2\kappa(d)$ (so that $e^{-\kappa d} \leq e^{-\kappa t_j/2} e^{-\kappa(d)t_j/2}$ for $d > t_j$), for every point z

$$\sum_{X \ni z, d_j(X) > t_j} |\mathbf{E}^{(j)}(X; \mathbf{1}, z)| \leq E_0 K_0 e^{-\kappa t_j/2}.$$

(γ) *Remainders.* The same argument applies, with Lemma 6.7(ii) in place of (i), with each domain anchored at any one of its points (every term is then counted at least once, and the terms are bounded in modulus), and with $|\mathbf{R}^{(j)}(X; \mathbf{1})| \leq e^{-\kappa d_j(X)}$; the bound per point is $K_0 e^{-\kappa t_j/2}$.

Summation. There are $(2L^{m+K-j})^4 = L^{4(K-j)} |\mathbb{T}_m|$ points of level j on $\mathbb{T}_m$, and q^4 times as many as on $\mathbb{T}_{m'}$. The remaining terms of both tori therefore contribute at most $2(E_0 + 1)K_0 e^{-\kappa t_j/2} L^{4(K-j)} |\mathbb{T}_m|$ to $|E_m - q^4 E_{m'}|$ at level j . Put $u := K - j \in \{0, \dots, K-1\}$. Then $t_j = L^{m'+u}/(2M) - 2 \geq \frac{1}{2} L^{11} L^u - 2$, because $L^{m'} \geq L^{11} M$. Summing over the levels and majorising by the series over all $u \geq 0$,

$$|E_m - q^4 E_{m'}| \leq |\mathbb{T}_m| \cdot 2(E_0 + 1)K_0 \sum_{u \geq 0} L^{4u} e^{-\frac{\kappa}{2}(\frac{1}{2} L^{11} L^u - 2)} = \bar{\delta} |\mathbb{T}_m|.$$

The series converges, since L^u in the exponent dominates $4u \log L$, and it does not involve K, g_0, g, m or m' . $\square$

Remark 6.9. (i) For the raw pressure $\pi_{m''} = |\mathbb{T}_{m''}|^{-1} \log Z_{m''}(0)$, (S) and (F) at $v = 0$ give $\pi_{m''} \in [e_{m''} - C_{\text{low}}, e_{m''} + E_+^\sharp]$; with Remark 6.5(ii) this gives the other half, $e_m - e_{m'} \leq \mathbf{B}$, which is not used. Conversely, given (S) and (F) on both tori, (6.9) gives, and up to $\mathbf{B}$ is equivalent to,

$$0 \leq \pi_{m'} - \pi_m \leq \mathbf{B} + \bar{\delta}, \quad \text{that is,} \quad \log Z_m(0) \geq q^4 \log Z_{m'}(0) - (\mathbf{B} + \bar{\delta}) |\mathbb{T}_m| : \quad (6.10)$$

a near super-multiplicativity of the raw zero-source partition function along the family, uniform in the cutoff. This is the half of the volume dependence of the free energy that reflection positivity does not give: it gives the sign of $\pi_{m'} - \pi_m$, and Remark 6.6 reduces the range of volumes to one adjacent pair, but neither bounds the first drop. (ii) Lemma 6.8 concerns field-independent numbers

at zero source and at the unit configuration; it involves no measure, no large-field label, no energy with sources and no limit of volumes. (iii) A torus-wide, non-extensive number $c_j \geq c > 0$ at each level would make $e_{m'} - e_m$ of order $cK/|\mathbb{T}_{m'}|$, and the threshold of Theorem 6.24 proportional to K . The bookkeeping above excludes this: gauge fixing and the averaging constraints act block by block, the numbers of integrated modes are exact counts, and the Gaussian normaliser is a count plus a sum of local terms.

Remark 6.10 (Another bookkeeping of the Gaussian normaliser). In the bookkeeping of [7, (1.3)–(1.4)], used in the companion account, the Gaussian normaliser of step k is split off as a ratio, and its logarithm at the unit configuration is one number per level and per torus. By [9, (3.33)] this number is the sum of the constants into which the first display of the paragraph on the Gaussian normaliser splits it, so $E_{\mathbb{T}}$ is the same in both bookkeepings. In that bookkeeping one uses instead that this number is a lattice count times a torus-independent constant plus $-\frac{1}{2} \log \det \mathbf{Q}_{\mathbb{T}}^{(k)}$, where the fluctuation operator $\mathbf{Q}_{\mathbb{T}}^{(k)}$ at the unit configuration is self-adjoint, bounded below by a positive constant, exponentially local and invariant under the translations of a block lattice, with constants independent of k , K and the torus (proved in the companion account), together with the estimate

$$|L^{-4(m-m')} \log \det \mathbf{Q}_{\mathbb{T}_m}^{(k)} - \log \det \mathbf{Q}_{\mathbb{T}_{m'}}^{(k)}| \leq C_{\det} n'^4 e^{-\delta_{\det} n'},$$

n' the number of blocks per side of $\mathbb{T}_{m'}$ at level k and $C_{\det}, \delta_{\det} > 0$ depending on the constants of locality, size and lower bound only (a Neumann series for the logarithm, a Combes–Thomas bound on the powers, and the decomposition of closed walks by their winding numbers). This gives (6.9) with $\bar{\delta} + \bar{\delta}_{\det}$ in place of $\bar{\delta}$, where $\bar{\delta}_{\det}$ is again a convergent series independent of K . This alternative is proved in the companion account; the proof of Lemma 6.8 does not use it.

6.4 The volume-transfer theorem

Let $R > 0$, let ℓ be the least power of L with $\ell \geq 16(R+1)$, and put

$$m_{\star}(g) := \max\{m_{\mathbb{T}}(g-(g)), \log_L \ell\}.$$

Lemma 6.11 (Placement). *Let B be a closed ball of radius R in $\mathbb{R}^4$. There is a $a \in \varepsilon\mathbb{Z}^4$ such that $B+a$ lies in the interior of a cell, at Euclidean distance at least $R+2$ from all the hyperplanes $\{x_{\mu} \in \ell\mathbb{Z}\}$.*

Proof. Let y be the centre of B . In direction μ , the ball $B + te_{\mu}$ comes within distance $R+2$ of $\{x_{\mu} \in \ell\mathbb{Z}\}$ if and only if $\text{dist}(y_{\mu} + t, \ell\mathbb{Z}) < 2R+2$, that is, if and only if t lies in a certain open arc of length $4R+4$ of the circle $\mathbb{R}/\ell\mathbb{Z}$. The complement is a closed arc of length $\ell - 4R - 4 \geq 12(R+1) > \varepsilon$; it contains a point of $\varepsilon\mathbb{Z}/\ell\mathbb{Z}$, because $\ell \in \varepsilon\mathbb{Z}$. Choose such a t_{μ} in each direction and put $a := (t_{\mu})_{\mu}$. $\square$

Lemma 6.12 (Schwarz–Cauchy). *Let $F_1, \dots, F_n$ be real Lipschitz functions, none identically zero, and $\mathfrak{r} \in \{1, 2\}$. Let $\mathbf{h}$ be holomorphic on $\mathbb{D}_{\mathfrak{r}} = \{z \in \mathbb{C}^n : \rho(z) < 1/\mathfrak{r}\}$, with $\mathbf{h}(0) = 1$ and $|\mathbf{h}| \leq Q$ there, $Q \geq 1$. Then $|\mathbf{h}(z) - 1| \leq (Q+1)\mathfrak{r}\rho(z)$ on $\mathbb{D}_{\mathfrak{r}}$; on $\{\rho(z) \leq 1/(2\mathfrak{r}(Q+1))\}$ the function $\mathbf{h}$ has no zero, its principal logarithm $\text{Log } \mathbf{h}$ is holomorphic, and $|\text{Log } \mathbf{h}| \leq \log 2$; and*

$$|\partial_{z_1} \cdots \partial_{z_n} \text{Log } \mathbf{h}(0)| \leq \log 2 \cdot (2Q+2)^n \left(\frac{n\mathfrak{r}}{\varrho}\right)^n \prod_{i=1}^n \|F_i\|_{\sharp}. \quad (6.11)$$

Proof. For $z \in \mathbb{D}_{\mathfrak{r}} \setminus \{0\}$ the function $\phi(\lambda) := \mathbf{h}(\lambda z/(\mathfrak{r}\rho(z))) - 1$ is holomorphic on $|\lambda| < 1$ (the point $\lambda z/(\mathfrak{r}\rho(z))$ has $\rho = |\lambda|/\mathfrak{r}$, vanishes at 0 and is bounded by $Q+1$). By Schwarz’s lemma

$|\phi(\lambda)| \leq (Q+1)|\lambda|$; at $\lambda = \mathfrak{r}\rho(z)$ this is the first claim. On the open set $\{(Q+1)\mathfrak{r}\rho < 1\}$ we have $|\mathfrak{h} - 1| < 1$, so $\text{Log } \mathfrak{h}$ is holomorphic there, and $|\text{Log}(1+u)| \leq -\log(1-|u|) \leq \log 2$ for $|u| \leq \frac{1}{2}$. The closed polydisc $\mathcal{P} := \prod_i \{|z_i| \leq \rho_i\}$, $\rho_i := \varrho/(2\mathfrak{r}(Q+1)n\|F_i\|_\#)$, lies in $\{\rho \leq 1/(2\mathfrak{r}(Q+1))\}$, and Cauchy's estimate for the coefficient of $z_1 \cdots z_n$ gives $|\partial_{z_1} \cdots \partial_{z_n} \text{Log } \mathfrak{h}(0)| \leq \log 2 / \prod_i \rho_i$, which is (6.11). $\square$

Theorem 6.13 (Volume transfer). *Let $g \leq \gamma_J$, $g_0 \in (0, g_-(g)]$, $K = K(g_0, g)$, $R > 0$, and $m \geq m' \geq m_*(g)$. For all real Lipschitz functions F_1, F_2 on $\mathbb{R}^4$ such that $\text{supp } F_1 \cup \text{supp } F_2$ is contained in one closed ball of radius R —with no condition on their signs, on the number of components of their supports, or on the distance or overlap of the supports—*

$$|S_{2;K,m}^T(F_1, F_2) - S_{2;K,m'}^T(F_1, F_2)| \leq C_{\text{VT}} \|F_1\|_\# \|F_2\|_\#, \quad (6.12)$$

$$C_{\text{VT}} := 2 \log 2 \cdot (2Q_* + 2)^2 \left(\frac{2\mathfrak{r}}{\varrho} \right)^2, \quad Q_* := \exp[(B + \bar{\delta})\ell^4 + 4C_{\text{face}}(R)],$$

with $B = E_+^\# + C_{\text{low}}$ (Proposition 6.1), $\bar{\delta}$ as in Lemma 6.8, $C_{\text{face}}(R)$ as in Proposition 6.2(ii), and $\mathfrak{r}$ as in §6.1. The constant C_{VT} depends only on L , Balaban's parameters, g , ϱ and R ; it depends neither on K , g_0 , m , m' , nor on the position of the ball, nor on the supports.

If $K < b_*$, the bound is elementary. The bare spacing is then $\varepsilon > L^{-b_*}$; for each of the six orientations the barycentres of the plaquettes form a translate of $\varepsilon\mathbb{Z}^4$, so at most $6(2RL^{b_*} + 1)^4$ plaquettes have their barycentre in the ball, and $|\mathcal{O}(F_i)| \leq 12(2RL^{b_*} + 1)^4 \|F_i\|_\infty$ because $0 \leq a_p \leq 2$. Hence $|S_{2;K,m''}^T(F_1, F_2)| \leq 2(12(2RL^{b_*} + 1)^4)^2 \|F_1\|_\infty \|F_2\|_\infty$ on both tori, and (6.12) holds with C_{VT} replaced by $\max\{C_{\text{VT}}, 4(12(2RL^{b_*} + 1)^4)^2\}$, again free of K , g_0 , m , m' and the position; we use this maximum below without changing the notation. In Theorem 6.24, $K \geq s(\mathbf{d}) + k_W \geq b_*(g)$ holds anyway, since $s_W(g) \geq b_*(g)$. In the proof we assume $K \geq b_*$.

Proof. Write $w_z := z_1 F_1 + z_2 F_2$, $\rho(z) = \varrho^{-1}(|z_1| \|F_1\|_\# + |z_2| \|F_2\|_\#)$ (norms on $\mathbb{R}^4$) and $\mathbb{D}_\mathfrak{r} = \{z \in \mathbb{C}^2 : \rho(z) < 1/\mathfrak{r}\}$. We may assume $F_1, F_2 \not\equiv 0$. Since $2R < \ell \leq L^{m'} \leq L^m$, both functions are transported to both tori, and their norms there are at most $\|F_i\|_\#$.

Step 0: placement and the mirror-periodic weight. By Lemma 6.11 there is $a \in \varepsilon\mathbb{Z}^4$ such that the ball, translated by a , lies in the interior of a cell c at distance at least $R+2$ from all faces. The grid of cells is common to $\mathbb{R}^4$, $\mathbb{T}_m$ and $\mathbb{T}_{m'}$ (since ℓ divides $L^{m'}$, which divides L^m), so this one translation serves both tori. It changes neither side of (6.12): transport commutes with translations, $\mathcal{O}(F(\cdot - a))(U) = \mathcal{O}(F)(U \circ \tau_a)$ for the lattice translation τ_a by a , and both bare Wilson measures are invariant under $U \mapsto U \circ \tau_a$. We assume this placement from now on. On $\mathbb{R}^4$ put

$$W_z := \sum_{\omega \in \mathfrak{G}_\ell} (w_z \circ \omega^{-1})^{\langle \text{sg}(\omega) \rangle}, \quad (6.13)$$

a sum of functions with pairwise disjoint supports, one in each cell. W_z is mirror-periodic for every $z \in \mathbb{C}^2$. For (M1): $\theta_{\mu,a}\mathfrak{G}_\ell = \mathfrak{G}_\ell$ and $\text{sg}(\theta_{\mu,a}\omega) = -\text{sg}(\omega)$, so, with $\omega' := \theta_{\mu,a}\omega$, $W_z \circ \theta_{\mu,a} = \sum_{\omega'} (w_z \circ \omega'^{-1})^{\langle -\text{sg}(\omega') \rangle} = \overline{W_z}$. For (M2): a plaquette containing a bond that crosses a face hyperplane has its barycentre on that hyperplane, where every copy vanishes. Hence W_z descends to $\mathbb{T}_{m''}$ for every $m'' \geq \log_L \ell$, where it equals $\sum_{\omega \in \mathfrak{G}_\ell(\mathbb{T}_{m''})} (w_z \circ \omega^{-1})^{\langle \text{sg}(\omega) \rangle}$, the translations by $2L^{m''}\mathbb{Z}^4$ having $\text{sg} = +1$. The supremum of W_z is that of w_z . Its Lipschitz constant, on $\mathbb{R}^4$ and hence on the tori, is at most that of w_z : for points in two different closed cells, the segment joining them leaves the first cell and enters the second at points of faces, where all copies vanish, so the increment is at most $\text{Lip } w_z$ times the length of the two end pieces. Thus

$$\|W_z\|_\# \leq |z_1| \|F_1\|_\# + |z_2| \|F_2\|_\# = \varrho \rho(z).$$

Step 1: the chessboard estimate on the large torus. The function $F := e^{\mathcal{O}(w_z)}$ depends only on the plaquettes whose barycentre lies in the ball; these lie within distance 2ε of the ball, hence their bonds lie in c° , and $F \in \mathfrak{A}_c$ on $\mathbb{T}_m$. For $\omega \in \mathfrak{G}_\ell(\mathbb{T}_m)$, $F(U \circ \omega) = \exp \sum_p w_z(x_p) a_{\omega p}(U) = e^{\mathcal{O}(w_z \circ \omega^{-1})(U)}$, and $\overline{e^{\mathcal{O}(v)}} = e^{\mathcal{O}(\bar{v})}$ because a_p is real. Hence $F^{[\omega c]} = e^{\mathcal{O}((w_z \circ \omega^{-1})^{\langle \text{sg}(\omega) \rangle})}$ and $\prod_{c'} F^{[c']} = e^{\mathcal{O}(W_z)}$. Proposition 6.2(i) gives

$$\frac{|Z_m(w_z)|}{Z_m(0)} = |\langle F \rangle_m| \leq \left(\frac{Z_m(W_z)}{Z_m(0)} \right)^{1/|C_m|}, \quad (6.14)$$

the ratio on the right being real and non-negative.

Step 2: from the large torus to the small one, by reflection positivity only. Let

$$\mathbf{n} := (2L^{m'}/\ell)(1, 1, 1, 1) \in (2\mathbb{N})^4$$

and $q := L^{m-m'}$, so that $\mathbb{T}[\mathbf{n}] = \mathbb{T}_{m'}$ and $\mathbb{T}[q\mathbf{n}] = \mathbb{T}_m$. Lemma 6.4 with $k = q$ gives $0 \leq Z_m(W_z) \leq Z_{m'}(W_z)^{q^4}$. As $|C_m| = q^4|C_{m'}|$ and $t \mapsto t^{1/|C_m|}$ is increasing on $[0, \infty)$,

$$Z_m(W_z)^{1/|C_m|} \leq Z_{m'}(W_z)^{1/|C_{m'}|}. \quad (6.15)$$

Step 3: the upper stability bound on the small torus, the lower one at zero source on the large torus, and the vacuum constant. Fix $z \in \mathbb{D}_r$. On $\mathbb{T}_{m'}$, (B) and (S) are available at the weight W_z . If they hold on all of $\mathcal{W}$, this follows from $\|W_z\|_\# \leq \varrho\rho(z) < \varrho$. In the form displayed in Proposition 6.1, write

$$W_z = \sum_{i=1,2} (z_i F_i^+ + \bar{z}_i F_i^-), \quad F_i^\pm := \sum_{\omega: \text{sg}(\omega)=\pm 1} F_i \circ \omega^{-1},$$

real Lipschitz functions with $\|F_i^\pm\|_\# \leq \|F_i\|_\#$ by the argument of Step 0; then W_z is the weight of the family $(F_1^+, F_2^+, F_1^-, F_2^-)$ at the parameter $(z_1, z_2, \bar{z}_1, \bar{z}_2)$, which lies in its polydisc whenever $2\rho(z) < 1$, in particular for $z \in \mathbb{D}_2$. Since $Z_{m'}(W_z) \geq 0$ (Lemma 6.4), the definition of $\tilde{Z}_{m'}$ and (S) give

$$Z_{m'}(W_z) = |e^{E_{m'} + \mathcal{E}_{m'}(W_z)} \tilde{Z}_{m'}(W_z)| \leq \exp [E_{m'} + \text{Re } \mathcal{E}_{m'}(W_z) + E_+^\# |\mathbb{T}_{m'}|];$$

this is an identity followed by one bound by moduli, and the modulus of $\mathcal{E}_{m'}$ alone is never taken. On $\mathbb{T}_m$ at zero source, (F) gives $Z_m(0) = e^{E_m} \tilde{Z}_m(0) \geq \exp[E_m - C_{\text{low}}|\mathbb{T}_m|]$; (F) holds on every torus of the family. Combining with (6.14), (6.15) and $|\mathbb{T}_{m'}|/|C_{m''}| = \ell^4$,

$$\frac{|Z_m(w_z)|}{Z_m(0)} \leq \exp \left\{ |C_{m'}|^{-1} \text{Re } \mathcal{E}_{m'}(W_z) + \mathbf{B}\ell^4 + \ell^4 [e_{m'} - e_m] \right\},$$

and by Lemma 6.8

$$\frac{|Z_m(w_z)|}{Z_m(0)} \leq \exp \left\{ |C_{m'}|^{-1} \text{Re } \mathcal{E}_{m'}(W_z) + (\mathbf{B} + \bar{\delta})\ell^4 \right\} \quad (z \in \mathbb{D}_r). \quad (6.16)$$

The two vacuum constants live on different tori, yet they enter only as ℓ^4 times the difference of their densities, that is, per cell: the trace inequality is homogeneous of the right degree, and the root $1/|C|$ of the chessboard estimate makes both members of the pair of stability bounds intensive. For $m = m'$ Step 2 is void and the two vacuum constants cancel exactly: (6.16) holds with $\bar{\delta}$ replaced by 0.

Step 4: one holomorphic counterterm, formed on the small torus. On $\mathbb{T}_{m'}$ split $\mathcal{D}_j = \sqcup_{c'} \mathcal{X}_{c'} \sqcup \mathcal{X}_\partial$ as in Proposition 6.2(ii), and put, with the numbers e_j of $\mathbb{T}_{m'}$,

$$\mathcal{E}_{c'}^{\text{bulk}}(v) := \sum_{j \leq K'} \sum_{X \in \mathcal{X}_{c'}} e_j(X; v), \quad \mathcal{E}^{\text{bulk}}(v) := \sum_{c'} \mathcal{E}_{c'}^{\text{bulk}}(v), \quad \mathcal{E}^\partial(v) := \sum_{j \leq K'} \sum_{X \in \mathcal{X}_\partial} e_j(X; v),$$

so that $\mathcal{E}_{m'} = \mathcal{E}^{\text{bulk}} + \mathcal{E}^\partial$. Define, for $z \in \mathbb{D}_1$,

$$\mathcal{E}''(z) := |\mathcal{C}_{m'}|^{-1} \sum_{\omega \in \mathfrak{G}_\ell(\mathbb{T}_{m'})} \left[\mathcal{E}_{\omega c}^{\text{bulk}}((w_z \circ \omega^{-1})^{\langle \text{sg}(\omega) \rangle}) \right]^{\langle \text{sg}(\omega) \rangle} + \mathcal{E}^\partial(w_z). \quad (6.17)$$

At $m = m'$ this is the re-centred counterterm of the per-ball bound of Section 3.

Holomorphy. The sum is finite. For $\text{sg}(\omega) = +1$ the summand is $\mathcal{E}_{\omega c}^{\text{bulk}}(z_1 F_1 \circ \omega^{-1} + z_2 F_2 \circ \omega^{-1})$, holomorphic on $\mathbb{D}_1$ by (A) applied to the family $(F_1 \circ \omega^{-1}, F_2 \circ \omega^{-1})$, which has the same norms. For $\text{sg}(\omega) = -1$, since the F_i are real, the summand is $\Phi_\omega(\bar{z})$ with $\Phi_\omega(\zeta) := \mathcal{E}_{\omega c}^{\text{bulk}}(\zeta_1 F_1 \circ \omega^{-1} + \zeta_2 F_2 \circ \omega^{-1})$ holomorphic on the conjugation-invariant domain $\mathbb{D}_1$; then $z \mapsto \overline{\Phi_\omega(\bar{z})}$ is holomorphic there, its power series having the conjugate coefficients. $\mathcal{E}^\partial(w_z)$ is holomorphic by (A). As $e_j(X; 0) = 0$, $\mathcal{E}''(0) = 0$. The distance of the supports plays no role here.

Real part. Since $\text{Re } \tilde{\xi} = \text{Re } \xi$,

$$\text{Re } \mathcal{E}''(z) = |\mathcal{C}_{m'}|^{-1} \sum_{\omega} \text{Re } \mathcal{E}_{\omega c}^{\text{bulk}}((w_z \circ \omega^{-1})^{\langle \text{sg}(\omega) \rangle}) + \text{Re } \mathcal{E}^\partial(w_z).$$

If $X \in \mathcal{X}_{\omega c}$, every plaquette of $\text{plaq}(X^+)$ lies within distance 2ε of $(\omega c)^\circ$, hence at distance more than R from every copy in (6.13) other than the one in ωc ; by (L), $e_j(X; W_z) = e_j(X; (w_z \circ \omega^{-1})^{\langle \text{sg}(\omega) \rangle})$. Summing over $X \in \mathcal{X}_{\omega c}$, over $j \leq K'$, and over ω —every cell is ωc for exactly one ω —we obtain the identity

$$\text{Re } \mathcal{E}''(z) = |\mathcal{C}_{m'}|^{-1} \text{Re } \mathcal{E}^{\text{bulk}}(W_z) + \text{Re } \mathcal{E}^\partial(w_z) \quad (z \in \mathbb{D}_\tau). \quad (6.18)$$

Faces. By Proposition 6.2(ii) on $\mathbb{T}_{m'}$, with one ball copy per cell for W_z and one ball for w_z , and with (B) at both weights, $|\mathcal{C}_{m'}|^{-1} |\mathcal{E}^\partial(W_z)| \leq C_{\text{face}}(R)$ and $|\mathcal{E}^\partial(w_z)| \leq C_{\text{face}}(R)$. With (6.18),

$$|\mathcal{C}_{m'}|^{-1} \text{Re } \mathcal{E}_{m'}(W_z) - \text{Re } \mathcal{E}''(z) = |\mathcal{C}_{m'}|^{-1} \text{Re } \mathcal{E}^\partial(W_z) - \text{Re } \mathcal{E}^\partial(w_z) \leq 2C_{\text{face}}(R). \quad (6.19)$$

The two normalised quotients. For $m'' \in \{m, m'\}$ put

$$\mathbf{h}_{m''}(z) := e^{-\mathcal{E}''(z)} \frac{Z_{m''}(w_z)}{Z_{m''}(0)}. \quad (6.20)$$

Both are holomorphic on $\mathbb{D}_\tau$ — $Z_{m''}(w_z)$ is the integral over a compact space of an integrand entire in z , and $Z_{m''}(0) > 0$ —and $\mathbf{h}_{m''}(0) = 1$ exactly, each generating function being divided by its own value at zero. By (6.16) and (6.19),

$$|\mathbf{h}_m(z)| = e^{-\text{Re } \mathcal{E}''(z)} \frac{|Z_m(w_z)|}{Z_m(0)} \leq \exp[(B + \bar{\delta})\ell^4 + 2C_{\text{face}}(R)] \leq Q_\star \quad (z \in \mathbb{D}_\tau),$$

and, by the case $m = m'$ of (6.16), $|\mathbf{h}_{m'}| \leq \exp[B\ell^4 + 2C_{\text{face}}(R)] \leq Q_\star$ on $\mathbb{D}_\tau$.

Dividing the generating function of the large torus by the counterterm formed on the small torus is legitimate: for every function $\mathcal{E}^\natural$ holomorphic near 0 with $\mathcal{E}^\natural(0) = 0$, near $z = 0$

$$\log Z_{m''}(w_z) = \log Z_{m''}(0) + \mathcal{E}^\natural(z) + \text{Log} \left[e^{-\mathcal{E}^\natural(z)} \frac{Z_{m''}(w_z)}{Z_{m''}(0)} \right],$$

both sides being holomorphic near 0, equal at 0, and with the same exponential. Nothing is asked of $\mathcal{E}''$ on $\mathbb{T}_m$: no covariance, no independence of the torus of energies with sources, no interpretation. The vacuum constants enter only (6.16), and only at zero source.

Step 5: Schwarz–Cauchy; the counterterm cancels. Near $z = 0$, $\log Z_{m''}(w_z) = \log Z_{m''}(0) + \mathcal{E}''(z) + \text{Log } h_{m''}(z)$ for $m'' \in \{m, m'\}$. Hence $S_{2;K,m''}^T(F_1, F_2) = \partial_{z_1} \partial_{z_2} \mathcal{E}''(0) + \partial_{z_1} \partial_{z_2} \text{Log } h_{m''}(0)$, with the same first term for both tori, and we have the identity

$$S_{2;K,m}^T(F_1, F_2) - S_{2;K,m'}^T(F_1, F_2) = \partial_{z_1} \partial_{z_2} \text{Log } h_m(0) - \partial_{z_1} \partial_{z_2} \text{Log } h_{m'}(0). \quad (6.21)$$

Lemma 6.12 with $n = 2$ and $Q = Q_\star$ bounds each term on the right by $\log 2 \cdot (2Q_\star + 2)^2 (2\mathfrak{r}/\varrho)^2 \|F_1\|_\# \|F_2\|_\#$, and (6.12) follows. $\square$

Remark 6.14 (Where each independence of C_{VT} comes from). *Cutoff.* $E_+^\#$, C_{low} do not depend on K , g_0 , m ; $C_{\text{face}}(R)$ and $\bar{\delta}$ are series over all levels; the quantities that diverge as $K \rightarrow \infty$ — E_m , $E_{m'}$, $\text{Re } \mathcal{E}_{m'}(W_z)$, $\partial_{z_1} \partial_{z_2} \mathcal{E}''(0)$ —occur only in exact cancellations: the vacuum constants through the difference of their densities, the bulk of $\mathcal{E}_{m'}(W_z)$ against $\text{Re } \mathcal{E}''$ by (6.18), and $\partial_{z_1} \partial_{z_2} \mathcal{E}''(0)$ against itself in (6.21). *Bare coupling.* It enters only the bare measures, and the chessboard estimate and Lemma 6.4 hold for every $x_0 > 0$. *Volumes.* $\mathbb{T}_m$ is read through exact statements (chessboard estimate, trace formula, $|\mathcal{C}_m| = q^4 |\mathcal{C}_{m'}|$), the constant C_{low} per unit volume, and the density $E_m/|\mathbb{T}_m|$; $\mathbb{T}_{m'}$ through constants per unit volume. *Position.* One bare translation serves both tori. *Separation and overlap of the supports.* They never enter: the one quantity that reads the separation, $\partial_{z_1} \partial_{z_2} \mathcal{E}''(0)$, cancels in (6.21). This is used in Lemma 6.18, whose arrays interleave; and it is why that lemma cannot be applied to S_2^T itself with the per-ball bound of Section 3, whose constant reads the separation through $\partial_{z_1} \partial_{z_2} \mathcal{E}''(0)$, where the two-point function lives. *Coupling.* g is read through B (and through ℓ if a cell side depending on g is preferred).

Remark 6.15 (How one-sided bounds give a two-sided bound). Nothing is differentiated through an inequality: (6.21) is an identity, because one holomorphic $\mathcal{E}''$ serves both volumes, and each of its terms is bounded in modulus by Schwarz’s lemma and Cauchy’s estimate from the upper bound $\sup_{\mathbb{D}_\mathfrak{r}} |h_{m''}| \leq Q_\star$ on a complex polydisc and the exact value $h_{m''}(0) = 1$. The maximum principle does the work that a lower bound would otherwise do. On a real segment the inference would be false: $h_A(t) = e^{-At^2}$ satisfies $0 < h_A \leq 1$ on $\mathbb{R}$ while $(\log h_A)''(0) = -2A$ is arbitrary; on $\{|z| \leq r\}$, however, $\sup |h_A| = e^{Ar^2}$, and a bound by Q forces $A \leq r^{-2} \log Q$. It would also be false with one counterterm for each torus: two Gaussian fields with different stiffnesses each satisfy a bound of the type of Section 3 with their own counterterm, and their two-point functions differ by a non-zero scale-invariant amount. The only lower bounds used are at $z = 0$ —(F) on $\mathbb{T}_m$ and Lemma 6.8, statements of ultraviolet stability at zero source—and all inequalities used (chessboard estimate, trace inequality, (S), (F) through $1/Z_m(0)$, Lemma 6.8, face bound) point in the direction of an upper bound on $|Z_m(w_z)|/Z_m(0)$. No lower bound on $|Z_m(w_z)|$ at $z \neq 0$, no sign of the difference and no second use of the trace inequality is needed; the absence of zeros of $h_{m''}$ near 0 is a conclusion of Lemma 6.12.

Remark 6.16 (Variants). (i) The proof gives (6.12) with $2C_{\text{face}}(R)$ in place of $4C_{\text{face}}(R)$ in $Q_\star$; the larger constant is kept in the definition. Using $\partial_1 \partial_2 \text{Log } h(0) = \partial_1 \partial_2 h(0) - \partial_1 h(0) \partial_2 h(0)$ and Cauchy’s estimate for h itself on the polydisc $\{|z_i| < \varrho/(2\mathfrak{r}\|F_i\|_\#)\} \subset \mathbb{D}_\mathfrak{r}$ gives the smaller admissible constant $2Q_\star(1 + Q_\star)(2\mathfrak{r}/\varrho)^2$. (ii) For $n \geq 2$ sources supported in one ball of radius R , the same proof gives $|S_{n;K,m}^T(F_1, \dots, F_n) - S_{n;K,m'}^T(F_1, \dots, F_n)| \leq C_{\text{VT},n} \prod_i \|F_i\|_\#$ with $C_{\text{VT},n} := 2 \log 2 \cdot (2Q_{\star,n} + 2)^n (n\mathfrak{r}/\varrho)^n$, $Q_{\star,n} := \exp[(B + \bar{\delta})\ell_n^4 + 2nC_{\text{face}}(R)]$, and ℓ_n the least power of L with $\ell_n \geq 8n(R + 1)$.

Remark 6.17 (Only reflection positivity beyond the next torus). Suppose that Balaban’s statements are used on $\mathbb{T}_{m'}$ and $\mathbb{T}_{m'+1}$ only: (L), (A), (B), (S) on $\mathbb{T}_{m'}$, (F) on $\mathbb{T}_{m'}$ and on $\mathbb{T}_{m'+1}$, and

Lemma 6.8 for the pair $(m', m' + 1)$. Then (6.12) holds for every $m \geq m'$, with $Q_\star$ replaced by $\exp[(1 + c_L)(B + \bar{\delta})\ell^4 + 4C_{\text{face}}(R)]$. Indeed, in Step 3 write

$$Z_{m'}(W_z)^{1/|C_{m'}|} Z_m(0)^{-1/|C_m|} = \left(\frac{Z_{m'}(W_z)}{Z_{m'}(0)} \right)^{1/|C_{m'}|} e^{\ell^4(\pi_{m'} - \pi_m)}.$$

The first factor is at most $\exp\{|C_{m'}|^{-1} \text{Re } \mathcal{E}_{m'}(W_z) + B\ell^4\}$ by (S) and (F) on $\mathbb{T}_{m'}$; and $\pi_{m'} - \pi_m \leq c_L(\pi_{m'} - \pi_{m'+1}) \leq c_L(B + \bar{\delta})$ by (6.6) and (6.10) for the pair $(m', m' + 1)$. Steps 4 and 5 are unchanged. On every torus $\mathbb{T}_m$ with $m \geq m' + 2$, then, nothing of Balaban's construction is used, only reflection positivity of the bare measure.

6.5 Averaging over translated arrays

Lemma 6.18 (Averaging). *Let D be a real bilinear form on the space of compactly supported real Lipschitz functions on $\mathbb{R}^4$ such that*

- (a) $D(F_1(\cdot - \tau), F_2(\cdot - \tau)) = D(F_1, F_2)$ for all $\tau \in \varepsilon\mathbb{Z}^4$;
- (b) $|D(F_1, F_2)| \leq C \|F_1\|_\# \|F_2\|_\#$ whenever $\text{supp } F_1 \cup \text{supp } F_2$ is contained in one closed ball of radius R .

Let f_1, f_2 be real Lipschitz functions with $\text{supp } f_1 \cup \text{supp } f_2 \subset \bar{B}(y, r_0)$, where $\varepsilon \leq r_0 \leq R/8$. Then

$$|D(f_1, f_2)| \leq C \min \left\{ 1, \left(\frac{32r_0}{R} \right)^4 \right\} \|f_1\|_\# \|f_2\|_\#. \quad (6.22)$$

Proof. Geometry and count. Let $r'_0 := \varepsilon \lceil r_0/\varepsilon \rceil$; then $r'_0 \in \varepsilon\mathbb{N}$ and $r_0 \leq r'_0 < r_0 + \varepsilon \leq 2r_0$. Let $\Gamma := \{\tau \in 4r'_0\mathbb{Z}^4 : |\tau|_\infty \leq R/4\} \subset \varepsilon\mathbb{Z}^4$. Then

$$\#\Gamma = (2\lfloor R/(16r'_0) \rfloor + 1)^4 \geq \max \{1, (R/(32r_0))^4\},$$

because $2\lfloor x \rfloor + 1 \geq x$ for $x \geq 0$ and $R/(16r'_0) > R/(32r_0)$. The closed balls $B_\tau := \bar{B}(y + \tau, r_0)$, $\tau \in \Gamma$, are pairwise at Euclidean distance at least $|\tau - \tau'|_\infty - 2r_0 \geq 4r'_0 - 2r_0 \geq 2r_0 > 0$, and all lie in $\bar{B}(y, R/2 + r_0) \subset \bar{B}(y, R)$, since $|\tau| \leq 2|\tau|_\infty \leq R/2$ in four dimensions and $r_0 \leq R/8$.

Gluing: an array has the norm of one copy. For $\sigma \in \{\pm 1\}^\Gamma$ put $F_i^\sigma := \sum_{\tau \in \Gamma} \sigma_\tau f_i(\cdot - \tau)$. The summands have supports in the disjoint balls B_τ , so $\|F_i^\sigma\|_\infty = \|f_i\|_\infty$. Since f_i is continuous and vanishes off $\bar{B}(y, r_0)$, it vanishes on the sphere $\partial B(y, r_0)$; so F_i^σ vanishes outside the open balls B_τ° . If x, x' lie in one closed ball B_τ , then $|F_i^\sigma(x) - F_i^\sigma(x')| = |f_i(x - \tau) - f_i(x' - \tau)| \leq \text{Lip } f_i |x - x'|$. Otherwise, up to exchanging x and x' , either neither point lies in an open ball, and both values vanish, or $x \in B_\tau^\circ$ and $x' \notin B_\tau$. In the latter case let x^* be the point where the segment $[x, x']$ leaves B_τ ; if moreover $x' \in B_{\tau'}^\circ$, let x'^* be the point where the segment enters $B_{\tau'}$, and otherwise $x'^* := x'$. Then x, x^*, x'^*, x' lie on the segment in this order, $F_i^\sigma(x^*) = F_i^\sigma(x'^*) = 0$, and

$$|F_i^\sigma(x) - F_i^\sigma(x')| \leq \text{Lip } f_i (|x - x^*| + |x'^* - x'|) \leq \text{Lip } f_i |x - x'|.$$

Hence $\text{Lip } F_i^\sigma \leq \text{Lip } f_i$ and $\|F_i^\sigma\|_\# \leq \|f_i\|_\#$.

Exact sign cancellation. For $\tau \neq \tau'$ the involution of $\{\pm 1\}^\Gamma$ that flips σ_τ alone reverses the sign of $\sigma_\tau \sigma_{\tau'}$; hence $2^{-\#\Gamma} \sum_\sigma \sigma_\tau \sigma_{\tau'} = \delta_{\tau\tau'}$, a finite average. By bilinearity and (a), which applies

because $\Gamma \subset \varepsilon\mathbb{Z}^4$,

$$\begin{aligned}
& 2^{-\#\Gamma} \sum_{\sigma} D(F_1^{\sigma}, F_2^{\sigma}) \\
&= \sum_{\tau, \tau' \in \Gamma} \left(2^{-\#\Gamma} \sum_{\sigma} \sigma_{\tau} \sigma_{\tau'} \right) D(f_1(\cdot - \tau), f_2(\cdot - \tau')) \\
&= \sum_{\tau \in \Gamma} D(f_1(\cdot - \tau), f_2(\cdot - \tau)) \\
&= \#\Gamma \cdot D(f_1, f_2).
\end{aligned}$$

The cross terms cancel exactly, and translation invariance makes the diagonal terms equal to the value on the given pair. On the other hand, for every σ the functions $F_1^{\sigma}, F_2^{\sigma}$ are supported in $\bar{B}(y, R)$ and $\|F_i^{\sigma}\|_{\#} \leq \|f_i\|_{\#}$, so (b) gives $|D(F_1^{\sigma}, F_2^{\sigma})| \leq C\|f_1\|_{\#}\|f_2\|_{\#}$. Therefore $\#\Gamma |D(f_1, f_2)| \leq C\|f_1\|_{\#}\|f_2\|_{\#}$, which with the count gives (6.22). $\square$

Corollary 6.19 (Volume transfer at short distance). *Let $g \leq \gamma_J$, $g_0 \in (0, g_-(g)]$, $K = K(g_0, g)$, $m \geq m' \geq m_{\mathbb{T}}(g_-(g))$ and $\vartheta \in (0, 1]$. For $0 < \mathbf{d} \leq 1/40$ with $\varepsilon \leq \mathbf{d}$, and for real Lipschitz functions f_1, f_2 on $\mathbb{R}^4$, of any signs and in any mutual position, with $\text{supp } f_1 \cup \text{supp } f_2 \subset \bar{B}(y, 5\mathbf{d})$ for some y and $\text{Lip } f_i \leq \|f_i\|_{\infty}/(\vartheta\mathbf{d})$,*

$$|S_{2;K,m}^T(f_1, f_2) - S_{2;K,m'}^T(f_1, f_2)| \leq C_T(g) \mathbf{d}^2 \|f_1\|_{\infty} \|f_2\|_{\infty}, \quad C_T(g) := 160^4 \left(\frac{40M}{\vartheta} \right)^2 C_{\text{VT}}(g), \quad (6.23)$$

where $C_{\text{VT}}(g)$ is the constant of Theorem 6.13 at $R = 1$. The constant $C_T(g)$ depends only on L , Balaban's parameters, g , ϱ and ϑ ; it depends neither on K , g_0 , m , m' , nor on $\mathbf{d}$ or y .

Proof. At $R = 1$ the cell side ℓ is the least power of L that is at least 32, so $\ell \leq 32L < L^{11} \leq L^{11}M \leq L^{m_{\mathbb{T}}(g_-(g))}$; hence $m_{\star}(g) = m_{\mathbb{T}}(g_-(g))$, and Theorem 6.13 applies to every pair $m \geq m' \geq m_{\mathbb{T}}(g_-(g))$. Let $D(F_1, F_2) := S_{2;K,m}^T(F_1^{(m)}, F_2^{(m)}) - S_{2;K,m'}^T(F_1^{(m')}, F_2^{(m')})$ for compactly supported real Lipschitz F_1, F_2 . It is real and bilinear, since each $S_{2;K,m''}^T$ is a covariance and transport is linear. It satisfies (a) of Lemma 6.18, since transport commutes with translations and both bare Wilson measures are invariant under the lattice translations by vectors of $\varepsilon\mathbb{Z}^4$. It satisfies (b) with $R = 1$ and $C = C_{\text{VT}}(g)$ by Theorem 6.13, for all real Lipschitz pairs in one closed ball of radius 1, signed or not, overlapping or not. Lemma 6.18 with $r_0 := 5\mathbf{d}$ —admissible, since $\varepsilon \leq \mathbf{d} \leq r_0$ and $r_0 \leq 1/8$ —gives $|D(f_1, f_2)| \leq C_{\text{VT}}(g)(160\mathbf{d})^4 \|f_1\|_{\#} \|f_2\|_{\#}$. Finally, $\|f_i\|_{\#} = \max\{\|f_i\|_{\infty}, 40M \text{Lip } f_i\} \leq \max\{1, 40M/(\vartheta\mathbf{d})\} \|f_i\|_{\infty} = (40M/(\vartheta\mathbf{d})) \|f_i\|_{\infty}$, because $40M \geq 1 \geq \vartheta\mathbf{d}$. Multiplying, $160^4 \mathbf{d}^4 (40M)^2 (\vartheta\mathbf{d})^{-2} C_{\text{VT}}(g) = C_T(g) \mathbf{d}^2$. $\square$

Remark 6.20 (The power of $\mathbf{d}$, and the order of the source norm). The arrays of Lemma 6.18 are signed, interleave, and have of the order of $\mathbf{d}^{-4}$ components; Theorem 6.13 bounds them with no condition on signs, components or separation, and their norm is that of one copy, of order $\mathbf{d}^{-1}$. The averaging gains $\#\Gamma \geq (160\mathbf{d})^{-4}$ and the two norms cost $(40M/(\vartheta\mathbf{d}))^2$: the net gain is $\mathbf{d}^2$. We do not claim that this power is optimal. For n sources (independent sign families for the first $n-1$ functions, their product for the last, and Remark 6.16(ii)) the method gives $\mathbf{d}^{4-n}$, a gain for $n = 2, 3$ only. With a source norm involving ν derivatives in place of $\|\cdot\|_{\#}$ it would give $\mathbf{d}^{4-2\nu}$, a gain only for $\nu < 2$, and this is intrinsic: since in four dimensions $\Delta(r^{-2} \log r) = -2r^{-4}$, $\Delta r^{-4} = 8r^{-6}$ and $\Delta r^{-6} = 24r^{-8}$ for $r > 0$, the translation-invariant bilinear form

$$D_{\star}(F_1, F_2) := \frac{1}{384} \iint \nabla \Delta F_1(x) \cdot \nabla \Delta F_2(y) |x - y|^{-2} \log |x - y| dx dy$$

is bounded on three times continuously differentiable functions supported in a ball, and equals $\iint F_1(x)F_2(y)|x-y|^{-8}dx dy$ on disjoint supports: a scale-invariant form, that is, a change of the short-distance coupling. Corollary 6.19 excludes a volume dependence of this kind because the bound of Theorem 6.13 involves the supremum and one Lipschitz constant only; accordingly (L), (A), (B), (S) are used on the class $\mathcal{W}$ defined by $\|\cdot\|_{\sharp}$, with constants depending on the weight through nothing else, as in Proposition 6.1.

6.6 Clause (iv) in every volume and on $\mathbb{R}^4$

The finite-volume witness. We recall the notation and the statement of Section 5. Let $b_0 := (N^2 - 1)/2$, and let $\zeta'_k := \partial_{\zeta}\zeta_k|_{\zeta=0} \in [\frac{2}{3}, \frac{4}{3}]$ be the derivative of the running shift of Section 3 at level k , with the modulus of continuity stated there. For a torus $\mathbb{T}_{m''}$ of the family and a depth $s \leq K$ let $\mathcal{K}_s^{(K, m'')}$ be the explicit free-field kernel of Section 5: with $\mathcal{A}$ the real free lattice gauge field at unit coupling on the bare lattice of $\mathbb{T}_{m''}$ (modulo gauge and harmonic modes), Q_{K-s} the linearised block averaging to level $K - s$, and F_B the lattice curvature of $\mathbb{E}[\mathcal{A} \mid Q_{K-s}\mathcal{A}]$,

$$\mathcal{K}_s^{(K, m'')}(f_1, f_2) := \sum_{p, q} f_1(x_p) f_2(x_q) (\mathbb{E}[F_B(p) F_B(q)])^2,$$

whose continuum counterpart, multiplied by b_0 , is $3(N^2 - 1)\pi^{-4} \iint f_1(x) f_2(y) |x - y|^{-8} dx dy$. The test class $\mathfrak{T}(\mathbf{d}; \vartheta)$ consists of the pairs of non-negative Lipschitz functions $f_1, f_2 \not\equiv 0$, each supported in a ball of radius $\mathbf{d}$, with $\int f_i \geq \vartheta \|f_i\|_{\infty} \mathbf{d}^4$, $\text{Lip } f_i \leq \|f_i\|_{\infty} / (\vartheta \mathbf{d})$ and $\mathbf{d} \leq \text{dist}(\text{supp } f_1, \text{supp } f_2) \leq 6\mathbf{d}$; this window is a condition on the test functions, not on the theory. We use the constants $\lambda_{\sharp}, B_{\sharp} > 0$ and c_5 of clause (0) (Section 2): $\lambda_{\sharp}(j - i) - c_5 \leq x_i - x_j \leq B_{\sharp}(j - i)$ for $0 \leq i \leq j \leq K$, and $g^{-2} < x_K \leq g^{-2} + B_{\sharp}$; in particular $g^{-2} + \lambda_{\sharp}s - c_5 < x_{K-s} \leq g^{-2} + B_{\sharp}(s + 1)$ for $0 \leq s \leq K$. The depth attached to a separation $\mathbf{d}$ is $s(\mathbf{d}) := \min\{s \in \mathbb{N} : C_W(s)L^{-s} \leq \mathbf{d}\}$, where $C_W(s) \geq 1$ is non-decreasing and polylogarithmic in $\bar{x}_s := g^{-2} + B_{\sharp}(s + 1)$.

Theorem 6.21 (Finite-volume witness; Section 5). *For every $\vartheta \in (0, 1]$ there are $\gamma_W^{\mathbb{T}} \leq \gamma_J$, a function C_W as above, a depth floor $s_W(g) \geq b_*(g)$ depending on g only, non-decreasing in g^{-2} and of order $\log_L \log g^{-2}$, an integer $k_W = k_W(L) \geq 1$, constants $c_K, C_K > 0$, an integer $m_V(g)$, and a depth floor $s_{\text{vol}}(g, m) \geq 0$, finite for every (g, m) and zero for $m \leq m_V(g)$ —none of them depending on $K, g_0, \mathbf{d}$ or the position—such that the following holds. For every $g \leq \gamma_W^{\mathbb{T}}$, every $g_0 \in (0, g_-(g)]$ with $K = K(g_0, g)$, every $m \geq m_{\mathbb{T}}(g_-(g))$, every $\mathbf{d}$ with $\max\{s_W(g), s_{\text{vol}}(g, m)\} \leq s(\mathbf{d})$ and $K - s(\mathbf{d}) \geq k_W$, $s := s(\mathbf{d})$, and every $(f_1, f_2) \in \mathfrak{T}(\mathbf{d}; \vartheta)$ on $\mathbb{T}_m$,*

$$S_{2, K, m}^T(f_1, f_2) = \zeta_{K-s}^{\prime 2} b_0 g_{K-s}^4 \mathcal{K}_s^{(K, m)}(f_1, f_2) (1 + R_{K, m}(f_1, f_2)), \quad |R_{K, m}| \leq \frac{1}{2}, \text{ indeed } |R_{K, m}| \leq \frac{1}{4},$$

$$\text{and } c_K \|f_1\|_{\infty} \|f_2\|_{\infty} \leq \mathcal{K}_s^{(K, m)}(f_1, f_2) \leq C_K \|f_1\|_{\infty} \|f_2\|_{\infty}.$$

The transfer threshold. With $C_T(g)$ from Corollary 6.19 and b_0, c_K, C_W from Theorem 6.21, let $s_T(g) = s_T(g; \vartheta, \varrho)$ be the least integer $s_0 \geq 0$ such that for every integer $s \geq s_0$

$$(T-a) \quad L C_W(s) L^{-s} \leq \frac{1}{40}, \quad (6.24)$$

$$(T-b) \quad 9 C_T(g) L^2 C_W(s)^2 L^{-2s} (g^{-2} + B_{\sharp}(s + 1))^2 \leq b_0 c_K. \quad (6.25)$$

Lemma 6.22 (The threshold is a floor). *$s_T(g)$ is finite and at least 1, and it depends only on L , Balaban's parameters, g, ϱ and ϑ ; it depends neither on K, g_0, m , nor on $\mathbf{d}$ or the position. Both (T-a) and (T-b) hold on the half-line $\{s \geq s_T(g)\}$. The condition $s(\mathbf{d}) \geq s_T(g)$ is equivalent to $\mathbf{d}$ lying in an interval with left end point 0 that depends on g only: a ceiling on the separation, equivalently a floor on the depth.*

Proof. $C_W(s)$ is polylogarithmic in $\bar{x}_s$, hence in s , while $C_T(g)$, b_0 and c_K do not depend on s . So the left side of (6.24) is L^{-s} times a polylogarithm of s , and that of (6.25) is L^{-2s} times a polylogarithm of s times a quadratic polynomial in s ; both tend to 0 as $s \rightarrow \infty$. The set of s at which (T-a) or (T-b) fails is therefore bounded, $s_T(g)$ exists, and both conditions hold for $s \geq s_T(g)$ by construction; no monotonicity in s is needed. At $s = 0$, $LC_W(0) \geq L > 1/40$, so $s_T(g) \geq 1$. The quantities entering are L , C_W (hence ϑ and g), $C_T(g)$ (hence M , ϑ , ϱ , $\mathfrak{r}$, ℓ , $\bar{\delta}$, $C_{\text{face}}(1)$, and g through $\mathbf{B}$), b_0 , c_K and $B_{\sharp}$; none of them involves K , g_0 , m , $\mathbf{d}$ or the position—for $C_T(g)$ this rests on the independence of $\bar{\delta}$ of the cutoff in Lemma 6.8. Finally, $s(\mathbf{d})$ is non-increasing in $\mathbf{d}$, so $\{\mathbf{d} : s(\mathbf{d}) \geq s_T(g)\}$ is an interval with left end point 0. $\square$

Remark 6.23 (Kind and size of the threshold). The condition $s(\mathbf{d}) \geq s_T(g)$ is of the same kind as the depth floor $s_W(g) \leq s(\mathbf{d})$ that clause (iv) carries in finite volume: a floor on the depth, equivalently a ceiling on the separation, depending on g only. It bounds none of L , M , Balaban's other parameters, K , m , or a number of steps or cells from above, and no coupling from below. The only upper limit on s in the clause is $s \leq K - k_W$, a floor on K at fixed $\mathbf{d}$: for $K < s_T(g) + k_W$ the clause is silent, and $K \rightarrow \infty$ as $g_0 \downarrow 0$. By (6.25) and $\log C_T(g) \leq C + 2(\mathbf{B} + \bar{\delta})\ell^4 + 8C_{\text{face}}(1)$, and since $\log C_W(s)$ and $\log \bar{x}_s$ grow only logarithmically in s , $s_T(g)$ is at most a constant multiple of $1 + \ell^4 \mathbf{B}(g) + \log_L g^{-2}$, up to terms logarithmic in $\log g^{-2}$. With the stability constants of Sections 2 and 3, $\mathbf{B}(g)$ is at most polylogarithmic in g^{-2} (the factor L^{4b_*} in $E_+^{\sharp}$ is a power of $\log g^{-2}$, and $E_+ \leq C(1 + \log g^{-2})$); so $s_T(g)$ is at most polylogarithmic in g^{-2} , and at most linear in $\log g^{-2}$ when $\mathbf{B} \leq C(1 + \log g^{-2})$. Nothing depends on this size; $s_T(g)$ may grow as $g \downarrow 0$, as $s_W(g)$ does.

Theorem 6.24 (Clause (iv) in every volume and on $\mathbb{R}^4$). *Let $\vartheta \in (0, 1]$; let $\gamma_W^{\mathbb{T}}$, C_W , s_W , s_{vol} , k_W , c_K , C_K be as in Theorem 6.21, $C_T(g)$ as in Corollary 6.19, and $s_T(g)$ as in (6.24)–(6.25). Put $s_{\text{vol}}^*(g) := s_{\text{vol}}(g, m_{\mathbb{T}}(g_{-}(g)))$, a depth floor depending on g only, and $\mathcal{K}_s^* := \mathcal{K}_s^{(K, m_{\mathbb{T}}(g_{-}(g)))}$, the kernel formed on the reference torus $\mathbb{T}_{m_{\mathbb{T}}(g_{-}(g))}$ (the pair transported there). Then for every $g \leq \gamma_W^{\mathbb{T}}$, every $g_0 \in (0, g_{-}(g)]$ with $K = K(g_0, g)$, every $m \geq m_{\mathbb{T}}(g_{-}(g))$, every $\mathbf{d} > 0$ with*

$$\max\{s_W(g), s_{\text{vol}}^*(g), s_T(g)\} \leq s(\mathbf{d}) \quad \text{and} \quad K - s(\mathbf{d}) \geq k_W, \quad (6.26)$$

$s := s(\mathbf{d})$, and every pair $(f_1, f_2) \in \mathfrak{T}(\mathbf{d}; \vartheta)$ on $\mathbb{T}_m$:

$$S_{2;K,m}^T(f_1, f_2) = \zeta_{K-s}'^2 b_0 g_{K-s}^4 \mathcal{K}_s^*(f_1, f_2) (1 + R_{K,m}(f_1, f_2)), \quad |R_{K,m}| \leq \frac{1}{2}, \quad (6.27)$$

with $\zeta_{K-s}' \in [\frac{2}{3}, \frac{4}{3}]$ and $c_K \|f_1\|_{\infty} \|f_2\|_{\infty} \leq \mathcal{K}_s^*(f_1, f_2) \leq C_K \|f_1\|_{\infty} \|f_2\|_{\infty}$. Hence, uniformly in K , g_0 , m and the position, $S_{2;K,m}^T(f_1, f_2) > 0$ and

$$\begin{aligned} \frac{2}{9} b_0 g_{K-s}^4 \mathcal{K}_s^*(f_1, f_2) &\leq S_{2;K,m}^T(f_1, f_2) \leq \frac{8}{3} b_0 g_{K-s}^4 \mathcal{K}_s^*(f_1, f_2), \\ \frac{2}{9} b_0 c_K \|f_1\|_{\infty} \|f_2\|_{\infty} &\leq S_{2;K,m}^T(f_1, f_2) \leq \frac{8}{3} b_0 C_K \|f_1\|_{\infty} \|f_2\|_{\infty} \\ &\quad \frac{1}{(g^{-2} + B_{\sharp}(s+1))^2} \leq S_{2;K,m}^T(f_1, f_2) \leq \frac{1}{(g^{-2} + \lambda_{\sharp}s - c_5)^2} \end{aligned} \quad (6.28)$$

(here $g^{-2} + \lambda_{\sharp}s - c_5 \geq g^{-2} - c_5 > 0$ by the choice of γ_H in (H3)). Moreover, for all $m \geq m' \geq m_{\mathbb{T}}(g_{-}(g))$ and all pairs as in Corollary 6.19, the transfer bound (6.23) holds. Finally, let $S_{2;\infty}^T$ be any limit of clause (ii) on $\mathbb{R}^4$, taken along $(g_0^{(j)}, K_j, m_j)$ with $K_j, m_j \rightarrow \infty$. For every $\mathbf{d}$ with $\max\{s_W(g), s_{\text{vol}}^*(g), s_T(g)\} \leq s(\mathbf{d})$, $s := s(\mathbf{d})$, and every pair $(f_1, f_2) \in \mathfrak{T}(\mathbf{d}; \vartheta)$ of continuously differentiable functions with compact support in $\mathbb{R}^4$, the two-sided bound in the second line of (6.28) holds for $S_{2;\infty}^T(f_1, f_2)$.

Proof. Fix g, g_0, K, m, d, s and (f_1, f_2) as in the statement, and put $m' := m_{\mathbb{T}}(g_{-}(g)) \leq m$.

(1) *The pair lies in one ball of radius $5d$, and $\varepsilon < d < 1/40$.* The supports are compact, so there are $a \in \text{supp } f_1$ and $b \in \text{supp } f_2$ with $|a - b| \leq 6d$; each support has diameter at most $2d$, so $\text{supp } f_1 \cup \text{supp } f_2 \subset \bar{B}(y, 5d)$ with $y := (a + b)/2$. Since $s = s(d) \geq s_{\mathbb{T}}(g) \geq 1$ is minimal and C_W is non-decreasing, $C_W(s)L^{-s} \leq d < C_W(s-1)L^{-(s-1)} \leq LC_W(s)L^{-s}$; so $d < 1/40$ by (T-a), and, since $K - s \geq k_W \geq 1$ and $C_W \geq 1$, $\varepsilon = L^{-K} \leq L^{-s-1} < L^{-s} \leq d$. The ball $\bar{B}(y, 5d)$ has radius less than $L^{m'}/2$; so the pair, lifted to $\mathbb{R}^4$, is transported to $\mathbb{T}_{m'}$, and since the conditions defining $\mathfrak{T}(d; \vartheta)$ are local to the ball, the transported pair lies in $\mathfrak{T}(d; \vartheta)$ on $\mathbb{T}_{m'}$.

(2) *The reference torus.* Theorem 6.21 applies on $\mathbb{T}_{m'}$: $g \leq \gamma_W^{\mathbb{T}}$; m' belongs to the family; its depth condition $\max\{s_W(g), s_{\text{vol}}(g, m')\} = \max\{s_W(g), s_{\text{vol}}^*(g)\} \leq s(d)$ holds by (6.26); and $K - s \geq k_W$. The integer K and the numbers g_{K-s}, ζ'_{K-s} are the same on both tori (clause (0) and Section 3). Theorem 6.21 gives $S_{2;K,m'}^T(f_1, f_2) = \mathfrak{M}^*(1 + R')$ with $|R'| \leq \frac{1}{4}$ and

$$\mathfrak{M}^* := \zeta_{K-s}'^2 b_0 g_{K-s}^4 \mathcal{K}_s^*(f_1, f_2) \geq \frac{4}{9} b_0 c_{\mathcal{K}}(g^{-2} + B_{\sharp}(s+1))^{-2} \|f_1\|_{\infty} \|f_2\|_{\infty} > 0,$$

by $\zeta'_{K-s} \geq \frac{2}{3}$, the lower bound on $\mathcal{K}_s^*$, and $g_{K-s}^{-2} = x_{K-s} \leq g^{-2} + B_{\sharp}(s+1)$ (clause (0)).

(3) *Transfer.* Corollary 6.19 applies: $m \geq m', g \leq \gamma_W^{\mathbb{T}} \leq \gamma_J$, the pair lies in a ball of radius $5d$, $\text{Lip } f_i \leq \|f_i\|_{\infty}/(\vartheta d)$, and $\varepsilon < d \leq 1/40$. With $d < LC_W(s)L^{-s}$ and (6.25),

$$\begin{aligned} |S_{2;K,m}^T(f_1, f_2) - S_{2;K,m'}^T(f_1, f_2)| &\leq C_T(g) d^2 \|f_1\|_{\infty} \|f_2\|_{\infty} < C_T(g) L^2 C_W(s)^2 L^{-2s} \|f_1\|_{\infty} \|f_2\|_{\infty} \\ &\leq \frac{1}{9} b_0 c_{\mathcal{K}}(g^{-2} + B_{\sharp}(s+1))^{-2} \|f_1\|_{\infty} \|f_2\|_{\infty} \leq \frac{1}{4} \mathfrak{M}^*. \end{aligned}$$

This is the one place where the main term, of the size of g_{K-s}^4 , is compared with the transfer error d^2 : an algebraic decay in s against a polylogarithmic growth.

(4) *Sum.* $S_{2;K,m}^T(f_1, f_2) = \mathfrak{M}^*(1 + R_{K,m})$ with $R_{K,m} := R' + (S_{2;K,m}^T - S_{2;K,m'}^T)(f_1, f_2)/\mathfrak{M}^*$, so $|R_{K,m}| \leq \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$. The two-sided bounds follow from $\zeta'_{K-s} \in [\frac{2}{3}, \frac{4}{3}]$, since $\frac{1}{2} \cdot \frac{4}{9} = \frac{2}{9}$ and $\frac{3}{2} \cdot \frac{16}{9} = \frac{8}{3}$, from the bounds on $\mathcal{K}_s^*$, and from the window $g^{-2} + \lambda_{\sharp}s - c_5 < x_{K-s} \leq g^{-2} + B_{\sharp}(s+1)$ of clause (0). For $m = m'$ the display is Theorem 6.21 itself. The transfer bound is Corollary 6.19.

(5) $\mathbb{R}^4$. For all large j we have $m_j \geq m_{\mathbb{T}}(g_{-}(g))$ and $K_j - s \geq k_W$, and the pair, transported to $\mathbb{T}_{m_j}$ (where $10d < L^{m_j}$), satisfies the hypotheses of the first part. So the number $S_{2;K_j,m_j}^T(f_1, f_2)$ lies in the closed interval given by the second line of (6.28), whose end points do not depend on j . By definition, $S_{2;\infty}^T(f_1, f_2)$ is the limit of these numbers, so it lies in the same interval. The limits $K \rightarrow \infty$ and $m \rightarrow \infty$ are taken jointly, as in clause (ii); no limits are exchanged. $\square$

Corollary 6.25 (Consequences in every limit). *Fix a continuously differentiable radial function $\phi \geq 0$ supported in the closed unit ball with $\phi(0) = 1 = \sup \phi$, let $0 < \vartheta \leq \min\{1, \int \phi, 1/\text{Lip } \phi\}$, and let $g \leq \gamma_W^{\mathbb{T}}$, the ceiling of Theorem 6.21 for this ϑ . In every limit of (ii-T-b), on every torus of the family, and in every limit on $\mathbb{R}^4$:*

- (a) *there is a centred curvature vector with strictly positive reflected norm: the Hilbert space $\mathcal{H}^{\text{curv}}$ of the reconstruction of Section 4 has dimension at least 2;*
- (b) *the limit is not scale invariant under any assignment of a scaling dimension to the curvature;*
- (c) *it is reproduced by no free Maxwell field, at any charge;*
- (d) *the coupling read off the two-point function at separation d is strictly positive and tends to 0 logarithmically as $d \downarrow 0$, at the two-sided rate of clause (0).*

Proof. We argue on $\mathbb{R}^4$, with the last part of Theorem 6.24; on a torus of the family the same argument applies to the limits $K \rightarrow \infty$ at fixed m , with the first part, where in (a) the Hilbert space is the one obtained, by the same construction, from the reflection positivity of the torus limit at the hyperplane $\{x^0 = 0\}$ (clause (ii)), or, equivalently for (a), the statement is that $S_{2;m}^T(f \circ \theta, f) > 0$. For $d > 0$ and $y \in \mathbb{R}^4$ the function $\phi((\cdot - y)/d)$ has integral $d^4 \int \phi \geq \vartheta d^4$, supremum 1, and Lipschitz constant $\text{Lip } \phi/d \leq 1/(\vartheta d)$. Call d admissible if $\max\{s_W(g), s_{\text{vol}}^*(g), s_T(g)\} \leq s(d)$; by Lemma 6.22 the admissible d form an interval with left end point 0.

(a) Let θ be the time reflection $x^0 \mapsto -x^0$, d admissible, and $f := \phi((\cdot - y)/d)$ with $y := (2d, 0, 0, 0)$. Then f is supported in $\{x^0 \geq d\}$, and $(f \circ \theta, f) \in \mathfrak{T}(d; \vartheta)$, the supports being at distance at least $2d$ and at most $4d$. In the reconstruction of Section 4, the centred curvature field smeared with f defines a vector v_f orthogonal to the vacuum, with $\|v_f\|^2 = S_{2;\infty}^T(f \circ \theta, f)$, since the smeared action density is invariant under θ . By Theorem 6.24 this is strictly positive; so $v_f \neq 0$ and $\dim \mathcal{H}^{\text{curv}} \geq 2$.

(b) Fix an admissible d_0 and a pair $(f_1, f_2) \in \mathfrak{T}(d_0; \vartheta)$ of continuously differentiable functions with $\|f_i\|_\infty = 1$ (for instance, two translates of $\phi(\cdot/d_0)$ at distance $4d_0$). For $\lambda \in (0, 1]$ let $f_i^{(\lambda)} := f_i(\cdot/\lambda)$. The conditions defining $\mathfrak{T}$ are invariant under this scaling, so $(f_1^{(\lambda)}, f_2^{(\lambda)}) \in \mathfrak{T}(\lambda d_0; \vartheta)$, and λd_0 is admissible. Put $S(\lambda) := S_{2;\infty}^T(f_1^{(\lambda)}, f_2^{(\lambda)})$ and $s_\lambda := s(\lambda d_0)$. Since $C_W \geq 1$ and C_W is polylogarithmic, $\log_L(1/(\lambda d_0)) \leq s_\lambda \leq \log_L(1/(\lambda d_0)) + O(\log \log(1/\lambda))$ as $\lambda \downarrow 0$, and (6.28) bounds $S(\lambda)$ above and below by positive constant multiples of $(\log(1/\lambda))^{-2}$ for small λ . A scaling law $S(\lambda) = \lambda^\alpha S(1)$, which is what any assignment of a scaling dimension to the curvature would give, is incompatible with this for every real α : $S(1) > 0$, and λ^α tends to 0 faster than any power of $1/\log(1/\lambda)$ if $\alpha > 0$, tends to infinity if $\alpha < 0$, and is constant if $\alpha = 0$, while $(\log(1/\lambda))^{-2} \rightarrow 0$.

(c) The two-point function of the action density of a free Maxwell field, at any charge, is a multiple of $\iint f_1(x)f_2(y)|x - y|^{-8} dx dy$ on separated supports; it is scale invariant, which (b) excludes.

(d) For $(f_1, f_2) \in \mathfrak{T}(d; \vartheta)$ with d admissible and $s = s(d)$, (6.28) gives

$$\left(\frac{2}{9}c_K\right)^{1/4} (g^{-2} + B_\sharp(s+1))^{-1/2} \leq \left(\frac{S_{2;\infty}^T(f_1, f_2)}{b_0\|f_1\|_\infty\|f_2\|_\infty}\right)^{1/4} \leq \left(\frac{8}{3}C_K\right)^{1/4} (g^{-2} + \lambda_\sharp s - c_5)^{-1/2},$$

and $s(d)$ grows like $\log_L(1/d)$: the coupling read off at separation d is strictly positive and tends to 0 like $(\log(1/d))^{-1/2}$, with the two-sided rate of clause (0). $\square$

Remark 6.26 (The kernel). Theorem 6.24 is stated with $\mathcal{K}_s^*$, the kernel of the reference torus, because the main term is imported from $\mathbb{T}_{m_\mathbb{T}(g_-(g))}$ and the transfer bounds only the difference of the two-point functions. The kernel $\mathcal{K}_s^{(K,m)}$ formed on $\mathbb{T}_m$ itself satisfies the same two-sided bounds as $\mathcal{K}_s^*$ and has the same continuum counterpart; the bounds (6.28) and Corollary 6.25 use only these. The display (6.27) with $\mathcal{K}_s^{(K,m)}$ in place of $\mathcal{K}_s^*$ would follow from the free-field estimate $|\mathcal{K}_s^{(K,m)}/\mathcal{K}_s^* - 1| \leq \frac{1}{16}$ on pairs of $\mathfrak{T}(d; \vartheta)$ beyond one more depth floor, together with 18 in place of 9 in (6.25). For the free, unconditioned kernels of the two tori the corresponding comparison is proved in the companion note on the free kernel; the passage to the block-conditioned kernels $\mathcal{K}_s^{(K,m)}$ and $\mathcal{K}_s^*$ is not carried out here; the estimate is not used.

Remark 6.27 (An alternative to the floor s_{vol}^*). Instead of the depth floor $s_{\text{vol}}^*(g)$ one may impose one more ceiling on the coupling, below which $m_\mathbb{T}(g_-(g')) \leq m_V(g')$ for every $g' \leq g$, so that $s_{\text{vol}}^* \equiv 0$. Such a ceiling exists, since $m_V(g)$ is of order g^{-2} (Section 5), while $m_\mathbb{T}(g_-(g))$ grows only like the logarithm of a power of $\log g^{-2}$; it is a ceiling on the coupling, chosen last, of the kind already present in (H7). We use the floor, which adds no condition on the coupling.

Remark 6.28 (What is not asserted). Theorem 6.24 asserts nothing beyond its displays. In particular it does not assert non-Gaussianity; $O(4)$ invariance; a mass gap or clustering; anything on block

observables or Wilson loops; statements for one source, for overlapping or signed sources in (6.27), for separations larger than $6d$, for $\vartheta \rightarrow 0$, or at depths outside the stated range; convergence of ζ'_{K-s} to 1 beyond its stated modulus; anything for $N \geq 3$; convergence or uniqueness of the limit as the volume grows; a volume-uniform estimate on the rarity of large-field regions, which is not used and is not proved in this paper; transfer bounds for four or more sources; or the optimality of the power d^2 . Corollary 6.19 passes to limits: an infinite-volume limit and a limit on the reference torus, taken along a common sequence of cutoffs, have two-point functions that differ by at most $C_T(g)d^2\|f_1\|_\infty\|f_2\|_\infty$ on pairs of scale d in one ball of radius $5d$. This is a statement at short distance, not uniqueness of the infinite-volume limit.

Remark 6.29 (Order of the constants; no circularity). The constants are chosen in the order L , Balaban's parameters, γ , γ_H , ϱ , γ_J , ϑ , $\gamma_W^{\mathbb{T}}$; then, for each g : B , ℓ , C_{face} , $\bar{\delta}$, $Q_\star$, C_{VT} , C_T ; then $s_T(g)$; then d , $s(d)$, and $K \geq s + k_W$. Nothing returns. The finite-volume witness of Section 5 uses none of the statements of this section; Theorem 6.13 uses nothing of Section 5; Lemma 6.18 uses nothing beyond its hypotheses. The existential constant L_0 of (H2) is untouched, and no condition introduced here bounds L , M , K , m or a depth from above, or a coupling from below. The proof of Theorem 6.24 is complete given Propositions 6.1 and 6.2, Theorem 6.21 and Lemma 6.7; clause (iv) in every volume therefore rests on exactly what the correlation bounds of clause (iii) and the finite-volume witness rest on.

6.7 What of the large volume is used

Remark 6.30. Of the large torus $\mathbb{T}_m$ the proof of Theorem 6.24 uses: link reflection positivity and the chessboard estimate of its bare Wilson measure, and the trace inequality—exact statements about the bare measure at every cutoff; one number, the lower stability bound $\tilde{Z}_m(0) \geq e^{-C_{\text{low}}|\mathbb{T}_m|}$ at zero source, obtained by restricting an unnormalised real integral to a neighbourhood of the trivial configuration, so that no large-field region has to be controlled; and the vacuum constant E_m , through Lemma 6.8, a statement about field-independent numbers at zero source. All large-field contributions on $\mathbb{T}_m$ are inside $Z_m(W_z)$ and $Z_m(w_z)$, which are never expanded: they are bounded, through reflection positivity, by the partition function of the small torus. On the small torus the upper stability bound (S) is a bound by moduli of unnormalised terms over all large-field histories; an upper bound on a total mass of this kind needs no estimate on the rarity of large-field regions. The price of the global pair of stability bounds, $e^{B|\mathbb{T}|}$, becomes $e^{B\ell^4}$ per cell, a constant depending on g only, and it is beaten by the power d^2 of the separation. Under the variant of Remark 6.17, which takes (F) on $\mathbb{T}_{m'+1}$ instead of on $\mathbb{T}_m$ at the price of the factor c_L in $Q_\star$, nothing of Balaban's construction is used on any torus beyond $\mathbb{T}_{m'+1}$: only reflection positivity of its bare measure.

In words: the two-point law of clause (iv) holds uniformly in the volume. For every torus of the family, below a separation threshold depending only on the renormalised coupling, the connected two-point function of the smeared action density equals $(N^2 - 1)/2$ times an explicit positive scale-free kernel times the fourth power of the running coupling at the separation scale, times the square of the derivative of the running shift, which lies in $[\frac{2}{3}, \frac{4}{3}]$, up to a relative error of at most one half, uniformly in the cutoff, the bare coupling, the volume and the position. Consequently every infinite-volume limit is non-trivial in the following precise sense: there is a centred curvature vector with strictly positive reflected norm, the limit is not scale invariant under any assignment of a dimension to the curvature, it is reproduced by no free Maxwell field, and the coupling read off this correlation at separation d is positive and tends to zero logarithmically as $d \downarrow 0$. The passage from one torus to all volumes uses, of the large volume, only reflection positivity of the bare Wilson measure and the lower stability bound at zero source: by a trace inequality for mirror-periodic partition functions and an averaging over translated copies of the test pair, the two-point functions

of any two volumes of the family differ, on test pairs of scale d , by at most a constant depending on the coupling times d^2 .

7 Clause (v): uniqueness given the renormalised coupling

Clauses (0)–(iii) fix the cutoff by first passage, and clause (ii-T-b) gives limits on a fixed torus only along subsequences. Clause (v) prescribes instead the number n of renormalisation steps (the *depth*) and tunes the bare inverse coupling so that the running inverse coupling at a fixed reference length is exactly g^{-2} . Along this tuned family the connected correlations of the smeared action density converge on each fixed torus without passing to subsequences; the limits form one real-analytic one-parameter family, labelled by an offset of the bare coupling or, equivalently, by the renormalised coupling at the reference length; equal renormalised couplings give equal limits; and, for $\varrho \geq 6(B_{\sharp} + 1)$ and g below a further threshold, every first-passage limit on the same torus lies on the family. Nothing is asserted uniformly in the volume, nor about uniqueness of the infinite-volume limit or of the vacuum (§7.6). Throughout, $N = 2$ and (H1)–(H6) hold.

Parts of the companion account. We cite: [T-corr], the part of the companion account whose structure Section 3 gives (the correlation theorem with sources, Theorem 3.1), including its form on a prescribed reference line; [T-U], the uniqueness theorem, which compares two cutoffs on one torus; [T-L], the same-cutoff Lipschitz step; [T-R], the renormalised coordinate; [T-B], the bridge between first-passage and tuned runs. “Proved in [T-x]” means proved there given the inputs it names; the supporting statements of [T-U] that we do not claim as proved are listed, each with its status, in §7.5.

7.1 The tuned family

Choices and units. After γ_H , in the order of (H3), a radius $\varrho > 0$ for constant shifts of the bare inverse coupling is fixed, $\varrho_2 := \varrho/2$; then a threshold $\gamma_U = \gamma_U(L, \mathbf{p}, \gamma; \varrho) \in (0, \gamma_J(\varrho)]$, where $\mathbf{p}$ denotes the parameters of (H3) and $\gamma_J(\varrho) \leq \gamma_H$ is the threshold of (H7) for clause (iii); this is (H7) for clause (v). Let $0 < g \leq \gamma_U$, $X := g^{-2}$. All further conditions are lower bounds on the depth and the torus, or upper bounds on g . With c_0, c_2 of the one-loop lemma (2.9) put $\lambda_U := c_0/2$; the ceilings on γ_J give $X \geq 8c_2 + 4$ [T-corr]. M is Balaban’s cube-side parameter, and p_0 the integer exponent of his large-field degree function $p_0(g) = A_0(\log g^{-2})^{p_0}$ [9], the exponent of Theorem 3.1; no confusion arises, since the function always carries its argument.

As in clause (ii-T-a), let $m_0 \geq m_{\mathbb{T}}(g_-(g))$ and $\mathbb{T} := \mathbb{T}_{m_0}$; lengths are measured in units of the *reference lattice* of the tuned runs, the lattice at which their running inverse coupling is pinned to X . *Depth* r refers to the lattice of spacing L^{-r} , the same for every $n \geq r$; the bare lattice at depth n has spacing L^{-n} . Balaban’s renormalisation group runs from depth n up to depth $w = w(g) \geq 0$, an integer depending only on $L, \mathbf{p}, g$ (the analogue of the integer b_* of Theorem 3.1); the last lattice has $N_w := 2L^{m_0+w}$ sites per side, and its integral is not expanded. We write $Q_w := \exp(c_{\text{vol}}N_w^4)$, with c_{vol} the volume constant of Theorem 3.1(d4), for the corresponding volume factor. The norm $\|\cdot\|_{\sharp}$ and distances between supports are computed in units of the last lattice, which affects only g -dependent constants.

Correlations at depth n . Let $\mathbf{p} \geq 2$ and let $\vec{f} = (f_1, \dots, f_{\mathbf{p}})$ be real Lipschitz functions on $\mathbb{T}$ with supports pairwise at distance at least $d > 0$. For a bare inverse coupling x (real, or complex as

below) and $z \in \mathbb{CP}$,

$$Z_n(x; z) := \int e^{-x A(U) + \sum_i z_i \mathcal{O}(f_i)} dU, \quad S^n(x; \vec{f}) := \partial_{z_1} \cdots \partial_{z_p} \log Z_n(x; z)|_{z=0}, \quad (7.1)$$

over the bare lattice of $\mathbb{T}$ at depth n with Haar measure. So $S^n(x; \vec{f})$ is the connected correlation S_p^T of Section 1 on that lattice at $g_0^{-2} = x$; the superscript is the depth, not a power. The exponent equals $-\sum_p (x - w(p))[1 - \text{Re tr } U(\partial p)]$, $w(p) := \sum_i z_i f_i(x(p))$ with $x(p)$ the barycentre of p : the sources are a local shift of the bare inverse coupling.

Definition 7.1 (runs on the reference line; tuned values). Let $\theta_r := X + c_0 r$ (the *reference line*, indexed by depth). The n -step run from real x_0 is the sequence $x_{(n)} := x_0, \dots, x_{(0)}$ given by Balaban's recursion (2.8) (so $x_{(r)}$ is the inverse coupling after $n - r$ steps), with the coefficients of the infinite lattice and with every coupling-dependent choice — thresholds for creating large-field regions, cube sizes, partitions, trees of the cluster expansions — read off θ instead of off the run. It is *admissible* if $|x_{(r)} - \theta_r| \leq \tau_r$ for $0 \leq r \leq n$, $\tau_r \leq \theta_r/2$ being the tube radius of [T-corr], at most of order $\theta_r^{1/2}(\log \theta_r)^{p_0}$; it *lies in the working tube* if $|x_{(r)} - \theta_r| \leq \tau_r + 3\varrho$ for $0 \leq r \leq n$ (at most $2\tau_r$ under the ceiling on γ_U), the tube on which the conclusions of [T-corr] hold. Its *reference inverse coupling* is $x_{(0)}$. A real a is a *tuned value at depth n* if the run from a is admissible and $x_{(0)} = X$.

The choices at depth r do not depend on n : runs of all lengths share one scaffold. A run is only a device for analysing the lattice measure, which depends on the lattice and the bare coupling alone.

Running shift. Let x_0 have an admissible n -step run; count steps from the bare level, $k = n - r$. For $|\hat{w}| < 2\varrho$ the run from $x_0 - \hat{w}$ with the same choices has inverse couplings $x_{(n-k)} - \zeta_k(\hat{w})$, where

$$\zeta_0(\hat{w}) = \hat{w}, \quad \zeta_{k+1}(\hat{w}) = \zeta_k(\hat{w}) + b_{k+1}(\hat{w}) - b_{k+1}(0), \quad (7.2)$$

$b_{k+1}(\hat{w})$ being the coupling increment of the $(k+1)$ -st step of the shifted run. We use:

(RS) ζ_k and b_{k+1} are holomorphic on $|\hat{w}| < 2\varrho$ and real for real $\hat{w}$;

$$|\zeta_k(\hat{w}) - \hat{w}| \leq |\hat{w}|/3; \quad \sum_{k < n} \text{Lip}_{\bar{D}(0, \varrho)} b_{k+1} \leq \frac{2}{3};$$

for real $|\hat{w}| < 2\varrho$ the shifted run lies in the working tube.

(RS) is proved in the companion account [T-corr], together with Theorem 3.1 for the runs on the reference line (structure in Section 3); its first three statements are Corollary 3.3 for these runs, with the same proof (the notation agrees with (3.4)), and the last says that shifted runs stay in the working tube. For real ζ it puts $\zeta_n(\zeta)$ between $\frac{2}{3}\zeta$ and $\frac{4}{3}\zeta$. For the tuned value a_n of Theorem 7.2(a) let $x_n^{\text{ref}}(\hat{w}) := X - \zeta_n(\hat{w})$, the reference inverse coupling of the run from $a_n - \hat{w}$, and $\mathcal{N}_n := \zeta'_n(0)$, which lies in $[\frac{2}{3}, \frac{4}{3}]$ by the Schwarz lemma.

7.2 Statement

Theorem 7.2 (Clause (v) of Theorem 0). *Let $N = 2$ and assume (H1)–(H6). Let $\varrho > 0$, let $\gamma_U = \gamma_U(L, \mathfrak{p}, \gamma; \varrho)$ be the threshold of §7.1, let $0 < g \leq \gamma_U$, $X = g^{-2}$, and let $\mathbb{T} = \mathbb{T}_{m_0}$, $\mathfrak{p} \geq 2$ and $\vec{f}$ be as in §7.1.*

- (a) (Exact pin.) For every $n \geq 0$ there is exactly one tuned value a_n ; it is also the only real bare value whose run lies in the working tube and has reference inverse coupling X . It depends only on $L, \mathbf{p}, g, n$, not on the torus, and

$$\lambda_U n - 2c_2 \leq a_n - X \leq 3\lambda_U n + 2c_2, \quad \lambda_U(n-l) - c' \leq a_n - a_l \leq 3\lambda_U(n-l) + c' \quad (0 \leq l \leq n), \quad (7.3)$$

with $c' = c'(L, \mathbf{p}, g)$; so the tuned bare coupling $a_n^{-1/2}$ is of order $n^{-1/2}$. No monotonicity of a_n in n is asserted.

- (b) (The one-parameter family.) Let $v_w := 3\rho/(256(Q_w + 1))$ and $\mathcal{D}_w := \{\hat{w} \in \mathbb{C} : |\operatorname{Re} \hat{w}| \leq \rho_2/2, |\operatorname{Im} \hat{w}| < v_w\}$. There are numbers $S^{(\hat{w})}(f)$, $\hat{w} \in \mathcal{D}_w$, a floor n_0 and a constant C with

$$\sup_{\hat{w} \in \mathcal{D}_w} |S^n(a_n - \hat{w}; \vec{f}) - S^{(\hat{w})}(\vec{f})| \leq C \prod_{i=1}^p \|f_i\|_{\#} n^{-1/2} (\log n)^{p_0+1} \quad (n \geq \max\{2, n_0\}), \quad (7.4)$$

so the convergence holds without passing to subsequences. $\hat{w} \mapsto S^{(\hat{w})}(\vec{f})$ is continuous on $\mathcal{D}_w$, holomorphic in its interior and real for real $\hat{w}$, hence real-analytic on $(-\rho_2/2, \rho_2/2)$; nothing is asserted for $|\operatorname{Im} \hat{w}| \geq v_w$. For real $\hat{w}$ the $S^{(\hat{w})}$ are multilinear and symmetric, invariant under all translations of $\mathbb{T}$ (for C^1 sources; Lemma 4.5) and under W_4 , reflection positive at every hyperplane $x^\mu = c$ on centred separated tensors, and obey the torus bound (1.12). There are functions $\mathbf{n}_n$, holomorphic on $|\hat{w}| < \rho$, real for real $\hat{w}$, $\mathbf{n}_n(0) = 0$, such that the runs from $a_{n+1} - \mathbf{n}_n(\hat{w})$ and from $a_n - \hat{w}$ have equal reference inverse couplings, and

$$|\mathbf{n}_n(\hat{w}) - \hat{w}| \leq \Upsilon_n |\hat{w}|, \quad \Upsilon_n \leq C_\Upsilon n^{-3/2} (\log n)^{p_0+1} \quad (n \geq 2); \quad (7.5)$$

$\mathcal{N}_{n+1} \mathbf{n}'_n(0) = \mathcal{N}_n$, and $(\mathcal{N}_n)$ converges. The objects $a_n, \mathbf{n}_n, \mathcal{N}_n, \Upsilon_n, C_\Upsilon$ do not depend on m_0 ; C depends on $L, \mathbf{p}, g, \mathbf{p}, \rho, \mathbf{d}$, the supports and m_0 , not on n ; and m_0 enters C, v_w, n_0 only through Q_w .

- (c) (Exhaustion.) (1) If (a'_n) is a real sequence with $|a'_n - a_n| \leq \rho_2/2$ for all n , then $S^n(a'_n; \vec{f}) - S^{(a_n - a'_n)}(\vec{f}) \rightarrow 0$; so sequences whose offsets $a_n - a'_n$ converge to the same $\hat{w}$ have the same limit $S^{(\hat{w})}(\vec{f})$, and such a sequence converges if and only if $S^{(a_n - a'_n)}(\vec{f})$ does. (2) (Bridge to first passage.) Let in addition $\rho \geq 6(B_{\#} + 1)$ and $g \leq \gamma_\varpi$, with γ_ϖ and $\varpi(g) = C_\varpi g^{\kappa_0-6} \leq 1$ as in Proposition 7.4, and consider the first-passage cutoffs of clauses (0)–(iii) on $\mathbb{T} = \mathbb{T}_{m_0}$, whose lattice at cutoff K is the bare lattice of $\mathbb{T}$ at depth K . Then for all $g_0 \in (0, g_-(g)]$ and $K = K(g_0, g)$

$$-\frac{3}{2}\varpi(g) \leq g_0^{-2} - a_K \leq \frac{3}{2}(B_{\#} + \varpi(g)) \leq \rho_2/2,$$

and every subsequential limit of the first-passage correlations of clause (ii) on $\mathbb{T}$ is $S^{(\hat{w})}$ for one real $\hat{w}$, $|\hat{w}| \leq \rho_2/2$. (3) The limits of $S^{n_j}(x_j; \cdot)$ along $n_j \rightarrow \infty$ with x_j real, $|x_j - a_{n_j}| \leq \rho_2/2$, are exactly the $S^{(\hat{w})}$, $\hat{w} \in [-\rho_2/2, \rho_2/2]$.

- (d) (The renormalised coupling as coordinate.) (1) (Exponential rate.) Re-file each remainder term left at level i (levels counted from the bare lattice) when the operation $\mathbf{R}$ of Section 2 removes (“heals”) a large-field region, among the regular terms of level $i'' := i + \log_L(M\check{R}_i)$, at which its partition cube, of side $M\check{R}_i$ in level- i units ($\check{R}_i$ the cube size of Section 2), has become one lattice cell; let $\hat{x}$ be the reference inverse coupling of the re-filed run. For $|t| \leq \rho_2$ there is exactly one real $\hat{a}_n(t)$ whose re-filed run is admissible with $\hat{x} = X - t$; with $\hat{\mathcal{N}}_n(t)$ the derivative of $\hat{x}$ in the bare value at $\hat{a}_n(t)$,

$$\left| \hat{\mathcal{N}}_{n+1}(t)^{-\mathbf{p}} S^{n+1}(\hat{a}_{n+1}(t); \vec{f}) - \hat{\mathcal{N}}_n(t)^{-\mathbf{p}} S^n(\hat{a}_n(t); \vec{f}) \right| \leq C n^c \rho_U^{n/2} \prod_i \|f_i\|_{\#}, \quad (7.6)$$

$\rho_U < 1$ as in Proposition 7.3, c and C independent of n : the t -labelled normalised family converges exponentially fast. The labels $\hat{w}$ and t describe the same family: at each depth t is real-analytic in $\hat{w}$ with derivative in $[\frac{2}{3}, \frac{4}{3}]$ at 0. No exponential rate is asserted in any labelling by the bare coupling. (2) (Equal renormalised coupling, equal limit.) Let $(x_n), (x'_n)$ be real sequences with $|x_n - a_n|, |x'_n - a_n| \leq \varrho_2/2$, and let $y_n := x_n^{\text{ref}}(a_n - x_n)$, $y'_n := x'_n^{\text{ref}}(a_n - x'_n)$ be the reference inverse couplings of their runs. If $y_n \rightarrow y$ and $y'_n \rightarrow y$, then $S^n(x_n; \vec{f})$ and $S^n(x'_n; \vec{f})$ converge to the same limit, $S^{(\hat{w}_*)}(\vec{f})$ with $\hat{w}_* \in [-\varrho_2/2, \varrho_2/2]$ the unique solution of $x_\infty^{\text{ref}}(\hat{w}_*) = y$, where $x_\infty^{\text{ref}} := \lim_n x_n^{\text{ref}}$ (Lemma 7.5). (3) (Change of units.) For $0 \leq s \leq n$ the correlations at depth n on $\mathbb{T}_{m_0}$ are those at depth $n-s$ on $\mathbb{T}_{m_0+s}$, at the same bare coupling, with sources rescaled by L^s (Lemma 7.6); the reference inverse coupling becomes the run's inverse coupling at depth s . So the lattice theories with data (m_0, y) (torus exponent, reference inverse coupling) are those with data $(m_0 + s, y_s)$, in units differing by L^s ; Corollary 7.7 carries this to the limit family. Along first-passage runs $x_{K-s} - x_K \in [\lambda_\# s - c_5, B_\# s]$ by (2.13), and $x_{K-s}^{-1/2} \rightarrow g_{\infty, s}$ along subsequences (Corollary 2.12). Apart from clause (v-f) of Theorem 0, nothing relates distinct reference couplings on one torus.

- (e) (Same-cutoff Lipschitz step.) On one lattice with one set of coupling-dependent choices, one complete renormalisation step of Balaban, the large-field operations included, applied to two input lists in the class of effective actions of the correlation theorem (complex sources allowed), produces outputs that differ by at most fixed multiples of the difference of the inputs in the weighted norms of that class: a negative power $\theta(g_k)$ of $\log g_k^{-2}$ for small-field components (g_k the running coupling of the step); for the coupling slot, a multiple of second order, $\theta(g_k)^2$ times the norm of the remainder terms; and $e^{-p_0(g_k)}$ -weighted sums over earlier levels for large-field components. No multiple depends on the cutoff, the level, the torus or the position. Each step of life of a large-field region costs a factor $q_* < 1$: old large-field regions are forgotten geometrically. This is the step iterated in the proof of (b).

Status. Part (a), the maps $\mathbf{n}_n$ and (7.5) are proved in the companion account [T-U]. The rest of (b) is deduced in §7.3 from the increment estimate, Proposition 7.3, proved in [T-U]. Parts (c), (d)(2), (d)(3) are proved in §7.3, (c)(2) from the bridge estimate, Proposition 7.4, proved in [T-B]. Part (d)(1) is proved in [T-R], (e) in [T-L]. The structure of these proofs is in §7.4. Clause (v) is thus proved as an implication from the uniqueness theorem [T-U] of the companion account; its supporting statements that we do not claim as proved are listed, each with its status, in §7.5. Part (a) is of the kind announced by Balaban in [7, Theorem 2] (a bare coupling with $g_K = g$ and all effective couplings in $(0, \gamma]$, with the proof deferred to a separate paper); the announcement is not used.

7.3 The deductions made here

Proposition 7.3 (increment estimate; proved in the companion account [T-U]). *There is a floor n_0 , depending on g and, through $\mathbf{Q}_w$, on m_0 , such that for $n \geq n_0$ and $\hat{w} \in \mathcal{D}_w$*

$$|S^{n+1}(a_{n+1} - \hat{w}; \vec{f}) - S^n(a_n - \hat{w}; \vec{f})| \leq (C_4 \Upsilon_n + \mathfrak{f}_n) \prod_i \|f_i\|_\#, \quad (7.7)$$

$$|\mathcal{N}_{n+1}^{-p} S^{n+1}(a_{n+1}; \vec{f}) - \mathcal{N}_n^{-p} S^n(a_n; \vec{f})| \leq (\frac{3}{2})^p \mathfrak{f}_n \prod_i \|f_i\|_\#, \quad (7.8)$$

where

$$\mathfrak{f}_n := C_1(\rho_U^{n/2} + \Pi_n) + C_2 e^{-c_d L^{n/2-w}/M} + C_3(\rho_U^{n-w} + \Pi_n), \quad \Pi_n \leq C_\Upsilon n^{-(\kappa_0-4)/2} (\log n)^{c_3}. \quad (7.9)$$

Here $\rho_U \in (0, 1)$ is a contraction base in a window $[\rho_*, 1)$, $\rho_* < 1$ depending only on L and the exponents fixed before L ; $c > 0$, c_3 , C_Υ depend only on L , $\mathbf{p}$, g ; κ_0 is Balaban's remainder exponent of Theorem 3.1, a lower bound only (here $\kappa_0 \geq 10$ is used); C_1 , C_2 depend on $\mathbf{p}$, ϱ and the sources but not on m_0 ; C_3 , C_4 depend also on m_0 through $\mathbf{Q}_w$. No constant depends on n .

Proof of (b), given (a), (7.5) and Proposition 7.3. Put $G_n(\hat{w}) := S^n(a_n - \hat{w}; \vec{f})$, continuous on $\mathcal{D}_w$, holomorphic inside and real for real $\hat{w}$, by the correlation theorem with a complex constant shift of the bare coupling [T-corr] (this is where v_w enters). By (7.7), $\sup_{\mathcal{D}_w} |G_{n'} - G_n| \leq \prod_i \|f_i\|_\# \sum_{j \geq n} (C_4 \Upsilon_j + f_j)$ for $n' > n \geq n_0$. By (7.5), $\sum_{j \geq n} \Upsilon_j \leq C' n^{-1/2} (\log n)^{p_0+1}$; as $\kappa_0 \geq 10$, $T_\Pi := \sup_{l \geq 1} l \sum_{j \geq l} \Pi_j < \infty$ and $\sum_{j \geq n} \Pi_j \leq T_\Pi/n$; the other tails are smaller. So G_n converges uniformly on $\mathcal{D}_w$ at the rate (7.4), and the limit is continuous, holomorphic inside (Weierstrass) and real on the real interval, hence real-analytic there. For real $|\hat{w}| \leq \varrho_2/2$ and $n \geq \varrho/(4\lambda_U)$, (7.3) and $X \geq 8c_2 + 4$ give $a_n - \hat{w} \geq X + \lambda_U n - 2c_2 - \varrho/4 > 0$, so the Wilson measure is reflection positive [23] and has the symmetries of the bare lattice; the properties of clause (ii) then pass to the limit as in Section 4 (Remark 4.9). Differentiating $\zeta_{n+1}^{[n+1]} \circ \mathbf{n}_n = \zeta_n^{[n]}$ at 0, where $\zeta^{[n+1]}$ is the running shift of the run from a_{n+1} and $\zeta^{[n]}$ that of the run from a_n , gives $\mathcal{N}_{n+1} \mathbf{n}'_n(0) = \mathcal{N}_n$; since $\mathbf{n}_n(\hat{w}) - \hat{w}$ vanishes at 0 and is at most $\Upsilon_n |\hat{w}|$, $|\mathbf{n}'_n(0) - 1| \leq \Upsilon_n$, which is summable, so $(\mathcal{N}_n)$ converges. $\square$

Proof of (c)(1) and (c)(3). (1) The offset $\hat{w}_n := a_n - a'_n$ lies in $\mathcal{D}_w$, and $|S^n(a'_n; \vec{f}) - S^{(\hat{w}_n)}(\vec{f})| \leq \sup_{\mathcal{D}_w} |G_n - S^{(\cdot)}(\vec{f})| \rightarrow 0$; the rest follows from the continuity of $S^{(\cdot)}$. (3) If $S^{n_j}(x_j; \vec{f})$ converges for every $\vec{f}$, pass to a subsequence with $a_{n_j} - x_j \rightarrow \hat{w} \in [-\varrho_2/2, \varrho_2/2]$; by uniform convergence and continuity the limit is $S^{(\hat{w})}(\vec{f})$, with one $\hat{w}$ for all $\vec{f}$. Conversely $S^{(\hat{w})} = \lim_n S^n(a_n - \hat{w}; \cdot)$. $\square$

Proposition 7.4 (bridge estimate; proved in the companion account [T-B]). *Assume $\varrho \geq B_\# + 1$. There are $C_\varpi > 0$, depending only on L and $\mathbf{p}$ (not on ϱ), and a threshold $\gamma_\varpi = \gamma_\varpi(L, \mathbf{p}; \varrho) > 0$, chosen after ϱ , such that for $0 < g \leq \min\{\gamma_U, \gamma_\varpi\}$ and $\varpi(g) := C_\varpi g^{\kappa_0-6}$ one has $\varpi(g) \leq 1$ and the following. On every torus $\mathbb{T}_m$, $m \geq m_\mathbb{T}(g_-(g))$, for every $g_0 \in (0, g_-(g)]$ and $K = K(g_0, g)$, the K -step run from g_0^{-2} in the sense of Definition 7.1 is admissible, and its reference inverse coupling $\tilde{x}$ satisfies $|\tilde{x} - x_K| \leq \varpi(g)$, where $x_K \in (X, X + B_\#)$ is the last inverse coupling of the first-passage run of clause (0).*

The two runs start from one bare coupling and differ only in where the choices are read: off the run, or off the reference line. In [T-B] the one-loop parts of their increments cancel level by level; the small-field parts are Lipschitz in the whole coupling history, by the memory bound of the correlation theorem, and see the choices only through where the fluctuation integral is cut; the large-field remainders are summed on each side separately; and a discrete Gronwall argument keeps the two runs within $\varpi(g)$ of each other at every level, uniformly in K . The estimate on the cut is proved there by a propagation argument.

Proof of (c)(2), given (a), (c)(1), (RS) and Proposition 7.4. Let $\delta := \tilde{x} - X$; by clause (0) and Proposition 7.4, $-\varpi \leq \delta \leq B_\# + \varpi$. By (RS) for the admissible run from g_0^{-2} , for real $|\zeta| < 2\varrho$ the run from $g_0^{-2} - \zeta$ lies in the working tube, with reference inverse coupling $\tilde{x} - \zeta_K(\zeta)$, $\zeta_K(\zeta)$ real, continuous, between $\frac{2}{3}\zeta$ and $\frac{4}{3}\zeta$. If $\delta \geq 0$, then $\zeta_K(0) \leq \delta \leq \zeta_K(\frac{3}{2}\delta)$ and some $\zeta \in [0, \frac{3}{2}\delta]$ has $\zeta_K(\zeta) = \delta$; if $\delta < 0$, some $\zeta \in [\frac{3}{2}\delta, 0]$ does. As $\varrho \geq 6(B_\# + 1)$, $|\zeta| \leq \frac{3}{2}(B_\# + \varpi) \leq \varrho/4 < 2\varrho$, so the run from $g_0^{-2} - \zeta$ lies in the working tube and has reference inverse coupling X , and $g_0^{-2} - \zeta = a_K$ by (a). Thus $g_0^{-2} - a_K = \zeta \in [-\frac{3}{2}\varpi, \frac{3}{2}(B_\# + \varpi)] \subset [-\varrho_2/2, \varrho_2/2]$. The correlations of clause (ii) at g_0 are $S^K(g_0^{-2}; \cdot)$, since the lattice measure depends only on the lattice and on g_0 ; along a convergent sequence, pass to a subsequence with $a_{K_i} - (g_0^{(i)})^{-2} \rightarrow \hat{w}$ and apply (c)(1). $\square$

Lemma 7.5 (the renormalised reference coupling). *For every n and $\hat{w}, \hat{w}' \in \bar{D}(0, \varrho)$,*

$$\frac{1}{3}|\hat{w} - \hat{w}'| \leq |x_n^{\text{ref}}(\hat{w}) - x_n^{\text{ref}}(\hat{w}')| \leq \frac{5}{3}|\hat{w} - \hat{w}'|, \quad (7.10)$$

and $|x_{n+1}^{\text{ref}}(\hat{w}) - x_n^{\text{ref}}(\hat{w})| \leq \frac{5}{3}\Upsilon_n|\hat{w}|$ for $|\hat{w}| \leq \varrho/2$ and $\Upsilon_n \leq 1$. Hence x_n^{ref} converges uniformly on $[-\varrho/2, \varrho/2]$ to a function x_∞^{ref} satisfying (7.10), a strictly monotone homeomorphism onto a compact interval.

Proof. By (7.2), $\zeta_n(\hat{w}) - \zeta_n(\hat{w}') - (\hat{w} - \hat{w}') = \sum_{k < n} [b_{k+1}(\hat{w}) - b_{k+1}(\hat{w}')]$, of modulus at most $\frac{2}{3}|\hat{w} - \hat{w}'|$ by (RS); since $x_n^{\text{ref}} = X - \zeta_n$, this is (7.10). By the definition of $\mathbf{n}_n$, $x_{n+1}^{\text{ref}}(\mathbf{n}_n(\hat{w})) = x_n^{\text{ref}}(\hat{w})$, and $|\mathbf{n}_n(\hat{w})| \leq 2|\hat{w}| \leq \varrho$ by (7.5); so (7.10) at depth $n+1$ gives $|x_{n+1}^{\text{ref}}(\hat{w}) - x_n^{\text{ref}}(\hat{w})| \leq \frac{5}{3}|\hat{w} - \mathbf{n}_n(\hat{w})| \leq \frac{5}{3}\Upsilon_n|\hat{w}|$. As $\sum_n \Upsilon_n < \infty$ the convergence is uniform; (7.10) passes to the limit, and its lower bound makes the continuous real function x_∞^{ref} injective. $\square$

Proof of (d)(2). With $\hat{w}_n := a_n - x_n$, Lemma 7.5 gives $x_\infty^{\text{ref}}(\hat{w}_n) - y_n \rightarrow 0$, so $x_\infty^{\text{ref}}(\hat{w}_n) \rightarrow y$; the image of $[-\varrho/2, \varrho/2]$ is compact and contains y , and $|\hat{w}_n - \hat{w}_*| \leq 3|x_\infty^{\text{ref}}(\hat{w}_n) - y| \rightarrow 0$. By (c)(1), $S^n(x_n; \vec{f}) \rightarrow S^{(\hat{w}_*)}(f)$; likewise for (x'_n) , with the same $\hat{w}_*$. $\square$

The same argument applies to the re-filed coupling $\hat{x}$ of (d)(1), given that $\hat{w} \mapsto \hat{x}$ is bi-Lipschitz uniformly in the depth and converges uniformly; for the re-filed run these two facts are proved in [T-R] (§7.5).

Lemma 7.6 (exact change of units). *Let $0 \leq s \leq n$ and $\mathbb{T}' := \mathbb{T}_{m_0+s}$. For every bare inverse coupling x and every $\vec{f}$,*

$$S_{\mathbb{T}}^n(x; f_1, \dots, f_p) = S_{\mathbb{T}'}^{n-s}(x; f_1(L^{-s} \cdot), \dots, f_p(L^{-s} \cdot)), \quad (7.11)$$

the subscript indicating the torus.

Proof. The dilation $y \mapsto L^s y$ maps $\mathbb{T}$ onto $\mathbb{T}'$ and the bare lattice of $\mathbb{T}$ at depth n onto that of $\mathbb{T}'$ at depth $n-s$; it maps plaquettes to plaquettes, multiplies barycentres by L^s , and preserves the Wilson action and the Haar measure. So $\mathcal{O}(f_i)$ on $\mathbb{T}$ is $\mathcal{O}(f_i(L^{-s} \cdot))$ on $\mathbb{T}'$, and the integrals (7.1) coincide. In the second reading the reference lattice is the depth- s lattice of the first; with (2.13) for $(i, j) = (K-s, K)$ and Corollary 2.12 this gives (d)(3). $\square$

Corollary 7.7 (change of units for the limit family). *Let $s \geq 0$, put $\mathbb{T}' := \mathbb{T}_{m_0+s}$, and let $\hat{w}$ be real with $|\hat{w}| + 3\lambda_U s + c' \leq \varrho/2$, c' as in (7.3). Then there is a real $\tilde{w}$ with $\hat{w} - \tilde{w} \in [\lambda_U s - c', 3\lambda_U s + c']$ such that*

$$S_{\mathbb{T}}^{(\hat{w})}(f_1, \dots, f_p) = S_{\mathbb{T}'}^{(\tilde{w})}(f_1(L^{-s} \cdot), \dots, f_p(L^{-s} \cdot))$$

for all separated Lipschitz $\vec{f}$, the subscript indicating the torus; every limit point of $a_{n-s} - a_n + \hat{w}$ serves as $\tilde{w}$.

Proof. By (7.11) with $x = a_n - \hat{w}$, $S_{\mathbb{T}}^n(a_n - \hat{w}; \vec{f}) = S_{\mathbb{T}'}^{n-s}(a_{n-s} - \tilde{w}_n; \vec{f}(L^{-s} \cdot))$ with $\tilde{w}_n := \hat{w} + a_{n-s} - a_n$, and $|\tilde{w}_n| \leq \varrho/2$ by (7.3). By (c)(1) on $\mathbb{T}'$, applied to the real sequence $a_{n-s} - \tilde{w}_n$, the right side differs from $S_{\mathbb{T}'}^{(\tilde{w}_n)}(\vec{f}(L^{-s} \cdot))$ by a quantity tending to 0; by (b) the left side tends to $S_{\mathbb{T}}^{(\hat{w})}(\vec{f})$. Along a subsequence $\tilde{w}_n \rightarrow \tilde{w}$, and the continuity of $\hat{w} \mapsto S_{\mathbb{T}'}^{(\hat{w})}$ gives the identity; the window for $\hat{w} - \tilde{w}$ is that of $a_n - a_{n-s}$ in (7.3). $\square$

Under the conditions of clause (v-f) of Theorem 0, $a_{n-s} - a_n$ converges to $-s\alpha_\infty$, and $\tilde{w} = \hat{w} - s\alpha_\infty$. Not asserted without them: convergence of $a_{n-s} - a_n$; a relation between $\tilde{w}$, or the reference coupling $x_\infty^{\text{ref}}(\tilde{w})$ on $\mathbb{T}'$, and the running couplings of the tuned runs at depth s ; the identification for larger s at fixed ϱ (it holds for every s after enlarging ϱ and lowering γ_U); an identification between families tuned at two different reference couplings, beyond the relation at short distance of (v-f)(3).

The intermediate couplings, by contrast, converge along the tuned family: for fixed $s \geq 1$ the inverse coupling $x_{(s)}$ at depth s of the tuned run of length n converges as $n \rightarrow \infty$, because the tuned runs of lengths n and $n+1$ have inverse couplings at depth s that differ by at most $C(\rho_U^{n-s} + (s+1)\Pi_n)$ (Steps 1–4 of §7.4, proved in [T-U]), which is summable in n since $\kappa_0 \geq 10$.

7.4 Structure of the proofs in the companion account

We describe the proofs in [T-U] of (a), of the properties of $\mathbf{n}_n$, and of Proposition 7.3; the division into steps is ours. $\mathcal{R}_n$ is the run of length n from $a_n - \hat{w}$, $\mathcal{R}_{n+1}$ the run of length $n+1$ from $a_{n+1} - \mathbf{n}_n(\hat{w})$, superscripts $[n]$, $[n+1]$ refer to them, and $e_r := \rho_U^{n-r} + \Pi_n$.

Steps 1–3: scaffold, pin, matching [T-U]. Reading the choices off θ gives $\mathcal{R}_n$ and $\mathcal{R}_{n+1}$ the same thresholds, cube sizes, partitions and trees at every common depth. Balaban’s sharp small-field tests read the finest lattice, which differs between the cutoffs; they are replaced by narrow ramps on the same test quantities, so that decompositions of unity remain identities and the two cutoffs’ tests differ only through the dependence of the test quantities on the added level. By (RS) the reference inverse coupling is strictly increasing in the bare value along admissible runs, and shooting from the reference line gives a_n [T-corr]. The first window in (7.3) follows from (2.9), each increment being within λ_U of its one-loop part; $a_n - a_l$ is the sum of the $n - l$ increments of the tuned run of length n between depths n and l , plus the differences at depth l between the tuned runs of lengths i and $i+1$, $l \leq i < n$, which Step 4 bounds by $C(\rho_U^{i-l} + (l+1)\Pi_i)$, summing to at most $C[(1 - \rho_U)^{-2} + 2T_\Pi]$. The tuned value is torus-free because the recursion uses infinite-lattice coefficients. Rouché’s theorem on one circle gives $\mathbf{n}_n$; the Schwarz lemma, with Step 4 for the runs matched at depth 0 without sources and a split of the sum over depths at $n - \lceil 2 \log n / |\log \rho_U| \rceil$, gives (7.5). Step 4 with sources comes after; there is no circularity.

Step 4: the two cutoffs, depth by depth [T-U]. At every depth r both runs live on one lattice. The families of $\mathcal{R}_{n+1}$ are transported to the localisation domains of $\mathcal{R}_n$ by a linear map preserving totals on regular backgrounds, hence coupling coefficients, and differences are measured component by component (small-field, source-carrying, large-field remainders) in the weighted norms of [T-corr]. Feeding the step of $\mathcal{R}_n$ with the transported inputs costs the Lipschitz bound (e) on the differences at coarser depths and on the coupling differences; the effect of the extra level beneath $\mathcal{R}_{n+1}$ is a source term of size $C_\sigma e_r$, with a constant C_σ [T-U]. The resulting system couples every depth to all others and is closed by the following lemma.

Lemma 7.8 (comparison lemma; [T-U]). *Let I be finite, $I_0 \subset I$, $D, \sigma \geq 0$ and $E > 0$ functions on I , and $M \geq 0$ a matrix on $I_0 \times I$, with $D_i \leq \sum_j M_{ij} D_j + \sigma_i$, $\sum_j M_{ij} E_j \leq \frac{1}{2} E_i$ and $\sigma_i \leq \frac{1}{2} E_i$ for $i \in I_0$, and $D_i \leq E_i$ for $i \notin I_0$. Then $D \leq E$.*

Proof. If $t := \max_i D_i/E_i > 1$, it is attained at some $i \in I_0$, where $tE_i \leq \sum_j M_{ij} E_j + \sigma_i \leq \frac{t}{2} E_i + \frac{1}{2} E_i$; so $t \leq 1$. $\square$

With envelopes proportional to e_r and smallness conditions on g , this gives the two-cutoff contraction estimate [T-U]: at depth r the three kinds of differences are at most $C_E e_r$, $4C_E e_r$

and $4C_E \hat{\mathbf{t}}_r e_r$, with a constant C_E free of n, r and the torus, with $\hat{\mathbf{t}}_r \leq \frac{1}{16}$ of order $\theta_r^{1-\kappa_0/2}$ up to logarithms, and the coupling differences at depth $r+1$ are at most $C(r+2+(1-\rho_U)^{-1})e_r$. The factor ρ_U^{n-r} is the geometric forgetting of the extra level; Π_n is a floor due to the marginal part of the large-field remainders, which (d)(1) removes.

Step 5: localisation [T-U]. As in Theorem 3.1(d), each run's partition function is $Z(w) = e^{\mathcal{E}(w)} Z(0) \mathbf{h}(w)$, where $\mathcal{E}(w) = \sum_{\mathbf{w} \leq r < n} \sum_Y e^{(r)}(Y; w)$ is the sum of the sourced vacuum energies of all localisation domains Y , wrapping ones included — each holomorphic, zero at $w = 0$, reading w only near Y , and at most $C_e e^{-(1-\delta)\kappa d(Y)}$, with κ, δ and the tree length $d(Y)$ as there — and $\mathbf{h}(0) = 1, |\mathbf{h}| \leq \mathbf{Q}_w$ by (3.6). By Step 4 the two runs' depth- r vacuum energies differ by a source-free constant plus $\sum_Y G_r(Y; z)$, where $G_r(Y; \cdot)$ reads the sources only near Y and is at most $C_\eta(1+r)e_r e^{-(1-\delta)\kappa d(Y)}$, with a constant C_η free of n, r and the torus.

Step 6: the last lattice, chart by chart [T-U]. The integral over the last lattice is $\tilde{Z} = \sum_h \mu_h(\mathcal{V}_w)$, a sum over the admissible histories h of large-field regions (labels) of complex measures on its gauge fields $\mathcal{V}_w$, and $\mathbf{h} = \tilde{Z}(w)/\tilde{Z}(0)$. On the common scaffold, the labels whose first large-field region is born at least one level above the bare lattice of $\mathcal{R}_n$ occur in both cutoffs. Such a label is compared in its own chart: on each live large-field region of Balaban's second class [11] the variables are written $V = V'V_\Lambda$, a Haar-preserving translation, and the delta functions of the construction act on the chart variables V' , which the two cutoffs share. (In the raw variables the two measures of such a label live on different graphs and are mutually singular.) In the chart, μ_h is one assembly formula applied to the numbers stored by the run — coefficients, couplings, weights. Interpolating between the two arrays with one complex dial per class of stored numbers, holomorphy, a bound on moduli uniform in the dial, and the age factor of (e) bound the difference by Step 4. Labels born within n_c levels of the bare lattice, n_c of order $\log \log n$, are not paired: each carries a large-field region of age at least $n - \mathbf{w} - n_c$, and the age factor bounds their total weight by $O(\rho_U^{n-\mathbf{w}})$; labels on which the ramps of the two cutoffs differ are paid by the ramp estimate of Step 1. In the normalisation of Step 5 the outcome is

$$|\mathbf{h}^{[n+1]} - \mathbf{h}^{[n]}| \leq \varepsilon_n \mathbf{Q}_w(1 + \mathbf{Q}_w), \quad \varepsilon_n \leq C_{\text{TC}} e_w, \quad (7.12)$$

for sources in a polydisc of radius proportional to $\varrho/\mathbf{Q}_w$, with C_{TC} depending on g and N_w : the rate is free of the volume, the constant is not.

Step 7: increments [T-U]. The sources being a local shift of the bare coupling, centre $\mathcal{R}_{n+1}$ at the real bare value $a_{n+1} - \mathbf{n}_n(u)$, $u := \text{Re } \hat{w}$, with the source $\mathbf{n}_n(u + i \text{Im } \hat{w} + w) - \mathbf{n}_n(u)$, $w = \sum_i z_i f_i$, and centre $\mathcal{R}_n$ at $a_n - u$ with the source $i \text{Im } \hat{w} + w$. At $z = 0$ these are the runs from $a_{n+1} - \mathbf{n}_n(\hat{w})$ and from $a_n - \hat{w}$, and, the supports being disjoint, $\mathbf{p}$ differentiations at $z = 0$ give $\mathbf{n}'_n(\hat{w})^{\mathbf{p}} S^{n+1}(a_{n+1} - \mathbf{n}_n(\hat{w}); \vec{f})$ and $S^n(a_n - \hat{w}; \vec{f})$. Write $Z(w) = e^{\mathcal{E}(w)} Z(0) \mathbf{h}(w)$ as in Step 5. At depths $\mathbf{w} \leq r \leq n/2$ a domain of Step 5 contributes to the mixed derivative only if it meets all $\mathbf{p} \geq 2$ supports, so its length is at least of order $dL^{r-\mathbf{w}}/M$; Cauchy's estimate on a polydisc of radii proportional to $\varrho/\mathbf{p}$ gives $C_1(\rho_U^{n/2} + \Pi_n)$. At depths $r > n/2$ each run is estimated alone, giving $C_2 e^{-cdL^{n/2-\mathbf{w}}/M}$. Since $|\mathbf{h}| \leq \mathbf{Q}_w$ and $\mathbf{h}(0) = 1$, the Schwarz lemma on complex lines gives $|\mathbf{h} - 1| \leq \frac{1}{2}$ on a polydisc of radius proportional to $\varrho/\mathbf{Q}_w$, and (7.12) with Cauchy's estimate gives $C_3(\rho_U^{n-\mathbf{w}} + \Pi_n)$. Replacing $\mathbf{n}_n(\hat{w})$ by $\hat{w}$ costs $C_4 \Upsilon_n$, by (7.5), the torus bound (3.7) with complex shifts, and Cauchy's estimate on the strip: this is (7.7); at $\hat{w} = 0$, $\mathcal{N}_{n+1}^{-\mathbf{p}} = \mathbf{n}'_n(0)^{\mathbf{p}} \mathcal{N}_n^{-\mathbf{p}}$ gives (7.8).

The Lipschitz step (e) [T-L]. One step of the correlation theorem is put in class form with an extra holomorphic dial between the two inputs, and homogeneity of the norms turns the bounds of [T-corr] into Lipschitz bounds [T-L]. An age lemma [T-L]: Balaban’s inductive estimate for the large-field operation [11, (1.79)–(1.89)], in the form proved in the companion account, holds for every value of its unnamed constant; run with a slightly larger value, it leaves a factor $q_* < 1$ per step of life of each large-field region, paid from his entropy constant, under the floor $p_0 \geq 6r + 1$ (r the cube-size exponent of (2.11)). The complete step is then estimated through moduli, the age factors making the large-field rows summable over the past uniformly in the cutoff [T-L].

The renormalised coordinate (d)(1) [T-R]. When Balaban heals a large-field region he leaves a local remainder term, subtracts its constant part, and keeps its next part — a multiple of the Wilson action, i.e. a shift of the coupling — inside the term. This is harmless for bounds, but the part does not decrease from scale to scale, so remainders created near the cutoff shift the coupling at all later scales by an amount that (2.8) does not register: this is the floor Π_n . At level i'' the partition cube of the remainder is one lattice cell, and by the alignment of Balaban’s nested partitions [7, 9] the term has every symmetry of an ordinary term [T-R]; re-filed there, it is renormalised like the ordinary terms, its coupling part entering the running coupling and the rest decaying geometrically [T-R]. The induction of [T-corr] tolerates the re-filing [T-R], and Step 4 holds with $\Pi_n \equiv 0$ [T-R], which with Step 7 gives (7.6). In a model satisfying all hypotheses of Step 4 the floor is attained in the unmodified labelling, and the bare labellings converge only polynomially because the normalisations $\mathcal{N}_n$ still run [T-R]; hence the coordinate t and the normalisation $\hat{\mathcal{N}}_n(t)$ in (d)(1).

Two blocking factors and injectivity: clause (v-f). Clause (v-f) of Theorem 0 is proved in the companion account as an implication from clauses (iii), (iv) in every volume, (v) and Corollary 1.2; it needs no new estimate. (1) Corollary 7.7 with $s = 1$ identifies $S^{(\hat{w})}$ on $\mathbb{T}_{m_0}$ with $S^{(\hat{w}-\alpha)}$ on $\mathbb{T}_{m_0+1}$, read in units larger by the factor L , for every limit point α of the steps $a_{n+1} - a_n$; with the two-sided bounds of clause (iv), which hold in every volume, this forces the steps to converge. (2) Theorem 0 holds for every odd $L \geq L_0$; it is used twice, for L and for $L + 2$, which are coprime, with a common ϱ . A bare inverse coupling x defines one lattice theory on one bare lattice; by (1.20) it is, in each of the two families, a tuned value up to an offset in the window, at depths n and n' ; and the inequality at finite cutoff behind Corollary 1.2, which reads a torus only through its size, compares the two- and three-point functions of that one lattice theory, on a common torus, with those of the two families read in their two systems of units, up to $O(d^{4-p})$. (3) A two-point function at scale r cannot agree with its dilate by a fixed factor $\rho \neq 1$ up to $O(r^2)$ for all small r , because by clause (iv) it tends to zero as $r \downarrow 0$, more slowly than r^2 . Keeping track of the two descriptions of one bare number—the ratio $L^n/(L + 2)^{n'}$ of the two units stays bounded above and below—this gives the equality of the slopes $\alpha_\infty/\log L$ of the two families and the convergence of the ratio of their units; applied to two offsets in one family, it gives (v-f)(3) and (v-f)(4). The correspondence between the two systems of units is an output of the argument, not an input; nothing about long distances enters.

7.5 What is complete here, and what the uniqueness theorem rests on

Complete in this section: the deduction of (b) from (7.5) and Proposition 7.3; (c)(1), (c)(3), and (c)(2) from Proposition 7.4; Lemma 7.5 and (d)(2); Lemma 7.6, (d)(3) and Corollary 7.7, with the convergence of the intermediate couplings stated after it, given the estimate of Step 4; Lemma 7.8.

Proved in the companion account (statements and structure given above): (a), $\mathbf{n}_n$ with (7.5), and Proposition 7.3 [T-U]; (RS) [T-corr], with Theorem 3.1 for the runs on the reference line that lie in the working tube; Proposition 7.4 [T-B]; (d)(1) [T-R]; (e) [T-L]. Among the statements on which these proofs rest, the following are proved in the companion account as well: index stationarity, object by object (when a lattice level is added beneath, Balaban’s background fields, level data, fluctuation covariances and one-loop coefficients change by a power of $L^{-\ell}$, ℓ the number of intervening levels, also beside large-field regions); the inductive clauses of [T-corr] for the re-filed run of (d)(1), with the bi-Lipschitz dependence of $\hat{x}$ on the bare value uniformly in the depth and its uniform convergence [T-R], given the reading listed below; inside the Lipschitz step, the dependence on the extra holomorphic parameter of the boundary terms stored from large-field components healed at earlier levels, with a factor geometric in their age, and the statement on the outcomes of the large-field operation that create a region of the second class (Section 5); the estimate of how a boundary term wrapped around a large-field region reads the charts of the two cutoffs (Step 6); and the treatment of a large-field component excluded in the middle of a run. Clause (v-f) of Theorem 0 is proved in the companion account as an implication from clauses (iii), (iv) in every volume, (v) and Corollary 1.2 (outline at the end of §7.4).

Proved in the companion account, among the statements on which the uniqueness theorem rests: (U2) the counting bound for decoupling families made of cubes of several sizes (the families of cubes that carry the interpolation parameters of the cluster expansions); and fifteen further statements: the two-cutoff comparison of the change-of-variables terms at a large-field component excluded in the middle of a run; the parts read over a region thrown back to an earlier level; the comparison at terms that meet the core of a large-field region left unintegrated by the large-field operation, with no age entering; the pointwise profile at complex decoupling parameters; the closure of compositions under one smallness condition; the two-cutoff stability of the small-field tests, in a corrected form: the smallness condition is tightened by fixed constants; four short lemmas; and five further statements of the two-cutoff comparison, named in the companion account.

Supporting statements of the uniqueness theorem. Besides the correlation theorem of Section 3 on the reference line and the statements just listed, the proof of the uniqueness theorem rests on the following statements of ours at $N = 2$; we do not claim them as proved.

- *Written; independent verification in progress* (three statements).

(U1) *Extension, in a corrected form.* As first formulated—every local configuration at which the two-cutoff comparison evaluates a term left over from an earlier level extends to an admissible configuration on the whole region, in the class of inherited domains—the statement is too strong: on a ring-shaped region there is an admissible flat configuration whose holonomy around the hole is set by the larger of two tolerance scales, and it has no admissible extension once the ratio of the two scales exceeds a fixed multiple of M , a ratio that the construction bounds from below only. The form used imposes the conditions of the inherited-domain class only near the domains where the proofs read them, and beyond these only the two regularity conditions [5, (3.35), (3.37)], with radii free of the coupling, together with one condition on the real background configuration, which is to be added to the definition of the class (a definition, not an estimate); in this form the extension holds for every configuration that the comparison meets.

(U3) *Un-integrated factors.* For the two factors that Balaban keeps un-integrated inside a region thrown back to an earlier level—the stored boundary terms and the quadratic fluctuation weight—a two-cutoff comparison, region by region and level by level of the region’s past: the difference of the two cutoffs’ factors is bounded linearly by the comparison’s own

accuracy at the level that created each contribution, the dependence on the age being carried by the un-normalised weight attached to every configuration label in which an old level is still present, under one smallness condition on the coupling. (The statement in the form of a smallness, at the current depth, of a single supremum over all such regions on an enlarged class of configuration pairs does not hold and is not used.) No statement on the rarity of large fields enters. Its argument uses, on the runs of the uniqueness theorem, the one-cutoff bounds for these two factors that are proved for the correlation theorem; the two proofs cut a term into pieces in different ways, and the companion account reduces this transfer to a convention on how the kept factors are stored and read and to the lemma (U5). The argument also uses (U4) and (U6).

(U5) *Background fields inside a healed region.* The lemma beneath (U3): in the interior of a region thrown back to an earlier level and healed there, the background field depends on the configuration outside the region only within the bounds stated in the companion account, at every box into which the uniqueness theorem cuts a term and with constants independent of the box. Given this lemma and the convention named under (U3), the one-cutoff bounds for the two factors of (U3), proved for the pieces into which the correlation theorem cuts a term, hold for the pieces of the uniqueness theorem. It is used only through (U3).

- *Not yet written* (two statements).

(U4) *Remainder terms of the large-field operation.* The two-cutoff comparison of the members of a family of correction terms of the large-field operation, localised in the strip around a region thrown back to an earlier level, which Balaban does not write out member by member [11]; the form of the bound is fixed in the companion account. It is used only through (U3).

(U6) *Gaussian factors of interrupted applications.* The two-cutoff comparison of the Gaussian factors kept by applications of the large-field operation that are stopped before completion and lie inside a region thrown back to an earlier level. It is used only through (U3).

- *Readings*, involving no estimate: the single-step lemmas of [T-corr] read with an extra holomorphic parameter, inside the Lipschitz step; and one reading of the re-filed run, used in [T-R].
- *For $N \geq 3$* : the group-specific step of the argument for $SU(N)$, which has not been carried out.

Clause (v) uses no statement on the rarity of large fields and nothing about regions removed at earlier levels; the statement of Balaban that it uses as he asserts it is item (Pr15) of Subsection 9.1. The complete list for Theorem 0 is in Section 9.

7.6 What clause (v) does not assert

Clause (v) asserts nothing about: $\mathbf{p} = 1$ or $\log Z_n$; coinciding or overlapping supports; Wilson loops, block-averaged observables, or the state on Balaban's block variables; $SU(N)$, $N \geq 3$; uniqueness in the cutoff on $\mathbb{R}^4$; convergence of the effective actions along the tuned family (the intermediate couplings $x_{(s)}$, $s \geq 1$, do converge: see the paragraph after Corollary 7.7); exponential rates in labellings by the bare coupling; a calibration of the parameter by a physical quantity beyond the relation at short distance of (v-f)(3) (under the conditions of (v-f) the parameter labels the limits injectively: (v-f)(4)); analyticity in the coupling on $\mathbb{R}^4$. Three further non-assertions concern the volume.

Uniformity in the torus. The exponents of the rates are torus-free, but C , v_w and n_0 depend on m_0 through Q_w , and nothing uniform in m_0 is asserted. This is a limitation of the method: every estimate used here sees the last-lattice integrals only through their total masses, via the upper and lower stability bounds of clause (i), whose extensive constants differ (their sum is c_{vol}), and the resulting slack is extensive in the volume and insensitive to separations. The volume transfer of Section 6 does not apply: it compares two volumes at one lattice action, whereas clause (v) compares two cutoffs on one torus. That $v_w \rightarrow 0$ with the volume reflects the open question whether the partition functions are free of zeros near the real coupling axis uniformly in the volume.

Uniqueness in the volume and of the vacuum. Clause (ii) gives limits on $\mathbb{R}^4$ along sub-sequences of volumes, and clause (v) does not identify them. Uniqueness of the infinite-volume limit, uniqueness of the vacuum and clustering are not asserted, and they do not follow from the properties that Theorem 0 asserts of the limits.

Remark 7.9 (five systems with every proved property). Let $(S_n)_{n \geq 2}$ be the connected Schwinger functions of a limit on $\mathbb{R}^4$ given by clause (ii). Each modification below is again a system of connected Schwinger functions with every property Theorem 0 asserts of such a limit — multilinearity, symmetry, invariance under $\mathbb{R}^4 \rtimes W_4$, reflection positivity and reconstruction, the per-ball bounds of clause (iii) and the two-sided bounds of clause (iv), existential constants adjusted where needed — and differs from (S_n) at unit separation: (1) a frozen random constant, $\mathcal{O}(f) \mapsto \mathcal{O}(f) + \xi \int f$ with ξ independent, with all moments and small positive variance, so that $S_n \mapsto S_n + \kappa_n(\xi) \prod_i \int f_i$, $\kappa_n(\xi)$ the n -th cumulant of ξ ; (2) a decoupled generalised free field of scaling dimension $\Delta \in [1, 3]$ and small amplitude, added to the field; (3) $\mathcal{O}(f) \mapsto \mathcal{O}(f) + \varepsilon : \psi^2 : (f)$, the Wick square of a decoupled massless free scalar ψ , which changes S_2 by $O(\varepsilon^2 d^4)$ for sources of size d ; the added free part is $O(4)$ invariant and clusters, so the modified system is $O(4)$ invariant, clusters, and satisfies the hypothesis of clause (vi) whenever the original one does; (4) the covariance of an independent Gaussian field, smooth and bounded on separated points, reflection positive and $\mathbb{R}^4 \rtimes W_4$ invariant, added to S_2 ; (5) a one-parameter family of reflection-positive Euclidean-invariant systems with the same short-distance behaviour and different mass gaps.

For (1): $\int f$ is Euclidean invariant; conditionally on ξ the field is shifted by a reflection-invariant constant, so reflection positivity holds for each value of ξ , hence on average; the added terms are at most $|\kappa_n(\xi)| \prod_i \|\text{supp } f_i\| \|f_i\|_\infty$, which keeps the per-ball bounds; for bumps of size d they change S_2 by at most $C \text{Var}(\xi) d^8 \|f_1\|_\infty \|f_2\|_\infty$, small against the lower bound $c g_{\infty, s(d)}^4 \|f_1\|_\infty \|f_2\|_\infty$ of clause (iv) on its bounded range of d , so the two-sided bounds persist with constants as close to the original ones as we please; and after reconstruction $\xi - \mathbb{E}\xi$ is a translation-invariant unit vector orthogonal to the vacuum. These examples do not show that uniqueness fails for Yang–Mills. They show that it does not follow from the properties in Theorem 0, and that a proof must use further, infrared, information on the lattice measures, such as a cluster expansion on the last lattice in infinite volume or a gap uniform in the volume. At strong bare coupling the infinite-volume lattice measure is controlled by the cluster expansion of [23]; that is the opposite regime.

Remark 7.10 (The short-distance corollary). Corollary 1.2 is proved in the companion account as a deduction from the volume-transfer bound of Section 6 (Corollary 6.19), applied on one fixed reference torus, from Lemma 7.6 with the flow bounds (2.13), and from parts (b) and (c) above. The five systems above are compatible with it, their admixtures being $O(d^2)$ or smaller. In this paper it enters only clause (v-f) (§7.4).

8 Clause (vi): rotation invariance under a flux hypothesis

The limits of clause (ii) are invariant under the translations of $\mathbb{R}^4$ and under the hyperoctahedral group W_4 of the 384 signed permutations of the coordinates (Theorem 4.10); these are the symmetries the lattice retains. This section concerns the remaining rotations. We display the flux hypothesis of clause (vi), with a sufficient condition in terms of the stress tensor (Hypothesis 8.2), state the conditional theorem (Theorem 8.4), and give its proof in three parts: an exact lattice Ward identity together with the classification of the local operators that can break rotations (part (A1), Subsection 8.3); a bound on the ultraviolet terms of that identity (part (A2), Subsection 8.4); and the deduction (Subsection 8.5). Without the hypothesis, Subsection 8.6 proves rotation invariance of the short-distance germ of the two- and three-point functions to all orders in the running coupling, and states, with an outline of the proof, that exact invariance fails at every scale on every torus. Throughout, the gauge group is $SU(2)$, hypotheses (H1)–(H7) are in force with $\varrho > 0$ fixed as in (H7), and nothing about long distances is assumed except the displayed hypothesis. Theorem 0 asserts nothing about the truth of that hypothesis (Subsection 8.7).

8.1 Notation

Rotations. Indices μ, ν, ρ, σ run over $\{0, 1, 2, 3\}$. For a real antisymmetric 4×4 matrix ω let $\xi(x) := \omega x$ be the Killing field of the rotations $R_t := e^{t\omega}$ about the origin, let $(R \cdot f)(x) := f(R^{-1}x)$ for $R \in O(4)$, and $\delta_\omega f := \frac{d}{dt}|_{t=0} R_t \cdot f = -\xi \cdot \nabla f$. For an $\mathbb{R}^4$ limit $S_n := S_{n;\infty}^T$ of clause (ii) and a separated n -tuple $\vec{f}$ of C^2 functions (Section 4), the *rotation defect* is

$$D_\omega(\vec{f}) := \sum_{i=1}^n S_n(f_1, \dots, \delta_\omega f_i, \dots, f_n). \quad (8.1)$$

Since the S_n are translation invariant, it is no loss to rotate about the origin.

Lattice densities. At cutoff K the bare lattice on $\mathbb{T}_m$ has spacing $\varepsilon = L^{-K}$ (Section 4). A link $\ell = (x, \mu)$ runs from x to $x + \varepsilon e_\mu$ and has midpoint x_ℓ ; a plaquette $p = \{x; \mu, \nu\}$ has barycentre $x(p)$, and $p \pm \varepsilon e_\rho$ is its translate; $a_p := 1 - \text{Re tr } U(\partial p)$ and $\mathcal{O}(f) = \sum_p f(x(p)) a_p$. Write $W \in SU(2)$ as $W = w^0 + i \vec{w} \cdot \vec{\sigma}$, with $\vec{\sigma}$ the Pauli matrices, so that $\text{tr } W = w^0$. For a link $\ell \subset \partial p$ let $W_{\ell,p}$ be the holonomy of ∂p that starts at the initial point of ℓ and traverses ℓ first, in its positive direction; then $a_p = 1 - w_{\ell,p}^0$ and $|\vec{w}_{\ell,p}|^2 = a_p(2 - a_p)$. For $\rho \neq \mu$ let $p(\ell, \pm \rho)$ be the plaquette spanned by ℓ and $\pm \varepsilon e_\rho$, and put

$$\vec{u}_{\ell,\rho} := -\frac{1}{2}(\vec{w}_{\ell,p(\ell,\rho)} - \vec{w}_{\ell,p(\ell,-\rho)}) \quad (\rho \neq \mu), \quad \vec{u}_{\ell,\mu} := 0, \quad \mathfrak{d}_{\ell,p,\rho} := \vec{u}_{\ell,\rho} \cdot \vec{w}_{\ell,p}. \quad (8.2)$$

For a plaquette p we define the *spin density* and the *orbital density*

$$\mathfrak{s}_{\sigma\rho}(p) := \sum_{\ell \subset \partial p} \varepsilon^{-1} (x_\ell - x(p))_\sigma \mathfrak{d}_{\ell,p,\rho}, \quad \mathfrak{r}_\rho(p) := \sum_{\ell \subset \partial p} \mathfrak{d}_{\ell,p,\rho} - \frac{1}{2}(a_{p+\varepsilon e_\rho} - a_{p-\varepsilon e_\rho}). \quad (8.3)$$

They are gauge-invariant polynomials in the links within distance 2ε of $x(p)$. Since $|\vec{u}_{\ell,\rho} \cdot \vec{w}_{\ell,p}| \leq \frac{1}{4}(|\vec{w}_{\ell,p(\ell,\rho)}|^2 + |\vec{w}_{\ell,p(\ell,-\rho)}|^2) + \frac{1}{2}|\vec{w}_{\ell,p}|^2$, both are bounded by $4 \sum_{p'} a_{p'}$, the sum running over p , the plaquettes sharing a link with p , and $p \pm \varepsilon e_\rho$. By definition $\mathfrak{s}_{\sigma\rho}(p) = 0$ unless e_σ lies in the plane of p . The following facts are proved in the companion account: under the symmetries of the lattice $\mathfrak{s}$ transforms as a 2-tensor and $\mathfrak{r}$ as a vector; and on smooth connections, with $\vec{F} \cdot \vec{G} := -\text{tr}(FG)$ for anti-Hermitian F, G and the field evaluated at $x(p)$, one has $\mathfrak{s}_{\nu\rho}(p) = \varepsilon^4 \vec{F}_{\rho\mu} \cdot \vec{F}_{\nu\mu} + O(\varepsilon^6)$ for $p = \{x; \mu, \nu\}$, so that summed over the six orientations the leading terms give $\varepsilon^4 \sum_\mu \vec{F}_{\sigma\mu} \cdot \vec{F}_{\rho\mu}$, symmetric in (σ, ρ) ; and $\mathfrak{r}_\rho(p) = O(\varepsilon^7)$.

Displacement field. For $\eta : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ let

$$V_\eta := \sum_{\ell} \sum_{b=1}^3 v_\ell^b \partial_\ell^b, \quad \vec{v}_\ell := \varepsilon^{-1} \sum_{\rho} \eta^\rho(x_\ell) \vec{u}_{\ell,\rho}, \quad (\partial_\ell^b G)(U) := \frac{d}{ds} \Big|_{s=0} G(\dots, e^{is\sigma^b} U_\ell, \dots). \quad (8.4)$$

V_η commutes with gauge transformations; on smooth fields $i\vec{u}_{\ell,\rho} \cdot \vec{\sigma} = \varepsilon^2 F_{\rho\mu}(x_\ell) + O(\varepsilon^4)$, so that V_η is the lattice form of the covariant displacement $\delta A_\mu = \eta^\rho F_{\rho\mu}$.

The lattice stress tensor. Let $x_0 := g_0^{-2}$ and $\mathfrak{s}_{(\nu\sigma)} := \frac{1}{2}(\mathfrak{s}_{\nu\sigma} + \mathfrak{s}_{\sigma\nu})$. For real Lipschitz h on $\mathbb{R}^4$ put

$$\Theta_{\nu\sigma}(h) := g_0^{-2} \sum_p h(x(p)) [\mathfrak{s}_{(\nu\sigma)}(p) - \delta_{\nu\sigma} a_p], \quad \mathsf{P}_\nu(h) := \sum_{p \parallel e_\nu} h(x(p)) a_p, \quad (8.5)$$

the second sum running over the plaquettes whose plane contains e_ν ; for a symmetric-matrix-valued h write $\Theta(h) := \sum_{\nu,\sigma} \Theta_{\nu\sigma}(h_{\nu\sigma})$. On smooth connections $\Theta(h) = \int h_{\nu\sigma} T_{\nu\sigma} dx + O(\varepsilon^2)$, where

$$T_{\nu\sigma} = g_0^{-2} \left(\sum_{\mu} \vec{F}_{\nu\mu} \cdot \vec{F}_{\sigma\mu} - \frac{1}{4} \delta_{\nu\sigma} \sum_{\alpha,\beta} \vec{F}_{\alpha\beta} \cdot \vec{F}_{\alpha\beta} \right)$$

is the Yang–Mills stress tensor. Thus Θ carries the factor x_0 and is traceless at tree level, while the orientation-weighted plaquette sums P_ν , the lattice form of the diagonal tensor $-\frac{1}{2} \sum_{\mu} \text{tr } F_{\nu\mu} F_{\nu\mu}$, carry no factor x_0 .

Cumulants. For an observable Y not depending on z we write $\langle Y; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_{K,m} := \partial_{z_1} \cdots \partial_{z_n} \langle Y \rangle_z \Big|_{z=0}$, where $\langle \cdot \rangle_z$ is the normalised expectation for $\exp[-g_0^{-2} A(U) + \sum_i z_i \mathcal{O}(f_i)] dU$; for $Y = \mathcal{O}(f_0)$ this is $S_{n+1;K,m}^T(f_0, f_1, \dots, f_n)$. Along a sequence $(g_0^{(j)}, m_j)$ as in Theorem 4.10, with $K_j = K(g_0^{(j)}, g)$ and $\varepsilon_j = L^{-K_j}$, we write $\langle \cdot \rangle_j$ for $\langle \cdot \rangle_{K_j, m_j}$.

The terms of the Ward identity; radial cut-offs; the flux. Let $\nabla_\rho^s \varphi(y) := (2\varepsilon)^{-1} [\varphi(y + \varepsilon e_\rho) - \varphi(y - \varepsilon e_\rho)]$, $\nabla_\rho^c \varphi(y) := \varepsilon^{-1} [\varphi(y + \frac{\varepsilon}{2} e_\rho) - \varphi(y - \frac{\varepsilon}{2} e_\rho)]$ and $\mathsf{L}_\xi \varphi := \sum_\rho \xi^\rho \nabla_\rho^s \varphi$. For a cut-off χ and a function c on plaquettes put

$$\mathfrak{A}_\chi[c] := \sum_p c(p) \chi(x(p)) \sum_{\rho,\sigma} \omega_{\rho\sigma} \mathfrak{s}_{\sigma\rho}(p), \quad \mathfrak{K}_\chi[c] := \sum_p c(p) \chi(x(p)) \varepsilon^{-1} \sum_\rho \xi^\rho(x(p)) \mathfrak{r}_\rho(p), \quad (8.6)$$

$$\mathfrak{h}_\chi(p) := \varepsilon^{-1} \sum_{\ell \subset \partial p} \sum_{\rho} (\chi(x_\ell) - \chi(x(p))) \xi^\rho(x_\ell) \mathfrak{d}_{\ell,p,\rho},$$

$$\mathfrak{B}_\chi^{\text{sh}} := g_0^{-2} \sum_p a_p (\mathsf{L}_\xi \chi)(x(p)) - g_0^{-2} \sum_p \mathfrak{h}_\chi(p) - \frac{3}{2} \sum_{p=\{x;\mu,\rho\}} a_p [\xi^\rho \nabla_\rho^c \chi + \xi^\mu \nabla_\mu^c \chi](x(p)). \quad (8.7)$$

The *spin term* $\mathfrak{A}_\chi$ sees only the antisymmetric part of $\mathfrak{s}$; the *orbital term* $\mathfrak{K}_\chi$ carries the weight $|\xi|/\varepsilon$; the *shell term* $\mathfrak{B}_\chi^{\text{sh}}$ is supported where χ is not locally constant, and its last sum carries no factor g_0^{-2} . When $\chi \equiv 1$ near $\text{supp } f$ we write $\mathfrak{A}[f] := \mathfrak{A}_\chi[f]$ and $\mathfrak{K}[f] := \mathfrak{K}_\chi[f]$, with f read as the plaquette function $p \mapsto f(x(p))$; these do not depend on χ . Fix a non-increasing $\chi_1 \in C^\infty(\mathbb{R})$ equal to 1 on $(-\infty, \frac{5}{4}]$ and to 0 on $[\frac{7}{4}, \infty)$, and put $\chi_{\mathcal{R}}(x) := \chi_1(|x|/\mathcal{R})$. A *radial cut-off admissible for $\vec{f}$* is a $\chi_{\mathcal{R}}$ such that $\bigcup_i \text{supp } f_i$ lies in the open ball of radius $\mathcal{R}$; for j large it satisfies the hypotheses of Theorem 8.7. The shell term $\mathfrak{B}_{\chi_{\mathcal{R}}}^{\text{sh}}$ is the flux of the lattice rotation (angular-momentum) current of the rotations generated by ω through the shell on which $\chi_{\mathcal{R}}$ varies; Theorem 8.7 below is the corresponding exact conservation law.

Definition 8.1 (Flux form). For a real antisymmetric $\omega \neq 0$, $(\text{IR-fl})_\omega$ is the statement: for every $n \geq 2$ and every separated n -tuple $\vec{f}$ of C^∞ functions,

$$\liminf_{\mathcal{R} \rightarrow \infty} \limsup_{j \rightarrow \infty, j \in J} |\langle \mathfrak{B}_{\chi_{\mathcal{R}}}^{\text{sh}}; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_j| = 0. \quad (8.8)$$

8.2 The hypothesis and the theorem

Hypothesis 8.2 (The flux hypothesis (H-IR) of clause (vi), and a sufficient condition). Along the sequence (K_j, m_j) , $j \in J$, considered: (H-IR) for one real antisymmetric $\omega \neq 0$, the flux form $(\text{IR-fl})_\omega$ of Definition 8.1 holds; this is (H-IR).

Sufficient condition. There is a sequence $\mathcal{R}_i \rightarrow \infty$ such that, for every $n \geq 2$ and every n -tuple $\vec{f}$ of real functions in $C_c^1(\mathbb{R}^4)$ with pairwise disjoint supports,

$$\lim_{i \rightarrow \infty} \limsup_{j \rightarrow \infty, j \in J} \sum_{\Delta \in \mathcal{C}_i} \sup_{Y \in \mathcal{Y}(\Delta)} |\langle Y; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_j| = 0, \quad (8.9)$$

where $\mathcal{C}_i$ is the set of unit cubes $\Delta = z + [0, 1)^4$, $z \in \mathbb{Z}^4$, that meet $\{\mathcal{R}_i \leq |y| \leq 2\mathcal{R}_i\}$, Δ^* is the cube of side 2 concentric with Δ , and $\mathcal{Y}(\Delta)$ consists of the following observables:

- (1) $\Theta_{\nu\sigma}(h)$, $\nu, \sigma \in \{0, 1, 2, 3\}$, for Lipschitz h supported in Δ^* with $\|h\|_\# \leq 1$;
- (2) the orientation-weighted plaquette sums $P_\nu(h)$ of (8.5), which carry no factor g_0^{-2} , for the same h ;
- (3) the link densities $g_0^{-2} \sum_p \sum_{\ell \subset \partial p} \sum_\rho k_{\ell,p,\rho} \mathfrak{d}_{\ell,p,\rho}$ of (8.2), for weights k vanishing unless $x(p) \in \Delta^*$, with $\varepsilon^{-1} \sup |k| \leq 1$.

The modulus is taken after the smearing over a cube, that is, after the sum over all plaquette orientations at a site. In particular the condition holds if, for some $\delta > 0$, every $Y \in \mathcal{Y}(\Delta)$ with Δ^* at distance $D \geq 1$ from $\bigcup_i \text{supp } f_i$ satisfies

$$\sup_{j \in J} |\langle Y; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_j| \leq C(\vec{f}) D^{-4-\delta}. \quad (8.10)$$

Remark 8.3 (Reading of the hypothesis). Hypothesis 8.2 is the hypothesis (H-IR) of clause (vi), with its sufficient condition (1.22) and the lattice densities made explicit. The flux form is what the proof uses (Proposition 8.15); given part (A2) it is necessary as well as sufficient for the conclusion (Remark 8.20). The sufficient condition implies it (Lemma 8.12). The pointwise bound (8.10) implies the cube-summed form (8.9): at most $C\mathcal{R}^4$ cubes meet the shell of radius $\mathcal{R}$, and for large $\mathcal{R}$ each contributes at most $C(\vec{f})(\mathcal{R}/2)^{-4-\delta}$.

(a) The decay asked for in the sufficient condition is strictly faster than $|x|^{-4}$ and uniform in the cutoff; a uniform decay slower than $|x|^{-4}$ does not suffice (Remark 8.21(b)). At tree level the correlation decays like $|x|^{-8}$; a mass gap uniform in the cutoff would give exponential decay. Theorem 0 asserts no decay at all.

(b) The factor x_0 multiplies only the combination $\mathfrak{s}_{(\nu\sigma)} - \delta_{\nu\sigma} a_p$, which is traceless at tree level. In (8.9) the modulus is taken after smearing over a cube, hence after the sum over all plaquette orientations at a site. A bound uniform in the cutoff on x_0 times a trace-type density alone—for instance on single plaquettes, where the diagonal entries of the summand of Θ equal $-x_0 a_p$ for the plaquettes whose plane does not contain the direction in question—would force the limiting correlations of that density with the sources to vanish identically, since $x_0 \rightarrow \infty$; the condition is not meant in that sense.

(c) The sufficient condition enters the proof only through Lemma 8.12, that is, through the decomposition (8.16) of the shell term of a radial cut-off. This is why it is required of the two further families (2), (3): besides the components of Θ , the shell term contains the x_0 -free sums P_ν (from the Jacobian of the displacement field) and link densities with the factor x_0 and weights of size $O(\varepsilon)$ (from the Taylor remainder of the cut-off). The link densities are admitted only with weights of size $O(\varepsilon)$; with weights of order one they would contain x_0 times trace-type densities, which (b) excludes.

(d) Through the words “uniformly in the cutoff” the sufficient condition also contains a boundedness statement for stress-tensor insertions, which is ultraviolet in nature. The flux form (H-IR), which is all the proof uses, does not contain it.

Theorem 8.4 (Clause (vi): rotation invariance under (H-IR)). *There is a threshold $\gamma_{\text{vi}} \in (0, \gamma_J]$, chosen last, after L , $\mathfrak{p}$, γ , ϱ and finitely many auxiliary exponents, such that for $0 < g \leq \gamma_{\text{vi}}$ the following holds. Let $S_n = S_{n;\infty}^T$, $n \geq 2$, be an $\mathbb{R}^4$ limit as in Theorem 4.10, along a sequence $(g_0^{(j)}, m_j)$ and an infinite set J satisfying the one-sided floor*

$$K_j \geq Q(m_j)^j \quad (j \in J), \quad (8.11)$$

where $Q(m) := \exp(c_{\text{vol}}(2L^{m+b_*})^4) \geq 1$ is the volume factor of the torus bound (3.7) of clause (iii); it depends on m and on L , $\mathfrak{p}$, γ , g , ϱ , not on K , g_0 or the positions. Assume the flux hypothesis (H-IR) of Hypothesis 8.2 along this sequence. Then for every $n \geq 2$, every $R \in O(4)$ and every separated n -tuple $\vec{f}$ on $\mathbb{R}^4$,

$$S_n(R \cdot f_1, \dots, R \cdot f_n) = S_n(f_1, \dots, f_n). \quad (8.12)$$

Consequently the S_n are invariant under $\mathbb{R}^4 \rtimes O(4)$, the centred Schwinger functions built from them are reflection positive with respect to every hyperplane, and the elements of $O(4)$ that fix the time vector e_0 act on the Hilbert space $\mathcal{H}$ of Theorem 4.17 by unitary operators that fix the vacuum vector Ω , commute with e^{-tH} for $t \geq 0$, extend the unitaries $U(w)$, $w \in W_3$, of that theorem, and depend strongly continuously on the group element.

Remark 8.5 (Status and scope). Clause (vi) is conditional, by design, on its displayed hypothesis, about which nothing is asserted. Given the hypothesis, the proof below is a deduction from part (A1), part (A2), and clauses (ii) and (iii). The identity of part (A1) is proved in full in Subsection 8.3; the classification is proved in the companion account. Part (A2) is proved in the companion account as an implication of the same kind as clauses (iii)–(v): beneath it lie Bałaban’s theorems as cited, the correlation theorem with sources (Section 3), the same-cutoff Lipschitz estimate and the two-cutoff comparison theorem of clause (v) as implications (Section 7), with the supporting statements listed under clause (v), and one proposition on the class form of the densities, used as a reading. Apart from the statements listed under clause (v), which it shares through the two-cutoff comparison, every statement of ours beneath (vi) is written out in the companion account. Two choices enter as definitions: the lattice stress tensor (8.5), and the place at which an old large-field term is localised at the collar of a large-field region. Given part (A2), the infrared half is proved: the shell flux tends to minus the rotation defect, so that it vanishes in the limit if and only if the defect does (Proposition 8.15), and the threshold $|x|^{-4}$ in the sufficient condition is sharp for this argument (Remark 8.21(b), proved in the companion account). The floor (8.11) and the ceiling $g \leq \gamma_{\text{vi}}$ are one-sided. The floor is met along every iterated limit $\lim_m \lim_K$ (choose $K_m \geq Q(m)^m$); it is used only to absorb the volume factor in the constants of part (A2), and it is not needed (Remark 8.6). The theorem holds for every limit along such a sequence; no uniqueness of limits is used. Clause (iv) is not used; of the material of clause (v), only the same-cutoff Lipschitz estimate, the two-cutoff comparison and the re-filing of healed large-field remnants are used (Section 7).

Remark 8.6 (The floor is not needed). The conclusion of Theorem 8.4 holds for every limit of $(ii-\mathbb{R}^4)$, without the floor (8.11). This is proved in the companion account as an implication from the volume-transfer bound of Section 6; we do not reproduce it here, and the statement of clause (vi) in Theorem 0 keeps the floor.

8.3 Part (A1): the exact Ward identity and the classification

Theorem 8.7 (Exact rotational Ward identity). *Fix K, m, g_0 , a real antisymmetric ω , and $\chi \in C^\infty(\mathbb{R}^4; [0, 1])$ supported in the open cube $(-L^m + 2\varepsilon, L^m - 2\varepsilon)^4$, so that $\chi\xi$ is a function on $\mathbb{T}_m$. Let $f_1, \dots, f_n$, $n \geq 1$, be real functions on $\mathbb{T}_m$ with $\chi \equiv 1$ on the 2ε -neighbourhood of $\bigcup_i \text{supp } f_i$. Then*

$$\begin{aligned} \sum_{i=1}^n S_{n;K,m}^T(f_1, \dots, \mathcal{L}_\xi f_i, \dots, f_n) &= -g_0^{-2} \langle (\mathfrak{A}_\chi + \mathfrak{K}_\chi)[1]; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_{K,m} \\ &\quad + \sum_{i=1}^n \langle (\mathfrak{A} + \mathfrak{K})[f_i]; (\mathcal{O}(f_l))_{l \neq i} \rangle_{K,m} + \langle \mathfrak{B}_\chi^{\text{sh}}; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_{K,m}. \end{aligned} \quad (8.13)$$

Proof. Put $w := \sum_i z_i f_i$, $c := g_0^{-2} - w$ and $S_w := \sum_p c(p) a_p$, so that $e^{-S_w} = \exp[-g_0^{-2} A(U) + \sum_i z_i \mathcal{O}(f_i)]$.

Step 1 (Schwinger–Dyson). For a vector field $V = \sum_{\ell,b} v_\ell^b \partial_\ell^b$ with smooth coefficients and $\text{div } V := \sum_{\ell,b} \partial_\ell^b v_\ell^b$, invariance of Haar measure under left multiplication gives $\int \partial_\ell^b (v_\ell^b e^{-S_w}) dU = 0$, hence $\langle V S_w \rangle_w = \langle \text{div } V \rangle_w$ for z near 0.

Step 2 (two link derivatives). From $(i\sigma^b)(w^0 + i\vec{w} \cdot \vec{\sigma}) = -w^b + i(w^0 \delta^{bd} - \epsilon^{bcd} w^c) \sigma^d$ one gets $\partial_\ell^b a_p = w_{\ell,p}^b$ for $\ell \subset \partial p$, and $\sum_b \partial_\ell^b w_{\ell,p}^b = 3w_{\ell,p}^0$. Hence $V_\eta a_p = \varepsilon^{-1} \sum_{\ell \subset \partial p} \sum_\rho \eta^\rho(x_\ell) \mathfrak{d}_{\ell,p,\rho}$ and $\text{div } V_\eta = \frac{3}{2\varepsilon} \sum_\ell \sum_\rho \eta^\rho(x_\ell) [a_{p(\ell,\rho)} - a_{p(\ell,-\rho)}]$. For $\eta = \chi\xi$ a summation by parts, using that ξ^ρ does not depend on x_ρ ($\omega_{\rho\rho} = 0$), turns the latter into the last sum in (8.7).

Step 3 (the split). Since ξ is affine, $\xi^\rho(x_\ell) = \xi^\rho(x(p)) + \sum_\sigma \omega_{\rho\sigma}(x_\ell - x(p))_\sigma$; writing $\chi(x_\ell) = \chi(x(p)) + (\chi(x_\ell) - \chi(x(p)))$, Step 2 gives

$$V_{\chi\xi} a_p = \chi(x(p)) \left[\varepsilon^{-1} \sum_\rho \xi^\rho(x(p)) \sum_{\ell \subset \partial p} \mathfrak{d}_{\ell,p,\rho} + \sum_{\rho,\sigma} \omega_{\rho\sigma} \mathfrak{s}_{\sigma\rho}(p) \right] + \mathfrak{h}_\chi(p).$$

Insert $\sum_{\ell \subset \partial p} \mathfrak{d}_{\ell,p,\rho} = \frac{1}{2}(a_{p+\varepsilon e_\rho} - a_{p-\varepsilon e_\rho}) + \mathfrak{r}_\rho(p)$, multiply by $c(p)$, which does not depend on U , and sum over p . Summing by parts on each translate lattice of plaquettes and using $\xi^\rho(x \pm \varepsilon e_\rho) = \xi^\rho(x)$,

$$\sum_p c(p) \chi(x(p)) \xi^\rho(x(p)) (2\varepsilon)^{-1} (a_{p+\varepsilon e_\rho} - a_{p-\varepsilon e_\rho}) = - \sum_p a_p \xi^\rho(x(p)) \nabla_\rho^s (c\chi)(x(p)).$$

Since $\chi \equiv 1$ on the 2ε -neighbourhood of the support of w , $c\chi = g_0^{-2}\chi - w$ at every plaquette read here, and $c = g_0^{-2}$ on the support of $\mathfrak{h}_\chi$. Therefore

$$V_{\chi\xi} S_w = \mathcal{O}(\mathcal{L}_\xi w) - g_0^{-2} \sum_p a_p (\mathcal{L}_\xi \chi)(x(p)) + \mathfrak{A}_\chi[c] + \mathfrak{K}_\chi[c] + g_0^{-2} \sum_p \mathfrak{h}_\chi(p),$$

and Step 1 with Step 2 gives $\langle \mathcal{O}(\mathcal{L}_\xi w) \rangle_w = -\langle \mathfrak{A}_\chi[c] \rangle_w - \langle \mathfrak{K}_\chi[c] \rangle_w + \langle \mathfrak{B}_\chi^{\text{sh}} \rangle_w$. Now apply $\partial_{z_1} \dots \partial_{z_n}$ at $z = 0$: mixed derivatives of $\log Z$ are joint cumulants, $\mathcal{O}(\mathcal{L}_\xi w) = \sum_i z_i \mathcal{O}(\mathcal{L}_\xi f_i)$, and $\mathfrak{A}_\chi[c] = g_0^{-2} \mathfrak{A}_\chi[1] - \sum_i z_i \mathfrak{A}[f_i]$, and likewise for $\mathfrak{K}$. This is (8.13). $\square$

Proposition 8.8 (Classical content and the orbital density). *Let A be a smooth compactly supported $\mathfrak{su}(2)$ -valued connection, $U_\ell := P \exp \int_\ell A$, $J_\mu := \sum_\nu D_\nu F_{\nu\mu}$, and $O_1 := \sum_{\mu,\nu} \text{tr}(D_\mu F_{\mu\nu} D_\mu F_{\mu\nu})$, the value μ occupying four index slots. As $\varepsilon \rightarrow 0$:*

- (i) $A(U) = \int (-\frac{1}{4} \sum_{\mu,\nu} \text{tr} F_{\mu\nu} F_{\mu\nu}) dx + \frac{\varepsilon^2}{24} \int O_1 dx + O(\varepsilon^4)$;
- (ii) the bulk part of the breaking terms of (8.13) is the rotation of O_1 up to field-equation terms:

$$\begin{aligned} \sum_p \left[\varepsilon^{-1} \sum_\rho \xi^\rho(x(p)) \mathfrak{r}_\rho(p) + \sum_{\rho,\sigma} \omega_{\rho\sigma} \mathfrak{s}_{\sigma\rho}(p) \right] &= V_\xi A(U) \\ &= \frac{\varepsilon^2}{24} \delta_\omega \int O_1 dx \\ &\quad + \varepsilon^2 \int \sum_\mu \text{tr}(J_\mu Y_\mu) dx + O(\varepsilon^4), \end{aligned}$$

where $\delta_\omega \int O_1$ is the derivative of $\int O_1 dx$ along the rotations $e^{t\omega}$ acting on A , and $Y_\mu = \frac{1}{6} \sum_\rho \xi^\rho D_\rho^2 F_{\rho\mu} - \frac{1}{12} \sum_\rho \omega_{\rho\mu} D_\mu F_{\rho\mu}$.

- (iii) without any limit $\varepsilon \rightarrow 0$, on every $\text{SU}(2)$ -valued configuration: $\mathfrak{r}_\rho(p) = \Psi_\rho(p) - \Psi_\rho(p - \varepsilon e_\rho) + \mathfrak{r}_\rho^\sharp(p)$, with Ψ_ρ an explicit local polynomial and $|\mathfrak{r}_\rho^\sharp(p)| \leq C(\varsigma_\rho(p)^{3/2} + \varsigma_\rho(p - \varepsilon e_\rho)^{3/2})$, where $\varsigma_\rho(p)$ is the sum of a over the four plaquettes joining p to $p + \varepsilon e_\rho$ (and $\mathfrak{r}_\rho^\sharp(p) = 0$ if the plane of p contains e_ρ): the orbital density is a lattice difference plus a term at least cubic in the plaquette variables.

Proposition 8.9 (Classification). *Let $\mathcal{V}_\delta$ be the space of gauge-invariant local polynomials of dimension δ in $F_{\mu\nu}$ and its covariant derivatives ($[F] = 2$, $[D] = 1$), with W_4 and $O(4)$ acting on the indices; let $V := \mathbb{R}^4$ and $\Lambda^2 := \Lambda^2 \mathbb{R}^4$.*

- (i) For $1 \leq \delta \leq 5$, every W_4 -invariant element of $\mathcal{V}_\delta$ is a multiple of $\text{tr} F_{\mu\nu} F_{\mu\nu}$ (for $\delta = 4$) or zero; in particular it is $O(4)$ -invariant.
- (ii) Λ^2 is W_4 -irreducible, and $\text{Hom}_{W_4}(\Lambda^2, \mathbb{R}) = \text{Hom}_{W_4}(\Lambda^2, \mathcal{V}_4) = \text{Hom}_{W_4}(\Lambda^2, \mathcal{V}_5) = 0$: no anti-symmetric tensor operator has dimension below six.
- (iii) There is no W_4 -vector of even dimension; the W_4 -vectors of dimension five form a three-dimensional space, and each is the divergence of a symmetric W_4 -tensor of dimension four.
- (iv) Under W_4 , $\text{Sym}^2 V = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{6}$ (trace, traceless diagonal, off-diagonal), and each summand occurs exactly once in $\mathcal{V}_4$: the equivariant symmetric tensors of dimension four are spanned by $\mathcal{T}_{\alpha\beta}^{(1)} = \sum_\kappa \text{tr} F_{\alpha\kappa} F_{\beta\kappa}$, $\mathcal{T}_{\alpha\beta}^{(2)} = \delta_{\alpha\beta} \text{tr} F_{\kappa\lambda} F_{\kappa\lambda}$ and $\mathcal{T}_{\alpha\beta}^{(3)} = \delta_{\alpha\beta} \sum_\kappa \text{tr} F_{\alpha\kappa} F_{\alpha\kappa}$, of which only the first two are $O(4)$ -equivariant.
- (v) Modulo total derivatives, the W_4 -invariant elements of $\mathcal{V}_6$ form a three-dimensional space with basis O_1 , $O_2 := \sum \text{tr}(D_\alpha F_{\mu\nu} D_\alpha F_{\mu\nu})$, $O_3 := \sum_\nu \text{tr}(J_\nu J_\nu)$, and $O_F := \sum \text{tr}(F_{\mu\nu} F_{\nu\rho} F_{\rho\mu})$ satisfies $O_2 \simeq 2O_3 - 4O_F$. O_2 , O_3 , O_F are $O(4)$ -invariant and O_1 is not, even modulo the field equation. Thus, modulo total derivatives, field-equation terms and $O(4)$ -invariants, O_1 is the only W_4 -invariant local operator of dimension at most six that breaks $O(4)$, and no F^3 term breaks it. As densities, the breaking part is spanned by O_1 and $Q := \sum_\alpha \partial_\alpha^2 \sum_\mu \text{tr} F_{\alpha\mu} F_{\alpha\mu}$.

Status of (A1). The identity (8.13) is exact at every cutoff and in every volume, and its proof is complete above. Propositions 8.8 and 8.9 are proved in the companion account; we give their statements and the structure of the proofs. Proposition 8.8(i)–(ii) is a Taylor expansion in the Fock–Schwinger gauge at the centre of each plaquette, and (iii) is a finite identity for 4-linear forms on the quaternions. Proposition 8.9 rests on a parity rule — by the sign changes in W_4 , a component

of a W_4 -invariant tensor vanishes unless every index value occurs an even number of times — and on explicit contractions with the Bianchi identity. The non-invariance of O_1 is seen on abelian configurations $A = a \cos(k \cdot x) T$: over a period, $\int O_1$ is proportional to $\sum_{\mu, \nu} k_\mu^2 (k_\mu a_\nu - k_\nu a_\mu)^2$, which equals t^4 at $(k, a) = (te_1, e_3)$ and $t^4/2$ at the rotated pair $(t(e_1 + e_2)/\sqrt{2}, e_3)$, while total derivatives integrate to zero and field-equation terms contribute multiples of the $O(4)$ -invariant O_3 . No decomposition of the lattice field equation into the divergence of a lattice stress tensor plus a small remainder is used or claimed: the three functionals of (8.13) are explicit, and no lattice stress tensor has to be chosen for the identity. Read with Proposition 8.8(ii), the identity says: rotating the sources equals inserting the rotation of $\frac{\varepsilon^2}{24} \int O_1$ with the factor g_0^{-2} of the action, up to field-equation terms, plus a flux through the region where χ varies.

8.4 Part (A2): the coefficient bound

For a Lipschitz $h : \mathbb{R}^4 \rightarrow \Lambda^2 \mathbb{R}^4$ (real antisymmetric matrices) put

$$\mathbf{I}(h) := g_0^{-2} \sum_p \sum_{\rho, \sigma} h_{\rho\sigma}(x(p)) \mathfrak{s}_{\sigma\rho}(p),$$

so that $g_0^{-2} \mathfrak{A}_\chi[1] = \mathbf{I}(\chi\omega)$ and $\mathfrak{A}[f] = \mathbf{I}(g_0^2 f\omega)$.

Proposition 8.10 (The coefficient bound; proved in the companion account as an implication, see “Status of (A2)” below). *Let $0 < g \leq \gamma_{\text{vi}}$.*

- (a) Spin channel. *Let $f_1, \dots, f_n$ be real Lipschitz functions with supports pairwise at distance at least $d > 0$, and let either $n \geq 2$, or $n \geq 1$ and $\text{dist}(\text{supp } h, \bigcup_i \text{supp } f_i) \geq d$. Then*

$$|\langle \mathbf{I}(h); \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_{K,m}| \leq C_n^{\text{vi}}(d) Q(m)^{n+1} \Gamma_K \|h\|_\# \prod_i \|f_i\|_\#, \quad (8.14)$$

where $C_n^{\text{vi}}(d)$ does not depend on K , g_0 , m or the positions of the supports, and $\Gamma_K \rightarrow 0$ as $K \rightarrow \infty$, bounded by a negative power of K times a power of $\log K$.

- (b) Vector channel: conservation of the lattice stress tensor. *Along every sequence as in Theorem 4.10 satisfying (8.11), for every $n \geq 2$, every separated n -tuple $\vec{f}$ of C^∞ functions, every ω and every radial cut-off χ admissible for $\vec{f}$,*

$$\lim_{j \rightarrow \infty, j \in J} \left[-g_0^{-2} \langle \mathfrak{A}_\chi[1]; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_j + \sum_{i=1}^n \langle \mathfrak{A}[f_i]; (\mathcal{O}(f_l))_{l \neq i} \rangle_j \right] = 0. \quad (8.15)$$

Consequently, along such sequences, the first two terms on the right of (8.13) tend to zero for every radial admissible cut-off.

For the consequence: the bulk spin term is $-\langle \mathbf{I}(\chi\omega); \mathcal{O}(f_1); \dots \rangle_j$, with $n \geq 2$; the slot terms are $\langle \mathbf{I}(g_0^2 f_i \omega); (\mathcal{O}(f_l))_{l \neq i} \rangle_j$, with $n - 1 \geq 1$ sources at positive distance from $\text{supp } f_i$. Under (8.11), $Q(m_j)^{n+1} \Gamma_{K_j} \rightarrow 0$ for every n , because Γ_K is bounded by a negative power of K up to logarithms. The orbital terms are (8.15).

On one fixed torus. For fixed m the factor $Q(m)^{n+1}$ in (8.14) is a constant, so that the right side of (8.14) tends to zero as $K \rightarrow \infty$; and part (b) holds, with the same proof, along every sequence of first-passage cutoffs $K_j \rightarrow \infty$ on one fixed torus $\mathbb{T}_m$, $m \geq m_{\mathbb{T}}(g_-(g))$, the floor (8.11) serving only to absorb the volume factor, which is then fixed (companion account). Hence on a fixed torus, too, the first two terms on the right of (8.13) tend to zero as $K \rightarrow \infty$, for every radial admissible cut-off. This is used only in Subsection 8.6.

Remark 8.11 (Rates; why not ε^2). By Proposition 8.8 the breaking terms are classically of order ε^2 , but neither part of Proposition 8.10 gives a power ε^2 , and none is needed: only convergence to zero is used. In (a), two features of the estimates limit Γ_K . Balaban’s bounds control the irrelevant remainder of a localised term with a gain of strictly less than one power of the scale ratio beyond dimension four, because second derivatives of the background field are not controlled, only Hölder norms of first derivatives, with an exponent below one [7, (4.17)–(4.18)]; and remainders of the large-field operations created at the first scales above the cutoff are carried by the induction with a weight that does not decay. This gives a negative power of K up to logarithms; with the re-filing of healed large-field remnants used for clause (v) one obtains a positive power of ε , never ε^2 . In (b), the proof gives a bound of order $g_0(\log x_0)^C$, and the terms are expected to be of order $g_0^2 \asymp 1/\log \varepsilon^{-1}$: they compensate the finite renormalisations of the naive lattice stress tensor in its traceless channels, which tend to their tree values because $g_0 \rightarrow 0$. The scalar operator O_1 itself behaves differently: a lattice transcription of O_1 inserted with a scalar weight is not small, since it mixes with $\text{tr } F_{\mu\nu} F_{\mu\nu}$ and shifts the bare coupling. The identity (8.13) involves only the rotation of O_1 , which lies in the antisymmetric channel, where by Proposition 8.9(ii) there is nothing of lower dimension to mix with.

Structure of the proof of (a). The insertion $t\mathbf{I}(h)$ is carried through Balaban’s renormalisation group as a first-order perturbation of the run of the correlation theorem with sources (Section 3): one differentiates the run at $t = 0$, and the derivative of the run is the run of the derivatives. (1) At the bare level, $|\mathbf{I}(h)| \leq C\|h\|_\infty g_0^{-2} \sum_{p'} a_{p'}$, the sum over the plaquettes near $\text{supp } h$; so the insertion is a bare-level term of the class of the correlation theorem, read through its size, range and analyticity only. (2) Every term that the steps generate from it transforms under the lattice symmetries as $\Lambda^2 \mathbb{R}^4$ times a weight. In Balaban’s renormalisation of localised terms the hypercubic symmetry is used once, to reduce the quadratic part of a localised term to one number multiplying $\text{tr } F_{\mu\nu} F_{\mu\nu}$ [9, (3.51)–(3.61)]. For terms of type Λ^2 this reduction gives zero, because $\text{Hom}_{W_4}(\Lambda^2, \mathcal{V}_4) = 0$ (Proposition 8.9(ii)): no counterterm is generated, and all these terms are irrelevant. (3) The same-cutoff Lipschitz estimate for Balaban’s complete step, large-field operations included (Section 7), gives a linear recursion for the sizes of these terms which closes with a geometric contraction, uniformly in the number of scales. Reading out the connected correlation at the last lattice gives (8.14), the volume entering only through the factor $Q(m)^{n+1}$.

Structure of the proof of (b). The content of (b) is that the lattice stress tensor is conserved in the limit. (1) *Exact input.* By Proposition 8.8(iii) the orbital density is a lattice difference plus a cubic term, so that in the bulk the orbital term is small already in the norms of the correlation theorem. And for every vector field η vanishing within distance $\varepsilon/2$ of the supports of $f_1, \dots, f_k$, the field V_η does not see the sources, and Step 1 of the proof of Theorem 8.7 gives the exact null identity

$$\langle g_0^{-2} V_\eta A(U) - \text{div } V_\eta; \mathcal{O}(f_1); \dots; \mathcal{O}(f_k) \rangle_{K,m} = 0 :$$

away from the sources, the lattice stress tensor is conserved at every cutoff. (2) *Three channels.* By Proposition 8.9(iv), the renormalisation of a symmetric-tensor insertion is described by three numbers, one for each of the channels **1**, **3**, **6**; the number of the trace channel is the derivative of the running shift of clause (iii). (3) *The weight.* The orbital term carries the weight $|\xi|/\varepsilon$, which costs one power of the scale ratio per scale, while the correlation theorem bounds an irrelevant term at scale ratio t by $t^{4+\hat{\alpha}}$, with $\hat{\alpha} < 1$ the Hölder exponent of [7, (4.18)]. An exact double subtraction — a lattice identity for bilinear vector-type functionals of closed 2-cochains against an affine weight — raises the gain of the weighted part to $t^{4+2\hat{\alpha}}$, which suffices once $\hat{\alpha} > \frac{1}{2}$. Balaban’s bounds allow any $\hat{\alpha} < 1$; $\hat{\alpha} > \frac{1}{2}$ is a one-sided choice made before γ_{vi} . With this, the orbital term

becomes, up to a remainder tending to zero, a first-order symmetric-tensor perturbation of the run. (4) *Transport*. A first-order correlation theorem for symmetric-tensor sources carries the three numbers channel by channel. Its one new datum is the one-loop drift of the two traceless numbers. In the traceless-diagonal channel this drift equals the one-loop drift of the inverse coupling, from Balaban's one-loop formulas [1, Prop. 1.1], [7, (2.2)–(2.12)] and the one-loop law of clause (0); hence traceless-diagonal insertions without the factor x_0 , at points separated from the sources, are $O(x_K/x_0) = O(K^{-1})$. (5) *Reduction*. Using the null identity at the sources and in the shell, the bracket in (8.15) equals

$$c_K \sum_i S_{n;K,m}^T(\dots, \xi \cdot \nabla f_i, \dots) + c'_K \langle \mathfrak{B}_\chi^{\text{sh}}; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle + c''_K Y^{\text{ct}} + c'''_K Y^{\text{sh}} + r_K,$$

where Y^{ct} and Y^{sh} are cumulants of traceless-diagonal insertions without the factor x_0 , smeared with the symmetrised gradients of $f_i \xi$ at the sources and of $\chi \xi$ in the shell; the numbers c_K, c'_K, c''_K, c'''_K do not depend on $\vec{f}, \chi, \omega, n$ or the positions, $|c_K| + |c'_K| + |c''_K| \leq C g_0 (\log x_0)^C$, $|c'''_K| \leq C g_0^{-1} (\log x_0)^C$, and $r_K \rightarrow 0$ under (8.11). By (4), $c'''_K Y^{\text{sh}} = O(g_0^{-1} K^{-1} (\log x_0)^C) \rightarrow 0$; the other terms tend to zero because the quantities they multiply stay bounded: the first by the per-ball bound (4.4), Y^{ct} by (4), and the shell cumulant because, by (8.13) and part (a), $1 + c'_K$ times it equals a sum of bounded terms and $c'_K \rightarrow 0$. Hence the bracket tends to zero.

Status of (A2). Both parts are proved in the companion account as implications (we have given the statements and the structure of the proofs): from the lemmas of the correlation theorem with sources, from the same-cutoff Lipschitz estimate and from the two-cutoff comparison of clause (v), as described in Sections 3 and 7 and listed in Section 9; from Balaban's papers [1, 7, 9, 10, 11], by statement; and from one proposition on the class form of the densities, used as a reading. Both parts also use verifications, involving no new estimate, that the inductive step of the correlation theorem of Section 3 (the run with sources), together with the large-field estimates of that step as stated there, applies to families of localised terms that transform under the lattice symmetries as $\Lambda^2 \mathbb{R}^4$, $\mathbb{R}^4$ or $\text{Sym}^2 \mathbb{R}^4$ times a weight; for the vector channel this includes the terms that straddle the boundary of a large-field region, treated by the device that the correlation theorem already uses for terms outside such a region, with the old large-field terms at the collar localised as fixed by definition in the companion account. No infrared input—decay, clustering, or uniqueness of a limit—enters either part, and no correlation theorem for the stress tensor in its physical normalisation is used.

8.5 The deduction

The hypothesis (H-IR) is the flux form (IR-fl) $_\omega$ of Definition 8.1, for one generator ω ; the sufficient condition of Hypothesis 8.2 implies it for every generator (Lemma 8.12).

Lemma 8.12 (The sufficient condition implies the flux form). *Assume Proposition 8.10(a) and the sufficient condition of Hypothesis 8.2 along a sequence satisfying (8.11). Then (IR-fl) $_\omega$ holds for every $\omega \neq 0$.*

Proof. Let $\chi = \chi_{\mathcal{R}}$ and $M_{\sigma\rho} := (\partial_\sigma \chi) \xi^\rho$. Since χ is radial and ξ is tangent to spheres, $\text{tr } M = \xi \cdot \nabla \chi = 0$. Let M^s and M^a be the symmetric and antisymmetric parts of M . Expanding $\chi(x_\ell) - \chi(x(p))$ and $\xi^\rho(x_\ell)$ about $x(p)$ in $\mathfrak{h}_\chi$, and using $\text{tr } M^s = 0$ in the definition of Θ , gives the exact decomposition

$$\mathfrak{B}_\chi^{\text{sh}} = -\Theta(M^s) - \mathbf{I}(m_\chi) - \frac{3}{2} \sum_\nu \mathbf{P}_\nu(\xi^\nu \nabla_\nu^c \chi) + g_0^{-2} \mathcal{O}(\mathbf{L}_\xi \chi) - g_0^{-2} \sum_p \sum_{\ell \subset \partial p} \sum_\rho r_{\ell,p,\rho} \mathfrak{d}_{\ell,p,\rho}, \quad (8.16)$$

where $(m_\chi)_{\rho\sigma} := M_{\sigma\rho}^a$ and $r_{\ell,p,\rho} := \varepsilon^{-1}(\chi(x_\ell) - \chi(x(p)))\xi^\rho(x_\ell) - \sum_\sigma \varepsilon^{-1}(x_\ell - x(p))_\sigma M_{\sigma\rho}(x(p))$. For $\varepsilon < \mathcal{R}/4$ all weights are supported in the shell $\{\mathcal{R} \leq |x| \leq 2\mathcal{R}\}$, and their $\sharp$ -norms are at most $C|\omega|$, uniformly in $\mathcal{R} \geq 1$, because $|\nabla\chi| \leq C\mathcal{R}^{-1}$, $|\nabla^2\chi| \leq C\mathcal{R}^{-2}$ and $|\xi| \leq 2|\omega|\mathcal{R}$ there; moreover $|r_{\ell,p,\rho}| \leq C\varepsilon|\omega|\mathcal{R}^{-1}$. Take the cumulant with $\mathcal{O}(f_1), \dots, \mathcal{O}(f_n)$ and let $j \rightarrow \infty$ at fixed large $\mathcal{R}$. The term $g_0^{-2}\mathcal{O}(\mathbb{L}_\xi\chi)$ contributes $g_0^{-2}S_{n+1;K_j,m_j}^T(\mathbb{L}_\xi\chi, f_1, \dots, f_n)$; since $\xi \cdot \nabla\chi = 0$, $\|\mathbb{L}_\xi\chi\|_\sharp \leq C|\omega|\varepsilon^2$, so by the per-ball bound (4.4) this is $O(g_0^{-2}\varepsilon_j^2) \rightarrow 0$. The term $\mathbf{I}(m_\chi)$ tends to zero by Proposition 8.10(a), with $n \geq 2$, and (8.11). The remaining three terms are made of the three families of densities of Hypothesis 8.2, with weights of the sizes admitted there: $\sharp$ -norms at most $C|\omega|$ for $\Theta(M^s)$ and for the sums $\mathbf{P}_\nu$, and $|r_{\ell,p,\rho}| \leq C\varepsilon|\omega|\mathcal{R}^{-1}$ for the link densities. Let (φ_Δ) be a smooth partition of unity on $\mathbb{R}^4$, indexed by the unit cubes Δ of the tiling of (8.9), with $\text{supp } \varphi_\Delta$ in the cube Δ^* of side 2 concentric with Δ and $|\nabla\varphi_\Delta| \leq C$. It splits each weight into pieces supported in the cubes Δ^* , of $\sharp$ -norm at most $C|\omega|$ (for the link densities, with $\varepsilon^{-1} \sup |k| \leq C|\omega|$). The weights vanish outside $\{\frac{5}{4}\mathcal{R} - 2\varepsilon \leq |x| \leq \frac{7}{4}\mathcal{R} + 2\varepsilon\}$, so once $\mathcal{R} \geq 8$ only cubes Δ that meet $\{\mathcal{R} \leq |y| \leq 2\mathcal{R}\}$ contribute. Hence, for every j , the cumulant of these three terms with $\mathcal{O}(f_1), \dots, \mathcal{O}(f_n)$ is at most $C|\omega|$ times the sum in (8.9) at the radius $\mathcal{R}$, and its $\limsup_j$ tends to zero along the sequence $\mathcal{R}_i$ of the sufficient condition. Since the other two terms tend to zero as $j \rightarrow \infty$ at every fixed large $\mathcal{R}$, (8.8) follows. $\square$

Lemma 8.13 (Differentiation and continuity). *Let S_n be an $\mathbb{R}^4$ limit as in Theorem 4.10. (a) For a separated n -tuple $\vec{f}$ of C^2 functions, $t \mapsto S_n(R_t \cdot \vec{f})$ is C^1 with derivative $D_\omega(R_t \cdot \vec{f})$. (b) For a separated n -tuple $\vec{f}$ (of C^1 functions), $R \cdot \vec{f}$ is separated and $R \mapsto S_n(R \cdot \vec{f})$ is continuous on $O(4)$.*

Proof. The S_n obey the per-ball bound (4.10). For $f \in C_c^2$, $h^{-1}(R_{t+h} \cdot f - R_t \cdot f) \rightarrow \delta_\omega(R_t \cdot f)$ uniformly together with first derivatives, because second derivatives of f are uniformly continuous; that is, the convergence holds in $\|\cdot\|_\sharp$, and all functions involved are supported in a fixed neighbourhood of $R_t \cdot \text{supp } f$, so that for small h the slots stay pairwise at distance at least half the distance $d(\vec{f})$ of Section 4. Telescoping the multilinear form and applying (4.10) gives the derivative. Its continuity in t follows in the same way from $\delta_\omega(R_t \cdot f) = R_t \cdot \delta_\omega f$ and $\|R_{t'} \cdot g - R_t \cdot g\|_\sharp \rightarrow 0$ as $t' \rightarrow t$ for $g \in C_c^1$. For (b), use $\|R' \cdot f - R \cdot f\|_\sharp \rightarrow 0$ as $R' \rightarrow R$, valid for $f \in C_c^1$. $\square$

Lemma 8.14 (Dichotomy). *A closed subgroup of $O(4)$ containing W_4 is finite or equal to $O(4)$.*

Proof. Let G be such a subgroup. It is a compact Lie group, and its Lie algebra $\mathfrak{g} \subset \mathfrak{so}(4)$ is invariant under $\text{Ad}(W_4)$. The diagonal sign matrices $\epsilon = \text{diag}(\epsilon_0, \dots, \epsilon_3) \in W_4$ act on the basis $E_{\mu\nu}$, $\mu < \nu$, of $\mathfrak{so}(4)$ by $\text{Ad}(\epsilon)E_{\mu\nu} = \epsilon_\mu\epsilon_\nu E_{\mu\nu}$; these six characters are pairwise distinct, so $\mathfrak{g}$ is spanned by the $E_{\mu\nu}$ that it contains, and the permutation matrices in W_4 permute the lines $\mathbb{R}E_{\mu\nu}$ transitively. Hence either $\mathfrak{g} = 0$, and G , being discrete and compact, is finite; or $\mathfrak{g} = \mathfrak{so}(4)$, so that G contains the identity component $SO(4)$, and since $W_4 \subset G$ contains a reflection, $G = O(4)$. $\square$

Proposition 8.15 (Defect equals flux). *Along a sequence satisfying (8.11), let $0 < g \leq \gamma_{\text{vi}}$, so that Proposition 8.10 applies. Then for every $n \geq 2$, every separated n -tuple $\vec{f}$ of C^∞ functions, every ω and every radial cut-off χ admissible for $\vec{f}$, the limit $\lim_{j \in J} \langle \mathfrak{B}_\chi^{\text{sh}}, \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_j$ exists and equals $-D_\omega(\vec{f})$. In particular it does not depend on χ , and $(\text{IR-fl})_\omega$ holds if and only if D_ω vanishes on all such tuples.*

Proof. For $f \in C_c^\infty$, $\nabla_\rho^s f - \partial_\rho f = \frac{\varepsilon^2}{6} \partial_\rho^3 f + O(\varepsilon^4)$ with all derivatives, so $\|\mathbb{L}_\xi f - \xi \cdot \nabla f\|_\sharp \leq C(f, \omega)\varepsilon^2$, with support in the ε -neighbourhood of $\text{supp } f$. By the per-ball bound (4.4) and multilinearity, $\sum_i S_{n;K_j,m_j}^T(\dots, \mathbb{L}_\xi f_i, \dots) - \sum_i S_{n;K_j,m_j}^T(\dots, \xi \cdot \nabla f_i, \dots) \rightarrow 0$, and by Theorem 4.10 the second sum tends to $\sum_i S_n(\dots, \xi \cdot \nabla f_i, \dots) = -D_\omega(\vec{f})$. By Proposition 8.10, the first two terms on the right

of (8.13) tend to zero. Hence the third converges, to $-D_\omega(\vec{f})$. If (IR-fl) $_\omega$ holds, then $|D_\omega(\vec{f})| = \lim_j |\langle \mathfrak{B}_{\chi_{\mathcal{R}}}^{\text{sh}}; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_j|$ for every large $\mathcal{R}$, and the lim inf over $\mathcal{R}$ is zero. Conversely, if D_ω vanishes, the limit is zero for every radial admissible cut-off. $\square$

Proof of Theorem 8.4. Let $\omega_0 \neq 0$ be a generator for which (IR-fl) $_{\omega_0}$ holds, as (H-IR) provides (under the sufficient condition every $\omega_0 \neq 0$ qualifies, by Lemma 8.12). By Proposition 8.15, $D_{\omega_0}(\vec{f}) = 0$ for every separated tuple $\vec{f}$ of C^∞ functions. By (4.10), $|D_{\omega_0}(\vec{f})| \leq C \sum_i \|\delta_{\omega_0} f_i\|_{\sharp} \prod_{l \neq i} \|f_l\|_{\sharp}$, and mollification approximates a separated tuple of C^2 functions in C^2 by separated tuples of C^∞ functions with supports in arbitrarily small neighbourhoods of the original ones; hence D_{ω_0} vanishes on separated tuples of C^2 functions. Since $e^{t\omega_0} \cdot \vec{f}$ is again such a tuple, Lemma 8.13(a) shows that $t \mapsto S_n(e^{t\omega_0} \cdot \vec{f})$ is constant; by density in $\|\cdot\|_{\sharp}$ and (4.10) the same holds for all separated tuples. The set Stab of those $R \in O(4)$ with $S_n(R \cdot \vec{f}) = S_n(\vec{f})$ for all $n \geq 2$ and all separated n -tuples is a group, since R maps separated tuples to separated tuples; it is closed by Lemma 8.13(b), and it contains W_4 (Theorem 4.10) and the one-parameter group $\{e^{t\omega_0}\}$. By Lemma 8.14, $\text{Stab} = O(4)$. This is (8.12). With the translation invariance of Theorem 4.10, the S_n are invariant under $\mathbb{R}^4 \rtimes O(4)$, and reflection positivity at $\{x^0 = 0\}$ (Section 4) is carried by these motions to every hyperplane. Let $R \in O(4)$ with $Re_0 = e_0$. Then R commutes with the time reflection θ and with the time translations, and preserves the half-space $\{x^0 > 0\}$. Acting on every argument of every component of the elements of the space $\mathcal{P}_+$ of Section 4, it maps $\mathcal{P}_+$ onto itself and preserves the form $(\cdot, \cdot)_{\text{OS}}$, since the centred Schwinger functions are built from the S_n ; hence it induces an isometry of $\mathcal{P}_+/\mathcal{N}$ onto itself, which extends to a unitary operator $U(R)$ on $\mathcal{H}$, fixing Ω and commuting with e^{-tH} . For $R \in W_3$ it is the unitary $U(R)$ of Theorem 4.17, and $R \mapsto U(R)$ is strongly continuous: on the dense image of $\mathcal{P}_+$ by Lemma 8.13(b), since $\|U(R')v - U(R)v\|^2 = 2\|v\|^2 - 2\text{Re}(U(R)v, U(R')v)$, and everywhere because the $U(R)$ are unitary. $\square$

8.6 Rotations at short distance, without the hypothesis

Without (H-IR), rotations act as a symmetry on the short-distance behaviour of the limits to all orders in the running coupling, but not exactly. This subsection proves the first statement for two and three points (Proposition 8.16), states it for every number of points (Proposition 8.18), and states, with an outline of its proof, that on every torus of the family exact invariance fails at every scale (Proposition 8.19, proved in the companion account); the last statement is the reason why clause (vi) carries a long-distance hypothesis. Throughout, $0 < g \leq \gamma_{\text{vi}}$.

Test tuples, rotations about a centre, and the coupling at a scale. For $d > 0$, $\vartheta \in (0, 1]$ and $n \geq 2$ let $\mathfrak{C}_n(d; \vartheta)$ be the set of n -tuples $\vec{f}$ of real C^1 functions on $\mathbb{R}^4$ with pairwise disjoint supports contained in one closed ball $\bar{B}(y, 5d)$ and with $\text{Lip } f_i \leq \|f_i\|_\infty / (\vartheta d)$ for every i ; for $d \leq \frac{1}{40}$ they are regarded as functions on every torus of the family through the quotient map. Rotations act about the centre y of the ball: $(R \cdot f)(x) := f(y + R^{-1}(x - y))$ for $R \in O(4)$, and $\delta_\omega f := -\omega(x - y) \cdot \nabla f$ for a real antisymmetric ω , with $|\omega|^2 := \sum_{\mu < \nu} \omega_{\mu\nu}^2$; the tuple $R \cdot \vec{f}$ again lies in $\mathfrak{C}_n(d; \vartheta)$, with the same suprema. For $0 < d \leq 1$ let $s_d := \lceil \log_L(1/d) \rceil$, and write $g(d)$ for the running coupling at depth s_d : $g(d)^{-2} := x_{K-s_d}$ at a cutoff $K \geq s_d$, or any limit of these numbers along the sequence of cutoffs that defines the limit considered. By clause (0) every such value lies in $[g^{-2} + \max\{0, \lambda_{\sharp}s_d - c_5\}, g^{-2} + B_{\sharp}(s_d + 1)]$, and by (0)(d) $g(d)^{-2} = \frac{11N^2}{12\pi^2} \log(1/d) (1 + o(1))$ as $d \downarrow 0$; replacing s_d by the depth $s(d)$ of clause (iv) changes only the constants in (c) below.

Proposition 8.16 (Rotation invariance of the short-distance germ to all orders: two and three points). *Let $n \in \{2, 3\}$ and $\vartheta \in (0, 1]$. There is a constant C_n , depending only on $L, \mathfrak{p}, \gamma, \varrho, g$ and*

ϑ , such that for every $\mathbf{d} \in (0, \frac{1}{40}]$, every $\vec{f} \in \mathfrak{C}_n(\mathbf{d}; \vartheta)$ and every $R \in \mathrm{O}(4)$ acting about the centre of the ball of $\vec{f}$ the following hold.

- (a) For every limit $S_{n;\infty}^T$ of (ii- $\mathbb{R}^4$), and for every limit $S_{n;m}^T$ of (ii- $\mathbb{T}$ -b) on every torus $\mathbb{T}_m$ of the family, S_n standing for either,

$$|S_n(R \cdot \vec{f}) - S_n(\vec{f})| \leq C_n \mathbf{d}^{4-n} \prod_{i=1}^n \|f_i\|_\infty. \quad (8.17)$$

- (b) Uniformly in the cutoff and the volume: $\limsup_{g_0 \downarrow 0} \sup_{m \geq m_{\mathbb{T}}(g_-(g))} |S_{n;K(g_0,g),m}^T(R \cdot \vec{f}) - S_{n;K(g_0,g),m}^T(\vec{f})| \leq C_n \mathbf{d}^{4-n} \prod_i \|f_i\|_\infty$.

- (c) In the running coupling: for every $\eta \in (0, 1)$ there is $A_{n,\eta}$, depending on the same quantities and on η , with

$$C_n \mathbf{d}^{4-n} \leq A_{n,\eta} \exp\left[-(1-\eta)(4-n) \frac{12\pi^2}{11N^2} g(\mathbf{d})^{-2}\right];$$

hence there are $A_n, c_n > 0$ with $C_n \mathbf{d}^{4-n} \leq A_n k! c_n^{-k} g(\mathbf{d})^{2k}$ for every integer $k \geq 0$: the defect is smaller than every power of the running coupling at the scale of the tuple.

- (d) For two points, relatively: if moreover $g \leq \gamma_W^{\mathbb{T}}$ and $\mathbf{d}$ satisfies (F-depth), then for every pair $(f_1, f_2) \in \mathfrak{T}(\mathbf{d}; \vartheta)$ of C^1 functions, every $R \in \mathrm{O}(4)$ acting about the centre of a ball of radius $5\mathbf{d}$ that contains both supports, and S_2 as in (a),

$$\left| \frac{S_2(R \cdot f_1, R \cdot f_2)}{S_2(f_1, f_2)} - 1 \right| \leq \frac{9C_2}{2b_0 c_K} \mathbf{d}^2 (g^{-2} + B_{\sharp}(s(\mathbf{d}) + 1))^2,$$

which is $O(\mathbf{d}^2 \log^2(1/\mathbf{d}))$. For three points the bound (8.17) is absolute; no relative form is asserted.

The constants do not depend on K , g_0 , m , $\mathbf{d}$, the position or the limit. The bound is a positive power of $\mathbf{d}$, not zero: exact invariance is not asserted at any scale, and on a torus it fails at every scale (Proposition 8.19).

The proof uses a cut-off at a fixed radius of order one on one reference torus, not at the scale of the tuple: the limit of the shell term does not depend on the radius (Proposition 8.15), and the bounds of Section 3 are not covariant under changes of scale, so a cut-off at radius of order $\mathbf{d}$ would give no power of $\mathbf{d}$. The power comes from packing many signed copies of the tuple into the unit ball. The one estimate used beyond Proposition 8.10 is the following.

Proposition 8.17 (The shell term at unit radius on one torus; proved in the companion account). *Let $m \geq m_{\mathbb{T}}(g_-(g))$, $n \geq 2$, ω real antisymmetric, $\xi(x) = \omega x$, and $\chi := \chi_2$, the radial cut-off of radius $\mathcal{R} = 2$ about the origin, equal to 1 on $B(0, \frac{5}{2})$ and to 0 off $B(0, \frac{7}{2})$. There is $C_{\mathrm{sh},n}$, depending only on L , $\mathfrak{p}$, γ , ϱ , g , n and m , such that for every n -tuple $\vec{F}$ of real C^∞ functions on $\mathbb{T}_m$ with pairwise disjoint supports in $B(0, 1)$*

$$\limsup_{g_0 \downarrow 0} \left| \langle \mathfrak{B}_\chi^{\mathrm{sh}}; \mathcal{O}(F_1); \dots; \mathcal{O}(F_n) \rangle_{K(g_0,g),m} \right| \leq C_{\mathrm{sh},n} |\omega| \prod_{i=1}^n \|F_i\|_{\sharp}. \quad (8.18)$$

The constant does not depend on the positions of the supports in $B(0, 1)$, nor on their mutual distances.

Structure of the proof. By the exact decomposition (8.16), with $\mathcal{R} = 2$, the shell term is a sum of five insertions whose weights are supported in $\{\frac{5}{2} - 2\varepsilon \leq |x| \leq \frac{7}{2} + 2\varepsilon\}$, at distance at least 1 from the supports of the F_i . The main one, $-\Theta(M^s)$, is the lattice stress tensor, with its factor x_0 , smeared with the traceless symmetric weight M^s ($\text{tr } M = \xi \cdot \nabla \chi = 0$). It is bounded channel by channel, in the traceless-diagonal and off-diagonal channels of Proposition 8.9(iv), by the first-order correlation theorem for symmetric-tensor sources of part (4) of the proof of Proposition 8.10(b), read in its far form, times the normalisation of these channels, which turns the factor x_0 into at most a constant times $x_K \leq g^{-2} + B_\#$. In its far form that theorem bounds the correlation of one first-order insertion whose weight is supported at distance at least D from all the sources by a constant depending on D and on the torus, times the $\sharp$ -norms of the weight and of the sources; its proof reads the supports of the sources only through this one distance, and not through their mutual distances—if it is stated for sources whose supports are pairwise at distance at least d , its proof does not read d . The other terms of (8.16), and the terms produced when the diagonal entries of Θ are reduced to the traceless-diagonal channel, tend to zero as $K \rightarrow \infty$ at fixed $\vec{F}$: the spin term $\mathbf{I}(m_\chi)$ by Proposition 8.10(a) on the fixed torus; $g_0^{-2} \mathcal{O}(\mathbf{L}_\xi \chi)$ because $\|\mathbf{L}_\xi \chi\|_\# \leq C|\omega|\varepsilon^2$ (proof of Lemma 8.12) and by the torus bound (1.12); the link densities, whose weights are of order $\varepsilon|\omega|$, because $x_0\varepsilon(\log x_0)^C \rightarrow 0$; and the sums $\mathbf{P}_\nu$, which carry no factor x_0 and whose weight $(\xi^\nu \nabla_\nu^c \chi)_\nu$ has trace of order ε^2 , by the normalisation of the traceless-diagonal channel without the factor x_0 , of order x_K/x_0 (part (4) of the proof of Proposition 8.10(b)). No long-distance input, no bound on correlations with two stress-tensor insertions, and no decay enter.

Proof of Proposition 8.16. By the translation invariance of the limits (Theorems 4.8(c) and 4.10(c)) we may take the centre of the ball of $\vec{f}$ to be the origin, so that R and δ_ω act about 0 and $\xi(x) = \omega x$; in the proof of (b) this reduction is made after passing to limits.

Step 0 (one reference torus). Let $m_\star := m_{\mathbb{T}}(g_-(g))$ and $\mathbb{T}_\star := \mathbb{T}_{m_\star}$; as $L^{m_\star} \geq L^{11} > 8$, the ball $B(0, 4)$ embeds isometrically in $\mathbb{T}_\star$. Let $\chi := \chi_2$ as in Proposition 8.17. Fix a limit $S_{n;\star} := S_{n;m_\star}^T$ of (ii-T-b) on $\mathbb{T}_\star$, along first-passage cutoffs $K_j \rightarrow \infty$, and put $D_\omega^\star(\vec{F}) := \sum_i S_{n;\star}(F_1, \dots, \delta_\omega F_i, \dots, F_n)$ for n -tuples $\vec{F}$ of C^2 functions with pairwise disjoint supports.

Step 1 (the flux on $\mathbb{T}_\star$). Let $\vec{F}$ be an n -tuple of real C^∞ functions with pairwise disjoint supports in $B(0, 1)$. For j large, χ satisfies the hypotheses of Theorem 8.7 on $\mathbb{T}_\star$. The first two terms on the right of (8.13) tend to zero (the paragraph “On one fixed torus” after Proposition 8.10); and, as in the proof of Proposition 8.15 with the torus bound (1.12) in place of the per-ball bound, $\sum_i S_{n;K_j,m_\star}^T(\dots, \mathbf{L}_\xi F_i, \dots) \rightarrow -D_\omega^\star(\vec{F})$. Hence

$$\lim_{j \rightarrow \infty} \langle \mathfrak{B}_\chi^{\text{sh}}; \mathcal{O}(F_1); \dots; \mathcal{O}(F_n) \rangle_{K_j, m_\star} = -D_\omega^\star(\vec{F}).$$

The radius of the cut-off is fixed and of order one; it is not adapted to the size of the supports.

Step 2 (the shell at unit radius). By Proposition 8.17 at $m = m_\star$, $|D_\omega^\star(\vec{F})| \leq C_{\text{sh},n} |\omega| \prod_i \|F_i\|_\#$, with a constant that does not see the positions of the supports in $B(0, 1)$. By approximation in C^2 , with (1.12) at a fixed positive separation, the same bound holds for n -tuples of C^2 functions with pairwise disjoint supports in $\bar{B}(0, \frac{5}{8})$.

Step 3 (translations). If $\vec{F}$ and $\vec{F}(\cdot - \tau)$ both have supports in $\bar{B}(0, \frac{5}{8})$, then $D_\omega^\star(\vec{F}(\cdot - \tau)) = D_\omega^\star(\vec{F})$. Indeed $\delta_\omega[F(\cdot - \tau)] = (\delta_\omega F)(\cdot - \tau) - ((\omega\tau) \cdot \nabla F)(\cdot - \tau)$; by the translation invariance of $S_{n;\star}$ the first parts give $D_\omega^\star(\vec{F})$, and the second parts give $-\frac{d}{dt}|_{t=0} S_{n;\star}(\vec{F}(\cdot + t\omega\tau)) = 0$ (differentiability as in Lemma 8.13(a), with (1.12)).

Step 4 (signed arrays: the factor d^4). Let $\vec{f} \in \mathfrak{C}_n(d; \vartheta)$ have C^2 components and centre 0, and put $r_0 := 5d \leq \frac{1}{8}$ and $\Gamma := \{\tau \in 4r_0\mathbb{Z}^4 : |\tau|_\infty \leq \frac{1}{4}\}$. As in the proof of Lemma 6.18 with $R = 1$, $\#\Gamma \geq (32r_0)^{-4} = (160d)^{-4}$, the balls $\bar{B}(\tau, r_0)$, $\tau \in \Gamma$, are pairwise at distance at least $2r_0$ and lie

in $\bar{B}(0, \frac{5}{8})$, and an array $\sum_{\tau} \sigma_{\tau} f_i(\cdot - \tau)$ with signs $\sigma_{\tau} = \pm 1$ has the supremum and the Lipschitz constant of f_i , hence $\sharp$ -norm at most $\|f_i\|_{\sharp}$. For $n = 2$ and $\sigma \in \{\pm 1\}^{\Gamma}$ put $F_i := \sum_{\tau} \sigma_{\tau} f_i(\cdot - \tau)$, $i = 1, 2$; for $n = 3$ and $\sigma, \sigma' \in \{\pm 1\}^{\Gamma}$ put $F_1 := \sum_{\tau} \sigma_{\tau} f_1(\cdot - \tau)$, $F_2 := \sum_{\tau} \sigma'_{\tau} f_2(\cdot - \tau)$ and $F_3 := \sum_{\tau} \sigma_{\tau} \sigma'_{\tau} f_3(\cdot - \tau)$. These tuples have pairwise disjoint supports in $\bar{B}(0, \frac{5}{8})$. Averaging over the signs, $2^{-\#\Gamma} \sum_{\sigma} \sigma_{\tau} \sigma'_{\tau'} = \delta_{\tau\tau'}$, and for $n = 3$ the average of $\sigma_{\tau_1} \sigma'_{\tau_2} \sigma_{\tau_3} \sigma'_{\tau_3}$ is $\delta_{\tau_1\tau_3} \delta_{\tau_2\tau_3}$; by multilinearity and Step 3 the average of $D_{\omega}^*(\vec{F})$ over the sign families is therefore $\sum_{\tau \in \Gamma} D_{\omega}^*(\vec{f}(\cdot - \tau)) = \#\Gamma \cdot D_{\omega}^*(\vec{f})$, while each $D_{\omega}^*(\vec{F})$ obeys the bound of Step 2 with $\prod_i \|f_i\|_{\sharp}$. Hence, with $\|f_i\|_{\sharp} \leq (40M/(\vartheta d)) \|f_i\|_{\infty}$ (proof of Corollary 6.19),

$$|D_{\omega}^*(\vec{f})| \leq (160d)^4 C_{\text{sh},n} |\omega| \prod_i \|f_i\|_{\sharp} \leq 160^4 \left(\frac{40M}{\vartheta}\right)^n C_{\text{sh},n} |\omega| d^{4-n} \prod_i \|f_i\|_{\infty}. \quad (8.19)$$

The power d^4 is the number of copies packed into a ball of fixed radius; each $\sharp$ -norm costs d^{-1} .

Step 5 (finite rotations on $\mathbb{T}_{\star}$). For $\vec{f}$ as in Step 4, $t \mapsto S_{n;\star}(e^{t\omega} \cdot \vec{f})$ is C^1 with derivative $D_{\omega}^*(e^{t\omega} \cdot \vec{f})$ (as in Lemma 8.13(a), with (1.12)), and $e^{t\omega} \cdot \vec{f} \in \mathfrak{C}_n(d; \vartheta)$ has the same suprema. Every $R \in \text{SO}(4)$ is e^{ω} with $|\omega| \leq \pi\sqrt{2}$, and $\text{O}(4) = \text{SO}(4) \cup \text{SO}(4)\theta_P$ with $\theta_P \in W_4$ the reflection in a coordinate hyperplane P through 0, under which $S_{n;\star}$ is invariant (Theorem 4.8(c)). By (8.19), and for C^1 tuples by approximation in $\|\cdot\|_{\sharp}$ (mollify after a slight contraction about the centre),

$$|S_{n;\star}(R \cdot \vec{f}) - S_{n;\star}(\vec{f})| \leq C_{\text{rot},n} d^{4-n} \prod_i \|f_i\|_{\infty}, \quad C_{\text{rot},n} := \pi\sqrt{2} \cdot 160^4 \left(\frac{40M}{\vartheta}\right)^n C_{\text{sh},n} \Big|_{m=m_{\star}}.$$

Step 6 (every torus, and $\mathbb{R}^4$). Let S_n be a limit of (ii- $\mathbb{R}^4$) along $(g_0^{(j)}, K_j, m_j)$, or of (ii- $\mathbb{T}$ -b) on $\mathbb{T}_m$ along $(g_0^{(j)}, K_j)$, in which case $m_j := m$. Passing to a subsequence, the correlations on $\mathbb{T}_{\star}$ at the same bare couplings converge to a limit $S_{n;\star}$ of (ii- $\mathbb{T}$ -b) on $\mathbb{T}_{\star}$ (Theorem 4.8). The volume-transfer bound for n sources in one ball of radius 1 (Theorem 6.13 for $n = 2$, Remark 6.16(ii) for $n = 3$; at radius 1 its cell side is at most $48L < L^{11}$, so it applies to every pair of tori of the family) holds for all signs, positions and overlaps of the sources. Applied to the difference of the two volumes with the arrays of Step 4 (independent sign families for $n = 3$, Remark 6.20), exactly as in Corollary 6.19, it gives for $\varepsilon_j \leq d$ and every $\vec{h} \in \mathfrak{C}_n(d; \vartheta)$

$$|S_{n;K_j,m_j}^T(\vec{h}) - S_{n;K_j,m_{\star}}^T(\vec{h})| \leq C_{T,n} d^{4-n} \prod_i \|h_i\|_{\infty}, \quad C_{T,n} := 160^4 \left(\frac{40M}{\vartheta}\right)^n C_{\text{VT},n},$$

with $C_{\text{VT},n}$ the constant of Theorem 6.13, resp. of Remark 6.16(ii), at $R = 1$. Apply this to $\vec{h} = \vec{f}$ and to $\vec{h} = R \cdot \vec{f}$, let $j \rightarrow \infty$, and use Step 5: (8.17) holds with $C_n := C_{\text{rot},n} + 2C_{T,n}$. This is (a). For (b), suppose that along some $g_0^{(j)} \downarrow 0$ and tori $m_j \geq m_{\star}$ the difference exceeds $(C_n + \eta)d^{4-n} \prod_i \|f_i\|_{\infty}$ for an $\eta > 0$; along a subsequence the correlations on $\mathbb{T}_{\star}$ at the same bare couplings converge, and the two applications of the transfer bound together with Step 5 contradict this.

Step 7 (the running coupling). Let $p := 4 - n \in \{1, 2\}$ and $s := s_d$, so that $d \leq L^{-(s-1)}$. By (0)(d), $x_{K-s} - x_K \leq s c_0 + \Delta(s)$ with $c_0 = \frac{11N^2}{12\pi^2} \log L$ and $\Delta(s)/s \rightarrow 0$, and $x_K \leq g^{-2} + B_{\sharp}$. For $x := g(d)^{-2}$ and $\eta \in (0, 1)$ this gives $s \geq (1 - \eta)(x - g^{-2} - B_{\sharp})/c_0$ once s exceeds a threshold depending on η ; below it, x is bounded by clause (0). Hence

$$d^p \leq L^p L^{-ps} \leq A'_{\eta} \exp\left[-(1 - \eta)p \frac{\log L}{c_0} x\right] = A'_{\eta} \exp\left[-(1 - \eta)p \frac{12\pi^2}{11N^2} g(d)^{-2}\right],$$

and the bound passes to limit values of $g(d)$. This is (c); the factorial form follows from $t^k e^{-ct} \leq (k/(ec))^k \leq k! c^{-k}$ for $t > 0$. For (d): a pair in $\mathfrak{T}(d; \vartheta)$ lies in $\mathfrak{C}_2(d; \vartheta)$, both supports lying in

the ball of radius $5\mathbf{d}$ about the midpoint of the centres of their balls; and the lower bound in (1.18), which holds in every limit considered (clause (iv), Consequences (1)), gives $S_2(f_1, f_2) \geq \frac{2}{9}b_0c_K\|f_1\|_\infty\|f_2\|_\infty(g^{-2} + B_{\frac{1}{2}}(s(\mathbf{d}) + 1))^{-2}$. Divide (8.17) by it. $\square$

The proof uses the exact identity of Theorem 8.7, Proposition 8.10 on one torus, Proposition 8.17, clauses (0)–(iii), the lower bound of clause (iv) for (d), and the volume transfer of Section 6; it uses no hypothesis and no information about long distances.

Proposition 8.18 (Every number of points; proved in the companion account). *Let $n \geq 2$ and $\vartheta \in (0, 1]$, and let g lie below a threshold in $(0, \gamma_{\text{vi}}]$ chosen last. There are $\mathbf{d}_*(g) > 0$, depending on g only besides L , $\mathbf{p}$, γ and ϱ , and for every $\epsilon > 0$ a constant $C_{n,\epsilon}$, depending only on L , $\mathbf{p}$, γ , ϱ , g , ϑ , n and ϵ , such that for every limit $S_{n;\infty}^T$ of (ii- $\mathbb{R}^4$), every $R \in \text{O}(4)$, every $\mathbf{d} \leq \mathbf{d}_*(g)$ and every n -tuple of real C^1 functions supported in one ball of radius $\mathbf{d}$, with supports pairwise at distance at least $\vartheta\mathbf{d}$ and $\text{Lip } f_i \leq \|f_i\|_\infty/(\vartheta\mathbf{d})$, rotated about the centre of that ball,*

$$|S_{n;\infty}^T(R \cdot \vec{f}) - S_{n;\infty}^T(\vec{f})| \leq C_{n,\epsilon} \mathbf{d}^{1-\epsilon} \prod_i \|f_i\|_\infty;$$

For every $\eta \in (0, 1)$, choose $\epsilon = \eta/2$ in the preceding bound. Then there is $A_{n,\eta}$ such that

$$|S_{n;\infty}^T(R \cdot \vec{f}) - S_{n;\infty}^T(\vec{f})| \leq A_{n,\eta} \exp \left[-(1-\eta) \frac{12\pi^2}{11N^2} g(\mathbf{d})^{-2} \right] \prod_i \|f_i\|_\infty.$$

Indeed, (2.20) gives $\frac{12\pi^2}{11N^2} g(\mathbf{d})^{-2} = \log(1/\mathbf{d})(1 + o(1))$. Since $1 - \eta/2 > 1 - \eta$, it converts the bound with this choice of ϵ into the displayed exponential estimate for sufficiently small $\mathbf{d}$. Increasing $A_{n,\eta}$ covers the remaining admitted separations, whose depths lie in a bounded range and whose inverse couplings are bounded by the flow estimates. This is an estimate on the rotation defect, not a comparison of every fixed- ϵ right side with every η -exponential. The same holds for the limits of (ii-T-b) on every torus of the family above a floor depending on g , for $\mathbf{d}$ below a further threshold depending on g and n . This is proved in the companion account as an implication of the same kind as clause (vi)—from part (A2), a bound obtained from reflection positivity at one fixed radius, the exact change of reference scale and clauses (0), (ii) and (iii)—with no long-distance input; it is used nowhere in this paper. Not asserted: exactness at any scale; a relative form for $n \geq 3$; a form at finite cutoff, uniform in the cutoff and the volume, for $n \geq 4$.

Proposition 8.19 (No exact rotation invariance on a torus). *Let $\phi \in C_c^\infty(B(0, 1))$ be real, non-negative and not identically zero, let $\vartheta_\phi := \min\{1, \int \phi / \|\phi\|_\infty, \|\phi\|_\infty / \text{Lip } \phi\}$, and let $g \leq \gamma_W^\mathbb{T}$ at $\vartheta = \vartheta_\phi$. On every torus $\mathbb{T}_m$ of the family, no limit $S_{2;m}^T$ of (ii-T-b) is rotation invariant at any scale: for every real antisymmetric $\omega \neq 0$ and every $\mathbf{d}_0 > 0$ there are real C^∞ functions h_1, h_2 on $\mathbb{T}_m$ with disjoint supports in one ball of diameter less than $\mathbf{d}_0$ such that $S_{2;m}^T(\delta_\omega h_1, h_2) + S_{2;m}^T(h_1, \delta_\omega h_2) \neq 0$, δ_ω acting about the centre of that ball.*

Outline; the proof is given in the companion account. Suppose the contrary for some $\omega \neq 0$ and $\mathbf{d}_0 > 0$. By translation invariance, $S_{2;m}^T(h_1, h_2) = \langle T, h_1 \star \check{h}_2 \rangle$ with $\check{h}_2(x) := h_2(-x)$ and T a distribution on $\mathbb{T}_m \setminus \{0\}$ in the separation of the two arguments; the assumption says that $(\omega u) \cdot \nabla T = 0$ on $B(0, \mathbf{d}_0) \setminus \{0\}$. Positivity of the lattice transfer operator in each coordinate direction makes the two-point function, smeared in the three transverse coordinates, real-analytic in the separation along that direction, away from the hyperplane where this coordinate of the separation vanishes; this separate analyticity spreads the identity from $B(0, \mathbf{d}_0)$ over the whole periodic cell. By periodicity and invariance under a coordinate reflection the identity then holds

also about a second centre $c \neq 0$, that is, $(\omega(u - c)) \cdot \nabla T = 0$; and two rotations about different centres differ by a translation, so that $(\omega c) \cdot \nabla T = 0$. With the hypercubic invariance, T is invariant under all translations of the separation, hence constant, and $S_{2,m}^T(f_1, f_2) = T \int f_1 \int f_2$ is of order $d^8 \|f_1\|_\infty \|f_2\|_\infty$ for the pairs in $\mathfrak{T}(d; \vartheta_\phi)$ made of two translates of $\phi(\cdot/d)$ at distance $3d$. This contradicts the lower bound of clause (iv), of order $g(d)^4 \|f_1\|_\infty \|f_2\|_\infty$, as $d \downarrow 0$. $\square$

Consequently no statement that holds for the limits on one torus of the family can imply exact rotation invariance of small clusters: an argument for exact invariance on $\mathbb{R}^4$ must use information that is false, or has no meaning, on every torus. This is why clause (vi) carries a long-distance hypothesis, and why Propositions 8.16 and 8.18 cannot be sharpened to exactness by arguments valid torus by torus.

8.7 What is not asserted; no fixed-radius version

Remark 8.20 (Without the hypothesis). Given part (A2) alone, Proposition 8.15 is a local Ward identity for the limits: for every radial cut-off around the supports, the limit of the shell term is $-D_\omega(\vec{f})$, independently of the radius; in the continuum reading, $D_\omega(\vec{f})$ is the total torque exerted on the enclosed sources. By Lemmas 8.13 and 8.14, the rotations that preserve a given limit form either a finite group containing W_4 or all of $O(4)$. So (IR-fl) is needed for one generator only, along one sequence of radii; and, given (A2), it is necessary as well as sufficient for the conclusion of Theorem 8.4.

Remark 8.21 (The hypothesis is not asserted; no fixed-radius version). (a) Theorem 0 asserts nothing about the truth of Hypothesis 8.2 or of (IR-fl). It asserts no decay of correlations, no clustering and no uniqueness of the vacuum; clause (vi) is a conditional statement.

(b) *The exponent four is sharp for this argument.* If the only information is a majorant $c(D)$ for the stress-tensor correlation at distance D with $\liminf_{D \rightarrow \infty} D^4 c(D) > 0$, then no choice of the shape of the cut-off region, of the order of the limits or of an averaging over radii yields $D_\omega = 0$: the Killing weight grows linearly with the radius and the area of the sphere like the cube of the radius. This is proved in the companion account. A correlation decaying like $|y|^{-4}$ is what a massless mode of spontaneously broken rotations would produce.

(c) *No fixed-radius version.* By Proposition 8.15, at every fixed radius the limit of the shell term is $-D_\omega(\vec{f})$. Conservation and symmetry of the limiting stress tensor — the content of part (A2) — make it independent of the radius; they do not make it vanish. Integrating the shell term by parts towards the sources returns the contact terms at the sources, that is, (8.13) itself; integrating it towards infinity requires the vanishing of the flux at infinity, which is (IR-fl). There is no third form: the symmetrised shell weight is not the symmetrised gradient of any compactly supported vector field vanishing near the sources, since such a field minus $\chi\xi$ would be a compactly supported Killing field, hence zero, so that the field would equal $\chi\xi$, which is $\xi \neq 0$ on the sources. Thus “the shell term vanishes at one radius” is equivalent to the conclusion of Theorem 8.4, and a version of clause (vi) at fixed radius without an infrared hypothesis is not available.

(d) *Examples.* The following are proved in the companion account. Let ϕ be the massless free field on $\mathbb{R}^4$, $h_4(x) := \sum_\mu x_\mu^4 - \frac{1}{2}|x|^4$ (harmonic, W_4 -invariant, not $O(4)$ -invariant), and $\mathcal{O}^\lambda(f) := \int f : \sum_I (\partial_I \phi + \lambda \partial_I h_4)^2 : dx$, where I runs over 4-tuples of indices. The connected correlations of the $\mathcal{O}^\lambda(f_i)$ at disjoint supports are invariant under $\mathbb{R}^4 \rtimes W_4$, reflection positive at the sixteen mirror hyperplanes of W_4 , bounded uniformly in the positions, and clustering like $|x|^{-10}$, with a unique vacuum and an $O(4)$ -covariant local dynamics; but the two-point function is not $O(4)$ -invariant, and the correlation of the stress tensor with the sources decays exactly like $|y|^{-4}$. The stress tensor of this example is not translation covariant. A second example, a free tensor field with a constant

W_4 -invariant mean, has a translation-covariant, local, symmetric and conserved stress tensor and satisfies all local translation and rotation identities; it is not $O(4)$ -invariant, and it is not reflection positive. So neither reflection positivity with clustering nor local conservation with symmetry can replace the hypothesis. Whether reflection positivity in a plane containing the rotation, together with a translation-covariant conserved stress tensor, forces $O(4)$ is open; it would be a statement about the state, not an identity at fixed radius.

Remark 8.22 (Not asserted). Clause (vi) does not assert: the hypothesis itself; covariance of tensor-valued fields (only the scalar field $\mathcal{O}$ is treated); rotation invariance of block observables or of Wilson loops; anything about rotations outside W_4 on a torus, where they do not act; the conclusion along the one-parameter families of clause (v) on $\mathbb{R}^4$; exact rotation invariance of the short-distance behaviour of the limits without the hypothesis (Proposition 8.16 gives invariance up to a positive power of the scale, and by Proposition 8.19 exactness fails on every torus); that reflection positivity and clustering imply the hypothesis; that conservation, symmetry and the hypercubic normalisations of the limiting stress tensor imply it.

9 What rests on what; what is not claimed; outlook

We list, clause by clause, the statements on which the proof of Theorem 0 rests without being proved in Sections 2–8; then what Theorem 0 does not claim, and why; then further work that is not part of Theorem 0.

The companion account. By *the companion account* we mean the detailed exposition of the construction on which this paper rests, together with the complete proofs to which it refers (Section 1); each of its parts is available on request. Its parts are cited in Sections 3–8 by the following keys.

- [T-corr] proves the correlation theorem with sources, Theorem 3.1, including its form on a prescribed reference line used in Section 7; it is written in full.
- [T-W] proves the decomposition of the renormalised insertions at the witness level and the bounds on the contributions of live large-field components used in Section 5; it is written in full.
- [T-GX] proves the Gaussian extraction with one and two inserted observables, with the localisation of the sourced step to a box, and the second-order extraction estimate (the room estimate, Proposition 5.17); it is written in full.
- [T-LF] proves the one-cell estimates for large-field components on a torus used in Section 5; it is written in full.
- [T-U] proves the uniqueness theorem, which compares two cutoffs on one torus; its supporting statements not proved there are listed under clause (v) below.
- [T-L] proves the same-cutoff Lipschitz step, Theorem 7.2(e); it is written in full, given one reading listed under clause (v).
- [T-R] proves the renormalised coordinate, Theorem 7.2(d)(1); it is written in full, given one reading listed under clause (v).
- [T-B] proves the bridge estimate between first-passage and tuned runs, Proposition 7.4; it is written in full.
- The companion note on the free kernel proves the continuum form, the positivity and the two-sided comparison on the torus of the free, unconditioned kernel of clause (iv) (Proposition 5.4); it is written in full.

Kinds of statements. *Used by citation:* a result of Balaban, used as stated and under its own hypotheses. *Used as Balaban asserts it* (marked *printed*): a statement printed by Balaban without a proof that covers our use, in print or in the companion account; we use it as he asserts it. All such statements are listed, with their places, in Subsection 9.1. Where his text is ambiguous we declare the reading we use; at two places we use a correction of his text: the sign of a pair of boundary terms in [9, (3.15)], on p. 269 (Remark 2.2(2)), which affects no estimate, and the treatment of the field tA_k asserted on [9, p. 276] (Section 3). *Proved in this paper:* proved in Sections 2–8, from the statements listed with it. *Proved in the companion account:* a complete proof is written there, from the statements listed with it; the statement and the structure of the proof are in the section indicated. *Deferred:* we state the item and defer its proof; a tag says what the companion account contains—*written in part* (a proof up to an identified step) or *statement only*. *Supporting statements of the uniqueness theorem:* statements of ours on which the uniqueness theorem behind clause (v) rests; each is marked *not yet written* or *written*; *independent verification in progress*, and we do not claim them as proved. An item used by several clauses is listed once. Verifications that involve no estimate (readings of definitions and of displays in the form used) are not listed unless their proofs are deferred. Each clause of Theorem 0 is thus deduced from Balaban’s theorems as cited—with the statements of Subsection 9.1 used as he asserts them, the declared readings and the two corrections—and from the companion account, together with the deferred items and the supporting statements of the uniqueness theorem listed below; no clause is claimed to hold without them, and clause (vi) is moreover conditional on its displayed hypothesis.

9.1 Statements of Balaban used as he asserts them

The following statements of Balaban’s papers are used as he asserts them, without a proof in print that covers our use. Each is listed with its place and with the item of Sections 2–9 where it enters. The volume-independence remark of [9, p. 278] is not among them: it is proved in the companion account (group (B) of Subsection 2.4).

- (Pr1) [11, (1.11), (1.23)–(1.48)]: the bounds along the chain of large-field operations, among them the lower bound $\exp(-O(1)|\Lambda|)$ for the normalising integrals of the large-field operation on a region Λ , (1.11), and the smallness, per component of a large-field region, of the terms coming from the Wilson terms of those integrals, (1.46)–(1.48) and p. 369 (“their sum is also small, although we need boundedness only”). Enters: (C14); clause (iii).
- (Pr2) [11, p. 375], the sentence “it is simple to see for all the old terms”: the old terms beside a new large-field region are analytic on the new class, read on the class of configurations regular with one power of the scale ratio (group (B) of Subsection 2.4). The inclusion of regularity classes across cube sizes asserted on the same page is not used (Remark 2.2(2)). Enters: (C14).
- (Pr3) [11, p. 376]: the resummed boundary term “satisfies the same bound with a different constant”. Enters: (C6).
- (Pr4) [11, p. 379], with the bookkeeping of new boundary terms and the resummation over hulls described on p. 390: the locality of the terms produced by the large-field operation. Enters: (C8); Lemma 6.7.
- (Pr5) [11, p. 360]: the estimate whose details are omitted there. Enters: (C5).
- (Pr6) [10, p. 191]: the remark on integrations interrupted by a new large-field region. Enters: (C13).

- (Pr7) [10, pp. 177–179]: the admissibility of the prepared large-field data, in the form used by the operator theorem at free current. Enters: (C11).
- (Pr8) [9, p. 275]: on the difference of two consecutive small-field regions the configuration of the next step coincides with one of the earlier type and has its regularity properties. Enters: (C2).
- (Pr9) [9, pp. 276–277]: the analysis of [7, §§3–5] and [8] carries over to the cover by cubes of the scales of the large-field layers (“We repeat the analysis of Sect. 3 [I] using the above cover”). Enters: (C9).
- (Pr10) [7, (1.17)], at complex configurations of the class regular with one power of the scale ratio. Enters: (C12).
- (Pr11) [7, p. 279]: the second inequality of the regularity condition verified before [7, Lemma 4], obtained there “in the same way” as the first. Enters: (C12).
- (Pr12) [7, (3.26)], one sentence of [3] and one sentence in one of the two papers [4, 5]: certain bounds are obtained “in the same way” as bounds proved there. Enters: (C7).
- (Pr13) [5, (3.35), (3.37)]: these inequalities at the two lowest levels of the tower, read on a large-field region (they are asserted there for the global field spaces). Enters: (C10).
- (Pr14) [6, (190)], derived there from a display (189) that is not computed. Enters: group (A) of Subsection 2.4; (C7).
- (Pr15) The regularity of the core of a large-field region that the large-field operation leaves unintegrated, as stated with the definition of that operation [10, 11]. Enters: clause (v) only, through the comparison at terms that meet such a core (Subsection 7.5).

9.2 What rests on what

Summary. At $N = 2$, the statements of ours on which Theorem 0 rests and which are not yet established are the following. Beneath clauses (0), (i), (iii) and (iv), and beneath clause (ii) apart from (ii-T-a) and the identification sentence of (ii-T-b): none. Beneath clause (v), and beneath those two parts of clause (ii), which are deduced in Section 7 together with clause (v): the supporting statements of the uniqueness theorem listed under clause (v) below—three statements written, with independent verification in progress, (U1), in a corrected form, (U3) and (U5), and two statements not yet written, (U4) and (U6)—together with one clause to be added to a definition. Beneath clause (vi): the same statements, through the two-cutoff comparison, and nothing else. For $N \geq 3$ see item (C15) below, the last item under clause (v), and Remark 1.4. The statements of Balaban that we use as he asserts them are listed in Subsection 9.1 and marked *printed* below.

All clauses. *Proved in the companion account:* that the order of choice of the parameters in (H3) has no cycle, and that it can be met: every window is non-empty by the condition that opens it, and the independence from the cube size M of the two decay constants and of the amplitude of the random-walk expansions of the fluctuation kernels, on the one family of cells over which the effective densities are decomposed, is proved for the whole lattice and, on that family, as an implication from the operator theorem at free current (Section 2), among whose inputs is the structural statement on the creation of large-field regions, item (C4) below, which is proved. No further item.

Clauses (0) and (i). The two clauses are deduced together, by one induction over the renormalisation steps (Section 2); the induction also uses the items listed under clause (iii).

Used by citation: [11, Theorem 1], whose one hypothesis, that all effective couplings lie in $(0, \gamma]$, is supplied by clause (0); [7, Theorems 1, 3] with [8] (if $0 < g_k \leq \gamma$ at every step, the small-field effective actions have the inductive form and bounds [7, (1.1)–(1.22)]; [7, (1.20)–(1.22)], [9, (3.61)] (the increment β_{k+1} of one step on the whole lattice, $x_{k+1} = x_k - \beta_{k+1}$); [6, Theorem 1] (the background field: the minimiser of the Wilson action with prescribed block averages, unique up to gauge, with its regularity bounds); [1, 2, 3, 4, 5] (propagators, also in a background field; renormalisation transformations; averaging operations; regular configurations under gauge fixing). The complete list, with equation numbers, is group (A) of Subsection 2.4. We use [11, Theorem 1] by statement. Of the intermediate assertions of [9, 10, 11] printed without proof, those we use are proved in the companion account (group (B) of Subsection 2.4), except the ones marked *printed* below and listed in Subsection 9.1; that his argument applies to the construction in the hereditary reading, the summed bounds on every history and the tree-length bound are proved there (Section 2). Of the stability bounds, only the lower bound on real fields and the upper bound after integration are asserted; the displayed pointwise upper bound [11, (0.1)] fails whenever the density carries the gauge-fixing factors that the large-field operation keeps on second-class regions, since these give it a singular part (Proposition 2.5).

Proved in this paper (Section 2): the bookkeeping of one step — the two-sided flow bounds, the first passage, the containment of all couplings in $(0, \gamma_H]$ —, the deduction of clause (0)(a)–(c) from the per-step bounds, the coupling windows at fixed depth, and the exact one-loop law (0)(d) from its two inputs.

Proved in the companion account (statements and structure in Section 2), as implications from the statements of Balaban cited above and from the items (C1)–(C15) below, as qualified there: the finite part of the one-loop law, $|\sum_{l=1}^k \beta_l^0 - kc_0| \leq c_2$, with β_l^0 the Gaussian part of β_l , $c_0 = \frac{11N^2}{12\pi^2} \log L$ and $c_2 = c_2(L, N)$; the existence and torus-independence of the increment; the bounds on the unmarked and the marked parts of the increment; the cube-size lemma; the one-step theorem in family form; the independence of the localised terms of a step from its small-field region, and their coincidence with the whole-lattice terms, asserted without proof in [9, p. 278]; the independence of the torus of the terms localised in domains that do not wrap around it, from which Section 6 derives the one-sided extensivity of the vacuum constant; the operator bounds [7, (3.9), (3.14), (3.31)–(3.32)] at a current not tied to the field; the class of configurations regular with one power of the scale ratio, replacing the inclusion of [8, pp. 9, 15], which cannot be used as printed; the summed bounds [9, Theorem 2] on every history, with the ratio of running couplings of clause (i-a); the representation [11, p. 390] and the bound [11, (1.99)] of the terms made by the large-field operation (bound half); the positivity of the factor kept by the large-field operation unless it vanishes identically (“obviously positive” in [11]); the admissibility of the data that the construction prepares as in [10, pp. 177–179], asserted there in general without proof, except in the form used by the operator theorem at free current, which is (C11) below; the two stability bounds, the equality of integrals, and the exact insertion identity, also for sources.

The statements (C1)–(C15) of Subsection 2.4, listed with the names used there and with their status. Those marked *printed* are statements of Balaban that we use as he asserts them. The statements of the companion account among them are proved there, or require no separate proof, as said for each; (C15), which concerns $N \geq 3$ only, is deferred.

- (C1) The regularity clause of the one-power class at source terms whose localisation domains meet two consecutive layers of the large-field regions. Requires no separate proof: the clause is part of the definition of the analyticity domains, and under the cumulative reading of those

domains the clause read after the large-field operation is the one already stated for the source term itself, about the unitary factor that the operation leaves untouched; at real fields it holds automatically.

- (C2) The sentence of [9, p. 275] that on the difference of two consecutive small-field regions the configuration of the next step coincides with one of the earlier type and has its regularity properties, used as printed on the intermediate layers; through it the widened reading of [7, (1.12)] on thin cubes is proved in the companion account. *[printed]*
- (C3) The analytic dependence of the quantities built along the towers at complex representatives of the one-power class. Proved in the companion account: holomorphy composes along the towers under the definition used there, with the inputs [6, Theorem 1, Proposition 9], [3, Proposition 2] and [7, (0.4)–(0.12), (1.8), (1.15)], used by statement; no estimate is involved.
- (C4) The creation of large-field regions: every component meets a creation set, and keeps doing so. Proved in the companion account, at the renormalisation steps and at the steps of the large-field operation, as a combinatorial lemma from Balaban’s definitions.
- (C5) The supports of the Wilson part of the densities at the regions prepared by the large-field operation. Proved in the companion account: this is bookkeeping of the supports in a localisation, which follows from [11, pp. 360–366], used by statement except the estimate on p. 360 whose details are omitted there (item (Pr5)) *[printed]*, under one further lower bound, on the waiting depth $\check{R}_k$ of Section 2, of the kind allowed by (H3).
- (C6) The statement of [11, p. 376] that the resummed boundary term “satisfies the same bound with a different constant”, used as Balaban asserts it; behind it stands an analyticity radius, free of the coupling, the level and the domain, for the recently created boundary-type constituents (a weaker bound, by the number of large-field levels present, is proved in the companion account). *[printed]*
- (C7) Three sentences, of [3], of one of the papers [4, 5] and of [7] (at (3.26)), stating that certain bounds are obtained “in the same way”, with the companion bounds read from the definitions at [7, p. 263] and in the third to fifth members of [6, (190)]. *[printed]*
- (C8) The locality of the terms produced by the large-field operation [11, p. 379], with the bookkeeping of new boundary terms and the resummation over hulls described in [11, p. 390]. *[printed]*
- (C9) The transfer, in [9, pp. 276–277], of the analysis of [7, §§3–5] and [8] to the cover by cubes of the scales of the large-field layers (“We repeat the analysis of Sect. 3 [I] using the above cover”, [9, p. 276]). *[printed]*
- (C10) The inequalities [5, (3.35), (3.37)], at the two lowest levels of the tower, on the regions of the one family of cells. *[printed]*
- (C11) The admissibility of the prepared large-field data [10, pp. 177–179] in the form used by the operator theorem at free current. *[printed]*
- (C12) The statements of [7] used as printed by the argument producing the one-power class: [7, (1.17)] at complex configurations (its printed derivation from [6, Proposition 9] does not apply there; the display is not refuted), and the second inequality of the regularity condition verified before [7, Lemma 4] [7, p. 279] *[printed]*; the uniqueness and locality of the minimiser that complete the proof of [7, Lemma 4] are proved in the companion account.

- (C13) The statements of the companion account on the representation and on one step: three statements on boundary terms whose domain holds a live second-class large-field region — a resummation bound, its form inside the single induction, a statement on tubes around the live regions — which give the decay of these terms in the relative length [11, (1.67)], or in a tree length not exceeding it; the inheritance statement for the representation; the containment statement used to sum boundary terms over finer cubes; the statements on terms with torus-wrapping supports, on terms created where a large-field region arises (the construction of the large-field regions of [10] in the form used), on the resummations and on the bounds for the large-field operation; the treatment of applications of the large-field operation that are stopped before completion, beside which the remark of [10, p. 191] on integrations interrupted by a new large-field region is used as he asserts it (item (Pr6)) [*printed*]; and the statement fixing the single family of densities on which the induction runs, with the exactness of the two-threshold decomposition at the step that creates a large-field region. All of these are proved in the companion account. For an application of the large-field operation that is stopped before completion, this includes the statement that every later configuration lies in the domain of every term stored earlier, also at those collar bonds of the stored domains that have never been averaged; Balaban’s definitions of the configuration spaces in [4, 7] cover such bonds, while those of his large-field papers [9, 10, 11] are silent there. The companion account gives these bonds a condition of the same kind as at all other bonds, attached to the finest scale and shrinking at each later application of the large-field operation by Balaban’s schedule continued to that scale. The response of the background field at such a bond to an operation several scales higher is exponentially smaller than the shrinkage, and over a history of any length the total shrinkage stays below one eighth of the initial size of the condition, because successive applications begin at strictly higher scales. This is a definition; it introduces no hypothesis and no upper bound on the number of steps.
- (C14) The steps, shared with the proof of the correlation theorem, by which clause (i) sums boundary terms beside large-field regions: its resummation lemma; the weight lemma for the factors that earlier large-field regions leave on later terms, with its membership statement (terms beside a treated region have nearby configurations in the required domain of analyticity) and its remainders — these are proved in the companion account; and the statements of [11] taken there as Balaban asserts them: that the old terms beside a new large-field region are analytic on the new class [11, p. 375] (“it is simple to see for all the old terms”, read on the one-power class of group (B); the inclusion of regularity classes asserted on the same page is not used), and the lower bound $\exp(-O(1)|\Lambda|)$ for the normalising integrals of the large-field operation on a region Λ [11, (1.11)] [*printed*].
- (C15) For $N \geq 3$: the form for general N of the constants of the stability bounds (i-b), and the reading for $SU(N)$ of every argument beneath clauses (0) and (i). [*written in part*]

None of these bounds L , M , K , m , a depth or a number of steps from above, or a coupling from below. For part (d) of clause (0) one more input is used: the bound on $|\beta_{k+1} - \beta_{k+1}^0|$ supplied by the correlation theorem (Theorem 3.1(f)), hence the items of clause (iii); (0)(d) is proved for $N = 2$.

Clause (iii). *Used by citation:* as for clauses (0) and (i), with complex sources. *Proved in this paper* (Section 3): the link reflection positivity of the Wilson measure (after [23]) and the chessboard estimate (after [17]); the torus bound (iii)(a) and the running shift (iii)(c) from the correlation theorem; the per-ball bound (iii)(b), free of the volume. *Proved in the companion account* (statement and structure of the inductive proof in Section 3): the correlation theorem with sources, as an implication from the inputs beneath clauses (0) and (i) and from finitely many sub-statements,

all proved there, together with its flow conclusions and the modulus of the running shift—among these sub-statements are the two inputs of the estimate for the terms in the layer, one block wide, around the region where the lowest-level small-field condition fails, namely the refined case of the lemma matching that layer to the first region inside it and one further family in the enumeration of configurations that the estimate distinguishes; and, as an implication from the containment, linearity, anchoring and geometric statements of the induction of clause (i) (item (C13)), the summation over the finer cubes of a coarser cube of the per-step bounds for the boundary terms stored next to large-field regions — it adds no item of its own.

Deferred: none. The run with sources uses the items (C13) and (C14) above, as qualified there.

Used as Balaban asserts them, beyond those of (C14): the smallness, per component of a large-field region, of the terms coming from the Wilson terms of the normalising integrals [11, (1.11), (1.46)–(1.48) and p. 369] (“their sum is also small, although we need boundedness only”; item (Pr1)). *[printed]* The random-walk expansions of the fluctuation kernels are used by statement, as in group (A) of Subsection 2.4, in particular [2, Propositions 2.3, 2.6, 2.7].

Clause (ii). *Used by citation:* nothing beyond the earlier clauses; the reconstruction follows [21], and reflection positivity of the Wilson measure is proved in Sections 3 and 4 after [23]. *Proved in this paper* (Section 4), from clause (iii)(a), (b) and clause (0): on each torus, the limits along subsequences of first-passage cutoffs, their symmetry, their invariance under translations and W_4 , their reflection positivity, and their extension to distributions on non-coincident arguments; on $\mathbb{R}^4$, the limits along one joint subsequence of cutoffs and volumes, invariance under $\mathbb{R}^4 \rtimes W_4$, reflection positivity, the regularity (E0) for each n , and the reconstruction of $\mathcal{H}$, Ω , $H \geq 0$ and the momenta. Convergence without subsequences on a fixed torus along the tuned family, and the identification of first-passage limits on a torus, are deduced in Section 7 together with clause (v) and rest on the items listed under clause (v). *Deferred:* nothing of its own; the other statements on tori and on $\mathbb{R}^4$ rest on the items of clause (iii).

Clause (iv). *Used by citation:* [7, §§3–4], [8, §§1–2] (the small-field fluctuation integral and its cluster expansion; their run with a bounded local source is the box localisation proved in the companion account), with the small-field windows [7, (2.9)], [8, (2.3)], [9, (3.16)]; [6, Theorem 1]; [11, (1.73)–(1.79)] (positivity of the kept densities; entropy factors); [9, (1.15), (3.15), (3.33)–(3.38)] (counts of the vacuum-energy terms); the decay of Balaban’s block propagators [1, Proposition 1.2], [2, Propositions 2.5–2.6], from which the two-sided comparison of the block-conditioned kernel $\mathcal{K}_s$ (Proposition 5.4) is derived; the complete list is item (b) of Subsection 5.8, together with the citations in Section 6. *Proved in this paper* (Sections 5 and 6): the exact total covariance, the colour factor and the absence of terms of order one and of single-line exchange; the reductions of the remainders in Steps 2–4 of Section 5, including the additive bound on live large-field labels near the supports from a one-cell estimate, and the assembly; the trace inequality, the one-sided extensivity of the vacuum constant, the volume transfer, the averaging lemma, and the passage to every torus and to every limit on $\mathbb{R}^4$. *Proved in the companion account* (structure in Section 5): the decomposition of the slopes and the bound on the terms that see both sources [T-W]; the Gaussian extraction with two inserted densities [T-GX]; the one-cell estimate [T-LF], with the bound on labels removed later, through a quotient that Balaban forms and bounds himself, and the bound on labels still live at the last lattice, by an age factor paid from Balaban’s entropy constant; the modulus of the running shift (Corollary 3.3); the room estimate (Proposition 5.17), that is, the second-order size estimate for the Gaussian extraction, as an implication from the correlation theorem, by the direct argument described in Section 5, under the lower bound (1.9) on the exponent q of [9, (2.28)];

the statement on second-class large-field regions used in the prepayment of age—each second-class outcome of Balaban’s large-field operation is marked by a plaquette of the block field of its level deviating from **1** by at least a fixed fraction of the small-field threshold, and the small factor so produced pays for the combinatorial entropy of the outcome—which is also used for clause (v); and, in the companion note on the free kernel, the statements on the free kernel without block conditioning: the continuum form $3(N^2 - 1)\pi^{-4} \iint f_1(x)f_2(y)|x - y|^{-8} dx dy$ of b_0 times that kernel, and its positivity and two-sided comparison on the torus.

Deferred: none. Clause (iv) uses the correlation theorem of clause (iii), read up to the scale of the supports, as qualified under clause (iii). The display of clause (iv) in every volume carries the free kernel of the reference torus; the kernel formed on the torus at hand obeys the same two-sided bounds and has the same continuum form, so every stated consequence is unaffected; the display with the latter kernel would follow from a free-field comparison stated in Section 6 as an optional variant and not used.

Clause (v). Clause (v) is proved as an implication from the uniqueness theorem of the companion account. *Used by citation:* as for clauses (0) and (i), on a scaffold of auxiliary choices fixed by the reference coupling; [11, (1.79)–(1.88)] (the factors attached to large-field regions). Balaban’s announced uniqueness statement [7, Theorem 2, (0.31)] is not used; its proof was not published [11, p. 355]. *Used as Balaban asserts it:* item (Pr15) of Subsection 9.1 [printed]. *Proved in this paper* (Section 7): the deduction of the one-parameter family, of its real-analyticity and of the exhaustion from the increment estimate; the bridge to first passage from the bridge estimate; the bi-Lipschitz dependence of the reference coupling on the bare offset and the identification of limits with equal renormalised reference coupling (Lemma 7.5); the exact change of units and its form for the limit family (Corollary 7.7), given the two-cutoff estimate of Step 4. *Proved in the companion account* (structure in Section 7): the correlation theorem on the prescribed reference line, with the running shift (RS) [T-corr]; the exact pin of the bare coupling at each depth and the depth-to-depth coupling maps [T-U]; the increment estimate [T-U]; the bridge estimate between tuned and first-passage runs [T-B]; the re-filing of healed large-field remnants that makes the renormalised coupling a coordinate with exponential rate [T-R]; the same-cutoff Lipschitz bound for the complete step, with the forgetting of old large-field regions at a geometric rate [T-L]; index stationarity, object by object; the treatment of a large-field component excluded in the middle of a run; (U2), the counting bound for decoupling families made of cubes of several sizes, and the fifteen further supporting statements listed in Subsection 7.5, one of them in a corrected form in which a smallness condition is tightened by fixed constants; and clause (v-f), as an implication from clauses (iii), (iv) in every volume, (v) and Corollary 1.2, under the lower bound (1.20) on ϱ .

Supporting statements of the uniqueness theorem (Subsection 7.5); we do not claim them as proved.

- *Written; independent verification in progress* (three statements): (U1) the extension statement, in a corrected form—as first formulated, in the class of inherited domains, it is too strong; the form used imposes the conditions of that class only near the domains where the proofs read them, and beyond these only the two regularity conditions [5, (3.35), (3.37)], with radii free of the coupling, and one condition on the real background configuration, which is to be added to the definition of the class (a definition, not an estimate); (U3) the two-cutoff comparison, region by region and level by level, of the two factors kept un-integrated inside a region thrown back to an earlier level (a linear bound by the comparison’s own accuracy at the creating level; the single-supremum smallness form does not hold and is not used); its argument uses (U4), (U5) and (U6); (U5) the lemma beneath (U3) on background fields in the interior of a region thrown

back to an earlier level and healed there: they depend on the configuration outside the region only within the bounds stated in the companion account, at every box into which the uniqueness theorem cuts a term and with constants independent of the box, so that the one-cutoff bounds for the two factors of (U3), proved for the pieces into which the correlation theorem cuts a term, hold for the pieces of the uniqueness theorem.

- *Not yet written* (two statements): (U4) the two-cutoff comparison of the members of a family of correction terms of the large-field operation, localised in the strip around a region thrown back to an earlier level, which Balaban does not write out member by member [11]; (U6) the two-cutoff comparison of the Gaussian factors kept by applications of the large-field operation that are stopped before completion and lie inside such a region. Both are used only through (U3).
- *Readings*, involving no estimate: the single-step lemmas of [T-corr] read with an extra holomorphic parameter, inside the Lipschitz step; one reading of the re-filed run, used in [T-R].
- For $N \geq 3$: the group-specific step of the uniqueness argument, which has not been carried out. *[statement only]*

Clause (v) uses no statement on the rarity of large fields and nothing about regions removed at earlier levels.

Clause (vi). Its hypothesis is a hypothesis of the clause; Theorem 0 asserts nothing about its truth. *Used by citation*: nothing beyond the earlier clauses. *Proved in this paper* (Section 8): the exact lattice rotational Ward identity at every cutoff and in every volume; the deduction of the clause from the identity, the coefficient bounds and the hypothesis (its infrared half: the far flux vanishes if and only if the rotation defect does; that the exponent four is sharp for this argument is proved in the companion account), with the dichotomy that the rotations preserving a limit form a finite group containing W_4 or all of $O(4)$; and the implication from the sufficient condition to the hypothesis. *Proved in the companion account* (structure in Section 8), as implications of the same kind as clauses (iii)–(v): the classical content of the identity and the classification of its breaking terms; the two parts of the coefficient bound, from the correlation theorem with sources, the same-cutoff Lipschitz step and the two-cutoff comparison of clause (v), and one proposition on the class form of the densities, used as a reading: the antisymmetric channel at the rate the construction gives, and the vector channel by a double subtraction and the transport of the correlation theorem to one symmetric-tensor insertion; and the removal of the floor (8.11) (Remark 8.6). *Definitions*: the lattice stress tensor (8.5), and the place at which an old large-field term is localised at the collar of a large-field region. *Deferred*: none of its own. Beneath clause (vi) lie, through the two-cutoff comparison, the supporting statements of the uniqueness theorem listed under clause (v); nothing beneath clause (iii), or beneath clause (ii) apart from (ii-T-a) and the identification sentence of (ii-T-b), neither of which clause (vi) uses, is deferred at $N = 2$, and clause (iv) is not used. Without the hypothesis, Section 8 proves rotation invariance of the short-distance germ of the two- and three-point functions (Proposition 8.16), from the exact Ward identity, Proposition 8.10 on one torus, Proposition 8.17 (proved in the companion account), clauses (0)–(iii), the volume transfer of Section 6 and, for its relative form, the lower bound of clause (iv); beneath it lie the same statements as beneath clause (vi). The corresponding statement for every number of points (Proposition 8.18), and the failure of exact invariance on every torus (Proposition 8.19), are proved in the companion account.

9.3 What is not claimed

Theorem 0 does not assert the following.

- *Uniqueness of the infinite-volume limit.* Our estimates see the volume only through the mismatch, a fixed amount per cell, between the upper and lower vacuum-energy densities of the stability bounds, which is overcome only at short distance (the transfer bound of Section 6 is of order d^2); the properties asserted in clauses (ii)–(iv) are shared by objects that differ from a given limit at fixed separation (e.g. the smeared action density plus an independent random constant of small variance), so uniqueness needs infrared input, such as a last-scale cluster expansion in infinite volume or a gap uniform in the volume.
- *Uniqueness of the vacuum.* It would follow from clustering, which we do not prove; nothing in Theorem 0 excludes several translation-invariant vectors.
- *The parameter of clause (v) as a physical quantity.* Under the conditions of (v-f) the parameter labels the torus theories injectively ((v-f)(4)), and at two values of it the two- and three-point functions at short distance are dilates of one another up to errors $O(d^{4-p})$ ((v-f)(3)); nothing more relates them, and the parameter is not calibrated by a physical quantity beyond this. Nor is any constant of clause (v) uniform in the size of the torus.
- *Clustering.* The volume-free bounds of clause (iii) do not decay with the distance between the supports, and the construction controls the integral at the last scales only through its total mass.
- *A mass gap.* The spectral statement obtained is that the Hamiltonian is non-negative and annihilates the vacuum vector; a gap is an infrared property of the infinite-volume theory, on which the construction gives no information.
- *Field operators.* The growth condition of [22] in the number of points is not asserted, and neither are field operators, Wightman functions or the spectral condition beyond the non-negativity of the Hamiltonian. A variant of the proof of the per-ball bound gives constants of factorial type in the number of points (see the Outlook); with it, the remaining obstruction to field operators is invariance under all rotations, which is clause (vi), under its hypothesis. One-point functions and coinciding points are not controlled.
- *Exact rotation invariance without the hypothesis of clause (vi).* Rotations outside the hypercubic group do not act on the tori, and on every torus of the family the continuum limit is not exactly rotation invariant at any scale (Proposition 8.19); on $\mathbb{R}^4$ exact rotation invariance is an infrared property of the limit; the hypothesis of clause (vi) is the infrared input our argument needs, and we do not derive it. At short distance, rotation invariance holds to all orders in the running coupling: up to errors of order d^2 and d at scale d for two and three points (Proposition 8.16), and of order $d^{1-\epsilon}$ for every number of points (Proposition 8.18).
- $SU(N)$, $N \geq 3$. Theorem 0 is stated for $N = 2$. Clauses (0)(a)–(c) and (i) are written for $SU(N)$, every $N \geq 2$, in the companion account, except two group-specific estimates of the stability bookkeeping, which are written in part (item (C15)); clause (0)(d) and clauses (ii)–(vi) need for $N \geq 3$ three further inputs, which we have not carried out (Remark 1.4).
- *Wilson loops and block observables.* Theorem 0 concerns the smeared action density only. Statements on Wilson loops of fixed physical size, on block-averaged observables or on the state on Balaban’s block algebra would need the comparison of the full measure with Balaban’s block laws at every depth, which we do not carry out here.

- *Non-Gaussianity.* Clause (iv) concerns the two-point function; nothing is asserted about connected functions with three or more arguments beyond the bounds of clause (iii).

9.4 Outlook

What follows is not part of Theorem 0 and is used nowhere in this paper.

The companion account contains a second proof of clause (iv) in every volume and in every infinite-volume limit, which does not use the volume-transfer theorem of Section 6; it is written, and its independent verification is in progress. It tilts the measure on the large torus by the exponential of the smeared action density, applies a chessboard estimate to that exponential, and concludes with a second-order remainder lemma and an entropy step; it uses the correlation theorem in a form in which the radius of the disc of sources may be as large as g^{-2} divided by a power of $\log g^{-2}$, obtained by following the dependence on that radius through its proof. A further argument, which combines reflection positivity of the bare measure on the large torus with a comparison argument of its own, gives the same conclusion as an implication from one further input, which is stated there and not proved. We present neither here.

A variant of the chessboard estimate of Section 3, with a grid independent of the number n of points and the test functions cut into pieces by multilinearity, gives volume-free constants of factorial type, at most $(n!)^c C^n$ with c and C independent of n ; this is proved in the companion account. With these bounds, and with the order of the seminorms in the regularity bound (4.14), which is linear in n , the growth condition of [22] reduces to a bookkeeping, and the remaining obstruction to Wightman field operators is invariance under all rotations, which is clause (vi), under its hypothesis.

For a small spatial torus $\mathbb{T}_\ell^3$ (of side $\ell = 2L^m$ in the units of Theorem 0, m as in (H5)), the companion account proves that the zero-temperature limit on $\mathbb{R} \times \mathbb{T}_\ell^3$ exists, that its Hamiltonian has purely discrete spectrum, a ground level of finite multiplicity and a gap above it, and that the two-point law of clause (iv) holds there for probe pairs separated in time; simplicity of the ground state is not asserted.

A finite-size criterion for a spectral gap is proved in the companion account. In each direction the lattice partition function on a torus is the trace of an integer power of a positive transfer operator; from this, a finite-size excess of the free energy obeys a quadratic inequality under doubling of the box. If the excess lies below a universal threshold in one box, uniformly in the cutoff, it decays doubly exponentially along the doubled boxes, and every larger box has a unique ground state with a gap above it. On $\mathbb{R}^4$ this gives a dichotomy: either the excess stays above the threshold at every scale, or the limit has a mass gap. The criterion does not decide between the two alternatives: its condition fails in every small box, and would have to be checked at large scales, which the construction does not reach.

An exact asymptotic law for the two-point function on $\mathbb{R}^4$ as $d \downarrow 0$, with the one-loop constant $\frac{11N^2}{12\pi^2}$ for the inverse square of the coupling and the constant $3(N^2 - 1)\pi^{-4}$ of the continuum kernel, follows as an implication from one further bookkeeping of the arguments above; it is written, and its independent verification is in progress. A bound on the large-field contributions uniform in the volume is proved in the companion account as an implication; it is not used here.

Finally, we expect the statements of Theorem 0 to extend to compact semisimple gauge groups; see Remark 1.4, which follows Theorem 0 in Section 1.

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